RECON — REVISION NOTES

– MCQ-focused · Orthobullets + OB/TO question explanations + TO lecture notes · 65 chapters (0 from TO notes) –

#	Chapter	Section index
1	Adult Hip Dysplasia 4	1 / 65
2	Adult Limb Deformity & Correction 14	2 / 65
3	Arthroplasty Preoperative Medical Optimization 22	3 / 65
4	High Tibial Osteotomy 26	4 / 65
5	Hip Anatomy 34	5 / 65
6	Hip Arthrodesis 40	6 / 65
7	Hip Biomechanics 43	7 / 65
8	Hip Hemiarthroplasty Periprosthetic Fracture 48	8 / 65
9	Hip Osteoarthritis 52	9 / 65
10	Hip Osteonecrosis 58	10 / 65
11	Hip Physical Exam - Adult 68	11 / 65
12	Hip Resurfacing 72	12 / 65
13	Idiopathic Transient Osteoporosis of the Hip (ITOH) 77	13 / 65
14	Knee Biomechanics 81	14 / 65
15	Knee Osteoarthritis 86	15 / 65
16	Ligaments of the Knee 94	16 / 65
17	Patellar Clunk Syndrome 100	17 / 65
18	Prosthetic Joint Infection 104	18 / 65
19	THA Acetabular Screw Fixation 120	19 / 65
20	THA Approaches 123	20 / 65
21	THA Aseptic Loosening 128	21 / 65
22	THA Dislocation 132	22 / 65
23	THA Iliopsoas Impingement 140	23 / 65
24	THA Implant Fixation 143	24 / 65
25	THA Leg Length Discrepancy 148	25 / 65
26	THA Other Complications 151	26 / 65
27	THA Periprosthetic Fracture 156	27 / 65
28	THA Postoperative Inpatient Management 166	28 / 65

#	Chapter	Section index
29	THA Prosthesis Design 169	29 / 65
30	THA Pseudotumor (Metal on Metal Reactions) 179	30 / 65
31	THA Rehabilitation 184	31 / 65
32	THA Revision 186	32 / 65
33	THA Sciatic Nerve Palsy 197	33 / 65
34	THA Stability Techniques 202	34 / 65
35	THA Templating 208	35 / 65
36	THA Trunnionosis 212	36 / 65
37	THA Vascular Injury & Bleeding 217	37 / 65
38	TKA Approaches 221	38 / 65
39	TKA Aseptic Loosening 227	39 / 65
40	TKA Axial Alignment 230	40 / 65
41	TKA Coronal Plane Balancing 234	41 / 65
42	TKA Extensor Mechanism Rupture 240	42 / 65
43	TKA Heterotopic Ossification 246	43 / 65
44	TKA in Patella Baja (Prior HTO) 248	44 / 65
45	TKA Instability 251	45 / 65
46	TKA Kinematic Alignment 257	46 / 65
47	TKA Metal Hypersensitivity 262	47 / 65
48	TKA Other Complications 264	48 / 65
49	TKA Patellar Maltracking 266	49 / 65
50	TKA Patellar Prosthesis Loosening 269	50 / 65
51	TKA Patellofemoral Alignment 271	51 / 65
52	TKA Periprosthetic Fracture 276	52 / 65
53	TKA Peroneal Nerve Palsy 284	53 / 65
54	TKA Polyethylene Wear & Manufacturing 287	54 / 65
55	TKA Postoperative Inpatient Management 292	55 / 65
56	TKA Postoperative Rehabilitation & Outpatient Management 296	56 / 65
57	TKA Prosthesis Design 298	57 / 65
58	TKA Revision 309	58 / 65
59	TKA Sagittal Plane Balancing 318	59 / 65
60	TKA Stiffness 325	60 / 65
61	TKA Templating 329	61 / 65
62	TKA Vascular Injury & Bleeding 332	62 / 65
63	TKA Wound Complications 335	63 / 65
64	Unicompartmental Knee Replacement 340	64 / 65

#	Chapter	Section index
65	Wear & Osteolysis Basic Science 351	65 / 65

Every chapter, same order: 1 Basics → 2 Mechanism → 3 Presentation → 4 Diagnosis & Classification → 5 Management → 6 Complications → X vs Y → How it's asked

Always included where sources have them: criteria (diagnostic, operative) · genes & lab markers · key radiology findings

Key: TO = Target Ortho fact (must-know) · ↑ increased · ↓ decreased · → leads to / then · ✗ Wrong options = MCQ distractors · ✓ correct answer · "TO — verify" = TO figure that differs from standard teaching

– – Active space – –

ADULT HIP DYSPLASIA

– Recon · 1 / 65 –

1 Basics

Epidemiology

Definition	abnormal hip development → **shallow acetabulum**, deficient **anterior + lateral** coverage
Incidence	US **3–5%** · **~10% of all THA** done for dysplasia · **48% of THA in patients <50 yr** are for dysplasia

Most common cause of THA <50 yr (excluding osteonecrosis) – **DDH**
✗ Wrong options: FAI · SCFE · Perthes/MED · femoral neck fracture

Demographics	**females > males** (2–4× relative risk)
Risk factors	**breech**, **female**, **primiparity**, **family history**

Pathophysiology

Mechanism	undercoverage → load concentrated on **superolateral acetabular rim** → ↑ contact stress → **degenerative labral tear**, cartilage delamination → **secondary OA**
Labrum	**hypertrophied**, intrasubstance tearing (labrum stabilises the hip → overloaded)
Arthritis timing	untreated → THA typically in **30s–40s**

Anatomic abnormalities

	Dysplastic hip
Acetabulum	**Shallow, lateralized, anteverted** · deficient **anteriorly + superiorly** (superolateral) · may be **retroverted** with superolateral bone loss (OB)
Femoral head	**Small**, aspherical
Femoral neck	**Short**, **↑ anteversion** (+5° to +16°, rises with subluxation) · **coxa valga** (↑ neck-shaft angle)
Greater trochanter	**Posteriorly placed**
Femoral canal	**Small, narrow, straight** · thick cortices
Contact area	**↓ head–acetabulum contact area**

"Most common abnormalities" → ✓ **excessive femoral anteversion + posterior greater trochanter**
✗ Wrong options: medialized acetabulum · large contact area · large head/long neck · ↓ neck-shaft angle

Femoral concern before THA → ✓ **excessive anteversion**
✗ Wrong options: osteopenia · varus · bowing · stove-pipe femur

Rotational deformity site	**diaphysis between lesser trochanter and isthmus** ✗ Wrong options: metaphyseal torsion
Normal values	acetabulum **anteversion 15°, abduction 45°** · femoral neck anteversion **15°** · neck-shaft angle **125°** · head centre at level of **greater trochanter**

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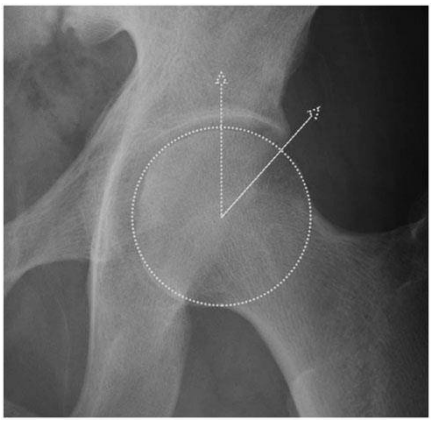
3 Presentation

Symptoms	insidious **groin/hip pain** · worse with hip flexion or ER in weight-bearing · **lateral hip pain + limp / Trendelenburg** (abductor fatigue)
Exam	↑ **internal rotation in flexion** (↑ femoral anteversion) · generalized laxity (**Beighton**) · ± FADIR pain (labral)
Anterior apprehension sign	lateral decubitus, hip extended, progressive **ER + adduction** → apprehension/pain
Prone external rotation test	anterior force on posterior greater trochanter

4 Diagnosis & Classification

Key radiology findings

Views	**AP pelvis**, lateral, **false-profile view**
Findings	femoral head **asphericity** · **coxa valga** · **narrow femoral canal**
CT	preoperative planning only (ordered by treating surgeon) · include **distal femur** if clinical anteversion · canal diameter **over-estimated on AP, under-estimated on lateral**



Lateral centre-edge angle of Wiberg

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Diagnostic criteria – radiographic parameters

Measure	View	Dysplastic	Borderline	Normal
LCEA (Wiberg) – vertical through head centre vs line to lateral acetabular edge	**AP**	**<20°**	**20–25°**	**25–39°**
Tönnis angle (acetabular roof / sourcil inclination)	**AP**	**>10°**	–	**<10°**
ACEA (Lequesne) – anterior coverage	**False-profile**	**<20°**	–	**25–40° · >40° = FAI**
FEAR index	AP	–	–	**<5° = stable hip, no treatment**
Extrusion index (uncovered head / head width)	AP	**>27%** suggests dysplasia (TO)	–	–

LCEA cut-off variants — Orthobullets **<20° dysplastic**, OB **<25° dysplastic** (LCEA or ACEA) · borderline/mild dysplasia **18–25°** (OB) · Wiberg (TO): **>25° normal, 20–25° uncertain, <20° pathological** · CE angle reliable only **>5 yr** (TO)
✗ Wrong options: <12 yr

Tönnis angle = acetabular index / acetabular roof angle — used in **hip dysplasia** (TO) · line teardrop-to-teardrop + line from medial to lateral edge of **sourcil** (weight-bearing subchondral condensation)
✗ Wrong options: genu varum · Blount · recurvatum

Paediatric acetabular index — Hilgenreiner's line + acetabular surface · **<25° after 6 months** (TO)

Other TO angles — **MCE** (head centre → medial sourcil) · **acetabular arc** = LCE + MCE

Acetabular version signs (TO)

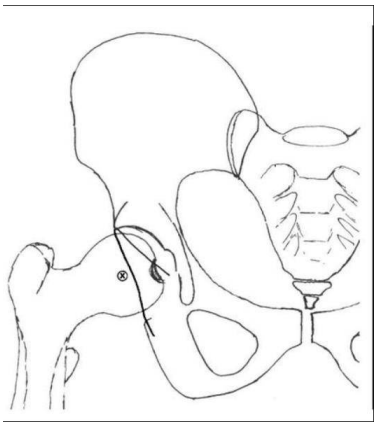
Normal — slight anteversion · **anterior wall projects medial** to posterior wall · posterior wall descends **through femoral head centre**

Retroversion — **anterior wall lateral to posterior wall** · signs: **crossover sign**, **posterior wall sign**, **ischial spine sign** (spine protrudes into true pelvis) · also narrow ischium, laterally protruding iliac wing

Crossover sign — anterior rim crosses posterior wall below superolateral margin → **anterior overcoverage OR whole-acetabulum retroversion OR posterior deficiency — any alone or combined** (TO)

Posterior wall sign — posterior wall runs **medial to head centre** → posterior deficiency

Posterior wall lateral to head centre — prominent posterior wall → **excessive anteversion**



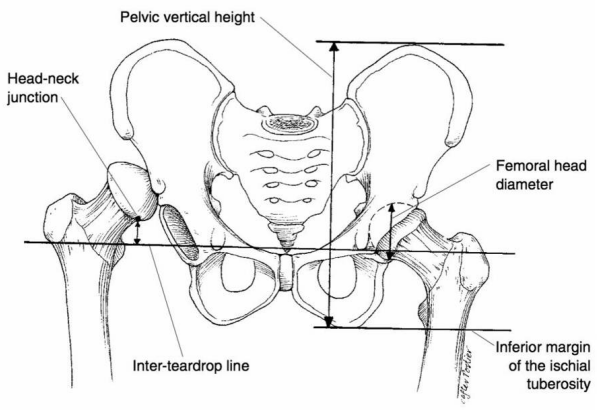
Posterior wall sign

Coxa profunda vs protrusio (TO)

	Coxa profunda	Protrusio acetabuli
Relation to **ilioischial line**	**Acetabular fossa** medial to line, head is **not**	**Femoral head** crosses medial to line
Meaning	Overcoverage **or normal variant (women)**	Excess depth, overcoverage, **pincer**

Crowe classification

Grade	Proximal displacement (% pelvic height)	Femoral head subluxation
I	**<10%**	**<50%** of head vertical diameter (head–neck junction from inter-teardrop line)
II	**10–15%**	**50–75%**
III	**15–20%**	**75–100%**
IV	**>20%**	**>100%** (high dislocation)



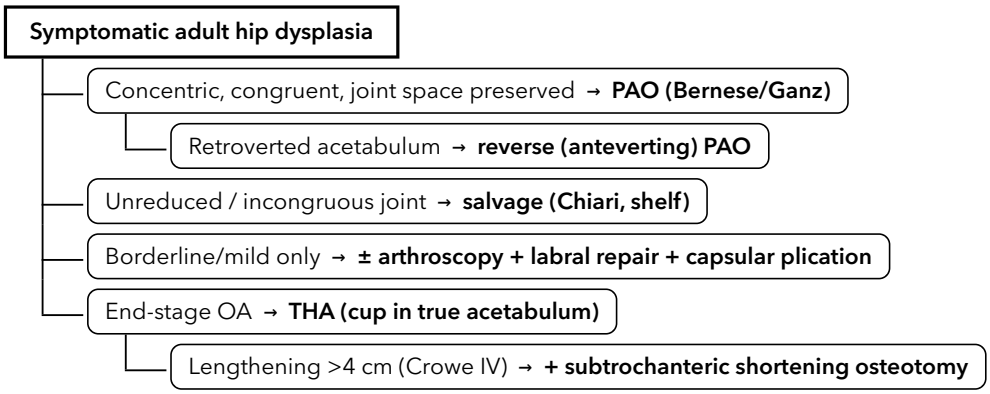
Crowe classification

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Hartofilakidis classification

Type	Head position	Acetabulum
A – Dysplasia	Within acetabulum despite subluxation	**Segmental superior wall deficiency** · inadequate depth
B – Low dislocation	**False acetabulum** superior to true	**Superior wall absent** · inadequate depth
C – High dislocation	Completely uncovered, migrated **superior + posterior**	**Complete deficiency** · excessive anteversion of true acetabulum

5 Management



Nonoperative

Supportive — weight loss, NSAIDs, activity modification, intra-articular injection · **limited role** (premature OA) · PT + steroid injection not for lasting relief in 20s

No remodelling — triradiate closes ~10 yr · at 18 yr **no pelvic growth left**
✗ Wrong options: "continued remodelling (observe")

Periacetabular osteotomy (PAO) indications

Indication — **symptomatic dysplasia**, adolescent/adult, **concentrically reduced hip + congruous joint space** · preserved ROM · **minimum 2 mm joint space** (OB) · best: **very low VCE 0–15°**, concentric through ROM

Typical stem — young woman, LCEA ~15°, ACEA ~14–18°, Tönnis 15°, labral tear, **no chondral lesion → ✓ Bernese PAO**
✗ Wrong options: surgical dislocation + osteoplasty (for FAI) · proximal femoral osteotomy (paediatric/abnormal NSA) · Salter · THA (needs advanced OA)

Intraop test — need **≥90° flexion + 15° IR** after correction, otherwise add femoral osteotomy (prevent FAI)

Advantages — **hyaline cartilage coverage** · **posterior column intact → WBAT** · large multidirectional corrections · preserves external rotators · delays THA

Outcome — **92% survivorship at 15 yr** avoiding THA · **>90% at 10–15 yr** · **better long-term survivorship than arthroscopy** (✓) · retroversion series: 10-yr survivorship **PAO 79% vs rim trimming 23%**

Secondary instability — after arthroscopic labral debridement, **not after PAO**

Technique	**modified Smith-Petersen** approach · osteotomies of **pubis (superior pubic root), ilium, ischium** · **avoids posterior column** (fracture → pelvic discontinuity → plate)
Later THA after PAO	↑ risk of **retroverted cup**
Reverse (anteverting) PAO	**acetabular retroversion**: ✓ **LCEA <20° + positive posterior wall sign** ✗ Wrong options: LCEA 21° + negative PWS · LCEA 23° + Tönnis 8 · LCEA 30°

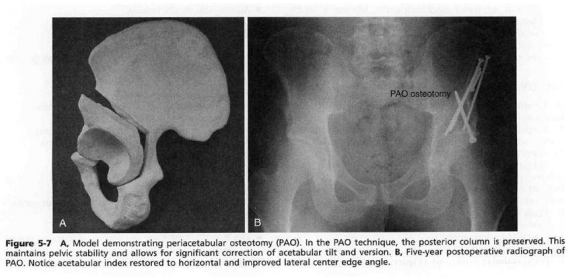


Figure 5-7 A, Model demonstrating periacetabular osteotomy (PAO). In the PAO technique, the posterior column is preserved. This maintains pelvic stability and allows for significant correction of acetabular tilt and version. B, Five-year postoperative radiograph of PAO. Notice acetabular index restored to horizontal and improved lateral center edge angle.

Bernese periacetabular osteotomy

Pelvic osteotomy descriptions

Osteotomy	Description	Patient
Bernese / Ganz PAO	**Multiple cuts: pubis, ilium, ischium** near acetabulum ✓	Skeletally mature
Salter	**Above acetabulum through ilium to sciatic notch** · opening wedge · hinges on **mobile symphysis / triradiate**	Immature
Pemberton	From **AIIS posteriorly** → ends at **ilioischial limb of triradiate**	Immature
Dega	Acetabular roof → triradiate · **incomplete** pericapsular cuts · **does not violate sciatic notch** · volume-reducing	**Neuromuscular** DDH
Chiari (salvage)	Cut **above acetabulum to sciatic notch**, **shift ilium laterally** · relies on **fibrocartilage metaplasia** · **shortens limb**	Incongruous joint
Staheli shelf (salvage)	**Extra-articular buttress** of bone over subluxed head · fibrocartilage cover	Incongruous joint

Salvage outcome	**84% survivorship at 17 yr** (advanced OA endpoint) · poor with advanced DDH, head asphericity

Hip arthroscopy failure in dysplasia

Role	controversial · **adjunct to PAO** (chondral, labral, cam lesions; abandon PAO if major chondral damage) · **contraindicated in moderate–severe dysplasia** · alone only for **borderline/mild** (with **labral repair + capsular plication**)
Prevalence	dysplasia in **~13%** of hip arthroscopy patients (mostly borderline/mild)
Outcome vs no dysplasia	✓ **greater failure rate** · **accelerated arthritis** (~$\frac{1}{3}$) · ↑ **femoral head subluxation/migration** · **reoperation ~½** · inferior function · **iatrogenic instability** ✗ Wrong options: decelerated arthritis · ↓ subluxation · ↓ reoperation · greater improvement

Missed DDH in FAI patients → poor outcomes after arthroscopic osteoplasty

– – Active space – –

Hip resurfacing

Indication	**Crowe I or II** · **cannot address LLD** · **10% revision at 6 yr** · higher revision/ complications than general population · posterior approach · complication: **femoral neck fracture**

THA in dysplasia – cup coverage, shortening osteotomy, nerve palsy

Indication	**end-stage OA** secondary to dysplasia ✓ ✗ Wrong options: hemiarthroplasty (acetabular arthritis) · redirectional osteotomies (best before arthritis)
Cup goal	**true acetabulum** (low + medial) · **small reamers 38–44 mm**, **small cup <50 mm** · no jumbo cup (reams away columns, medial wall blow-out)
Coverage rule	**uncemented cup OK if <30% uncovered** → ✓ accept **25% superolateral uncoverage** and proceed · insufficient coverage = **<60–70%** covered ✗ Wrong options: ↑ abduction to 60° (↓ coverage, ↑ wear) · breach medial wall · elevate hip centre · cement augments
Options for deficient coverage	downsize cup · **medialize** · elevate hip centre · **trabecular metal / cement augments** · superolateral bone graft · tantalum revision shells
TO lecture notes	**Hartofilakidis preferred for surgical planning** · Crowe by subluxation **< 50% · 50–75% · 75–100% · > 100%** · posterior approach ± trochanteric osteotomy · **true acetabulum found by inferior margin + acetabular notch (teardrop)** · **anterior wall thin** in DDH · cementless cup preferred, **cemented acceptable if poor bone stock** (TO) vs cemented → high loosening (OB) — read the options
Cup / head size (TO)	**cups < 40 mm**, heads **≤ 28 mm** · anatomical stems with anterior bow for narrow canals · centre checked with **Köhler + Shenton lines** under C-arm · cup **< 50 mm** (Orthobullets) — read the options

Hartofilakidis type	Key step (TO)
Dysplasia	Ream medially, deepen · protrusion **≤ 5 mm beyond quadrilateral plate** (Chandler)
Low dislocation	Recognise true acetabulum · **≥ 20% cup uncovered → graft / augment** · often cotyloplasty
High dislocation	Soft-tissue release + **femoral shortening** + **GT distalization** · anatomic cup

Cotyloplasty (TO)	**deliberate controlled perforation of medial wall** → medialization + centralised cup · femoral-head autograft chips packed beneath cup
Dunn & Hess shortening (TO)	osteotomy **just distal to lesser trochanter**, segment removed · **GT distalized** (attached laterally) → abductor tension
Pseudoacetabulum	in **ilium**, poor support · cup there → **loosening, malposition, ilium fracture** (avoid)
High hip centre	gains bony support but **↓ abductor tension, ↑ JRF, ↑ loosening, ↑ dislocation** · bone poorer higher in ilium
Cemented cup	PMMA poor in **shear** → high loosening (avoid)
Bearing	avoid **MoM** in young · **ceramic-on-PE** longest survivorship, fewest complications (OB)
Femur	**modular stems** correct rotational deformity · trochanteric osteotomy aids exposure in **Crowe III/IV**

Femoral shortening & sciatic nerve palsy

– – Active space – –

Risk — restoring true hip centre in high dislocation → **over-lengthening** → **sciatic palsy (peroneal division)** → **foot drop** (TO)
✗ Wrong options: hip centre too inferior (concern is too superior) · over-medialization (over-lateralization is the concern) · acetabular fracture (less likely)

Threshold — lengthening **>4 cm** → palsy (Orthobullets, OB) · **2–4 cm** associated with palsy (OB)

Muscle at risk — ✓ **EHL** (peroneal)
✗ Wrong options: quadriceps (femoral) · lateral gastrocnemius/tibialis posterior/FHL (tibial) · adductor magnus · semitendinosus

Prevention — ✓ **subtrochanteric femoral shortening osteotomy** · also corrects **angular/rotational deformity** · obviates GT osteotomy · proximal + **diaphyseal-engaging** fit, no extra hardware · alternatively **downsize components**
✗ Wrong options: upsizing

Not preventive — hip extension + knee flexion during exposure (helps exposure only) · **anterior approach** (no ↓ palsy; ↑ femoral nerve risk) · ↑ offset (does not change length) · navigation · ↑ medialization · anterior retractor placement · EMG · hamstring/quadriceps lengthening · GT osteotomy with progressive shortening (↑ nonunion, needs hardware)

Reconstruction Options Based on Severity of Hip Dysplasia*

Crowe Type[2]	Acetabulum	Femur	Approach
I	Uncemented component in true acetabular region with slight medialization	Cemented or uncemented stem based on patient age, bone quality, and bone geometry	Anterolateral or posterolateral based on surgeon preference
II or III	Uncemented component at or near true acetabular region, if necessary with autograft augmentation; or high hip center; or medialization	Cemented or uncemented stem based on patient age and bone geometry	Anterolateral, posterolateral, transtrochanteric, or subtrochanteric approach based on reconstructive technique and need for femoral shortening
IV	Extra-small uncemented acetabular component in true acetabular region	Greater trochanteric osteotomy with sequential proximal shortening and cemented "DDH stem," or shortening subtrochanteric osteotomy and uncemented stem	Transtrochanteric or posterior approach with shortening subtrochanteric osteotomy

Crowe classification and reconstruction options

6 Complications

Complication	Fact
Sciatic palsy	**10× ↑** incidence (**5–15%**) · lengthening **>4 cm** · peroneal division **80%** · only **35–40%** recover fully
GT osteotomy nonunion	**29%** · ↓ with **subtrochanteric osteotomy** + trochanteric advancement
Dislocation	**5–10%**, esp. **high hip centre**
Loosening	↑ with **high hip centre** cup · Crowe III/IV revision rates > non-dysplastic
Others	Periprosthetic femur fracture · infection · neurovascular injury

Long-term — higher THA revision rate in dysplastic hips · Crowe I–II short-term = primary OA

Sciatic palsy risk factors — **dysplasia**, revision, **female**, post-traumatic OA, **lengthening** · causes: posterior-approach trauma, stretch, haematoma, cement heat

– – Active space – –

X vs Y

	Adult hip dysplasia	FAI
LCEA	**<20°**	Normal/↑ (pincer)
ACEA	**<20°**	**>40°** = overcoverage
Labrum	**Hypertrophied**, rim overload	Crushed/sheared
Surgery	**PAO**	Arthroscopy / surgical dislocation + osteoplasty
Arthroscopy alone	**Fails → instability**	Indicated

	PAO	Salvage (Chiari / shelf)
Joint	**Concentric, congruous**	**Incongruous / unreduced**
Coverage	**Hyaline cartilage**	**Fibrocartilage** (metaplasia)

How it's asked

"Young woman, LCEA 16°, Tönnis 15°, labral tear, cartilage intact — best surgery?"	→ **Bernese PAO** ✗ Wrong options: surgical dislocation + osteoplasty · Salter · THA
"Which statement about osteotomies is true?"	→ **Bernese = multiple cuts in pubis, ilium, ischium** ✗ Wrong options: Salter from AIIS (that is Pemberton)
"Dysplasia, 18 yr, labral tear — true statement?"	→ **PAO long-term survivorship > arthroscopy** ✗ Wrong options: secondary instability after PAO · remodelling continues · Dega for retroversion
"Reverse PAO — imaging that led to decision?"	→ **LCEA 18° + positive posterior wall sign**
"Hip arthroscopy with dysplasia vs without?"	→ **greater failure rate**
"Most common femoral abnormality in DDH?"	→ **excessive anteversion + posterior greater trochanter**
"Femoral concern on THA templating in DDH?"	→ **excessive anteversion**
"Crowe IV THA — clinical scenario most likely?"	→ **over-lengthening the extremity** (TO)
"Crowe IV THA — nerve supply of which muscle at risk?"	→ **EHL** (TO)
"Foot drop after THA with large lengthening — what would have prevented it?"	→ **subtrochanteric femoral shortening osteotomy** ✗ Wrong options: anterior approach · ↑ offset · upsizing stem
"Template shows 25% superolateral uncoverage — next step?"	→ **accept and proceed as templated** (<30% OK)

"Crowe IV with pseudoacetabulum — preop planning?"	→ **subtrochanteric osteotomy to avoid foot drop** ✗ Wrong options: cup in pseudoacetabulum · jumbo cup · MoM · cemented cup
"Most common cause of THA <50 excluding osteonecrosis?"	→ **DDH**
"52-yr-old with dysplasia and secondary OA?"	→ **THA** ✗ Wrong options: PAO · hemiarthroplasty
"Positive crossover sign indicates?"	→ **anterior overcoverage, retroversion or posterior deficiency — any alone or combined** (TO)
"Medial acetabular fossa medial to ilioischial line, head not?"	→ **coxa profunda** (TO) ✗ Wrong options: protrusio
"CE angle of Wiberg — true?"	→ **<20° abnormal** (TO) ✗ Wrong options: only reliable <12 yr · Hilgenreiner–Perkin angle
"Tönnis angle used in?"	→ **hip dysplasia** (TO)
"Deliberate medial wall perforation to medialize the cup in DDH?"	→ **cotyloplasty** (TO)
"DDH classification preferred for surgical planning?"	→ **Hartofilakidis** (TO)

– – Active space – –

ADULT LIMB DEFORMITY & CORRECTION

– Recon · 2 / 65 –

1 Basics

Epidemiology & aetiology

Causes	childhood growth impairment, metabolic bone disease, severe OA, **trauma / malunion / nonunion**
Demographics	**no gender predilection** · young → trauma / childhood growth abnormality · upper limb rare (elbow, wrist malreduction)
Pathoanatomy	altered anatomic axis → **mechanical axis deviation (MAD)** → abnormal contact pressure + ligament strain → degeneration → worsening alignment
Childhood causes	**Blount's**, focal fibrocartilaginous dysplasia, proximal tibial physeal injury, **Cozen's phenomenon**, SED, femoral anteversion, **LLD**
Medical causes	renal osteodystrophy, **rickets** (vitamin D deficient, hypophosphataemic), OI, CP, neurofibromatosis
Site rule	adult **genu varum → mostly tibial** (CORA proximal tibia) · adult **genu valgum → mostly femoral** (CORA distal femur)

Anatomy

Proximal femur	neck-shaft angle **130 ± 7°** · anteversion **10 ± 7°** · **GT tip level with head centre** (start of anatomic femoral axis)
Femoral anatomic vs mechanical axis	differ by **5–7°** · tibia: **anatomic = mechanical** if no deformity
Tibial tubercle	**anterolateral** · triangular shaft
sMCL	just posterior to medial epicondyle → **6 cm** below joint line · LCL – posterior to lateral epicondyle → fibular head
JLCA >2°	**soft-tissue (ligamentous) laxity**

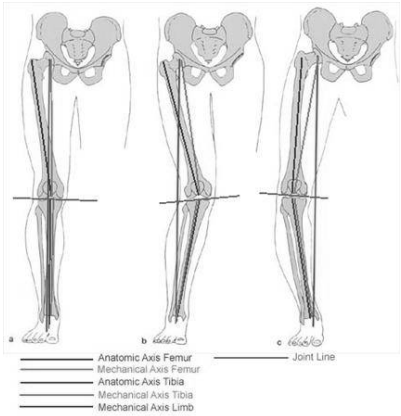
3 Presentation

Pain by deformity	**varus → medial knee pain** · **valgus → lateral knee pain**
Exam	coronal/sagittal deformity · **femoral anteversion → IR >70°** · residual ↑ external tibial torsion · LLD · hip/knee/ankle ROM, laxity
Intermalleolar distance >8 cm	**valgus** (OB)

4 Diagnosis & Classification

Key radiology findings

Views	**weight-bearing full-length AP + lateral** · **patellae facing directly anterior** (rotation) · tube **10 ft** away at knee level · plus WB views of each segment
CT	**rotational malalignment** · axial cuts for femoral version
Mechanical axis (limb)	**centre of femoral head → centre of ankle**
Anatomic axes	**bisect medullary canal** (femur from GT tip, exits just medial to distal femoral midpoint)
MAD	**lateral to knee centre = valgus** · **medial = varus**
LLD	**top of femoral head → centre of ankle**, compare sides · femur/tibia measured separately



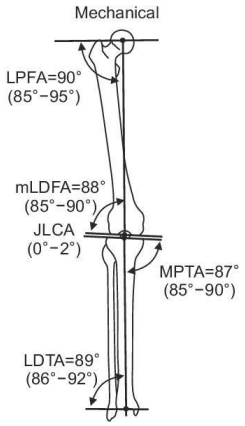
Mechanical vs anatomic axes

Normal joint orientation angles

Angle	Axis	Normal
aMPFA	Anatomic	**84°** (80–89)
aLDFA	Anatomic	**81°** (79–83)
mLPFA	Mechanical	**90°** (85–90)
mLDFA	Mechanical	**87°** (85–90)
MPTA	Both	**87°** (85–90)
LDTA	Both	**89°** (86–92)
JLCA	–	**0–2°** (femoral/tibial joint lines parallel)
PDFA (sagittal)	–	**79–83°**
PPTA (sagittal)	–	**77–84°**

Notation lowercase **a** = anatomic · **m** = mechanical

– – Active space – –



Mechanical joint orientation angles

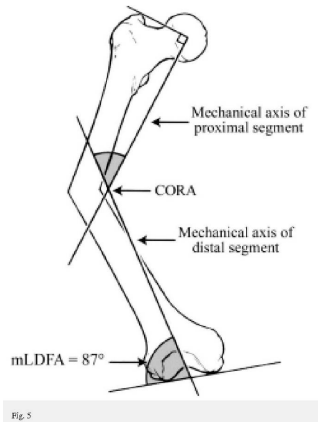
Deformity analysis – CORA

CORA	intersection of **proximal + distal mechanical axes** when deformity is purely angular
Angulation + translation	CORA **moves** from the axis intersection
Multi-level deformity	unrecognized intermediate segment → CORA may lie **outside bone**
Osteotomy not at CORA	induces **translation** in addition to angulation
Proximity rule	deformity **closer to joint → greater effect** on that joint's angulation
Correction order	**angulation → translation → length → rotation**
Mechanical vs anatomic axis (TO)	mechanical axis **always straight** · anatomic axis may be **curvilinear** (femur, sagittal) · **tibial mechanical axis lies lateral to anatomic axis** (TO)
Normal MAD (TO)	**1–15 mm** medial · **> 15 mm** = malalignment
Malalignment test (Paley) (TO)	MAD → **mLDFA** (abnormal = femoral) → **mMPTA** (abnormal = tibial) → **JLCA > 2°** (laxity / subluxation) → condylar non-parallelism (condylar deficiency)
Functional CORA (TO)	any point on the **bisector** of the deformity angle

Paley osteotomy rule (TO)	Result
Rule 1 – CORA = correction axis = osteotomy level	**Pure angulation**, no translation
Rule 2 – CORA = correction axis, osteotomy elsewhere	Angulation + **relative translation** (bone realigns)
Rule 3 – correction axis away from CORA	Angulation + **iatrogenic translation**

Open vs closed wedge (TO)	open: simple, **lengthens**, needs **graft**, unstable, NV stretch · closed: **stable, no graft, safer for nerves**, **shortens** · dome: always some translation
Oblique-plane deformity (TO)	magnitude = **$\sqrt{(AP^2 + lateral^2)}$**
Scanogram set-up (TO)	hip–knee–ankle on one film · **patella forward** regardless of foot · **level pelvis with shoe raise** in LLD

– – Active space – –



CORA

Diagnostic criteria – varus vs valgus by angle

	Varus (femoral source)	Varus (tibial source)	Valgus (femoral source)
Threshold	**mLDFA >90°** or **aLDFA >83°**	**MPTA <85°**	**mLDFA <85°** or **aLDFA <79°** · **MPTA >90°** also valgus
MAD	Medial	Medial	**Lateral**

OB trap values LDFA **88° = normal** · LDFA **91° = slight varus** · MAD **5–7° from femoral anatomic axis = normal**

5 Management

Adult limb deformity

- Mild / minimal deficit / unfit → **bracing (KAFO), poorly tolerated**
- Varus, CORA tibia (MPTA <85°) → **valgus HTO (MOWHTO or LCWHTO)**
- Varus, CORA femur (mLDFA >90°) → **valgus DFO (lateral closing wedge)**
- Valgus >12°, CORA femur → **varus DFO (lateral opening / medial closing)**
- LLD >2 cm → **gradual lengthening ~1 mm/day**
- Multicompartment OA → **TKA ± extra-articular osteotomy**

Nonoperative

Indications subtle deformity within normal parameters · minimal functional deficit · unfit for surgery

KAFO AFO + metal uprights + mechanical knee joint + two thigh bands · bracing **poorly tolerated** · little data on progression

– – Active space – –

Operative criteria

Procedure	Indication
Derotational femoral osteotomy	No absolute · **excess femoral anteversion with <10° ER** · patellar maltracking/ instability · gait (difficulty running)
Valgus-producing HTO	Varus MAD + **MPTA <85°**, CORA in tibia · **no arthritis in other 2 compartments** · young, not TKA candidate
Valgus-producing DFO	Varus MAD + **mLDFA >90° / aLDFA >83°**, CORA in femur · **joint-line obliquity after HTO** in combined deformity
Varus-producing DFO	Valgus MAD + **mLDFA <85° / aLDFA <79°** · valgus **>12°** · **no medial compartment arthritis** · **age <50** · isolated **lateral** OA · relative: genu valgum with ↑ Q angle, MCL stress
Gradual lengthening	LLD **>2 cm** (not absolute)
TKA	Deformity with **multicompartment** arthritis

Derotational femoral osteotomy

Sites proximal (intertrochanteric/subtrochanteric, blade/locking plate — **also corrects flexion/extension + varus/valgus**) · diaphyseal (IM nail) · distal supracondylar (medial subvastus, locking plate)

Diaphyseal IM technique derotate to **45° ER + 45° IR** with patella in line/slightly ER relative to ASIS

Outcome **all forms improve outcomes** · ideal correction debated · may alter coronal alignment

Complications undercorrection · overcorrection → retroversion + **out-toeing** (rare)

Valgus-producing tibial osteotomy (HTO)

	Medial opening wedge	Lateral closing wedge
Approach	Anteromedial	Anterolateral
Steps	Biplanar guide · hinge pin medial → lateral just below lateral plateau · protect patellar tendon · pack wedge	**Resect anterior fibular head** · cut proximal to tuberosity · **medial cortex = hinge** · staples/plate
Slope	**↑ tibial slope**	**↓ tibial slope**
Correction	May **lose correction** over time	–

HTO outcome **90–95% excellent at 5 yr → 50–70% at 10 yr · 15–25% need TKA within 10 yr** · worse: **smokers, age >60**, other-compartment OA, obesity, prior failed arthroscopy, insufficient correction

Distal femoral osteotomy – indications & outcomes

Valgus-producing DFO **lateral closing wedge** · K-wires aimed at far medial cortex just above MFC · **hinge through medial cortex without perforating** → intact medial cortex = ↑ axial + rotational stability

Varus-producing DFO	✓ **lateral opening wedge** (leave **1–2 cm medial hinge**, spacer + lateral plate) **or medial closing wedge** (metaphyseal–diaphyseal junction, medial locking plate) ✗ Wrong options: lateral closing / medial opening (those make valgus)
Candidate for lateral opening wedge DFO	✓ **young (35-yr) runner, isolated lateral joint-line pain, mechanical axis through fibular head** ✗ Wrong options: PF/anterior pain · tricompartmental OA → TKA · prior MPFL/TTT anterior pain · varus/medial pain

37-yr-old, 15° valgus, early lateral OA, failed NSAIDs + unloader brace → ✓ lateral opening wedge DFO
✗ Wrong options: lateral closing DFO + MOWHTO (varus) · HTO (varus) · TKA (too young)

Survival	✓ **15-yr survival ~50%** (failure **48.5%**) · high mid/long-term deterioration + conversion to TKA
Better outcome	**higher preoperative function** ✗ Wrong options: lower function
TKA after varus DFO	**conventional components** if bone stock adequate ✗ Wrong options: stemmed/constrained in most



Medial opening wedge HTO

Gradual limb lengthening

Distraction osteogenesis	**latency 5–7 days**, then **~1 mm/day**
Devices	external fixator (uniplanar, circular/hexapod) · **lengthening IM nail** ± ex-fix (for angular correction)
Outcome	LLD to **<1 cm** · can correct small diaphyseal/metaphyseal angulation
Complications	ex-fix: **pin-site infection**, scarring, cosmesis · IM: **poor regenerate**, joint contracture/dislocation, nail failure

TKA with deformity

Goal	neutral mechanical alignment via distal femoral + proximal tibial cuts · **valgus cut angle 4–7°** (IM guide on AAF) · tibial IM = EM guides (AAT = MAT)
Extra-articular deformity	locate CORA · **CORA further from knee → easier** intra-articular correction · **severe or close-to-joint deformity → extra-articular osteotomy + TKA** · osteotomy better tolerated on **tibia** (equal effect on flexion/extension gaps)
Constraint	↑ constraint / **stems** if collaterals compromised (laxity, prior osteotomy)

– – Active space – –

Outcome consistent improvement · results **less reliable after prior extra-articular osteotomy**

6 Complications

Complication	Fact
Under/ overcorrection	Unfavourable compartment loading · **opening wedge** may **lose correction**
Nonunion	**~5%** · risk: **smoking, age >60, obesity, far-cortex hinge fracture**
Patella baja	Proximal tibial osteotomy · **supra-tubercle > infra-tubercle** · pain, harder future TKA
Tendon injury	**Quadriceps** → femoral osteotomy · **patellar tendon** → tibial osteotomy
Infection	**1–4%** · ↑ nonunion · risk: smoking, inpatient admission
DFO-specific	Lateral opening: quadriceps tendon injury, **intercondylar fracture (too distal)**, lateral hinge fracture (↓ stability) · closing: medial hinge fracture

X vs Y

	Varus-producing DFO	Valgus-producing DFO
Treats	**Genu valgum**, lateral OA	Genu varum with femoral CORA
Wedges	**Lateral opening** or **medial closing**	**Lateral closing** (medial hinge)
Criteria	mLDFA **<85°**, valgus >12°, age <50	mLDFA **>90°**

	Genu varum (adult)	Genu valgum (adult)
Usual CORA	**Proximal tibia**	**Distal femur**
Pain	Medial	Lateral
Osteotomy	HTO	DFO

How it's asked

"Young man, 15° valgus, early lateral OA, failed brace – best surgery?"	→ **lateral opening wedge DFO** ✗ Wrong options: HTO · TKA
"Varus-producing DFO – true?"	→ **15-yr survival ~50%** ✗ Wrong options: lateral closing/ medial opening · age >50 medial OA · needs constrained TKA later
"Lateral opening wedge DFO – which patient?"	→ **35-yr-old runner, isolated lateral pain, axis through fibular head**
"16-yr-old, intermalleolar distance 10 cm, lateral opening DFO planned – alignment?"	→ **MAD lateral to knee centre** ✗ Wrong options: LDFA 88° normal · LDFA 91° varus
"Normal anatomic–mechanical femoral axis difference?"	→ **5–7°**
"Valgus HTO indication?"	→ **MPTA <85°, CORA in tibia, other compartments spared**

"Order of correcting deformity?"	→ **angulation, translation, length, rotation**
"Osteotomy away from CORA causes?"	→ **translation**
"Opening wedge HTO changes tibial slope?"	→ **↑ slope** (closing → ↓)
"Distraction osteogenesis protocol?"	→ **5–7 day latency, 1 mm/day**
"Osteotomy at CORA with correction axis through CORA?"	→ **pure angulation (Paley rule 1)** (TO)

– – Active space – –

– – Active space – –

ARTHROPLASTY PREOPERATIVE MEDICAL OPTIMIZATION

– Recon · 3 / 65 –

1 Basics

Overview

Five domains	**medical optimization** · **preoperative counselling** · **VTE prophylaxis** · **anaesthesia plan** · **pain management plan**
Goal	identify modifiable **risk factors** for postoperative complications

Genes & lab markers

HbA1c	screen hyperglycaemic patients · **HbA1c >6.7 → ↑ wound complications** · target roughly **<8** (varies by surgeon/hospital)
Nicotine testing	preoperative, to confirm cessation
Toxicology screen	failed screen → ↑ mortality, readmission, reoperation

4 Diagnosis & Classification

Cardiac & vascular preoperative evaluation

Noninvasive cardiac testing	**≥3 risk factors in a sedentary patient**: CAD/MI · heart failure · prior CVA · diabetes mellitus or chronic renal disease
No stress test	if patient can do **moderate activity (climb a flight of stairs)**
Typical stem	✓ **67-yr-old for TKA with MI + CVA + DM** ✗ Wrong options: spine patient with hypercholesterolaemia/HTN/DVT · healthy drinker · hip fracture with CRF/cancer/obesity · 81-yr-old who climbs stairs
PVD + ↓ pulses before TKA	next step ✓ **ankle-brachial index** of affected limb → vascular surgeon decides revascularization vs clearance ✗ Wrong options: angiography first · venous Dopplers · MRI popliteal fossa · radiographs for calcified plaque

5 Management

Medical optimization – risk factors

Risk factor	Threshold / action
Diabetes	**HbA1c >6.7** ↑ wound complications · target ~**<8**
Obesity	**BMI >40** used in studies (continuous risk, no clear cut-off) · **>5% weight loss** may ↓ risk · bariatric surgery may help · ↑ AKI, CV, wound, infection
Cardiac stents	Delay elective surgery if DAPT stopped within **30 days of BMS** / **12 months of DES**
Clopidogrel	Discuss with cardiologist · **stop 7 days preop** → ↓ bleeding/transfusion, no ↑ CV events · restart ASAP
Transfusion	Perioperative transfusion → ↑ complications
Dialysis	**10–20× ↑ complications** · ESRD may do better with **kidney transplant before arthroplasty** (weigh dialysis septicaemia vs immunosuppression)
MRSA	Screen + decolonize (**mupirocin** nasal / **chlorhexidine** wipes) · universal decolonization may be cost-effective
Smoking	↑ complications + infection · formal cessation programme · **6 weeks cessation** · nicotine testing
Illicit drugs	**5× ↑ mortality** · ↑ infectious + non-infectious complications
Inflammatory arthritis	**Hold biologics** before surgery

Smoking cessation

Minimum cessation before elective THA — **6 weeks** (Orthobullets) · TO (FNB 2024) asked this with options 3/4/6/8 weeks — answer not provided by the source

Preoperative counselling

Expectations — satisfaction driven more by **expectations** than function · patients **over-optimistic** · set expectations for pain, function, complications

Social support — affects **LOS, readmission, non-home discharge** · psychological distress worsens postoperative pain · absence of anxiety → lower pain, better function

VTE prophylaxis

Body	Recommendation
ACCP	**Mechanical compression + one agent**: warfarin, LMWH (enoxaparin), **aspirin**, factor Xa inhibitor (apixaban, rivaroxaban), fondaparinux, dabigatran
AAOS	**Pharmacologic, mechanical, or both** if no increased VTE risk

Aspirin — favoured: oral, compliance, cost, **low bleeding**

Anaesthesia

Neuraxial (spinal) — ↓ stress vs GA → ↓ surgical time, **infection**, CV events, **transfusion**, LOS

– – Active space – –

Multimodal pain management

Preoperative orals	**NSAIDs (COX-2)** + **pregabalin** → ↓ opioid use, ↓ pain, ↑ ROM, **no ↑ bleeding**
Postoperative	continue above + judicious opioids · **opioid adverse events up to 50%** of post-arthroplasty complications
Periarticular injection	local anaesthetic + opioid + NSAID · TKA: **= femoral nerve block**, cheaper/safer/easier · THA: ↓ opioid, better pain + function
Not effective	✓ **intra-articular pain catheter** ✗ Wrong options: acetaminophen · opioids · gabapentin · NSAIDs (all valid components)

Block	Joint	Note
Lumbar plexus / psoas, femoral, sciatic	THA	Good analgesia, may affect muscle function
Femoral nerve	TKA	**Quadriceps weakness**
Adductor canal	TKA	**Similar pain control to femoral block, no quad weakness**
Sciatic	TKA	Posterior knee pain · significant motor effect

X vs Y

	Bare-metal stent	Drug-eluting stent
Delay elective arthroplasty if DAPT stopped within	**30 days**	**12 months**

	Femoral nerve block	Adductor canal block
Analgesia after TKA	Good	**Similar**
Quadriceps	**Weak**	**Spared**

How it's asked

"Which patient needs preoperative noninvasive cardiac testing?"	→ **sedentary, ≥3 of CAD/MI, CHF, CVA, DM/CKD** ✗ Wrong options: able to climb stairs
"PVD with ↓ pulses before TKA – next evaluation?"	→ **ankle-brachial index** ✗ Wrong options: angiography · venous Doppler
"Not an effective part of a multimodal regimen?"	→ **intra-articular pain catheter**
"Smoking cessation before elective THA?"	→ **6 weeks** (Orthobullets; TO answer not given)
"HbA1c associated with wound complications?"	→ **>6.7**
"Delay elective surgery after DES?"	→ **12 months** (BMS **30 days**)
"Clopidogrel before TKA?"	→ **stop 7 days preop**

"Dialysis patient for THA — complication risk?"	→ **10–20× ↑**
"TKA block sparing the quadriceps?"	→ **adductor canal**

– – Active space – –

– – Active space – –

HIGH TIBIAL OSTEOTOMY

– Recon · 4 / 65 –

1 Basics

Overview

Purpose	correct angular deformity → unload one compartment → prevent/slow **unicompartmental OA**
Commonest indication	primary/secondary **medial knee arthrosis** with **varus** · isolated lateral OA much less common
History (TO)	introduced **Jackson & Waugh (1961)** · popularised **Coventry (1965)** – lateral closing wedge · medial opening wedge = **Hernigou** · **medial opening wedge = gold standard for joint preservation** (TO)
Combined with	**cartilage restoration** procedures (better mechanical environment) · ligament reconstruction (ACL/PCL/PLC) when malaligned

Anatomy – mechanical axis

Mechanical axis	centre of femoral head → centre of ankle · normally passes **just medial to medial tibial spine**

Fujisawa point (TO) – **62% of tibial plateau width measured from the medial edge** · slightly **lateral to lateral tibial spine** · = mechanical axis **3–5° valgus** · correction angle = lines from this point to femoral head + ankle centres · TO technique text also cites **62.5%**
✗ Wrong options: 62% from lateral · 67%



Fujisawa point

Normal tibial slope (OB)

Posterior tibial slope	**9–11° medial**, **6–8° lateral**
Varus types – TO notes (HTO lecture)	**double varus** = medial OA + **partial PLC laxity** (lateral opening **5–15 mm**) · **triple varus** = **PLC completely deficient** (lateral opening **> 15 mm**) → staged procedures · ACL-based definition above (TO MCQ) – read the options
High tibial slope (TO)	slope **> 15°** with ACL strain / recurrent ACL injury → **slope-correcting HTO + ACL reconstruction**

3 Presentation

Symptoms	medial (varus) or lateral (valgus) knee pain · localized
Exam	malalignment · **varus thrust** (LCL laxity, lateral opening in stance — predicts medial OA progression)

4 Diagnosis & Classification

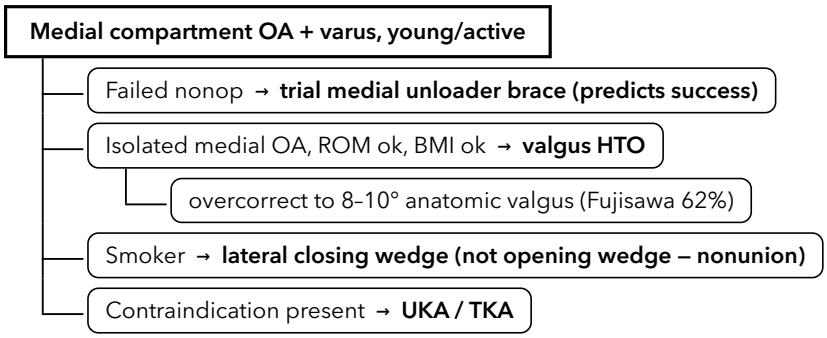
Key radiology findings

Views	**standing hip-to-ankle alignment films** → mechanical axis · plus knee AP/ lateral
Stress radiographs	narrow **lateral** compartment cartilage space → contraindicates valgus HTO
Unloader brace test	**medial unloader brace** = **therapeutic + diagnostic** · benefit → likely benefit from valgus HTO ✗ Wrong options: steroid injection · NSAIDs · patellar/strap brace

Varus types (TO)

Type	Components	HTO
Primary varus	Bony varus, **no ligament instability**	**Best indication**
Double varus	Bony varus + **ACL** instability	Combined/staged ligament + HTO
Triple varus	Varus + **ACL + PCL** insufficiency + **PLC failure** · **varus thrust**	Combined/staged

5 Management



Indications & patient selection

Ideal candidate	**young, active <50 yr** (arthroplasty would wear out) · **good vascular status** · **non-obese** · pain/disability · **one compartment only** · compliant
Overcorrection	✓ **overcorrect into valgus**: anatomic axis **8–10° valgus** (Orthobullets, OB) · mechanical **3–5°** valgus (TO) · least recurrence
Return to impact sport	**HTO allows faster return to sport/impact work** than UKA (activity levels equal) · running discouraged after UKA (PE wear, loosening)

– – Active space – –

Typical stems 30–48-yr-old labourer/athlete, isolated medial OA, varus 5–9°, failed nonop → ✓ **medial opening wedge HTO**
✗ Wrong options: UKA · TKA · arthroscopic debridement/microfracture · OCA · meniscal transplant · lateral opening DFO (for valgus) · unispacer

HTO vs UKA with ACL tear (TO) – 45-yr-old, ACL torn, lateral meniscus intact, grade 4 medial OA, 10° passively correctable varus → ✓ **HTO** · ACL deficiency **contraindicates UKA**, only relative for HTO (reconstruct simultaneously) · UKA preferred if varus <10°

HTO vs UKA with thrust + lateral meniscal tear (TO) – 48-yr-old marathoner, **varus thrust**, 8° varus, functional ACL, stable lateral meniscal tear, no lateral chondral loss → ✓ **medial opening wedge HTO**
✗ Wrong options: fixed/mobile UKA · TKA

Finding (TO)	UKA	HTO
Varus thrust	Relative/strong contraindication (loosening)	**Strong indication** – corrects cause
Lateral meniscal tear	**Absolute contraindication**	Relative (depends on extent + cartilage)
Goal	Resurfacing → pain relief	**Axis realignment + stability**

Note – varus thrust conflict Orthobullets/OB: **varus thrust = contraindication** to HTO · TO (2025 case): thrust **favours HTO over UKA** – read the stem (OB: thrust + lateral laxity → ✓ "contraindication")

Contraindications

Contraindication	Orthobullets / OB	TO
Inflammatory arthritis	✓ (psoriatic – high failure)	✓
BMI	**>35**	**>30**
Flexion contracture	**>15°** (10° OK)	**>10°** (extension lag)
Knee flexion	**<90°**	**<90°**
Correction needed	**>20°**	–
Varus deformity	**>10°** (valgus HTO)	**>15°** · >10° relative only
Lateral tibial subluxation	**>1 cm**	**>1 cm** (✓ 2 cm subluxation)
Medial bone loss	**>2–3 mm**	Medial compartment bone loss
Lateral compartment	Narrow on stress films · **loss of lateral meniscus** · lateral OA	Lateral narrowing (bi/tricompartmental)
PF arthritis	✓	✓
Ligament instability / varus thrust	✓	**Relative** – (✗ Wrong options: cruciate insufficiency alone)
Smoking	✓ (OB) · esp. opening wedge (nonunion)	High smoker
Age	Best <50 · **>60** worse	**Age >60 relative**
Others (OB)	Nonconcordant pain (PF pain + medial OA) · unwilling to accept cosmetic angular change	–

Ideal candidate (TO notes)	age **< 60**, active · **BMI < 30 (< 90 kg)** · **Ahlbäck grade I–II** · intact ACL + lateral compartment · flexion **> 100°**
Extra contraindications (TO notes)	**concave medial plateau ("teeter-totter" effect — Kettelkamp)** · bone loss **> 15 × 15 mm** · lateral subluxation **> 10 mm** · uncorrected slope **> 10°** · symptomatic PFJ OA · flexion **< 90°** or contracture **> 10°** · RA · nicotine · obesity
Not contraindications	**ACL tear** (TO) · **varus 10°** (TO) · **flexion ROM 0–100°** (TO) · chondral lesion size · BMI 22–24 · age 38–42 · 10° flexion contracture · postmenopausal

Varus-producing tibial osteotomy

Use	valgus knee + lateral OA · **less common than DFO** (creates **oblique joint line**) · deformity must be **<12°** · contraindicated: medial OA, loss of medial meniscus · **DFO better if lateral femoral condyle hypoplasia**

Opening vs closing wedge

	Medial opening wedge	**Lateral closing wedge**
Approach	Anteromedial · pes + sMCL release	Anterolateral · **fibular osteotomy / PTFJ disruption**
Tibial slope	**↑ slope** (anterior plate) + anterior tibial translation	**↓ or no change** + posterior translation
Patellar height	**↓ → patella baja**	**↑ → patella alta** (TO)
Stability / WB	Less stable · **longer NWB** · may lose correction	**More stable · earlier WB · faster rehab**
Graft	Defect **≥10 mm → graft**	**No graft**
Peroneal nerve	**Minimal risk** · avoids PTFJ	**↑ risk** (fibular osteotomy)
Nonunion / plate failure	**↑** · avoid in **smokers**	**↓** · choose in smokers, diabetics, steroids
Limb length	Lengthens	Shortens
Later TKA	Preserves tibial shape	Harder (stem impingement, lateral bone loss)
Favoured with (TO)	**PCL injury**, patella alta, MCL laxity, LLD	**Patella baja**, ACL reconstruction, nonunion risk

Exclusive to closing wedge	✓ **proximal tibiofibular joint disruption** ✗ Wrong options: compartment syndrome · plate breakage · neurologic injury (seen in both)

Smoker → favours **lateral closing wedge**
✗ Wrong options: age · BMI · meniscectomy history

Lateral closing vs medial opening	✓ **shorter time to weight bearing** ✗ Wrong options: ↑ baja · ↑ popliteal injury · less stable · ↓ peroneal risk
Outcomes	opening vs closing: both appropriate ✗ Wrong options: "opening superior"

– – Active space – –

Planning & technique (TO)

Target	weight-bearing line at **62.5% of tibial width (Dugdale)** · hip–knee–ankle scanogram
Wedge size	base (mm) = **tibial width × 0.02 × correction (°) → 1 mm ≈ 1°** for a **57 mm** tibia
Slope	keep **< 10°** · opening wedge: **anterior gap ≈ ½ posterior gap** · patella baja common post-op
Technique (opening wedge)	**5 cm** vertical incision between tuberosity + posteromedial border · expose MCL, detach pes · cut **3.5–4 cm below joint line** aimed at **fibular head** · **lateral hinge 7–10 mm** · **TomoFix / Puddu** plate · iliac crest graft if opening **≥ 10 mm**
Rehab (TO)	early ROM + isometrics → WB at **6 wk** → full WB at **3 mo**

Proximal fibular osteotomy – PFO (TO)

Origin	**Xia et al. 2014 (China)**
Rationale — non-uniform settlement	medial tibia **cancellous → settles early** · lateral **fibula = rigid cortical support** → varus progression · removing fibular support → even plateau loading → pain relief
Technique	remove **1–2 cm** fibular segment **7–8 cm below fibular head** · **posterior approach between peroneus and soleus** · keep incision **posterior to coronal plane** → protect **common peroneal nerve**

	PFO	**HTO**
Technique	Simple, quick, cheap	Demanding
Implant / graft	**None**	Plate ± graft
Recovery	Early	Delayed WB
Patient	**Elderly / high-risk, mild varus OA**	Young, active
Correction	**Mild, 5–10°**	**Up to 20°**
Risks	**Variable results, CPN injury**	Nonunion, hinge fracture, CPN

Other osteotomies (TO)

Osteotomy	**Feature**
Focal dome	Concavity proximal · centre at **CORA** · **least translation**, minimal shortening · for **large corrections (≥20°, 18–20 mm)** e.g. Blount, trauma · cut below tubercle → **no change in slope or patellar height**
Hemicallotasis	Gradual distraction, ex-fix · below tubercle · minimal change in slope/height
Chevron	Inverted-V · no bone loss/graft · technically complex



Lateral closing wedge HTO

Tibial slope & ligament instability

Rule **↑ slope → anterior tibial translation** (stress ACL) · **↓ slope → protects ACL**

Deficiency	Slope change in HTO
ACL-deficient	**Decrease** slope
PCL / PLC-deficient	**Increase** slope ✓ (✗ Wrong options: decrease · neutral)

ACL + varus + medial OA, young athlete (pain + instability) → ✓ HTO + ACL reconstruction at the same time
✗ Wrong options: either alone · vibration rehab

Recurrent PCL + PLC (dial >10° asymmetry at 30° + 90°) + varus, 1A Lachman → ✓ PCL + PLC reconstruction + HTO increasing slope
✗ Wrong options: ACL reconstruction · PLC alone

Native slope 14° ignored during HTO → ✓ positive Lachman (anterior translation persists)
✗ Wrong options: medial pseudolaxity · quadriceps active test/sag (PCL) · varus thrust

Decrease slope if slope **>13°** + anterior translation **≥10 mm** vs contralateral · **1 mm anterior closure ≈ 2° slope decrease**

– – Active space – –

6 Complications

Complication	Fact
Recurrence of varus	✓ **Most common** (TO) · **60% failure at 3 yr** if undercorrected or overweight · prevented by overcorrection
Patella baja	**Most patients** after proximal tibial osteotomy · **opening wedge** (raises joint line) · scarring, tendon contracture, immobilization · ✓ **HTO most likely procedure** to cause baja (✗ Wrong options: ACL/PCL/MPFL recon · TKA)
Patella alta	**Lateral closing wedge** (TO)
Loss of posterior slope	Closing wedge
Peroneal nerve palsy	**Lateral** opening/closing wedges · **fibular osteotomy** · minimal in medial opening
Popliteal artery	Rare, devastating
Nonunion / malunion	↑ opening wedge, smokers
Compartment syndrome	Any osteotomy

Prognosis varus-producing HTO **87% success at 10 yr** · valgus-producing **50–85% at 10 yr** (Orthobullets) · good results after later conversion to TKA

Conversion to TKA

Risk factors for TKA failure after HTO ✓ **BMI >35** · **male** · **young age** at TKA · laxity · preoperative malalignment
✗ Wrong options: female · stiffness · advanced age

X vs Y

	HTO	UKA
Patient	**<50, high-impact**, labourer	Older, lower demand
ACL deficiency	Relative – reconstruct together	**Contraindication**
Varus	Up to ~10–15° (TO)	**<10°**, passively correctable
Return to impact sport	**Faster**	Discouraged (running)

	Valgus HTO	Varus DFO
Deformity	Varus, tibial CORA	Valgus >12°, femoral CORA
Compartment	Medial OA	Lateral OA

How it's asked

"Young labourer, isolated medial OA, varus – best treatment?"	→ **medial opening wedge HTO** ✗ Wrong options: UKA · TKA · arthroscopy
"Diagnostic + therapeutic test before HTO?"	→ **medial unloader brace**

"True of HTO?"	→ **overcorrect into valgus** ✗ Wrong options: BMI irrelevant · contraindicated >40 · patella alta common · opening superior
"Contraindication to HTO?"	→ **inflammatory arthritis** (TO) ✗ Wrong options: ACL deficiency · varus >10° · age >60
"Contraindication to HTO?"	→ **lateral tibial subluxation 2 cm** (TO) ✗ Wrong options: varus 10° · ACL tear · ROM 0–100°
"Postmenopausal smoker, BMI 22, isolated medial OA – contraindication?"	→ **smoking**
"Varus thrust, BMI 24, 8° flexion contracture – contraindication?"	→ **varus thrust**
"Psoriatic arthritis, 10° flexion contracture, 10° varus – contraindication?"	→ **inflammatory arthropathy**
"ACL tear, intact lateral meniscus, grade 4 medial OA, 10° correctable varus?"	→ **HTO** (TO)
"Marathoner, varus thrust, stable lateral meniscal tear, functional ACL?"	→ **medial opening wedge HTO** (TO)
"Fujisawa point?"	→ **62% of plateau width from medial edge** (TO)
"Complication exclusive to lateral closing wedge?"	→ **proximal tibiofibular joint disruption**
"Smoker – what favours lateral closing wedge?"	→ **smoking history**
"Lateral closing wedge vs medial opening – true?"	→ **shorter time to weight bearing**
"PCL-deficient knee – slope effect?"	→ **increased slope ↑ stability**
"Slope 14° not addressed – postop finding?"	→ **positive Lachman**
"Most common complication of HTO?"	→ **recurrence of varus** (TO)
"Which osteotomy can cause patella alta?"	→ **lateral closing wedge** (TO)
"Patella baja most likely after?"	→ **HTO**
"Risk factor for TKA failure after HTO?"	→ **BMI >35**
"33-yr-old, wants fastest return to impact activity?"	→ **medial opening wedge HTO** ✗ Wrong options: UKA · TKA · BMAC alone · varus DFO
"Theory behind proximal fibular osteotomy?"	→ **non-uniform settlement** (TO)
"Concave medial plateau contraindicating HTO?"	→ **teeter-totter effect (Kettelkamp)** (TO)

– – Active space – –

HIP ANATOMY

– Recon · 5 / 65 –

1 Basics

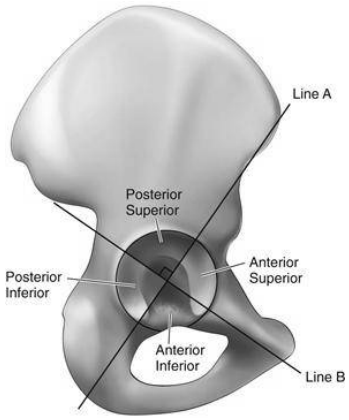
Osteology

Joint	ball-and-socket diarthrodial · stability from **bony architecture**, augmented by **labrum + capsule**
Acetabulum	3 ossification centres: **ilium, ischium, pubis** · **posterosuperior** cartilage thickest (weight-bearing)
Acetabular orientation	**anteverted 15°** · **abducted 45°** (caudal tilt)
Femoral neck	**anteverted 15°** (relative to condyles) · **neck-shaft angle 125°**
Femoral head centre	level with **tip of greater trochanter**

Acetabular quadrants – screw safety

Construction line **ASIS → acetabular centre** (anterior/posterior) + perpendicular bisector (superior/inferior)

Quadrant	Structures at risk	Screws
Posterior-superior	**Superior gluteal nerve + vessels**, **sciatic nerve**	**Safest**
Posterior-inferior	**Inferior gluteal nerve**, **internal pudendal vessels**	Second safest
Anterior-inferior	**Obturator vessels** (and nerve)	Unsafe
Anterior-superior	**External iliac artery + vein**	Unsafe



Acetabular quadrant system

Capsule & ligaments

Capsule	anterior: **intertrochanteric line** · posterior: **intertrochanteric crest** · basicervical + intertrochanteric regions **extracapsular**
Iliofemoral (Y ligament of Bigelow)	**strongest** · AIIS → intertrochanteric line · **taut in full extension** · prevents **anterior dislocation / hyperextension**
Ischiofemoral	reinforces **posterior** capsule · **pubofemoral** – inferior/anterior

Ligamentum teres cotyloid notch → fovea

Transverse acetabular ligament **most inferior** labrum · landmark for **acetabular version** during THA reaming

Femoral head blood supply

Age	Main supply
Birth–4 yr	**MFCA + LFCA + ligamentum teres**
4 yr–adult	**Posterosuperior + posteroinferior retinacular vessels** from MFCA
Adult	**Medial femoral circumflex artery** ✓ (deep branch)

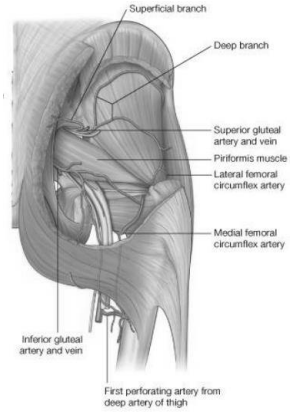
MFCA branch of **profunda femoris** · ✓ **major supply to femoral head** · **at risk in psoas tenotomy** · **avoid transecting quadratus femoris** in posterior approach (non-arthroplasty)
✗ Wrong options: internal pudendal · IGA · SGA · LFCA

Piriformis-start nails damage **posterosuperior retinacular vessels** → AVN

Ligamentum teres artery **posterior division of obturator** · insignificant in adults

Inferior gluteal artery internal iliac branch · exits greater sciatic notch **between piriformis and coccygeus**, runs **inferior to / along piriformis** · supplies **short ER + gluteus maximus** · **primary femoral head supply in up to 17%** → preserve in surgical dislocation, posterior dislocation, acetabular fracture · ✓ **found along border of piriformis tendon**

Vessel-location distractors (OB) between pectineus + iliopsoas = **MFCA** · beneath TFL = **LFCA** portion · under gluteus medius = **superior gluteal** · between quadratus femoris + upper adductor magnus = **deep branch of MFCA**



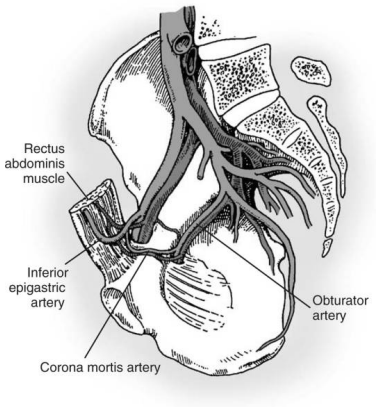
Pelvic vessels

– – Active space – –

Vessels at risk

Vessel	Origin	At risk
External iliac → common femoral	Aorta → external iliac	**Anterosuperior** quadrant screws
LFCA ascending branch	Profunda	**Direct anterior approach**
MFCA	Profunda	**Psoas tenotomy**, quadratus transection
Obturator (posterior branch)	Internal iliac	**Anteroinferior** screws + acetabular retractors · supplies TAL
Superior gluteal	Internal iliac	Supplies **abductors + TFL**
Internal pudendal	Internal iliac	Exits greater, **re-enters via lesser sciatic notch**
Femoral perforators	Femoral	Supply vastus lateralis

Corona mortis anastomosis between **inferior epigastric** (external iliac) and **obturator** vessels



Corona mortis

Nerves

Lumbosacral plexus **T12–S4 · posterior to psoas**

Sciatic division	Branches
Posterior	**Peroneal** · superior gluteal · inferior gluteal · nerve to piriformis · posterior femoral cutaneous
Anterior	**Tibial** · nerve to quadratus femoris · nerve to obturator internus · pudendal · coccygeus

Peroneal division **lateral** to tibial · runs deep to long head of biceps · **most commonly injured nerve in THA**

Obturator medial compartment · all adductors **except hamstring part of adductor magnus**

Femoral **between iliacus and psoas**

LFCN **L2–L3** · deep to inguinal ligament near ASIS

Genitofemoral **L1–L2** · femoral + genital branches · pierces iliopsoas, runs on its **anteromedial** surface

Muscles by action

Action	Muscles
Flexors	Iliopsoas, rectus femoris, sartorius, pectineus, iliacus
Adductors	Adductor magnus (adductor part), longus, brevis, minimus, gracilis
External rotators	Gluteus maximus, piriformis, obturator externus + internus, gemelli, quadratus femoris
Abductors	Gluteus medius, gluteus minimus, TFL
Extensors	Gluteus maximus, semitendinosus, semimembranosus, long head biceps, adductor magnus (hamstring part)

MRI anatomy (TO)

Piriformis — muscle **just below the end of the sacrum** · **sciatic nerve between piriformis and obturator internus** · piriformis syndrome: oedema, accessory head, sciatic impingement

LFCN on MRI — small **fat triangle just below ASIS, between sartorius and TFL** → entrapment = **meralgia paraesthetica** (injection site)

Rectus femoris — **direct head from AIIS** · **indirect (reflected) head from supra-acetabular ridge**

Gluteus medius — **two parts** (anterolateral + posterior) · **fat plane between them is normal**, not a tear · **gluteus minimus** inserts anteriorly on GT

Gluteus maximus — ends in **ITB with TFL**, "never inserts directly on bone" (TO — verify: deep lower fibres insert on the gluteal tuberosity)

Quadratus femoris — between **GT and ischial tuberosity** · narrow space + oedema → **ischiofemoral impingement**

Hamstrings — semimembranosus + **conjoint tendon (long head biceps + semitendinosus)** · semitendinosus fleshy early

Psoas — tendon insinuates into iliacus · oedema without tear in dancer → **snapping psoas** · iliopsoas anterior to joint → **iliopsoas bursitis**

Normal variants — **lateral femoral tubercle** (not FAI) · physeal scar · small plica · little hip fluid

![Axial T1 pelvis — glutei, piriformis, obturator internus, iliopsoas](../../TO notes/attachments/TOn-notes-mri-of-hip-and-thigh-mar-2026-p03-f006.jpg)

Approaches

Five — anterior **Smith-Petersen** · anterolateral **Watson-Jones** · medial · direct lateral **Hardinge** · posterior **Southern / Moore**

- - Active space - -

4 Diagnosis & Classification

MRI sequences (TO)

Sequence	Bright	Best for
T1	**Fat only** (bladder urine dark)	Marrow replacement, **AVN line**, trabeculae, healed fractures, muscle atrophy, scarring
T2	**Fat + fluid**	General
Fat-suppressed (STIR, PDFS, T2FS)	**Fluid**, fat dark	**All acute conditions** – oedema, effusion, tears, strains
GRE	–	**Bone detail** – cortex, small calcified / bony lesions

Sequence trick look at the **urinary bladder** (TO)

Request "**pelvis with right / left hip**" → adds dedicated **oblique axial** (along neck axis) + oblique sagittal sequences · orthogonal axials **never show the neck in continuity** · labrum / FAI need oblique axials (TO)

Hamstring / rectus femoris tear request **hip AND thigh** MRI (pelvis scan stops at lesser trochanter) (TO)

Labral tear on MRI **cleft (sulcus)** through labrum ± **paralabral cyst** (TO)

5 Management

Surgical hip dislocation – capsulotomy (TO)

Ganz safe dislocation ✓ **anterolateral + anteroinferior capsulotomy, hip dislocated anteriorly** (flexion + ER)
✗ Wrong options: posterolateral/posteroinferior · dislocate posteriorly · anteromedial/anterosuperior

Why first incision **anterolateral along neck axis** avoids **deep branch of MFCA** · anteroinferior limb kept **anterior to lesser trochanter** (main MFCA lies **just superior + posterior to LT**) · extended to rim, turned **posteriorly parallel to labrum** to retracted piriformis

Preserve **IGA** along piriformis border (OB)

X vs Y

	MFCA	LFCA
Femoral head supply (adult)	**Major**	Minor
At risk	Posterior approach (quadratus), psoas tenotomy	**Ascending branch – direct anterior approach**

How it's asked

"Main blood supply of femoral head in adult (femoral neck fracture)?" → **medial femoral circumflex artery**

"Surgical dislocation — inferior gluteal artery reliably found?"	→ **along border of piriformis tendon**
"Safe surgical dislocation — capsulotomy?"	→ **anterolateral + anteroinferior, dislocate anteriorly** (TO)
"Safest quadrant for acetabular screws?"	→ **posterosuperior** (then posteroinferior)
"Anterosuperior screw injures?"	→ **external iliac vessels**
"Anteroinferior screw / retractor injures?"	→ **obturator vessels**
"Ascending branch of LFCA at risk in?"	→ **direct anterior approach**
"Most commonly injured nerve in THA?"	→ **peroneal division of sciatic**
"Strongest hip ligament?"	→ **iliofemoral (Y of Bigelow)**
"Fat plane between two parts of gluteus medius on MRI?"	→ **normal** (TO)
"MRI sequence in which only fat is bright?"	→ **T1** (TO)
"Landmark for cup version?"	→ **transverse acetabular ligament**

– – Active space – –

– – Active space – –

HIP ARTHRODESIS

– Recon · 6 / 65 –

1 Basics

Biomechanics

Gait efficiency	**↓ ~50%**
Energy	**↑ O_2 consumption · 30% more energy** for ambulation
Adjacent joints	↑ contralateral pelvic rotation · ↑ stress on **lumbar spine, ipsilateral knee, contralateral hip**

5 Management

Hip arthrodesis decision

- Young labourer, unilateral painful ankylosis (infection/trauma) → **primary fusion**
- Failed THA (most common) → **salvage fusion**
- Painful fused hip (back > knee > contralateral hip) → **convert to THA**
  - Abductors nonfunctional on EMG → **constrained cup**

Operative criteria – primary arthrodesis

Indications	Contraindications
Salvage for failed THA (most common)	**Active infection**
Young active labourer, painful unilateral ankylosis after infection or trauma	**LLD >2.0 cm**
Neuropathic arthropathy	**Bilateral** hip arthritis
Tumour resection	**Adjacent joint degeneration** – lumbar spine, contralateral hip, ipsilateral knee
Missed SCFE with severe arthritis, 18-yr-old (OB)	Severe osteoporosis
	Contralateral THA (THA failure **40%** with contralateral fusion)

Optimal fusion position

Plane	Orthobullets / OB	TO
Flexion	**20–35°**	**20–30°**
Adduction	**0–5°**	**0–5°**
External rotation	**5–10°**	**0–15°**

Avoid abduction pelvic obliquity → ↑ back pain

OB answer	✓ **5° ER, 5° adduction, 20° flexion** ✗ Wrong options: any abduction · 0° flexion · 15° ER with 0° adduction
TO answer	✓ **20–30° flexion, 0–5° adduction, 0–15° ER** ✗ Wrong options: 10° flexion · abduction

Technique

Goals	apposition of surfaces · **rigid internal fixation** · early mobilization
Approach	**lateral with trochanteric osteotomy** preferred · preserve **abductors** + **superior gluteal nerve** · anterior approach also popular
Fixation	**cobra plate** · anterior plating



Hip arthrodesis

Conversion of fusion to THA

Indications	**severe low back pain (most common)** · ipsilateral knee pain + instability · contralateral hip pain
Preop	**EMG of gluteus medius** · **nonfunctional abductors → constrained acetabular component**
TO lecture notes	also **painful pseudarthrosis** · **fusion in improper position → corrective osteotomy, not THR** · abductors assessed **clinically (palpation in lateral decubitus, knee flexed) – no role of EMG / MRI** (TO) vs EMG gluteus medius (Orthobullets) – read the options
Technique (TO, ankylosis / fusion)	**double femoral neck osteotomy** (divergent cuts) · ream under C-arm after drilling for depth · centre from Shenton + Köhler lines
Prognosis (TO)	**poorest of all difficult THR scenarios** · ↑ **HO** + infection · abductor recovery up to **2 years**
Outcome	depends on **abductor function** · weakness → prolonged rehab, **severe lurching gait**

6 Complications

Low back pain	improved by takedown, but depends on **abductor function**
Ipsilateral knee	degeneration + **laxity**
Contralateral hip	degeneration
Adjacent joint degeneration	**60%** · LBP + ipsilateral knee pain most common · within **25 yr**

– – Active space – –

How it's asked

"Ideal hip arthrodesis position?"	→ **20–30° flexion, 0–5° adduction, 0–15° ER** (TO)
"18-yr-old, missed SCFE, hip fusion — optimum position?"	→ **5° ER, 5° adduction, 20° flexion**
"Most common indication for hip arthrodesis?"	→ **salvage of failed THA**
"Most common reason to convert fusion to THA?"	→ **low back pain**
"Before converting fusion to THA?"	→ **EMG of gluteus medius** (nonfunctional → constrained cup)
"Gait efficiency after hip fusion?"	→ **↓ ~50%**, energy ↑ 30%
"Contraindication to hip fusion?"	→ **ipsilateral knee / lumbar / contralateral hip arthritis, LLD >2 cm, active infection**
"Position to avoid in hip fusion?"	→ **abduction**
"Hip fused in malposition, painful back — next?"	→ **corrective osteotomy, not THR** (TO)

HIP BIOMECHANICS

– Recon · 7 / 65 –

1 Basics

Definitions

Joint reaction force (JRF)	force within a joint in response to forces acting on it · hip: balances **body-weight moment vs abductor moment → level pelvis**
Joint congruence	fit of surfaces · ↑ congruence → ↑ contact area
Instant centre of rotation	point of rotation, shifts with translation · body **centre of gravity just anterior to S2**
Friction	**not a function of contact area** · lubrication ↓ friction · **human joint μ = 0.002–0.04** · **metal-on-PE TJA μ = 0.05–0.15**

Degrees of freedom

Hip	ball-and-socket → **3 rotational DOF**: flexion–extension (sagittal), abduction–adduction (frontal), axial rotation (transverse) · **constrains all translations** ✗ Wrong options: 2 DOF · translations · 6 DOF

Free-body physics

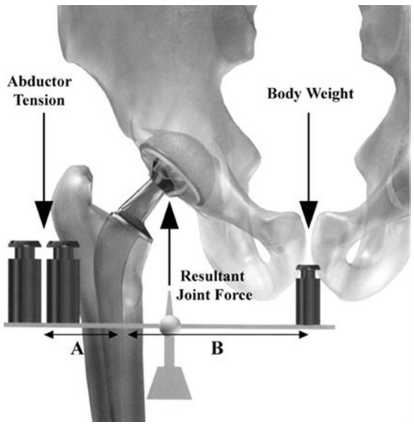
Quantity	Equation
Newton 1st	No net force → constant velocity
Force (Newton 2nd)	**F = ma** · 1 N = 1 kg × 1 m/s²
Newton 3rd	**$F_2 = -F_1$**
Vector	Magnitude + direction (scalar = magnitude only)
Moment (torque)	**Force (perpendicular) × distance**
Work	Force (parallel to displacement) × distance
Potential energy	**m × g × h**
Kinetic energy	**½ mv²**

2 Mechanism

Hip free-body analysis

W	body weight minus stance limb = **5/6 body weight**
M	abductor force · **R** – JRF = **3–6× body weight**
Step 1 (moments)	**A × My = B × W** · A = 5 cm (abductor arm), B = 12.5 cm (body-weight arm) → **My = 2.5 W**
Step 2	**Ry = My + W = 3.5 W**
Step 3	**R = Ry / cos 30° ≈ 4 W**

– – Active space – –



Hip free-body diagram

Lever-arm ratio (TO)

Normal ratio	body-weight lever arm : abductor lever arm ≈ **2.5 : 1** → abductors ≈ **2.5 × BW** to keep pelvis level ✗ Wrong options: 1:2 · 2:1 · 1:2.5
FNB 2024	"abductor lever arm ratio in single-leg stance" with options 0.5/1.5/2.5/3.5 — answer not given by source; matches **2.5** above
Arthritic hip	**abductor lever arm shortened** (head lost, neck short) · body-weight arm **unchanged** · ✓ **ratio ↑ (up to 4 : 1)** ✗ Wrong options: abductor arm lengthened · body-weight arm lengthened · unchanged
Surgical aim	ratio toward **1 : 1** → total hip load ↓ **~30%** (theoretical)

5 Management

Normal values & offsets (TO)

Parameter	Value (TO)
Femoral anteversion	**~15°** (neck tilted anterior to coronal plane)
Acetabular anteversion	**15–20°**
Acetabular inclination	**~45°**
Vertical offset	Head centre → collar / lesser trochanter = **leg length**
Horizontal offset	Head centre → stem (shaft) axis · neck-shaft-angle dependent

Joint reactive force – cane, medialization, offset

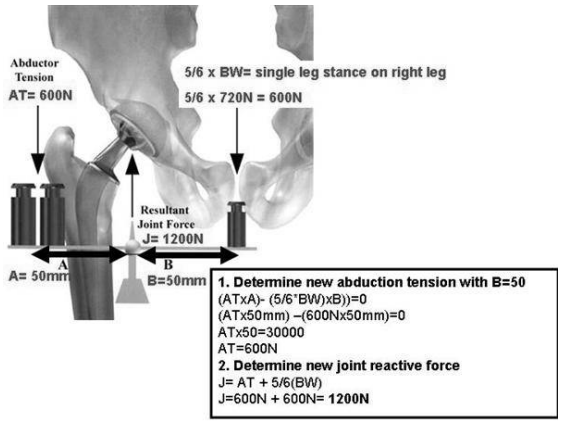
↓ JRF (↑ A/B ratio)	↑ JRF
Medialize cup (also inferior + anterior) → ↓ body-weight arm	**Valgus** neck-shaft angle (↓ shear)
↑ femoral offset → lengthens abductor arm	Lateralized / high hip centre
Long-stem prosthesis · **lateralize GT**	
Varus neck-shaft angle (↑ **shear**)	
Lean over affected hip (Trendelenburg lurch)	
Cane in contralateral hand	
Load carried in **ipsilateral** hand	

Offset question

✓ **↓ JRF; ↑ offset lengthens abductor moment arm**
✗ Wrong options: ↑ JRF · shortens arm · only hip centre matters

Medialization worked example (OB)

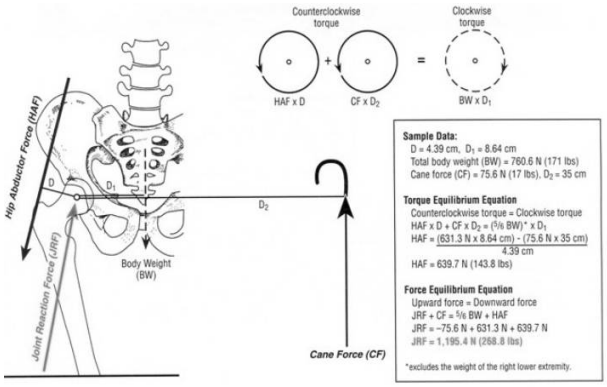
J = 1800 N; body-weight arm 100 → 50 mm → ✓ **J = 1200 N** · equilibrium: **(AT × A) − (5/6 BW × B) = 0**
✗ Wrong options: 1000 · 1800 · 2200 · 3600



Medialized acetabulum — free-body worked answer

Contralateral cane

↓ hip JRF **up to ~50%** by ✓ **reducing abductor pull** · long moment arm from opposite side · needs upper-limb + trunk strength
✗ Wrong options: ↑ flexor pull · moving centre of rotation · ↑ congruence · moving COG posterior to S2



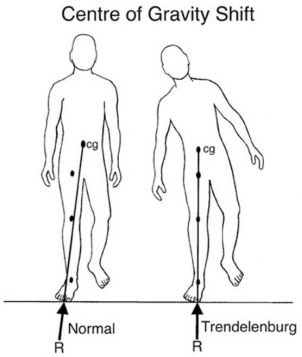
Cane in contralateral hand

– – Active space – –

3 Presentation

Trendelenburg gait

Cause	✓ **hip abductor weakness** ✗ Wrong options: adductors · flexors · extensors · knee extensors
Pattern	✓ **pelvis on swing side drops → ↑ adduction of affected hip in stance** ✗ Wrong options: swing side elevates · ↓ adduction · circumduction
Compensation (lurch)	trunk leans **over affected hip** → COG closer → ↓ moment arm → ↓ abductor demand + ↓ JRF



Trendelenburg gait

X vs Y

	Varus neck-shaft	Valgus neck-shaft
JRF	↓	↑
Shear across joint	↑	↓

	Medialize cup	↑ Femoral offset
Lever arm changed	**Body-weight arm ↓**	**Abductor arm ↑**
JRF	↓	↓

How it's asked

"Arthritic hip – true?"	→ **ratio of body-weight lever arm to abductor arm ↑** (TO)
"Ratio of body-weight lever arm to abductor arm?"	→ **2.5 : 1** (TO)
"JRF after medializing cup (1800 N, arm 100 → 50 mm)?"	→ **1200 N**
"Contralateral cane reduces JRF by?"	→ **reducing abductor pull**
"High-offset stem – effect on JRF?"	→ **↓ – lengthens abductor moment arm**
"Trendelenburg gait compensates for weak?"	→ **hip abductors**

"Trendelenburg gait pattern?"	→ **swing-side pelvis drops, ↑ adduction of stance hip**
"Hip degrees of freedom?"	→ **3 — flexion–extension, abduction–adduction, axial rotation**
"JRF in single-leg stance?"	→ **~3–6 × BW** (worked example ≈ 4 W)
"Coefficient of friction, human joint vs metal-on-PE?"	→ **0.002–0.04 vs 0.05–0.15**

– – Active space – –

– – Active space – –

HIP HEMIARTHROPLASTY PERIPROSTHETIC FRACTURE

– Recon · 8 / 65 –

1 Basics

Epidemiology

Incidence	**1–4%** of hemiarthroplasties · **uncemented 1–7.4%** vs **cemented 0.2–1.3%**
Risk factors	↑ age · **female** · **osteoporosis** · **uncemented stem** (RCTs + registries) · **undersized** component · **slip-taper stem** in cemented hemi
Associated	stress shielding, peri-implant osteopenia, infection, **osteolysis**, subsidence
Mechanism	**ground-level fall**

3 Presentation

History	fall · **start-up pain → antecedent loosening** · progressive pain, constitutional symptoms, wound problems → suspect **PJI**
Exam	pain, cannot bear weight · **shortened, externally rotated** limb · pain on log roll · check prior **scars**

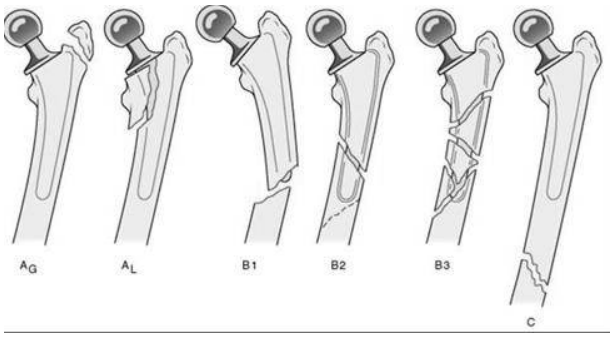
4 Diagnosis & Classification

Key radiology findings

Radiographs	**AP + lateral femur** (pattern, displacement, implant position) · **AP pelvis** if revising
CT	implant stability, fracture morphology · **best for cement-mantle integrity**
Stable stem signs	proximal cement mantle intact · **no subsidence** · always **test stem stability intraoperatively**

Vancouver classification & treatment

Type	Description	Treatment
AG	Greater trochanter	ORIF if **>2 cm displaced** (abductor escape) · else nonop
AL	Lesser trochanter	**Nonoperative**
B1	Around stem, **well fixed**	**ORIF**
B2	Around stem, **loose**, good bone stock	**Revision** (diaphyseal-engaging) ± ORIF
B3	Around stem, loose, **poor bone stock**	**Proximal femoral replacement**
C	**Distal** to stem	**ORIF**



Vancouver classification

5 Management

Hemiarthroplasty periprosthetic fracture

- Trochanteric: AL or AG <2 cm → **protected WB, no active abduction 6–12 wk**
  - AG >2 cm → **ORIF (claw plate, cables, TBW)**
- Stem well fixed (B1) or distal (C) → **ORIF (locking plate)**
- Stem loose, good bone (B2) → **revise to diaphyseal-engaging stem ± ORIF**
- Stem loose, poor bone (B3) → **proximal femoral replacement**

Treatment by stem stability

Nonoperative (AG <2 cm, AL)	protected WB · early walker WB in elderly · **restrict active abduction 6–12 weeks** (AG)
ORIF (B1, C)	as soon as medically optimized · lateral subvastus / **MIPO** · **polyaxial locking plate** · proximal fixation: cables, cerclage, screws · **drilling into cement ↑ pull-out strength but may compromise mantle** · allograft struts controversial · **15% complications**
B2 ORIF alone (controversial)	select **elderly, low-demand** · favourable: **simple pattern, anatomic reduction, no prior loosening, intact bone–cement interface**
B2 revision	✓ **uncemented diaphyseal-engaging stem bypassing fracture by 2 cortical diameters** ± ORIF · monoblock cylindrical or **modular fluted tapered** · ± trochanteric bolt/claw · **up to 25% complications** (revision or ORIF) · revision → ↑ blood loss
Long cemented stem (controversial)	cement trapped in fracture → impaired healing · cement-in-cement may equal tapered stems in selected patients
Metaphyseal-engaging components	controversial in fracture setting
B3	resect fracture · cemented/uncemented **PFR** · reattach trochanter (suture/cable/wire)

– – Active space – –



Vancouver B2 around a loose hemiarthroplasty stem

OB answer pattern B2 (6–8 wk post hemi, loose stem): ✓ **fracture fixation + diaphyseal-engaging revision**
✗ Wrong options: protected WB (Vancouver A) · cerclage alone (B1) · locking plate ± strut (B1/C) · PFR (B3) · reuse original stem · **proximally coated stem** (needs press-fit in fractured zone) · bed rest

B1 (5 yr post hemi, fracture at stem tip, proximal mantle intact) → ✓ ORIF alone
✗ Wrong options: long cemented revision (cemented stems not implant of choice in revision) · long uncemented ± ORIF (for B2) · PFR (salvage)

6 Complications

List infection · dislocation · refracture · **stem subsidence** · implant failure · **trochanteric escape**

Mortality **~25% at 1 yr**, up to **50% at 2 yr**

Prognostic **early mobilization** critical · **surgical delay → ↑ mortality**

X vs Y

	Cemented hemi	**Uncemented hemi**
Periprosthetic fracture rate	**0.2–1.3%**	**1–7.4%**
Special risk	**Slip-taper** stem	Undersized stem

	B1	**B2**	**B3**
Stem	Fixed	Loose	Loose
Bone stock	–	Good	**Poor**
Treatment	**ORIF**	**Diaphyseal revision ± ORIF**	**PFR**

How it's asked

"Hemi 6 weeks ago, fall, loose stem – optimal treatment?" → **fracture fixation + revision with diaphyseal-engaging stem** (B2)

"Hemi 8 weeks, fall — best approach?"	→ **ORIF + diaphyseal-engaging stem** ✗ Wrong options: original stem · proximally coated stem
"Hemi 5 years, fracture at stem tip, mantle intact proximally?"	→ **ORIF alone** (B1)
"Vancouver AG — ORIF threshold?"	→ **>2 cm displacement**
"Revision stem must bypass fracture by?"	→ **2 cortical diameters**
"Risk factor for hemi periprosthetic fracture?"	→ **uncemented stem** (cemented: slip-taper)
"Best imaging for cement mantle?"	→ **CT**
"Mortality at 1 year?"	→ **~25%**

– – Active space – –

– – Active space – –

HIP OSTEOARTHRITIS

– Recon · 9 / 65 –

1 Basics

Epidemiology

Incidence (symptomatic)	**hip 88 / 100,000 / yr** · knee 240 / 100,000 / yr
Modifiable risks	**articular trauma** · muscle weakness · heavy physical work · high-impact sport
Non-modifiable	**female** > male · ↑ age · genetics · **DDH, SCFE, Perthes**

Cartilage pathophysiology

Feature	Osteoarthritis	Ageing
Water content	**↑** (matrix disruption) ✓	**↓**
Proteoglycan	**↓** quantity	**↓** content + size
Collagen	Disorganized · **↓ in severe OA** · **↓** cross-linking	**Normal** quantity · **↑** stiffness, **↑** cross-linking
Chondrocyte activity	**↑** (proliferation)	Unchanged
Subchondral bone	**↑ thickness**	Unchanged
Chondroitin : GAG	–	**↓**

Moderate OA cartilage → ✓ ↑ water
✗ Wrong options: ↓ subchondral thickness · ↓ chondrocyte activity · ↑ proteoglycan · ↑ collagen

Synovium early mild inflammation → middle **hypervascular** → late **thick + vascular**

Subchondral bone remodelling → lytic lesions with **sclerotic edges** (unlike RA cysts) → late **cysts**

	Aging	Osteoarthritis
Water Content	Decreased	Increased
Collagen	Same	Disorganized
Proteoglycan Content	Decreased	Decreased
Proteoglycan Synthesis	Same	Increased
Chondrocyte Size	Increased	Same
Chondrocyte Number	Decreased	Same
Modulus of Elasticity	Increased	Decreased

OA vs ageing cartilage

Genes & lab markers

MMPs (degrade matrix)	**stromelysin**, plasmin, **aggrecanase-1 (ADAMTS-4)** · zinc co-factor · stromelysin secretion **induced by PDGF**, **inhibited by TGF-β**
TIMPs	inhibit MMPs · **MMP–TIMP imbalance** in OA

Cytokines (from synoviocytes, ↑ MMPs)	**IL-1** (catabolic), IL-6, **TNF-α**
Cartilage degeneration implicated	✓ **stromelysin** ✗ Wrong options: TGF-β (anabolic) · aggrecan (compressive proteoglycan) · osteoprotegerin (RANKL decoy) · IGF-1 (anabolic)
Inheritance	**non-Mendelian** · candidate genes: **vitamin D receptor**, **oestrogen receptor 1**, IL-1, IL-4, matrilin-3, BMP-2, BMP-5
Histology	**loss of superficial chondrocytes** · **tidemark replication + breakdown** · fissuring · eburnation

Rapidly progressive OA (RPOA)

Causes	**steroid injections** (high dose **>80 mg**, multiple) · cytokine/immune · autoimmune · **subchondral insufficiency fracture**

3 Presentation

Pain	function-limiting · **groin** commonest · referred to lateral hip, **anterior knee**, thigh, buttock, low back · night/rest pain · stiffness · mechanical symptoms
Exam	gait, LLD, scars · **flexion contracture >5°** · **flexion <90–100°** · **↓ internal rotation** · **SLR negative**
Intra-articular signs (OB)	**Stinchfield** (pain with resisted hip flexion) · **C-sign** (groin cupping)

Hip pain referred to knee

Concomitant hip pathology	**2–29%** of knee-pain patients
Suspect hip when	disability **disproportionate** to knee findings · fails cortisone/PT · **no improvement after knee surgery** · Trendelenburg gait
Next step	✓ **hip radiographs** (✗ Wrong options: ESR/CRP first · more PT · revision TKA · pain specialist) · benign knee exam + pain on hip motion → assess for ✓ **hip OA** (✗ Wrong options: meniscus · occult fracture · septic knee · LLD)

4 Diagnosis & Classification

Key radiology findings

Views	**standing AP pelvis** + **AP and lateral hip** · optional **false-profile** (dysplasia)
OA signs	joint-space narrowing · osteophytes · subchondral sclerosis · subchondral cysts
Pelvic obliquity	spinal deformity → LLD issues
Acetabular retroversion (crossover)	makes cup positioning harder

– – Active space – –

Tönnis classification

Grade	Findings
0	Normal
1	Sclerosis of head + acetabulum · slight JSN · slight lipping
2	**Small cysts** · **moderate JSN** · moderate loss of head sphericity
3	**Large cysts** · obliterated / severe JSN · severe head deformity vs AVN



Tönnis grade 2

Protrusio acetabuli (TO)

Definition ✓ **femoral head migrates past ilioischial (Köhler) line** · **CE angle of Wiberg >40°**
✗ Wrong options: iliopectineal line · CEA >25° · ↓ CEA

Köhler line most lateral pelvic brim → most lateral obturator foramen

Unilateral causes	Bilateral causes
OA, septic + **TB** arthritis, AVN, **central fracture-dislocation**, fibrous dysplasia, **THA** (migrating prosthesis)	**RA**, **osteomalacia**, **ankylosing spondylitis**, **Paget**, renal osteodystrophy, rickets, osteoporosis, **Marfan**

Secondary causes (TO notes) RA, Paget, post-traumatic, **old hemiarthroplasty (Austin Moore / Thompson)**

Otto pelvis (arthrokatadysis) **primary** protrusio · **bilateral** · **younger women** · early pain + stiffness · ✓ **not caused by osteomalacia** (that is secondary)

5 Management

Symptomatic hip OA

- First line → **NSAIDs + weight loss/exercise (BMI >25) + contralateral cane**
- Short-term relief → **IA corticosteroid (also diagnostic)**
- Young, dysplasia, concentric, mild–moderate OA → **PAO ± femoral osteotomy**
- Tönnis 2–3 → **no arthroscopy (high conversion)**
- End-stage, >50 yr → **THA (≥3 months after ipsilateral injection)**

Nonoperative management

Recommendation	Modality
First line	**NSAIDs** · **weight loss (BMI >25), activity modification, exercise/PT** · walking stick in **contralateral** hand
Strongly supported (2013 AAOS CPG)	**IA corticosteroid** short-term (therapeutic + diagnostic) · weight loss · low-impact exercise · NSAIDs
Controversial	Acupuncture · viscosupplement · glucosamine/chondroitin
Against	**Opioids incl. tramadol** · **IA hyaluronic acid** (strong against, 2017 CPG) · needle/arthroscopic lavage

Steroid mechanism ✓ **binds nuclear receptors → interrupts inflammatory cascade via mRNA**
✗ Wrong options: hypothalamic antipyretic (acetaminophen) · cannabinoid receptor · synovial viscosity (HA) · Na^+ channel block (lidocaine)

OB note 2019 OB lists **tramadol** among supported options; Orthobullets lists tramadol as **recommended against** — read the stem

Contraindication to joint preservation (Tönnis 2)

Arthroscopy controversial · degenerative labral tears with little radiographic OA · ✓ **Tönnis ≥2 → high conversion to THA** · also **age >50**, obesity

Not contraindications prior steroid injections · age >45 · large cam / 2B subspine · acetabular dysplasia (OB) · degenerative labral tear · alpha angle 52° · male sex · ligamentum teres tear

PAO ± femoral osteotomy **symptomatic dysplasia**, adolescent/young adult, **concentric** hip, mild–moderate OA · mixed results, may delay THA

Femoral head resection pathological lesions · painful subluxation

Resurfacing **young (<50), active, male** · or severe proximal femoral deformity · ↓ use (MoM concerns)

THA indications

Indication end-stage/severe symptomatic OA · **preferred >50 yr** + advanced change · most predictable pain relief + function

Typical stem 67-yr-old golfer, end-stage OA, failed NSAIDs/PT → ✓ **cementless THA**
✗ Wrong options: steroid (transient + delays surgery 3 months) · HA · resurfacing · cemented hemi (acetabular disease)

Hip–spine syndrome 1 week relief after IA injection + Stinchfield + C-sign → ✓ **recommend THA**
✗ Wrong options: repeat injection · PT · lumbar fusion · osteochondroplasty (no cam/pincer)

Injection timing wait **≥3 months** after ipsilateral IA injection (↑ PJI)

Contraindications (TO) **absolute**: active infection (hip or elsewhere), unstable medical illness · **relative**: **morbid obesity**, **severe dementia**, tobacco, severe osteoporosis, untreated skin disease (psoriasis), **abductor insufficiency**

Not a contraindication ✓ **asymptomatic bacteriuria** (TO) · HbA1c unreliable for predicting infection · Charnley: THA possible with chronic low-grade infection of **opposite** hip

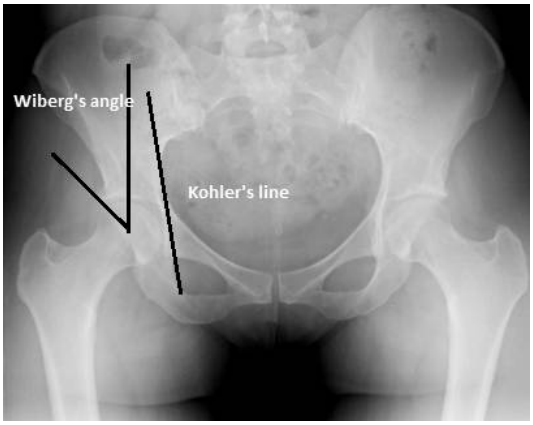
– – Active space – –

Combined hip & knee deformity – sequence (TO)

Flexion deformity of hip + knee	✓ **hip first, then knee** ✗ Wrong options: knee first · simultaneous · knee then hip 3 weeks later
Reasons	knee FFD ↓ after hip FFD corrected · knee easier with mobile hip · referred hip pain resolves · knee aligned to **new hip centre** · faster rehab

Protrusio acetabuli – reaming (TO)

Precaution	✓ **peripheral reaming only** ✗ Wrong options: one size larger midway · one size smaller throughout · horizontal reaming
Technique	ream **periphery first** with larger reamers (1–2 sizes below final cup), **preserve subchondral bone** · medial floor: curette geodes, drill sclerotic bone with K-wires · **morselized graft** impacted on floor (none at rim) · cup **1–2 mm larger** than last reamer · ≥**50% host-bone contact**
THA strategy (TO)	difficult dislocation (head incarcerated) → remove posterior wall osteophytes · **double femoral neck osteotomy** (divergent cuts) to extract head · restore centre with **Köhler + Shenton** lines · rebuild medial wall: **morselized head autograft ± wire mesh / tantalum chips** · **antiprotrusio cage** for large defects · **reverse reaming** over graft · **err toward lateralization** (keep horizontal offset, avoid over-medialization)



Protrusio reaming technique

X vs Y

	Coxa profunda	Protrusio acetabuli
Medial to ilioischial line	**Acetabular fossa** only	**Femoral head**
CE angle	–	**>40°**

	OA subchondral cyst	RA cyst
Margin	**Sclerotic**	Non-sclerotic (periarticular erosion)

How it's asked

"Characteristic cartilage change in moderate OA?" → **↑ water content**

"Implicated in cartilage degeneration?"	→ **stromelysin** ✗ Wrong options: TGF-β · IGF-1 · aggrecan · OPG
"Mechanism of drug strongly recommended for short-term hip OA pain (2013 CPG)?"	→ **corticosteroid — nuclear receptor, mRNA change**
"Contraindication for THR — all except?"	→ **asymptomatic bacteriuria** (TO)
"67-yr-old golfer, end-stage OA, tournament in 5 months?"	→ **cementless THA** ✗ Wrong options: steroid injection (delays surgery 3 months)
"Brief relief from hip injection, Stinchfield + C-sign — next?"	→ **recommend THA**
"What precludes arthroscopy for joint preservation?"	→ **Tönnis grade 2**
"Persistent knee pain after TKA, Trendelenburg gait — next?"	→ **hip X-rays**
"Benign knee exam, pain with hip motion — imaging looks for?"	→ **hip OA**
"Flexion deformity of hip and knee — sequence?"	→ **hip first, then knee** (TO)
"Protrusio acetabuli defined by?"	→ **femoral head past ilioischial line** (TO)
"Reaming precaution in protrusio?"	→ **peripheral reaming only** (TO)
"Not true about Otto pelvis?"	→ **osteomalacia** (it is bilateral, primary, young women) (TO)
"Steroid dose linked to RPOA?"	→ **>80 mg**, multiple injections
"THA in protrusio — cup position?"	→ **err toward lateralization** (TO)

– – Active space – –

HIP OSTEONECROSIS

– Recon · 10 / 65 –

1 Basics

Epidemiology

Incidence	**20,000 / yr** (US) · **10% of all THAs**
Demographics	**male > female** · age **35–50** (Orthobullets) · **30–50** (TO) – read the options
Bilateral	**80%** · **multifocal** (≥3 joints) **3%** → image contralateral hip
Bilateral rate	**80%** (Orthobullets) · **50–75%** (TO) – read the options · MRI opposite hip even if X-ray normal (TO)
Synonyms (TO)	osteonecrosis · **Chandler's disease** (non-traumatic) · "**coronary artery disease of the hip**"
History (TO)	**Phemister** – traumatic AVN + concept of **bone grafting** for AVN
Commonest cause (TO)	**idiopathic > steroid > alcohol**

Etiology & risk factors

Direct causes	Indirect causes
Irradiation	**Alcoholism**
Trauma (MFCA injury)	**Hypercoagulable states**
Haematologic – leukaemia, lymphoma	**Steroids** (endogenous / exogenous)
Dysbaric – Caisson disease, "the bends"	**SLE**
Marrow replacement – **Gaucher**	Transplant
Sickle cell	Viruses – CMV, hepatitis, **HIV**, rubella, rubeola, varicella
	Protease inhibitors · **idiopathic**

Definite vs probable causes – Kulkarni (TO)

Definite	Probable
SLE · **irradiation** · **post-traumatic** · **steroids** · **Gaucher** · haemoglobinopathies · dysbaric · transplant	**Alcohol** · **idiopathic** · **gout**

Indirect mechanism (TO)	steroid / alcohol → **fat accumulation** → ↑ intraosseous pressure → vascular compression
Also (OB)	**antiphospholipid syndrome** ✓ (20% asymptomatic ON without steroids) · antiseizure drugs · **collagen II mutation** (idiopathic ON) ✗ Wrong options: collagen I disease · haemochromatosis · achondroplasia · Paget

Pathophysiology

Idiopathic final common pathway — **intravascular coagulation**: microcirculation coagulation → **venous thrombosis** → retrograde arterial occlusion → **intraosseous hypertension** → ↓ flow → AVN → chondral fracture + collapse

Trauma — injury to **medial femoral circumflex artery**

Blood supply (TO) — **MFCA → retinacular vessels** (main) · **artery of ligamentum teres** (obturator branch, secondary)

Watershed (TO) — **anterosuperolateral subchondral** zone → **first to necrose** = commonest site of initial necrosis

Natural history (TO) — necrosis → fragmentation → collapse → secondary OA · some → spontaneous revascularization

Injury	AVN rate
Femoral head fracture	**75–100%**
Basicervical fracture	**50%**
Hip dislocation	**2–40%** (**2–10% if reduced <6 h**)
Femoral neck fracture (OB)	**10–45%** (even non-displaced)
Intertrochanteric	Rare

↑ risk — greater displacement, poor reduction · **↓ risk** – quick reduction, capsular decompression of haematoma

Paediatric hip fracture AVN (TO)

Best predictors — ✓ **fracture type + patient age** (meta-analysis, 360 cases)
✗ Wrong options: open reduction + rigid fixation · surgery <36 h + capsular decompression · displacement + associated injuries

Delbet type	Level	AVN
IA	Transphyseal **with dislocation**	**~100% (highest)**
IB	Transphyseal, no dislocation	High
II	Transcervical	Intracapsular
III	Cervicotrochanteric	Intracapsular
IV	Intertrochanteric	**Extracapsular – lowest**

3 Presentation

Symptoms — insidious **anterior hip/groin** pain · worse with **stairs, inclines, impact**

Exam — normal early · late like OA (↓ **IR**)

Order of movement loss (TO) — **IR → abduction → flexion**

Sectoral sign / axis deviation / Caffey's sign (TO) — IR ↓ **more in flexion than extension** · seen in **AVN and Perthes**

Gait (TO) — antalgic ± Trendelenburg

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

Radiographs	**AP + frog-lateral** hip · **AP + lateral of contralateral hip**
MRI	**sensitivity 99%, specificity 99%** (best test) · order when X-rays negative but suspicion persists · **double-line sign**: **T1 dark band**, **T2 bright inner line** (marrow oedema / granulation) · serpentine low-signal border with central fat-like signal · wedge-shaped subchondral T1 low signal
Double-line sign meaning (TO)	**active transition zone** dead bone ↔ granulation tissue = ongoing, often failing repair ✗ Wrong options: "repair complete, lesion stable"
Bone marrow oedema on MRI	✓ **best predictor of worsening pain + collapse** ✗ Wrong options: steroid aetiology · effusion · age <30 · Kerboul 180°
Histology	**wedge-shaped infarct** with head collapse ✗ Wrong options: OA · Perthes (no wedge) · TB (more cartilage destruction)
Bone scan	cold spot early (Ficat 0–1)
Earliest X-ray sign (TO)	**crescent sign** (subchondral fracture under intact cartilage) · late: sclerosis, lysis, collapse, loss of sphericity
MRI classic (TO, radiology)	**serpiginous line** in head marrow ± collapse + effusion · serpiginous / dark marrow signal = AVN **even before collapse**
Early AVN on X-ray (TO)	AP may show only faint sclerosis → **frog-leg view shows sclerosis + heterogeneity better**

![AVN — serpiginous marrow signal, left head collapse with effusion](../../TO notes/attachments/TOn-notes-mri-of-hip-and-thigh-mar-2026-p04-f010.jpg)

Bone scan later (TO)	**donut sign** — peripheral uptake + central photopenic zone
MRI (TO)	**investigation of choice**, T1 hypointense band · T2 bright **double-line**
Functional marrow tests (TO)	rarely used · **bone marrow pressure normal 20–30 mmHg** · O2 saturation, intramedullary venography · **core biopsy** → cause (lipid infiltration, marrow disease)

Genes & lab markers

Blood work-up (TO)	CBC · **lipid profile** · **uric acid** · LFT/KFT · **Hb electrophoresis** (sickle) · **ANA** (SLE) · HIV / hepatitis markers · **coagulation profile**



Osteonecrosis — wedge-shaped infarct

Steinberg classification (modified Ficat)

Stage	Radiograph	MRI / bone scan
0	Normal	Normal
I	Normal	**Abnormal**
II	**Cystic / sclerotic** change	Abnormal
III	**Crescent sign** (subchondral collapse)	Abnormal
IV	**Flattening** of femoral head	Abnormal
V	**Joint narrowing**	Abnormal
VI	**Advanced degenerative** change	Abnormal



Steinberg III – crescent sign

Ficat–Arlet staging (TO)

Stage	Radiograph
0	None (bone scan ↓ uptake)
1	Normal (bone scan **cold spot**)
2A	Diffuse/localized osteoporosis, cyst, sclerosis
2B	**Crescent sign – subchondral collapse with maintained sphericity** ✓
3	**Loss of sphericity**
4	Joint-space narrowing, arthritis

Ficat (TO notes) stage **0** = preclinical, **positive functional marrow tests** · stage **1** = X-ray normal, **MRI positive** (oedema) · stage **4** = arthritic change on **acetabular side** · **most used classification** (TO)

Steinberg / University of Pennsylvania 1984 (TO) 6 stages · MRI-based · each subdivided by extent

Steinberg subgroup (TO)	Extent of head involvement / collapse
A	**< 15%**
B	**15–30%**
C	**> 30%**

– – Active space – –

Note crescent sign = **Steinberg III** (Orthobullets) = **Ficat 2B** (TO) · OB uses Ficat III for crescent, IV flattening, V acetabular change — read which system the stem uses

Prognosis of collapse – lesion size, Kerboul angle

Most important factor ✓ **lesion size** (>⅓ of head)
✗ Wrong options: age · sex · BMI/activity · alcohol

Modified Kerboul combined necrotic angle **arc on mid-coronal + mid-sagittal MRI** added (e.g. 126° + 94° = **220°**)

Combined angle	Orthobullets / OB risk	TO grade
<160°	Low	**Grade 1** — low; CD works
160–190°	Low	**Grade 2** — moderate
190–240°	**Moderate** (✓ 220° = moderate)	**Grade 3 — high**
>240°	**High**	**Grade 4 — very high**, preservation often contraindicated

TO high-risk thresholds Kerboul **>200°** · **combined necrotic index >33 → >80% collapse** untreated · lesions **>30% volume** — bisphosphonates don't prevent collapse

Kerboul 3-tier (TO notes) **< 190° low · 190–240° moderate · > 240° high → early collapse → THR likely**

Bad prognostic factors (TO)

Poor prognosis Kerboul **> 240°** · **subchondral collapse** · **anterosuperolateral** site · **marrow oedema** on MRI · **age > 50**

Not a bad prognostic factor ✓ **bilateral disease** (TO)

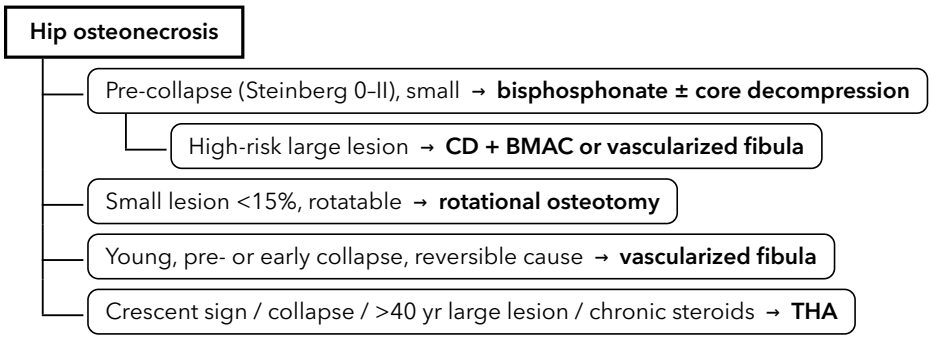
Sickle cell asymptomatic MRI ON → **>75% progress to collapse** (vs **33%** non-sickle) · screen contralateral hip · some treat both



Fig. 1
The calculation of the combined necrotic angle from magnetic resonance imaging scans. A: The angle of necrotic area in the midcoronal image. B: The angle of necrotic area in the midsagittal image. The combined necrotic angle = A+B.

Modified Kerboul angle

5 Management



Medical treatment – bisphosphonates

Indication	**pre-collapse** (Ficat/Steinberg 0–II) · **alendronate** prevented collapse with subchondral lucency · later RCTs (zoledronate) – **no benefit** in medium/large lesions
Answer	✓ **bisphosphonate** ↓ collapse ✗ Wrong options: cyclic PTH · RANKL/RANK therapy · SERM
Mechanism	✓ **↓ osteoclast resorption by flattening ruffled border** + ↑ osteoclast apoptosis ✗ Wrong options: vitamin D · denosumab (RANKL antibody, no benefit) · teriparatide · COX inhibition
TO notes	**alendronate 70 mg/week** = bisphosphonate of choice · **statins** in steroid / alcohol AVN · **anticoagulants (enoxaparin)** in documented hypercoagulable states · adjuncts **hyperbaric O2**, **ESWT** · goal: retard progression, protect contralateral hip (TO)
Risks	atypical femur fracture · ONJ · gastric ulcer · oesophagitis

Core decompression ± grafting

Indication	**early, pre-collapse** · **reversible** aetiology · not after collapse
Indication (TO)	**Ficat I–IIA** · CD + bone grafting for **Ficat IIB** · preferred: **multiple 3.2 mm drill holes** (↓ fracture risk), aimed at **anterosuperolateral** necrosis under fluoroscopy
Graft options with CD (TO)	**non-vascularized fibula** (strut) · vascularized fibula · **muscle pedicle grafts (TFL, sartorius)**
Graft alternatives (TO)	**BMP** · **BMAC** · PRP · **porous tantalum rods** · **MSCs** (**10^6 CFU → ~75% 1-yr success**, Zhou)
Technique	traditional **8–10 mm** core through subchondral necrosis · alternative **3.2 mm pin passed 2–3×** · relieves **intraosseous hypertension** → ↓ pain · stimulates angiogenesis · augment with BMP / **BMAC**
High-risk pre-collapse (TO)	28-yr-old, double-line sign, Kerboul 210°, CNI 45 → ✓ **core decompression alone insufficient → vascularized fibula or BMAC** ✗ Wrong options: isolated multiple small-diameter CD · bisphosphonates + protected WB · THA (still young + pre-collapse)

Rotational osteotomy

Indication	**small lesions (<15%)** rotatable off weight-bearing zone · intertrochanteric

– – Active space – –

Direction	**medial** disease → **varus** rotational · **anterolateral** disease → **valgus flexion** osteotomy
TO criteria	Kerboul **< 200°** · **< 30%** head involvement · age **< 40** · **full ROM** · no acetabular pathology · **small lesion <15%** (Orthobullets) · **<30% / Kerboul <200°** (TO) – read the options
Valgus-flexion osteotomy	**Sugioka** (TO)

Transtrochanteric osteotomy for AVN (TO)	Required arc of viable bone
Rotational (TRO, Sugioka)	**> 120°** on mid-sagittal MRI
Curved varus (TCVO)	**> 150°** on mid-coronal MRI

Sugioka	**transtrochanteric rotational osteotomy** (TO, osteotomies lecture) · valgus-flexion osteotomy (TO, line above) – read the options
Intertrochanteric "very good" indications (TO)	trochanteric malunion · **neck nonunion with viable head** · congenital coxa vara · small localised AVN · malpositioned arthrodesis · "good": coxa valga with retained abduction, chronic SCFE 25–50°, excess anteversion, Perthes
Outcome	**60–90%** success (Japan) · distorts head → harder THA · not after collapse

Head-preserving grafting

Techniques ✓ **core track**, **trapdoor**, **light bulb** · **"tramtrack"** – not a technique (TO)

Technique	Route	Note
Core track	Standard core	Wide debridement, simple, no dislocation, low complications
Trapdoor (Mont)	**Articular surface** flap via **surgical dislocation**	Direct view · cartilage damage, iatrogenic ON risk
Light bulb (Merle d'Aubigné)	**2 × 2 cm window at head–neck junction**	Allograft packed · **↑ neck fracture risk**
Retrograde grafting (TO)	**Trocar from lateral cortex**	Curettage + backfill with graft or substitute

Vascularized free fibula

Indication	**pre-collapse and collapsed** AVN in **young** patient · reversible aetiology preferred
Technique	large core removes necrosis · fibular strut under subchondral bone (prevents/tamps collapse)
Outcome	**80% success at 5–10 yr** (some centres) · **less predictable >40 yr**
Donor complications	sensory deficit · motor weakness · **FHL contracture** · **tibial stress fracture**
Tibial stress fracture after graft	X-ray normal → ✓ **MRI** (or bone scan) → protected WB ✗ Wrong options: compartment pressures · CT (inferior) · US · biopsy

Surgical hip dislocation (TO)

Artery to preserve	✓ **deep branch of MFCA** ✗ Wrong options: deep/ascending LFCA · descending MFCA

Deep MFCA course	inferior border of **obturator externus**, between **pectineus (medial) and iliopsoas (lateral)** → intertrochanteric crest → terminal branches under reflected capsule, **posterosuperior neck** → enter bone **2–4 mm distal to head–cartilage junction** · **intact obturator externus protects it**
Ganz	✓ **posterior (Kocher–Langenbeck/Gibson) approach → trochanteric flip osteotomy (≤1.5 cm, 5 mm medius cuff + piriformis left on bed) → anterior arthrotomy (Z-capsulotomy) → anterior dislocation** · capsulotomy stays **anterior to lesser trochanter** ✗ Wrong options: posterior arthrotomy · posterior dislocation

THA – outcomes & complications

Indications	**crescent sign or worse** (any age) ± acetabular DJD · **irreversible** cause (chronic steroids) · **>40 yr with large lesion** · most reliable pain relief + return of function
Typical answers	✓ **27-yr-old with subchondral collapse** (✗ Wrong options: IT fracture → nail · non-displaced neck → fixation · compression stress fracture → NWB/ screws · young displaced neck → ORIF) · 70-yr-old, chronic steroids, bilateral collapse → ✓ **THA** (✗ Wrong options: fibular graft >40 · osteotomy (collapse) · CD · resurfacing (poor bone)) · 68-yr-old, large MRI necrosis → ✓ **THA** (✗ Wrong options: ORIF · biopsy · resection · arthrodesis)
Technique	**cementless cup + stem** · **cemented THA → higher failure** in AVN · **hemiarthroplasty abandoned** (poor in young)
Sickle cell THA	✓ **uncemented MoP THA** best · most common intraop complication ✓ **femoral canal perforation (4.9–18.2%)** – marrow hyperplasia + patchy sclerosis → **ream over guidewire** ✗ Wrong options: fracture distal to implant · pelvic discontinuity · fat embolism arrest · sciatic injury
Risk vs OA patients	✓ **higher post-op dislocation** · **canal perforation** · medial wall breach · **higher medical + surgical complications** · poor bone + soft-tissue laxity (steroids)
Young AVN vs elderly OA	✓ **↑ PE wear + osteolysis** ✗ Wrong options: sciatic palsy · ↑ longevity · ↓ ROM · ↓ revision

Stage-wise treatment (TO)

Ficat–Arlet	Preferred treatment
0–I	Medical ± bisphosphonates
IIA	**Core decompression** ± osteotomy (Kerboul < 200°)
IIB	**CD + bone grafting** (fibula)
III	Osteotomy (Kerboul < 200°) / **THR** if extensive
IV	**THR**

Resurfacing contraindication

TO	**THR preferred over resurfacing** in AVN · Birmingham resurfacing largely abandoned (poor long-term results)
Indication	advanced DJD with **small isolated** focus · needs good bone

– – Active space – –

Contraindicated	**chronic steroids** / underlying disease (poor bone) · **renal disease** (metal ions) · **large necrotic segment**
Higher failure	✓ **osteonecrosis of femoral head** · best results: **young men <55 with OA** ✗ Wrong options: BMI 32 · acetabular changes (acetabulum resurfaced too) · age <55 · male
Outcome	acetabular erosion + pain at medium term

Other options

Hip arthrodesis	only very young **labour-intensive** worker

Sequelae after femoral neck fracture fixation

AVN collapse after ORIF (72-yr-old) → ✓ THA
✗ Wrong options: Girdlestone · hardware removal ± revision ORIF

Hemiarthroplasty for post-traumatic AVN, groin pain, normal labs, aspirate <500 cells → ✓ acetabular protrusio (erosion) → convert to THA · THA gives better function than hemi for AVN
✗ Wrong options: infection · radiculopathy · impingement · psoas tendinitis

X vs Y

	Hip AVN	ITOH	Hip OA
MRI	**Double-line**, wedge/serpentine	**Diffuse oedema**, no band	Joint-based change
Course	Progressive collapse	Self-limited	Progressive

	Hip AVN	Subchondral insufficiency fracture (TO)
Patient	Young–middle-aged, risk factors	**Middle-aged / older woman** (e.g. 47 yr), sudden pain
MRI	**Serpiginous line**, double-line	**Widespread head + neck oedema** · **thin curvilinear subchondral T1 line** · no serpiginous line
Rx	Stage-based	**Protected WB + calcium** → settles · not infection / inflammation

	Steinberg III	Ficat 2B (TO)
Sign	**Crescent**	**Crescent, sphericity kept**

How it's asked

"Tibial pain 6 months after free fibula, X-ray negative – confirm?"	→ **MRI**
"Wedge-shaped infarct with head collapse on histology?"	→ **osteonecrosis**
"Serpentine low-signal T1 band with central fat signal?"	→ **osteonecrosis**
"Associated with ON of large joints?"	→ **antiphospholipid syndrome**
"28-yr-old, Kerboul 210°, necrotic index 45, no collapse?"	→ **CD alone insufficient – vascularized fibula or BMAC** (TO)
"Sickle cell, asymptomatic MRI ON – % collapsing?"	→ **>75%**

"Kerboul 126° + 94° — angle and risk?"	→ **220°, moderate**
"Most important factor in contralateral asymptomatic hip?"	→ **lesion size**
"MRI finding most linked to pain progression?"	→ **bone marrow oedema**
"Pre-collapse with subchondral lucency — drug shown to ↓ collapse?"	→ **bisphosphonate**
"Bisphosphonate mechanism?"	→ **flattens osteoclast ruffled border**
"41-yr-old steroid AVN THA vs 80-yr-old father — risk?"	→ **↑ PE wear + osteolysis**
"Sickle cell AVN with collapse — best intervention?"	→ **uncemented MoP THA**
"Most common intraop complication of THA in sickle cell?"	→ **femoral canal perforation**
"THA for AVN vs OA — complication to consider?"	→ **higher dislocation risk**
"MoM resurfacing — higher failure because of?"	→ **osteonecrosis of head**
"Best predictor of AVN after paediatric hip fracture?"	→ **fracture type + age** (TO)
"Hemi for AVN 5 yr ago, groin pain, normal aspirate?"	→ **acetabular protrusio**
"Subchondral collapse with maintained sphericity — Ficat–Arlet stage?"	→ **2B** (TO)
"Artery preserved in safe surgical dislocation?"	→ **deep branch of MFCA** (TO)
"True about safe dislocation?"	→ **anterior dislocation + anterior arthrotomy** (TO)
"Not a bone-grafting technique in AVN?"	→ **tramtrack** (TO)
"Earliest X-ray sign of AVN?"	→ **crescent sign** (TO)
"Not a bad prognostic factor in AVN hip?"	→ **bilateral disease** (TO)
"Bone scan sign in later AVN?"	→ **donut sign** (TO)
"IR more restricted in flexion than extension?"	→ **sectoral (Caffey) sign — AVN or Perthes** (TO)

– – Active space – –

HIP PHYSICAL EXAM - ADULT

3 Presentation

Inspection

Skin	discoloration, wounds, deformity
Bone	length vs contralateral · rotation (IR/ER) · flexion contracture
Gait	stride, foot rotation, pelvic rotation, stance phase
Antalgic gait	**shortened stance phase** on painful side
Trendelenburg gait	**abductor weakness** → stance on affected hip → **contralateral hip drops**

Palpation

Site	Pain suggests
Greater trochanter	Bursitis, tendinitis, infection, fracture · **snapping ITB** over GT
ASIS	**Sartorius** avulsion
Ischial tuberosity	**Hamstring** avulsion / tendinopathy
Iliac crest	Oblique avulsion / **hip pointer**

Neurovascular

Motor	Nerve
Hip adduction	**Obturator**
Thigh abduction	**Superior gluteal**
Hip flexion	**Femoral**
Hip extension	**Inferior gluteal**

Sensory area	Nerve
Proximal anteromedial thigh	**Genitofemoral**
Inferomedial thigh	**Obturator**
Lateral thigh	**LFCN**
Anteromedial thigh	**Femoral**
Posterior thigh	**Posterior femoral cutaneous**

Pulse femoral · **reflexes** – none at hip

Range of motion

– – Active space – –

Motion	Normal
Flexion	**120–135°**
Extension	**20–30°**
Abduction	**40–50°**
Adduction	**20–30°**
Internal rotation	**30°**
External rotation	**50°**

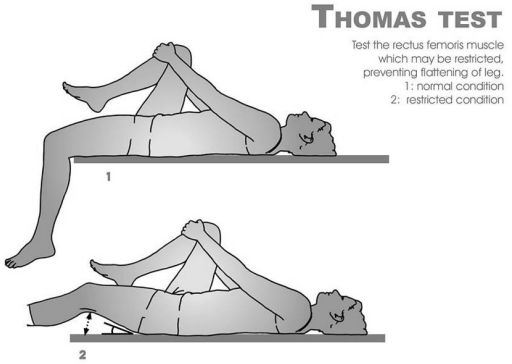
4 Diagnosis & Classification

Special tests

Test	Technique	Positive →
FADIR	Flex 90°, **adduct + IR**	Groin pain → **labral tear / FAI**
FABER (Patrick)	Flex 90°, **abduct + ER**	Hip/back pain, ↓ ROM → intra-articular, **iliopsoas**, **SI joint** (posterior pain)
Log roll	Passive max IR/ER supine	Click → **labral tear** · ↑ total ROM → **capsular laxity**
Thomas	Supine, fully flex one hip	Contralateral thigh lifts → **fixed flexion deformity**
Modified Thomas (TO)	Same at table edge, leg hangs	Measures peak hip extension angle in all
Ober	Lateral, affected side up, hip slightly extended, abduct then drop	Cannot adduct → **tight ITB**
Stinchfield	Supine, knee straight, resist hip flexion at **30–45°**	Pain → **intra-articular** pathology
Staheli (TO)	**Prone** at table edge, hand on pelvis, extend thigh until pelvis rises	**Hip flexion contracture** · measure thigh–horizontal angle
Ely (TO)	Prone, flex knee to buttock	Pelvis/hip flexes → **short rectus femoris**

Staheli conflict (TO) one version: ✓ **psoas contracture** (psoas mainly causes hip flexion contracture) · FNB 2020 version without psoas option: ✓ **rectus femoris contracture** – read the options
✗ Wrong options: quadriceps · hamstring · gastrocnemius

– – Active space – –



Thomas test

Clinical tests – Allis, Trendelenburg (TO)

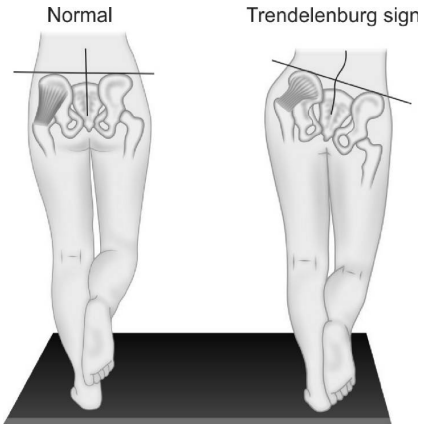
Galeazzi / Allis test — supine, **hips 45°, knees 90°**, malleoli together · normal knees level · affected knee **behind → femur short** · affected knee **lower → tibia short**

Allis not done for — ✓ **polio of lower limb**
✗ Wrong options: DDH · Perthes · SCFE

Trendelenburg test – opposite (non-tested) limb — ✓ **foot neutral off the ground, hip slight extension** (flexion may activate iliopsoas)
✗ Wrong options: plantigrade/hip neutral · hip 30–45° flexion

Positive sign — stance on weak side → **opposite pelvis dips** · patient **lurches to the weak (same) side** · bilateral = **waddling gait**

Abductor mechanism defect (TO)	Causes
Fulcrum (hip joint)	Dislocation, advanced **TB/septic** hip, **DDH**, Perthes, advanced AVN
Lever arm (head/neck)	Head/neck fracture, **coxa vara**, trochanteric fracture
Power (abductors)	**Polio**, **superior gluteal nerve** injury, TFL/ITB palsy, **L5 radiculopathy**



Trendelenburg test

Key radiology findings – MRI for occult fracture

Occult femoral neck fracture within 24 h — ✓ **MRI** (NICE 2011) · **CT** second line if MRI contraindicated / not available in 24 h (TO)
✗ Wrong options: PET-CT · repeat X-ray

Fracture signal — ✓ **dark on T1, bright on T2** (free water/haemorrhage)
✗ Wrong options: bright both · dark both · gadolinium

How it's asked

"Investigation of choice for occult femoral neck fracture within 24 h?"	→ **MRI** (TO)
"Limited MRI for occult hip fracture — signal?"	→ **dark T1, bright T2**
"Allis test is not done for?"	→ **polio** (TO)
"Trendelenburg test — position of the limb not being tested?"	→ **foot off ground, hip slight extension** (TO)
"Staheli test is done for?"	→ **psoas (hip flexion) contracture**; if psoas absent → **rectus femoris** (TO)
"Thomas test detects?"	→ **fixed flexion deformity**
"Ober test detects?"	→ **tight ITB**
"Stinchfield positive means?"	→ **intra-articular pathology**
"FADIR positive suggests?"	→ **labral tear / FAI**
"Antalgic gait?"	→ **short stance phase** on painful side

– – Active space – –

HIP RESURFACING

– Recon · 12 / 65 –

1 Basics

History & design

Early failures	large head on **polyethylene** → ↑ **volumetric wear** → osteolysis
Modern (FDA 2006, Birmingham – McMinn)	**metal-on-metal** + **large femoral head** · popular in young (less femoral resection) · cemented femoral cap



Birmingham hip resurfacing

Advantages & disadvantages

Advantages	Disadvantages
Preserves proximal femoral bone ✓ (only proven advantage)	**No modularity** – cannot adjust **length or offset**
Better stability vs 22–32 mm THA · **dislocation <1%** (0.75% at 3 yr)	**Larger exposure** / soft-tissue release
Restores biomechanics · **lower LLD risk** · physiologic stress transfer	**Femoral neck fracture** (unique)
Easier revision than IM THA	↑ **metal ions**, metallosis, pseudotumour
Suits **proximal femoral deformity**	↑ AVN of head · ↑ reoperation

vs THA	✓ **preservation of proximal femoral bone** ✗ Wrong options: better compliance · more sport · more mobility · better pain relief
vs large-head MoM THA	✓ **femoral bone preservation** ✗ Wrong options: lower loosening · acetabular preservation · lower reoperation · lower wear (**same bearing**)
Disadvantages EXCEPT	✓ **higher dislocation rate** (it is lower) · true disadvantages: ↑ periprosthetic fracture, ↑ metal ions, ↑ osteonecrosis, larger incision

5 Management

– – Active space – –

Indications & contraindications

Indications (controversial) advanced arthritis + **good proximal femoral bone** · best in **young, large males with OA** · proximal femoral deformity making THA hard

TO indications ✓ **active men <60 with OA** (or post-traumatic) · male labourers who squat · prerequisite: near-normal proximal femur, **¾ of head intact** after contouring
✗ Wrong options: osteopenia · AVN >30% head · malunited trochanter

Absolute	Relative
Femoral head/neck bone-stock deficiency – **cysts** (1.5 cm)	**Coxa vara** (↑ neck fracture)
Small / bone-deficient **acetabulum**	**Significant LLD** (cannot correct)
	Women of child-bearing age (ions cross placenta; ↑ complications)
	Renal failure (cannot excrete ions)
	Osteoporosis (TO) · **ON >20–30% of head** (TO)

Contraindications EXCEPT ✓ **age <50**
✗ Wrong options: head cysts · acetabular dysplasia · coxa vara · neck bone deficiency

Osteonecrosis – conflict OB: **not a contraindication** if proximal bone stock preserved · OB (large collapse): **higher failure** → THA safer · TO: ON **>20–30%** head = contraindication

38-yr-old woman, AVN with mild collapse – true? → ✓ **female sex → ↑ complications** · Canadian series: **96.4% 5-yr survival**, failures = neck fracture + femoral loosening
✗ Wrong options: ↑ ions = well-functioning · ON absolute contraindication · ↑ LLD · <50% success (success **>90%**)

Outcomes **79–98%** success · better in young large males with **OA** than dysplasia/ON · some products withdrawn · recent RCTs: few differences vs THA

Tönnis grading (TO)

Used for ✓ **arthritis grading for hip-preservation surgery** · grade 0 none · 1 sclerosis, minimal JSN · 2 small cysts, moderate JSN · 3 large cysts, obliteration, head deformity
✗ Wrong options: infection · FAI grading · resurfacing

6 Complications

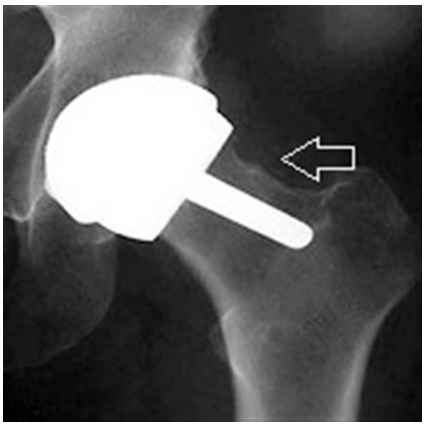
Femoral neck fracture – early reoperation

Incidence **0–4%** (~1%) · **> THA**

Timing **most common early complication** (first 3 yr) · ✓ **most common cause of revision <20 weeks / first 6 months** · unique vs THA → ✓ **transcervical neck fracture**
✗ Wrong options: abductor rupture · dislocation · HO · stiffness · loosening · infection · pseudotumour

– – Active space – –

Mechanism	**osteonecrosis** of retained head · vertical line neck → lesser trochanter · groin pain
Risk factors	**femoral neck notching** (superior) · **varus positioning** ✓ · **osteoporotic** bone · large preexisting **AVN / cystic bone loss** · **incomplete seating** · neck impingement from malaligned cup (e.g. **excessive retroversion**) · **female** · older age · obesity ✗ Wrong options: sclerotic
Not a risk factor	✓ **excessive acetabular inclination** (EXCEPT question)
Prevention	femoral component in **slight valgus** (~10° relative valgus ↑ load to fracture) · avoid notching
Treatment	**convert to primary THA** · **cerclage above lesser trochanter** before stem insertion



Femoral neck notching

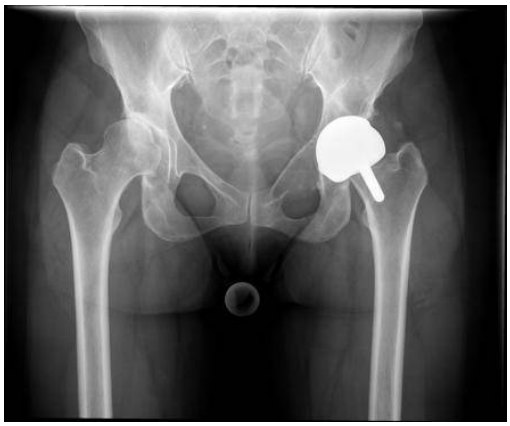
Metal ions, pseudotumour & ARMD

Metallosis	↑ **cobalt, chromium**, Co:Cr ratio in blood + urine · mostly **edge loading** · cup in **less abduction** than standard THA
Causes of metallosis	✓ **edge loading** · malposition · third-body wear · impingement · cobalt sensitivity ✗ Wrong options: activity-related wear · psoas tendinitis · PJI · stress fracture
Mimics infection	synovial WBC falsely ↑ by metal debris/corrosion → ✓ **manual cell count** next · also MARS MRI, ion levels, repeat aspiration · PJI + metallosis can coexist ✗ Wrong options: 2-stage revision · isolated cup/stem revision · arthrodesis
Pseudotumour	solid/fluid periprosthetic mass (TO) · pain, swelling, DVT, abductor destruction · risk: ✓ **women ≤40**, **small components**, **dysplasia** · may be asymptomatic · symptomatic → **revision to non-MoM** bearing ✗ Wrong options: large components · men ≥40 · primary OA
Pseudotumour revision	**poor prognosis** ✓ · soft-tissue/nerve destruction, high re-revision ✗ Wrong options: neck fracture · ON · femoral loosening (revise easily to stem)

Revision with pseudotumour, abductors intact, cup well placed → ✓ revise femoral component to stem + non-MoM bearing
✗ Wrong options: I&D + spacer · PE liner exchange (no PE) · MoM THA · constrained liner

ALVAL	**aseptic lymphocyte-dominant** vasculitis-associated lesion ✗ Wrong options: "macrophage-dominant"

– – Active space – –



Pseudotumour around resurfacing

Other complications

Complications EXCEPT	✓ **polyethylene debris** (MoM – no PE) · true: DVT, neck fracture, aseptic loosening, dislocation
Aseptic loosening	early loosening of **cemented femoral component** → convert to THA
Heterotopic ossification	↑ **vs THA** (wider exposure)
Hemi-resurfacing	acetabulum not resurfaced → **acetabular erosion / groin pain**

X vs Y

	Hip resurfacing (MoM)	THA (CoP / MoP)
Femoral bone	**Preserved**	Resected
Dislocation	**<1%**	Higher
Early reoperation	**Neck fracture**	Dislocation / infection
Unique risks	Neck fracture, **metal ions, pseudotumour, ALVAL**	PE wear (MoP)
Length/offset	**Fixed**	Adjustable
Best patient	**Young large male, OA**	Older, women, poor bone

How it's asked

"Indication for hip resurfacing?"	→ **active man <60 with OA** (TO)
"Contraindications EXCEPT?"	→ **age <50**
"Disadvantages of resurfacing EXCEPT?"	→ **higher dislocation rate**
"Advantage vs THA?"	→ **proximal femoral bone preservation**
"Advantage vs large-head MoM THA?"	→ **femoral bone preservation**
"38-yr-old woman with AVN considering resurfacing – true?"	→ **female sex ↑ complications**
"Most common cause of revision <20 weeks?"	→ **periprosthetic (femoral neck) fracture**

– – Active space – –

"Neck fracture 2 months after resurfacing — risk factor?"	→ **varus positioning**
"Uniquely at risk vs father's CoP THA?"	→ **transcervical femoral neck fracture**
"Neck fracture risk EXCEPT?"	→ **excessive acetabular inclination**
"Masses around trochanter on MRI after resurfacing?"	→ **pseudotumour** (TO)
"Complications EXCEPT?"	→ **polyethylene debris**
"Poor prognosis after revising failed resurfacing?"	→ **pseudotumour**
"High risk of pseudotumour?"	→ **women ≤40**
"Synovial WBC 15,000, ↑ Co/Cr — next step?"	→ **manual cell count**
"Runner with ↑ Co/Cr after BHR — cause?"	→ **edge loading**
"Pseudotumour, abductors intact — surgery?"	→ **revise femoral component, non-MoM bearing**
"Tönnis grading is used for?"	→ **arthritis grading before hip preservation** (TO)

IDIOPATHIC TRANSIENT OSTEOPOROSIS OF THE HIP (ITOH)

– Recon · 13 / 65 –

1 Basics

Epidemiology

Incidence	rare · **men > women (3 : 1)** · rare in Asians
Two groups	**middle-aged men (40–55)** (OB: sixth decade) · **women in 3rd trimester of pregnancy** · also non-pregnant women (TO)
Laterality	**unilateral** · may **recur**

Pathogenesis

Mechanism	local **hyperaemia** + impaired **venous return → marrow oedema → ↑** intramedullary pressure
Synonyms	bone marrow oedema syndrome · regional migratory osteoporosis · migratory osteolysis

3 Presentation

Symptoms	progressive **atraumatic hip/groin pain** over weeks · sudden onset in some (TO) · may not bear weight · pain with WB + rest
Exam	local tenderness · ✓ **hip ROM often normal** (preserved except at extremes) · antalgic gait

Genes & lab markers

ESR	**elevated** (Orthobullets) · OB stem: WBC, ESR, aspiration **normal** → still ITOH

4 Diagnosis & Classification

Key radiology findings

Radiographs (AP + lateral)	lag clinical signs **4–8 wk** · **diffuse osteopenia of head + neck** · subchondral cortical loss · effusion · **joint space always preserved** · may be normal early
MRI (modality of choice)	**diffuse marrow oedema of head + neck** · **T1 ↓ signal** (loss of fatty marrow) · **T2 / STIR ↑ signal** · effusion · **no focal subchondral defect**, no double line
Bone scan	**homogeneous ↑ uptake** of head + neck · **precedes X-ray change**
Mimic	often mistaken for **metastasis** (no focal mass on MRI) · early subchondral bone formation favours ITOH (not seen in ON)

– – Active space – –



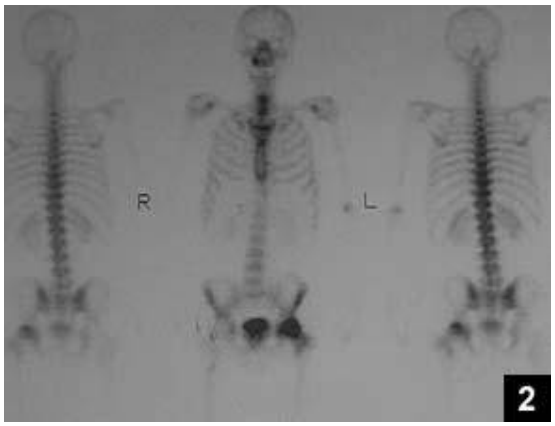
ITOH – T2 marrow oedema

Diagnosis of exclusion

Exclude	femoral neck **stress fracture** · **infection** · **malignancy** · **AVN**
Subchondral insufficiency fracture (TO)	middle-aged woman, sudden pain · widespread head + neck oedema **plus thin curvilinear subchondral T1 line** · no serpiginous AVN line · **protected WB + calcium** → settles

ITOH vs avascular necrosis

	ITOH	AVN
Sex	**Male 3:1**	**Equal** (ITOH table) · male > female (AVN note)
Age	**3rd-trimester women, middle-aged men**	**20–40** (ITOH table) · 35–50 (AVN note)
Laterality	**Unilateral**	**Bilateral >50%** (80% per AVN note)
Risk factors	**Pregnancy**	**Alcohol, steroids**
Symptoms	Acute, improves with PWB	Insidious, progressive
Exam	Preserved except extremes	**Marked ↓ ROM, esp. IR**
Radiographs	**Diffuse osteopenia** head + neck at 4–6 wk	**Localized sclerosis, crescent, collapse**
Bone scan	**Homogeneous** ↑ head + neck	**Localized**
MRI	**Diffuse oedema, no focal/subchondral defect** · ↓ T1, ↑ T2	**Focal anterosuperior subchondral defect**, double line · ↓ T1 + T2 band
Treatment	**Protected WB, symptomatic**	Early surgery
Prognosis	**Self-limiting**	Progressive



ITOH — bone scan

5 Management

Diagnosis & protected weight bearing

First line	✓ **protected / partial / toe-touch weight bearing** + supportive care + **NSAIDs**, mild analgesics → prevents **stress fracture**
Natural history	**resolves spontaneously in 6–8 months**
Avoid	core decompression · vascularized fibula · intraosseous steroid (that is for simple bone cysts) · oral steroids · hemiarthroplasty · THA · PAO · biopsy · pinning · resection/megaprosthesis (malignancy) · radiation · antibiotics (osteomyelitis)

X vs Y

	ITOH	Femoral neck stress fracture
MRI	**Diffuse** head + neck oedema	**Fracture line** with oedema
Treatment	Protected WB	NWB / fixation by site

How it's asked

"37-yr-old woman, sudden groin pain, MRI marrow oedema, no collapse?"	→ **toe-touch WB + supportive care** (TO)
"58-yr-old man, diffuse STIR signal through head and neck?"	→ **protected WB + NSAIDs** ✗ Wrong options: core decompression · fibula · hemi/THA
"Normal labs + aspirate, diffuse ↓ T1 / ↑ T2?"	→ **protected WB + observation** ✗ Wrong options: biopsy · pinning
"63-yr-old woman, normal X-ray/CT, marrow replaced on T1?"	→ **partial WB + observation** ✗ Wrong options: megaprosthesis · radiation · antibiotics · core decompression

– – Active space – –

"Pregnant (30 weeks), groin pain, ITOH – true?"	→ **hip ROM often normal** ✗ Wrong options: bilateral · needs surgery after birth · focal subchondral defects · late joint degeneration
"Time to resolution?"	→ **6–8 months**
"Imaging of choice?"	→ **MRI**
"Joint space in ITOH?"	→ **always preserved**

– – Active space – –

KNEE BIOMECHANICS

– Recon · 14 / 65 –

1 Basics

Joint type (TO)

Knee ✓ **ginglymus (hinge)** synovial joint · two joints: **tibiofemoral + patellofemoral**
✗ Wrong options: condyloid · saddle · pivot

Synovial type	Example
Ball & socket	Hip, glenohumeral
Hinge (ginglymus)	**Elbow, knee, IP joints**
Saddle	Thumb CMC
Plane	ACJ
Pivot	Atlantoaxial, proximal radioulnar
Condyloid (ellipsoid)	MCP, wrist

Fibrous sutures, gomphoses, syndesmoses (middle radioulnar/tibiofibular) · **cartilaginous** – synchondroses (physis, 1st rib), symphyses (pubis, disc)

Patellofemoral mechanics

Function transmits quadriceps tension to patellar tendon · **↑ extensor lever arm** · **patellectomy ↓ extension force 30%**

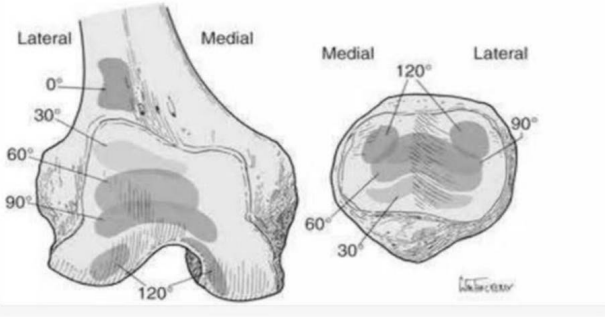
Lever arm (TO) ✓ **greatest at 20–30° flexion → quadriceps force needed is minimum** · force ↑ markedly in **last 20° of extension** (less patella in trochlea)

Motion sliding · patella moves **7 cm caudally** in full flexion · **maximum PF contact at 45° flexion**

PFJRF	Orthobullets	TO
Walking	–	**0.5–1 × BW**
Vertical jump	–	**2.4–4.6 × BW**
Stairs	**Descending 2–3 × BW**	**3.5–6 × BW** (descent > ascent)
Running	–	**4.5–7 × BW**
Squat	**up to 7 × BW**	**7 ×**; **deep squat up to 20 × BW**

TO ascending order ✓ **walking < jumping < stair ascent < stair descent < running < deep squat** · PFJRF **↑ with ↑ flexion**
✗ Wrong options: more going up stairs · less with more flexion

– – Active space – –



Extensor lever arm vs flexion angle

Patellar stability

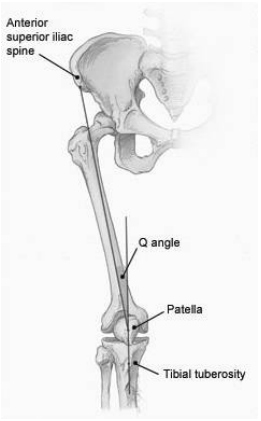
Restraint to lateral translation	Share
MPFL – primary at **20° flexion**	**60%**
Medial patellomeniscal ligament	**13%**
Medial retinaculum	**10%**
Dynamic	Quadriceps

Q angle

Definition ASIS → centre of patella → tibial tuberosity

Normal Orthobullets: extension **male 13°, female 18°**, flexion **8°** · TO: **male 14°, female 17°** (standing)

↑ Q angle (TO) ✓ **↑ femoral anteversion** · **external tibial torsion** · **laterally displaced tubercle** · **genu valgum** · ↑ lateral pull → PF instability
✗ Wrong options: medialized tubercle · internal tibial torsion · genu varum



Q angle

Tibiofemoral mechanics

TFJRF **3 × BW walking · 4 × BW climbing**

ROM **3° hyperextension → 155° flexion** · limited by thigh–calf contact · **normal gait needs 0–70°**

2 Mechanism

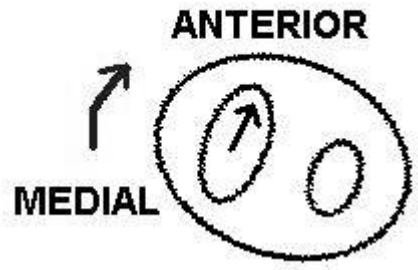
– – Active space – –

Knee kinematics – rollback, rotation

Instant centre of rotation	contact point · **polycentric**
Medial pivot	medial femoral contact **relatively stationary** · lateral contact **moves posteriorly** with flexion (**rollback**) → avoids impingement, ↑ flexion · ✓ **medial condyle translates much less than lateral** ✗ Wrong options: equal · lateral less
Beyond 120°	**both** condyles roll back
Tibial rotation with flexion	✓ **tibia internally rotates** · initial IR in first **~20°**, maintained, then **more IR past 90°** · **tibia ER with extension** ✗ Wrong options: femur IR
Paradoxical motion	✓ **contact point moves anteriorly during flexion** (ACL-deficient CR TKA fails to roll back)

Screw-home mechanism

Orthobullets	**tibia ER 5° in last 15° of extension** · cause: **medial tibial plateau longer** than lateral · locks knee → ↓ quadriceps work standing
TO	**open chain** (foot in air): tibia **ER ~10° in last 30°** of extension · **closed chain** (foot planted): **femur IR ~10°** · factors: **medial femoral condyle longer** (30° lateral curve → 10–15° tibial ER in last 20°) + quadriceps lateral pull tensions ACL
Unlocking	**popliteus** · open chain: ✓ **tibia IR** to allow flexion · closed chain: ✓ **femur ER** during flexion
TO answers	✓ "tibia must IR slightly to flex in open chain" · ✓ "with flexion normal tibia rotates internally" · ✓ "foot plantigrade – femur ER during flexion" ✗ Wrong options: tibia IR in last 30° of extension · "brought by cruciates" (mainly bony anatomy)



Screw-home mechanism

– – Active space – –

Ligament restraints

Load	Primary restraint
Varus	**LCL**
Valgus	**Superficial MCL**
Anterior translation	**ACL** – AM bundle **tight in flexion**, PL bundle **tight in extension** · also axial rotation
Posterior translation	**PCL** – AL bundle **tight in flexion**, PM **tight in extension**
External rotation	**Posterolateral corner**
Internal rotation	**ALL + Kaplan fibres** (TO)

ACL origin **posteromedial lateral femoral condyle** (semicircular) → insertion **anterior to / between tibial eminences**

PCL origin **anterolateral medial femoral condyle** → **tibial sulcus** below articular surface

Kaplan fibres (TO) deep ITB → distal femur · proximal bundle **53.6 mm**, distal **31.4 mm** above lateral epicondyle (distal near **superior lateral genicular artery**) · ✓ **restrain internal rotation** · sectioning ↑ IR at **15–90°** (esp. 60–90°) vs ALL (15–30°) · **co-dominant with ALL** in anterolateral stability · control pivot shift
✗ Wrong options: ER · anteromedial stability · recurvatum

Injury mechanism – valgus collision (TO)

Foot planted, hit on lateral distal leg → ✓ forced flexion + valgus → ACL tear
✗ Wrong options: isolated valgus MCL · varus LCL

ACL **non-contact ~70%** · flexion-valgus + femoral adduction + tibial ER + ankle eversion (cutting) · stiff landing (<20° flexion) · deceleration · hyperextension · risk: poor core, weak hip abduction, ↑ valgus, heel-strike landing

MCL **direct blow to lateral knee (valgus)** – commonest

PCL **dashboard** (posterior force on flexed tibia) · **fall on flexed knee with foot plantarflexed** · hyperflexion · hyperextension (± ACL)

Loading & rehabilitation forces

Landing (basketball) ✓ **females → greater total valgus knee loading**

Anterior tibial translation ✓ **terminal non-weight-bearing (open-chain) extension 60° → 0°** → avoid early after ACLR
✗ Wrong options: WB extension/flexion · NWB flexion

Terminal extension – VMO activity (TO)

Open chain, terminal extension (15–20°) + tibia medially rotated → ✓ VMO maximal
✗ Wrong options: VL · VI · RF

Closed chain **squats with external rotation** → max VMO

How it's asked

"Knee joint is an example of?"	→ **ginglymus** (TO)
"True about screw-home mechanism?"	→ **tibia must internally rotate to flex in open chain** (TO)
"Screw home — with flexion the normal tibia?"	→ **rotates internally** (TO)
"Foot plantigrade — during flexion?"	→ **femur rotates externally** (TO)
"Normal tibiofemoral kinematics?"	→ **tibia internally rotates with flexion**
"Paradoxical motion?"	→ **contact point moves anteriorly with flexion**
"Tibial rotation through flexion?"	→ **initial IR, maintained, more IR in deep flexion**
"Medial vs lateral condyle translation?"	→ **medial translates much less**
"Kaplan fibres restrain?"	→ **internal rotation** (TO)
"Gender and landing?"	→ **females — greater valgus loading**
"Exercise causing anterior tibial translation?"	→ **terminal NWB knee extension**
"Foot planted, hit on lateral distal leg?"	→ **flexion + valgus → ACL tear** (TO)
"Patellofemoral — at 20–30° flexion?"	→ **lever arm greatest, quadriceps force minimum** (TO)
"Q angle increased by?"	→ **femoral anteversion** (TO)
"Ascending PFJRF?"	→ **walking < jumping < stair ascent < descent < running < deep squat** (TO)
"Terminal extension + tibial IR — max activity?"	→ **VMO** (TO)
"Primary restraint to lateral patellar translation at 20°?"	→ **MPFL (60%)**
"Patellectomy reduces extension force by?"	→ **30%**

– – Active space – –

KNEE OSTEOARTHRITIS

– Recon · 15 / 65 –

1 Basics

Epidemiology

Incidence (symptomatic)	**knee 240 / 100,000 / yr** · hip 88 / 100,000
Modifiable	**articular trauma** · occupation (repetitive bending) · muscle weakness · **large body mass** · **metabolic syndrome**
Non-modifiable	**female** > male · age · genetics · race
Metabolic syndrome	**central obesity**, **dyslipidaemia** (↑ TG, ↑ LDL), **hypertension**, **↑ fasting glucose** · ✓ **PVD is NOT part** · strong link to **knee** (not hip) OA · **BMI most predictive** · no CRP link

Disparities in arthroplasty

Race	✓ **African-American males have lower TJA rates than white or Hispanic males** · Hispanics < whites · not explained by symptom perception

Pathophysiology

Cartilage	**↑ water**, ↑ **permeability**, ↓ proteoglycan (GAG), disorganized + ↓ collagen, ↑ **chondrocyte activity** incl. ✓ **↑ aggrecanase production** ✗ Wrong options: ↑ GAG · ↓ permeability · ↓ IL-6 · ↓ water movement
Synovium	mild → hypervascular → thick + vascular
Bone	lytic lesions with **sclerotic edges** (vs RA) · cysts late · subchondral marrow oedema with **~97% ↑ intraosseous pressure**
Osteophytes	pathologic **endochondral ossification** via ✓ **Indian hedgehog (Ihh)** · Ihh inactive in normal adult cartilage ✗ Wrong options: PPARγ (adipocytes) · RANKL · OPG · sclerostin (Wnt)
Histology	loss of superficial chondrocytes · **tidemark** replication + breakdown · fissuring · eburnation

Genes & lab markers

MMPs	stromelysin, plasmin, **aggrecanase-1 (ADAMTS-4)** · **TIMPs** inhibit · **MMP/TIMP imbalance**
Cytokines	**IL-1** (catabolic), IL-6, **TNF-α**
Genetics	**non-Mendelian** · vitamin D receptor, oestrogen receptor 1, IL-1, IL-4, matrilin-3, BMP-2/5
DMOADs (TO)	monoclonal antibodies against **ADAMTS (aggrecanase, ADAMTS-5 > 4)**, **nerve growth factor (NGF)**, **VEGF** · ✓ **incorrect: "target fibroblast growth factor"** · anti-NGF: **tanezumab, fasinumab**, fulranumab (stopped 2016) · IV + SC routes (not oral) · AE: **abnormal peripheral sensation** (paraesthesia)

3 Presentation

– – Active space – –

Symptoms	function-limiting pain (walking distance) · night/rest pain · activity swelling · stiffness · instability, locking, catching
Exam	habitus · **limb alignment** · effusion · **flexion contracture >5°** · **flexion <110°** · ligaments

Biomechanics – adduction moment & gait

Knee adduction moment	surrogate of **medial tibiofemoral contact force** · ✓ **↑ in medial compartment OA** (varus – GRF passes medial to knee) · large peak → ↑ pain + progression · ↓ by **offloader braces** · ↑ flexion moment also ↑ medial load
Not linked to ↑ adductor moment	cervical myelopathy · hip OA (slow, leans forward, pelvis drops) · ankle OA (↑ double support) · scoliosis
Antalgic gait in knee OA	knee held in **slight flexion** (quadriceps avoidance, ↓ capsular tension) · **toe walking** · ↓ stride · ↓ stance time · ✓ **not seen: walking predominantly on the heel**

4 Diagnosis & Classification

Key radiology findings

Views	**weight-bearing** AP · optional **sunrise** + **PA 30° flexion (Rosenberg)**
Findings	JSN · osteophytes · eburnation · subchondral sclerosis · cysts · pattern (medial, lateral, PF)



Rosenberg / notch view

Kellgren & Lawrence (AP weight-bearing)

Grade	Findings
0	No JSN or reactive change
1	**Doubtful** JSN + possible osteophytic lipping
2	**Definite osteophytes** + possible JSN
3	**Moderate multiple osteophytes + definite JSN** + some sclerosis + possible bone-end deformity
4	**Large osteophytes + marked JSN + severe sclerosis + definite deformity**

– – Active space – –

Original coal miners, **1952** · grade ≤2 none/possible OA, >2 definite



Kellgren–Lawrence grade 3

Loose bodies & radiographic signs (TO)

Most common source of loose body in knee
✓ **osteophytes (OA)** · young → **OCD** · multiple → **synovial chondromatosis**
✗ Wrong options: damaged menisci

Grand piano sign
✓ **knee**: anterior femoral resection surface in TKA → marker of **correct femoral rotation**



Grand piano sign

5 Management

Symptomatic knee OA

- All → **exercise + education + oral/topical NSAIDs (strong)**
- BMI >25 → weight loss · medial OA → **unloader brace (moderate)**
- Persisting → **IA steroid (moderate) · RFA (genicular) · PRP > HA**
- Young, unicompartmental + malalignment → **HTO (varus) / DFO (valgus)**
- Failed nonop, end-stage → **TKA (UKA if isolated compartment)**

AAOS guidelines – nonoperative management

Strength	2021/2022 CPG (OB)
Strong FOR	**Exercise** (supervised, self-managed, aquatic) · **patient education** · **oral + topical NSAIDs** · **oral acetaminophen**
Moderate FOR	**Weight loss** (BMI >25) · **IA corticosteroid** · **unloader brace** · cane
Limited FOR	**PRP** · acupuncture, massage · supplements (turmeric, ginger, glucosamine, chondroitin, vitamin D) · **denervation (RFA)**
Moderate AGAINST	**IA hyaluronic acid** (NNT 17) · **arthroscopic lavage/debridement**
Strong AGAINST	**Oral narcotics incl. tramadol** · **lateral wedge insoles**

Modality	Older CPG positions
HA	2009 inconclusive → **2013 strong against** (weakest evidence) → 2021 moderate against
IA steroid	2013 **inconclusive** (for/against) → 2021 moderate for
Tramadol	2013 **recommended** → 2021 **strong against**
2008 CPG	**Low-impact aerobic exercise "A"** · glucosamine/chondroitin "A" against · arthroscopic debridement without mechanical symptoms "A" against · **lateral heel wedge "B" against** · valgus brace inconclusive
Orthobullets list	Acupuncture strong against · glucosamine/chondroitin strong against · viscosupplement moderate against · needle lavage moderate against · lateral wedge moderate against · **arthroscopic debridement/lavage strong against** · meniscal debridement inconclusive
Acetaminophen	Max dose **3000 mg/24 h** (was 4000)

Not recommended (TO)
✓ **HA injection**
✗ Wrong options: NSAIDs · weight loss · self-management + exercise

Exercise
supervised + home best · **benefit lost 6 months after stopping**

Physio (TO)
quadriceps strengthening (static + dynamic) · hamstring conditioning · avoid **squatting, stairs** · weight reduction ↓ joint load

Medial OA brace
✓ **medial unloader (valgus)** for correctable varus
✗ Wrong options: varus brace · Cho-Pat (patellar tendon) · ACL brace · sleeve

Injections & orthobiologics

IA steroid
moderate for · short-term relief · **avoid repeat soon** (last 3 weeks ago, 10-day relief → discuss TKA)

IA steroid (TO)
gold standard for early pain relief · relief **4–6 weeks** · **max 3 injections in 6 months** (early OA, up to KL 3) · anti-inflammatory, **not disease-modifying** (neither regenerates nor degrades cartilage)

– – Active space – –

Viscosupplementation – hyaluronic acid (TO)

HA — **anionic, non-sulfated glycosaminoglycan** · synthesized by **plasma membrane enzymes** · major ECM component · viscoelasticity (TO)

Synovial HA (TO)	Concentration	Change
Normal knee	**2.5–4.0 mg/mL**	Normal viscosity
OA knee	**↓ 33–50%**	↓ molecular size → ↓ viscosity + elasticity · **fluid volume ↑ but quality ↓**

Mechanisms (TO) — **viscoinduction** (chondrocytes + synoviocytes make native HA) · **chondroprotection** · **viscosupplementation** (lubrication, cushioning) · **binds CD44 → ↓ IL-1β + MMP**

Anti-inflammatory effect — none (current chapter) · possible, via CD44 → ↓ IL-1β (TO) – read the options

Indication (TO) — early–moderate OA (**KL II–III**) · unresponsive to other therapy or unfit for surgery

Evidence (TO) — AAOS 2013 **strong against** · network meta-analysis (Campbell et al. 2015) → HA **most effective** vs NSAIDs, steroid, PRP, placebo in early OA, **26–32 weeks** · average relief **24–26 weeks**

Safety (TO) — minor arthralgia / effusion **0–14%** · **repeat injections safe**

HA brand (TO)	Injections	Source
Synvisc / Synvisc-One	3 / **1 (6 mL)**	**Avian**
Supartz · Hyalgan	3–5	**Avian**
Euflexxa · Orthovisc	3 · 3–4	**Bacterial**
Durolane	**1 (3 mL)**	**Bacterial**

HA — rooster-comb preparations → **contraindicated with chicken/egg allergy** ✓ · also joint/skin infection · no anti-inflammatory effect

PRP — ✓ **better clinical outcomes than HA** (**level 1** RCTs, meta-analyses) · best in **KL 2–3** (not KL 4) · **leukocyte-poor > leukocyte-rich** (more effective, safer) · injection site irrelevant · AAOS **limited** (downgraded)

Orthobiologics – AAOS definition (TO) — biologic substances naturally found in the body that enhance repair / regeneration of bone, cartilage, tendon, ligament · PRP, **autologous conditioned serum**, BMAC, adipose-derived stem cells, amniotic + cord blood concentrates, prolotherapy, ozone

PRP mechanism (TO) — supraphysiological platelets → cytokines + growth factors → chemotaxis of chondrocytes + MSCs, ↑ proteoglycan · **inhibits MMP-13 + NF-κB**

PRP preparation (TO) — **differential centrifugation** → top **PRP** · middle **buffy coat (WBC)** · bottom RBC · platelets **200–1000 × 10³/µL** (no fixed ideal) · **LP-PRP yellow** · **LR-PRP red** (↑ inflammation)

PRP evidence (TO) — Mayock 2016: **better at 12 months, not at 6** · AAOS 2013 **inconclusive** · may delay surgery, **not disease-modifying**

BMAC (TO) — **iliac crest = best harvest site** (also distal femur, proximal tibia, humerus, calcaneus) · Shapiro: **no difference vs saline at 12 months** · safe, needs RCTs

Adipose-derived MSCs (TO) — **lipoaspiration** (abdomen / flank / buttock) or arthroscopic **infrapatellar fat pad** · **stromal vascular fraction (SVF)** · microfragmentation or enzymatic digestion · phase I safe · **no FDA approval**, no RCTs

Therapy (TO)	Evidence	Safety
Steroid	Short-term relief	High
HA	Moderate – early OA	High
PRP	Moderate – early OA	Moderate
BMAC	Low, under research	High
ADMSC	Very low, experimental	High

BMAC — ✓ **highest IL-1 receptor antagonist** concentration (> LP-PRP, LR-PRP, HA)

Genicular RFA — ablates **superior lateral, superior medial, inferior medial** genicular nerves · ✓ **superior to corticosteroid** (RCTs) and to HA · safe · OK in **diabetes**, renal, CV, GI risk

Glucosamine — ✓ **substrate for proteoglycan** (chondroitin sulfate + hyaluronate)
✗ Wrong options: COX-2 · viscosity · pain mediator · anti-apoptosis

Arthroscopy

Lavage/debridement vs placebo — ✓ **no significant benefit** in chronic OA (1 wk–2 yr) · only for **mechanical** symptoms (meniscal tear, loose body)

Surgical options – osteotomy, TKA

HTO — young, **medial** unicompartmental OA · AAOS limited evidence · ideal: **thin, active, stable, <60 → ✓ outcomes approach TKA**
✗ Wrong options: HTO faster recovery · more complete relief · better in old

DFO — 25-yr-old, **valgus + lateral JSN → ✓ varus-producing DFO** · most delay TKA ≥10 yr
✗ Wrong options: varus HTO (joint-line obliquity) · lateral UKA · TKA · meniscectomy

UKA — isolated unicompartmental disease · **TKA has lower revision rates than UKA** · not tricompartmental

TKA — symptomatic OA + failed nonop · 70-yr-old with meniscal tear + tricompartmental change → ✓ **TKA** (✗ Wrong options: meniscectomy · HTO · UKA) · 62-yr-old, diffuse pain, no mechanical symptoms → ✓ **arthroplasty** (✗ Wrong options: PRP · meniscectomy · chondroplasty · allopurinol)

TKA technique facts — **CR = PS outcomes** · **patellar resurfacing**: no difference in pain/function, ↓ **reoperation** · **drains not recommended** · **sensor-balancing device**: ✓ **no added benefit** in ROM/PROMs at 2 yr, ↑ time + cost

– – Active space – –

X vs Y

	OA cartilage	Ageing cartilage
Water	↑	↓
Chondrocyte activity / aggrecanases	↑	Unchanged
Collagen	↓, disorganized	Normal

	PRP	HA	BMAC
AAOS 2021	Limited for	**Moderate against**	–
Evidence	**Level 1 > HA**	Rooster comb (egg allergy)	**Highest IL-1ra**

How it's asked

"Not recommended by AAOS for knee OA?"	→ **HA injection** (TO)
"Weakest evidence (2013 CPG)?"	→ **IA hyaluronic acid**
"Arthroscopic lavage vs placebo?"	→ **no significant benefit**
"48-yr-old, early medial OA, normal X-ray – initial treatment?"	→ **low-impact aerobic exercise**
"61-yr-old, BMI 31, early medial OA?"	→ **weight loss + low-impact aerobic exercise**
"Least evidence (2021)?"	→ **lateral wedge insoles**
"Strong evidence (2021)?"	→ **supervised exercise programme** ✗ Wrong options: tramadol · HA
"Most strongly recommended (5° varus, moderate–severe OA)?"	→ **NSAIDs**
"BMI 38, failed glucosamine + home exercise – strongly recommended next?"	→ **NSAIDs**
"Recommended and strongly supported (2022)?"	→ **exercise**
"Recommended AGAINST (2021)?"	→ **arthroscopic lavage + debridement**
"AAOS on HA?"	→ **moderate recommendation against**
"Incorrect about DMOADs?"	→ **target fibroblast growth factor** (TO)
"Glucosamine works as?"	→ **substrate for proteoglycan**
"Systemic reaction to HA – also allergic to?"	→ **chicken or eggs**
"Highest IL-1ra?"	→ **BMAC**
"Hyaluronic acid is a?"	→ **anionic non-sulfated glycosaminoglycan** (TO)
"Best harvest site for BMAC?"	→ **iliac crest** (TO)
"Maximum IA steroid injections in knee OA?"	→ **3 in 6 months** (TO)
"Highest level of evidence PRP > HA?"	→ **level 1**

"RFA – true?"	→ **superior to corticosteroid**
"PRP vs HA – true?"	→ **PRP better clinical outcomes**
"Metabolic syndrome EXCEPT?"	→ **PVD**
"Osteophyte signalling molecule?"	→ **Indian hedgehog**
"OA pathophysiology?"	→ **↑ chondrocyte aggrecanases**
"Antalgic gait not typical?"	→ **heel walking with knee OA**
"↑ adductor moment associated with?"	→ **medial compartment knee OA**
"Medial OA biomechanical change?"	→ **↑ knee adduction moment**
"25-yr-old, valgus, lateral narrowing?"	→ **distal femoral osteotomy**
"HTO vs TKA?"	→ **HTO in thin, active, young approaches TKA**
"70-yr-old, meniscal tear + tricompartmental OA?"	→ **TKA**
"Correctable varus, medial OA – brace?"	→ **medial unloader**
"Failed NSAIDs, steroid 3 weeks ago lasted 10 days?"	→ **discuss TKA**
"Sensor-balancing device RCT?"	→ **no added benefit**
"KL grade 3?"	→ **moderate osteophytes + definite JSN + some sclerosis**
"Racial disparity?"	→ **African-American males lower than white/Hispanic**
"Most common source of knee loose body?"	→ **osteophytes** (TO)
"Grand piano sign?"	→ **knee – femoral rotation in TKA** (TO)

– – Active space – –

– – Active space – –

LIGAMENTS OF THE KNEE

– Recon · 16 / 65 –

1 Basics

Ligament function overview

Ligament	Primary	Secondary
ACL	Resists **anterolateral** tibial displacement	**Varus** at 0° (full extension)
PCL	Resists **posterior** tibial displacement, esp. **90°**	**Varus** at 0°
LCL	Resists **varus at 30°**	Posterolateral rotation at flexion **<~50°**
PLC / popliteofibular	Resists **posterolateral rotation**	Varus + posterior displacement
MCL	Resists **valgus**	Axial rotation (with ACL)

Rotational stabilizers (TO)

Rotation	Main stabilizer
Internal rotation 60–90°	Anterolateral structures — **ITB** most significant
Internal rotation near full extension	✓ **Posterior oblique ligament** (✗ Wrong options: LCL · popliteus/PFL · ACL+MCL)
External rotation 60–90°	**PLC — LCL, popliteus, popliteofibular** · ✓ **LCL = primary stabilizer to ER** (TO) (✗ Wrong options: ACL · PCL · MCL)

Layered anatomy

Lateral layer	Contents
1	**ITB, biceps femoris**
between 1 and 2	***Common peroneal nerve***
2	Patellar (lateral) retinaculum
3 superficial	**LCL**, fabellofibular ligament, **ALL**
between superficial and deep 3	***Lateral geniculate artery***
3 deep	Arcuate ligament, coronary ligament, **popliteus tendon, popliteofibular ligament**, capsule

– – Active space – –

Medial layer	Contents
1	**Sartorius** + fascia (medial retinaculum)
between 1 and 2	***Gracilis, semitendinosus, saphenous nerve***
2	**Superficial MCL**, **MPFL**, **semimembranosus**, POL
3	**Deep MCL**, capsule, coronary ligament

Medial quadruple complex (TO) — **MCL, semimembranosus, pes anserinus (ST, gracilis, sartorius), POL** · ✓ **not medial meniscus**

Not attached to medial femoral condyle (TO) — ✓ **popliteus** (lateral condyle, popliteal notch → posterior tibia above soleal line) · MFC attachments: MCL, MPFL, adductor magnus, PCL

ITB (TO) — **moves forward in extension, backward in flexion**, **tense in both** · in flexion **ITB, popliteus, LCL cross**; **ITB + biceps stay parallel**
✗ Wrong options: lax in flexion · ITB/LCL parallel

ACL

Anatomy — **extrasynovial, intracapsular** · origin **lateral femoral condyle** (medial wall, behind **resident's ridge**) · PL origin **posterior + distal** to AM · insertion broad, **anterior to / between intercondylar eminences** · **33 × 11 mm**

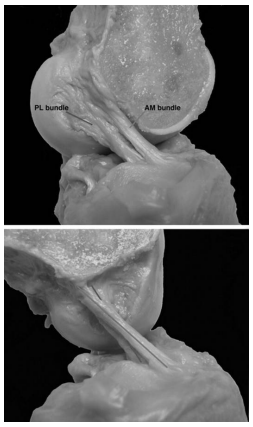
AM bundle — **tight in flexion**, loose in extension (Orthobullets/OB) · TO: **tight in both** flexion and extension · fibres parallel in extension, ER in flexion · **longer** · tibial attachment **anterior** to PL

PL bundle — **tight in extension**, loose in flexion · **prevents pivot shift** + IR near extension · shorter · isolated PL cut → ✓ **↑ translation + rotation at 30°**

Blood supply — **middle geniculate artery** · **innervation** – posterior articular branches of **tibial nerve** · mechanoreceptors (Ruffini, Pacini, GTO, free endings) → **proprioception**, quadriceps modulation

Composition — **90% type I, 10% type III** collagen

Tensile strength — **native ACL 2200 N < BPTB 3000 N < quadrupled hamstring 4000 N** · TO false statement: ✓ "native ACL > BPTB"



ACL bundles in flexion and extension

PCL

Anatomy — **extrasynovial, intracapsular** · origin **medial femoral condyle** · insertion **tibial sulcus** · **38 × 13 mm** · strength **2500 N** · middle geniculate artery
✗ Wrong options: intrasynovial

– – Active space – –

AL bundle — **shorter, thicker, stronger** · **tight in flexion** · double-bundle: tension **mid-flexion**

PM bundle — longer, thinner, weaker · **tight in extension** · tension in extension + high flexion (protects against **hyperextension**)

Ridges — **medial intercondylar ridge** = proximal border of femoral footprint · **medial bifurcate ridge** separates AL/PM

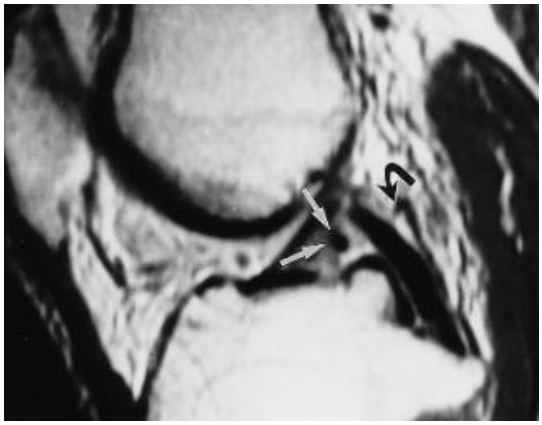
	ACL	PCL
Tight in flexion	**AM**	**AL**
Tight in extension	**PL**	**PM**

Meniscofemoral ligaments

Course — **posterior horn lateral meniscus → medial femoral condyle** (lateral wall), into PCL substance

Humphrey — **anterior** to PCL · **Wrisberg** – **posterior** to PCL (more common) · one present in **71–100%**

MRI — sagittal: **small ovoid hypointense structures** just anterior (Humphrey) / posterior (Wrisberg) to PCL · coronal: hypointense band parallel to PCL · ✓ arrow = **ligament of Humphrey**
✗ Wrong options: PCL · bucket-handle tear · ACL · intermeniscal ligament



Meniscofemoral ligaments on MRI

LCL

Anatomy — origin LFC **posterior + superior to popliteus** origin · runs **superficial to popliteus** · inserts **fibula anterior to popliteofibular ligament** · cord-like

Biomechanics — **tight in extension**, lax in flexion · **750 N**

Posterolateral corner

Function — with PCL controls **external rotation + posterior translation**

Structures — **LCL (295 N)** · **popliteus (680 N)** · **popliteofibular ligament (229 N)** · lateral capsule · variable: **arcuate ligament**, ITB, fabellofibular ligament

Dial test — ✓ **↑ ER at 30° only → PLC** · ↑ at **30° + 90° → PLC + PCL**

Posterior capsule — closest structure ✓ **popliteal artery** · superficial → deep (lateral → medial): **tibial nerve → popliteal vein → popliteal artery** · CPN posterolateral · sural subcutaneous

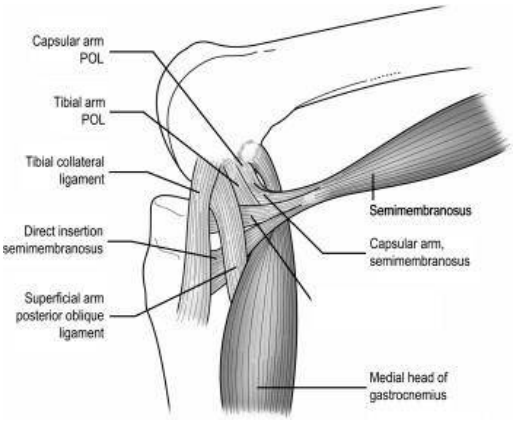
MCL

Superficial MCL (tibial collateral)	origin **3.2 mm proximal + 4.8 mm posterior** to medial epicondyle ✓ · inserts **periosteum of proximal tibia deep to pes**, ~6 cm below joint · layer 2, deep to gracilis + ST · **primary valgus restraint at all angles** · **57% at 5°, 78% at 25°** of medial stability · **4000 N**
Deep MCL (medial capsular)	separated by bursa · **meniscofemoral + meniscotibial** parts · attaches **medial meniscus** (coronary) · with posteromedial capsule = **secondary valgus restraint at full extension**

Valgus opening 5 mm at 30°, other tests normal → sMCL → ✓ origin **slightly posterior + proximal to medial epicondyle**
✗ Wrong options: layer 1–2 · Gerdy's (ITB) · popliteal hiatus (popliteus) · Schöttle point (MPFL)

Posteromedial corner – posterior oblique ligament

Components	**semimembranosus** insertion · **POL** · **oblique popliteal ligament** · posterior capsule · posterior horn medial meniscus
POL	**adductor tubercle** (just distal/posterior) → posteromedial tibia near articular margin · **posterior to MCL** · three arms (superficial, **central** – most important, capsular) · **tense near full extension**, lax with flexion
Function	✓ **resists valgus + tibial internal rotation near full extension** · secondary restraint to posterior translation if PCL injured · mostly **nonoperative**; ↑ cruciate graft strain in multiligament ✗ Wrong options: at 60° · ER · anterior translation (ACL) · varus/ER (PLC)



Posteromedial corner – POL

MPFL

Function	restrains **lateral patellar translation 0–30°** · low tension (**2–10 N**) · **isometric 0–90°**, slack >90° · fails at **200 N** (ACL **1725 N**)
Anatomy	**layer 2** · bundles: short oblique (superior pole), inferior straight · femoral: **distal to adductor tubercle, proximal to MCL** · **Schöttle's point** – **1.3 mm anterior** to posterior cortex line, **2.5 mm distal** to posterior origin of MFC, proximal to posterior Blumensaat line · patella: junction proximal–middle thirds of superomedial border
Tears	femoral > patellar · 2nd dislocation risk **13%**, 3rd after 2nd **50%**

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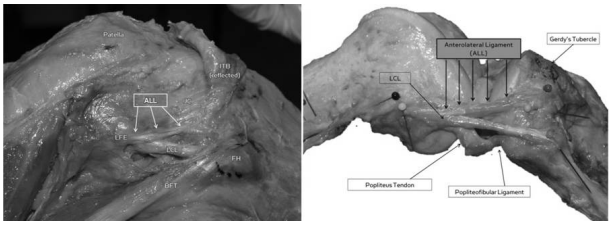
Anterolateral ligament

Anatomy — **layer 3** with LCL (TO ✓) · femoral **lateral epicondyle** → tibia **midway between Gerdy's tubercle and fibular head** · length **59 mm**, width 7 mm · attached to **lateral meniscus body** (meniscofemoral + meniscotibial) · **lateral inferior genicular vessels** between meniscus and ALL · **no ITB connection** · rotational stability

4 Diagnosis & Classification

Avulsion fractures – Segond, arcuate

Fracture	Structure	Associated
Segond	**ALL** (meniscotibial part), lateral tibial plateau	**ACL tear 75%** · **IR + varus**
Arcuate sign	**Arcuate complex** – proximal fibula (LCL/biceps on fragment)	**Cruciate injury ~90%**
Reverse Segond	**Deep MCL** capsular component (medial)	PCL
Gerdy's tubercle	ITB	–



Segond fracture

Rotatory instabilities (TO)

Instability	Disrupted	Test
Anteromedial	Medial capsule, MCL, **POL**, ACL	Anterior drawer in 15° ER · valgus laxity at 30°
Anterolateral (90°)	ALL, ITB, ACL	Pivot shift
Anterolateral (extension)	ACL (primary) · ITB/Kaplan, lateral meniscus, ALL	**Jerk test, Slocum, MacIntosh pivot shift**
Posterolateral	LCL, arcuate, popliteus, lateral gastrocnemius ± PCL	**ER-recurvatum**, **reverse pivot shift** · varus thrust + hyperextension gait
Posteromedial	MCL, medial capsule, POL, PCL, posteromedial capsule, semimembranosus ± ACL	–

How it's asked

"False about knee ligaments?" → **native ACL stronger than BPTB** (TO)

"Isolated PL bundle transection?"	→ **↑ translation + rotation at 30°**
"Bundle A anterior on tibia, tight in flexion?"	→ **AM bundle; PL loose in flexion**
"Main stabilizer of IR near full extension?"	→ **POL** (TO)
"Valgus laxity at 30°, injured structure — true?"	→ **origin slightly posterior + proximal to medial epicondyle**
"Resists internal tibial rotation in full extension?"	→ **POL**
"POL — true?"	→ **static valgus restraint in full extension**
"Structure A (posteromedial) — function?"	→ **valgus + IR near full extension**
"Primary stabilizer to ER?"	→ **LCL** (TO)
"Closest to posterior capsule?"	→ **popliteal artery**
"↑ ER at 30° only?"	→ **posterolateral corner**
"Arrow anterior to PCL on MRI?"	→ **ligament of Humphrey** (TO)
"Coronal T1 band from lateral meniscus to MFC?"	→ **meniscofemoral ligament**
"Two avulsions — player A and B?"	→ **A = ALL (Segond), B = arcuate complex**
"True about knee ligaments?"	→ **ALL in layer 3 of lateral structures** (TO) ✗ Wrong options: PCL intrasynovial · AL tight in extension · AM prevents pivot shift
"Medial quadruple complex EXCEPT?"	→ **medial meniscus** (TO)
"True about ITB?"	→ **forward in extension, backward in flexion** (TO)
"Not attached to MFC?"	→ **popliteus** (TO)
"Nerve between lateral layers 1 and 2?"	→ **common peroneal nerve**

– – Active space – –

– – Active space – –

PATELLAR CLUNK SYNDROME

– Recon · 17 / 65 –

1 Basics

Epidemiology

Prevalence	**3.5%** (mean) · up to **18% (12–21%)** of **PS TKAs** (TO) · rarer with newer designs
Timing	average **12 months** (Orthobullets) · **7–11 months** (OB) · **3–12 months**, occasionally up to 3 yr (TO) · more with patellar resurfacing
First described	**Hozack 1989** (TO)

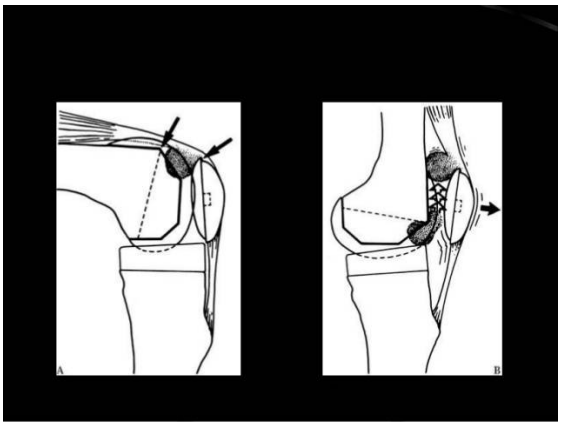
Pathophysiology

Lesion	**fibrous nodule at superior pole of patella / posterior surface of distal quadriceps tendon** (suprapatellar, articular side) ✗ Wrong options: inferior pole
Mechanism	nodule **entrapped in intercondylar box during flexion → jerked out during active extension** → painful clunk at **30–45°** (~40°)
Cause	trauma to quadriceps tendon at patellar resection · impingement of tendon on **sharp anterosuperior edge of the box** · abrupt transition ridge between box and trochlea

Risk factors

Category	Factors
Preoperative	**Patella baja** / short patellar tendon · **valgus** alignment · fibrosynovial proliferation at quadriceps insertion (resect it) · **previous knee surgery** / revision
Patellar component	**Small** button · placed **low (distal)** – exposes superior pole · **over-resection / thin button** (↓ quadriceps offset) · uncovered cut bone (resect lateral facet) · implant overhang at distal pole
Femoral component	**PS design** · **↑ posterior condylar offset** · **smaller (undersized) femur** · **flexed** femoral component (posterior IM entry point – anterior femoral bow) · **high intercondylar box ratio** (trochlea extends proximal + anterior) / **larger box**
Other	**Thick PE insert** (raises joint line → relative baja) · excessive distal femoral resection · **anterior tibial tray** · forced flexion

Intercondylar box ratio (ICBR)	box height ÷ AP height of femoral component · **high ICBR** → earlier quadriceps contact → ↑ clunk · **low ICBR** favourable
Prevention	low-ICBR design · side-specific femur (7° oblique groove) · trochlear deepening + lengthening · exact sizing · avoid downsized/flexed femur (**anterior IM entry point**) · avoid anterior tibial tray · patellar button **as proximal as possible, not beyond superior border** · avoid over-resection/ thin button · **neutral or posterior tibial tray**



Patellar clunk — suprapatellar nodule

3 Presentation

Symptoms	patellofemoral pain · **popping, catching** · may be painless
Exam	palpable **"pop"/"catch" as knee extends from flexion at ~40°** (30–45°)

4 Diagnosis & Classification

Key radiology findings

Diagnosis	mostly **clinical** · **ultrasound** shows suprapatellar fibrous tissue · **MRI** can confirm (TO)

Takahashi classification (TO)

Type	Finding
I	**Isolated suprapatellar fibrous nodule** (true PCS)
II	Impinging **hypertrophic synovitis**, no nodule
III	Nodule **+** generalized hypertrophic synovitis

Related entities

Entity	Feature
Patellar clunk	Discrete nodule · **clunk on extension 30–45°**
Patellofemoral crepitus	Fibrosynovial hyperplasia **without discrete nodule** · grinding at **30–60°** with loading · 2 months–2 yr
Patellar crunch	Scar **around the patellar component** · crunching going **extension → flexion**
Intra-articular fibrous band	Transverse band prevents patellar seating

- - Active space - -

5 Management

Patellar clunk after PS TKA

- Mild → **observe / quadriceps exercises (20–60% improve, TO)**
- Severe or failed nonop → **arthroscopic nodule resection**
- Failed arthroscopy / component problem → **open excision ± patellar revision**

Causes & management

Conflict	OB: **nonsurgical management rarely successful → surgical debridement** best ✓ · TO: **conservative quadriceps exercises help 20–60%**, so "conservative has no role, arthroscopy for all" is ✓ **incorrect** · Orthobullets: observe mild, resect severe ✗ Wrong options: rest/NSAIDs · patellectomy · patellar revision · lateral release/realignment
Arthroscopic resection	good outcomes, **very low recurrence**, ↑ Knee Society scores · recurrence rare after arthrotomy + button revision
Open excision	failed arthroscopy · planned component revision / realignment · limited lateral incision under local

Component rotation & patellar maltracking

Patellar pain/ subluxation/ dislocation	✓ **internal rotation of tibial component** (lateralizes tubercle) · also **IR of femoral component** ✗ Wrong options: lateralized tibial tray (moves tubercle medially, helps) · lateralized femur (moves trochlea laterally, helps) · femoral ER
Correct rotation	femur **parallel to epicondylar axis** at 90° · tibial tray centred on **medial third of tibial tubercle**

X vs Y

	Patellar clunk	CAM jump	Midflexion instability
Cause	Suprapatellar nodule in PS box	**Large flexion gap** in PS knee	Elevated joint line, multiradius design, MCL laxity

How it's asked

"True about PCS?"	→ **undersized femoral component ↑ risk** (TO) ✗ Wrong options: more common in CR · inferior pole · grinding in flexion
"PCS occurs?"	→ **superior pole, knee extending from flexion** (TO)
"PCS results from?"	→ **scar at superior pole impinging on femoral component during extension** (TO)

"Incorrect statement on PCS?"	→ **conservative treatment has no role** (TO)
"PS TKA, ↑ posterior offset, small patellar button — complication?"	→ **patellar clunk** ✗ Wrong options: maltracking · midflexion instability · CAM jump
"Best management (OB)?"	→ **surgical debridement**
"Clunk on extension after PS TKA — associated?"	→ **larger femoral box** ✗ Wrong options: posterior tibial tray · oversized button · proximal button · ↓ offset
"Patellar pain/dislocation from component orientation?"	→ **internal rotation of tibial component**
"Angle of clunk?"	→ **30–45° (~40°) on extension**

– – Active space – –

PROSTHETIC JOINT INFECTION

– Recon · 18 / 65 –

1 Basics

Epidemiology

Procedure	Orthobullets / OB	OB (alternative)
Primary THA	**0.3–1.3%**	1.7%
Primary TKA	**1–2%**	1–3% · 2.5%
Revision THA	**3–4%**	3–5%
Revision TKA	**5–6%** (highest)	5–10%

Order of risk ✓ **primary hip < primary knee < revision hip < revision knee** · knee higher (mobile, superficial) · most PJIs in first **1–2 yr** · aseptic revision ≪ revision for infection

Early TKA revision (<2 yr) infection, instability, stiffness · ✓ **infection > stiffness** · late (>2 yr): **aseptic loosening**, infection (> instability) · PE wear now uncommon, occurs **late**

Pathologic fracture salvage (PFR for metastasis) ✓ **deep infection = most common complication**

Risk factors

Category	Factors
Preoperative	**Active infection** (skin, deep, joint, septicaemia) · **previous surgery / prior infection** · active skin lesions (erysipelas, ulcer, folliculitis – contraindication until healed)
Immunosuppressant drugs	**Anti-TNF** (infliximab, etanercept, adalimumab, certolizumab, golimumab) · leflunomide · corticosteroids
Diabetes	**HbA1c >7** or **fructosamine ≥293 µmol/L** (Orthobullets) · OB: ✓ **HbA1c >8%** ↑ infection · fructosamine **≥292**
Other medical	CKD · acute liver failure · **malnutrition (albumin <3.5**, lymphocytes <800) · **HIV CD4 <400** · cancer/metastasis · depression
Inflammatory arthropathy	**RA** (seropositive men), psoriasis (controversial), AS
Lifestyle	**Morbid obesity (BMI >40)** · smoking · alcohol · IVDU · poor oral hygiene
TO / OB extras	**Male sex** · **BMI ≥30** · **allogeneic transfusion** · **hinged / constrained / PS** implant (vs CR) · indwelling catheter **>48 h** (UTI) · anticoagulation (haematoma) · trauma diagnosis
Lower risk	Uncemented TKA, **PFJ, UKA** vs cemented TKA · OA diagnosis · **age >65** (risk ↓ 1.2%/yr after 65 – Kaye)

Not a risk factor ✓ **asymptomatic bacteriuria** (TO) – no screening/treatment/delay needed · TO text elsewhere: a marker of risk but urine is not the source

Not associated (TO) ✓ **age >65**
✗ Wrong options: BMI ≥30 · transfusion · male

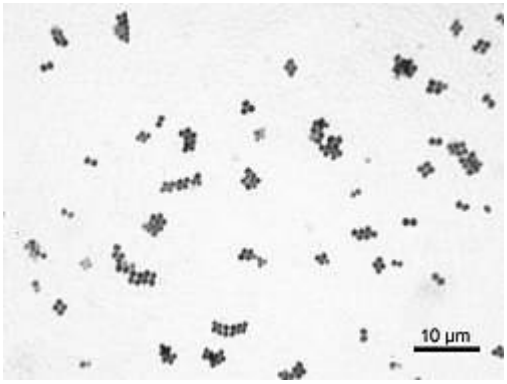
RA THA (OB) ↑ **PJI** (e.g. 16,000 WBC, 88% PMN at 6 wk) · ↑ intraop fracture, revision, dislocation, wound infection · **no ↑ mortality or DVT**

Stem-cell transplant arthroplasty ≥1 yr after (TO)

Organisms

Setting	Organism
Acute (<4 wk)	**S. aureus** ✓ (GPC in **clusters**) · β-haemolytic strep · gram-negatives
Chronic / late (>3 mo) THA	**S. epidermidis (CoNS)** ✓ · S. viridans · P. acnes
Fungal	**Candida** (C. albicans)
OB series, acute	S. aureus **43.8%**, CoNS **17.7%**, Enterococcus 6.3%
OB series, chronic	S. aureus **35.8%**, CoNS **21.2%**, Strep **11.6%**

Organism	Lab identity
S. aureus	**Catalase +, coagulase +**, GPC clusters · MRSA gene **mecA**
S. epidermidis	Catalase +, **coagulase −**, novobiocin-sensitive
S. pyogenes	**Catalase −**, β-haemolytic, bacitracin-sensitive



S. aureus – gram-positive cocci in clusters

Cutibacterium (Propionibacterium) acnes (TO)

Organism **anaerobic gram-positive rod** (diphtheroid) · skin follicles · **shoulder**, men · ~**10%** of PJI (underestimated)

Course ✓ **slow onset** · delayed (3–24 mo) / late · **CRP and WBC often normal**
✗ Wrong options: high WBC

Culture needs **anaerobic media, 7–15 days** (hold **≥14 days**) · sonication, molecular tests help
✗ Wrong options: 48 h

Antibiotics **penicillin G, ceftriaxone** first choice · **rifampin** active against biofilm · **resistant to metronidazole + fosfomycin** · aminoglycosides weak

– – Active space – –

Biofilm & quorum

Biofilm	forms **within 4 weeks** · **15% cells, 85% polysaccharide (glycocalyx)** · adheres to implant, shields from immunity · infection invades **prosthesis–bone interface** · cannot be safely removed → **explant if >4 weeks**
Antibiotics alone fail because	✓ **biofilm** · **sessile** bacteria more resistant than planktonic ✗ Wrong options: resistance · polymicrobial · diabetes · planktonic bacteria
Quorum	✓ **critical bacterial density needed to establish biofilm** → colony shifts **from replication to virulence/biofilm** ✗ Wrong options: MIC · LD50 · empiric options index

Prevention & prophylaxis

Screening	optimize risk factors · **nasal mupirocin** decolonization (MSSA/MRSA) · **no routine urine culture** unless UTI symptoms · **stop DMARDs 4–6 wk** · revision: **normal ESR/CRP off antibiotics** · reported penicillin/cephalosporin allergy → **allergy testing** (cephalosporins ↓ morbidity/mortality vs alternatives)
Timing	**within 30 min** of incision (Orthobullets) · **within 1 h** (OB ✓ cefazolin ↓ PJI) · **>10 min before tourniquet** · continue **24 h**
Weight-based cefazolin	**1 g <60 kg · 2 g 60–120 kg · 3 g >120 kg** (>120 kg often under-dosed)
Aseptic revision	✓ **single weight-based dose + 3 more weight-based doses over 24 h** (q8h) (130 kg → 3 g) ✗ Wrong options: 48 h (no benefit) · single dose · fixed 2 g
Dilute povidone-iodine	✓ **cost-effective ↓ infection** · **0.35% for 3 min** · must be **rinsed out** (cytotoxic) · not an antibiotic (no resistance) · WHO/CDC/ICM support
OR	**vertical laminar airflow** · limit traffic · antiseptic skin wash · glove changes
Antibiotic-loaded cement in primary TKA – conflict	OB (2023): low-dose ALBC (1 g vanc + 1 g tobra / 40 g, <3%) → ✓ **↓ PJI** (no loosening, no AKI, ↑ cost) · OB (2021): routine ALBC in primary TKA **not shown to ↓ PJI** – read the stem
No effect	**staples vs sutures** · absorbable monofilament vs staples
Dental prophylaxis	depends on **host risk factors**

3 Presentation

History	bacteraemia · multiple surgeries · skin breach (IVDU, colonoscopy, dental work, ulcer, wound problem)
Persistent pain + stiffness	infection in **>90%** of such patients
Acute	swelling, tenderness, **drainage** · **chronic** – subtle, deteriorating function
Sinus tract to joint	**definite infection**
Surgical site infection (with implant)	within **1 year** of index surgery · 2 yr post-TKA sinus with pus → ✓ **"SSI" least accurate** term (✓ PJI, sinus, purulence, chronic)

Classification by timing

System	Categories
Orthobullets	**Acute <3–6 wk** (CDC **<90 days**) – joint space only, no biofilm · **Chronic >3–6 wk** (CDC >90 days) – biofilm, bone–implant interface
OB (DAIR)	**Early postop ≤30 days** · **acute haematogenous** >30 days with symptoms **<3–4 wk** · **chronic** >30 days with symptoms **>1 month**
Fitzgerald (OB)	**I** acute postop, radiographically silent · **II** 6–24 mo, indolent, new bone formation · **III** >2 yr, **haematogenous**
TO (P. acnes)	Early <3 mo · delayed 3–24 mo · late >24 mo
Source	**Direct** (sinus, dehiscence) · **haematogenous** (dental, gallbladder)

Tsukayama (modified Coventry, Segawa) – TO

Type	Definition	Treatment
1	**Positive intraoperative cultures** (≥2)	**IV antibiotics (6 wk)**
2	**Early postop** (<4 wk)	**Debridement, prosthesis salvage**
3	**Acute haematogenous**	**Debridement, prosthesis salvage**
4	**Late chronic** (>4 wk)	**Prosthesis removal**

Tsukayama numbering – TO lecture notes — **I early post-op (< 1 mo)** → debridement + lavage + PE exchange, retain stable components · **II late chronic (> 1 mo)** → **two-stage** (spacer, e.g. Prostalac) · **III acute haematogenous** → debridement / single-stage (Hansen) · **IV positive intra-op culture** → tailored antibiotics ± staged · numbering above (TO MCQ) puts positive cultures as type 1 – read the options

Timing (TO notes) — acute **< 1 month**, chronic **> 1 month** (Campbell's definition, cited in TO)

Key radiology findings

Radiographs — **periosteal reaction** (lacey new bone, scalloping) · **patchy osteolysis** · **generalized resorption without wear** · transcortical sinus · loosening · TO: lucency **>2 mm** at bone–cement interface, component shift, cement fracture, motion on stress views · progressive lucency in **first year** → infection work-up

Bone scan (Tc-99m) — Orthobullets **99% sens, 30–40% spec** · TO **93% sens, 65% spec** (PPV 63%, NPV 94%) · hot for **years** after THA / at 6–7 months after TKA → not useful early · triple phase shows loosening, not sepsis

Labelled WBC (In-111 / Tc-WBC) — better to rule out · TO 2023: **Tc-99m WBC + bone scan = gold standard** for septic vs aseptic loosening (spec **86%**)

FDG-PET — **98% sens, 98% spec** (Orthobullets) · OB: no more accurate than ESR + CRP, no organism · normal scan excludes PJI

MRI — limited by metal (✓ TO "not sensitive due to metal") · MARS MRI knee **92% sens, 99% spec** · not first-line

– – Active space – –

Septic vs aseptic loosening — conflict (TO)

one TO item: ✓ **hip aspiration** (identifies + guides antibiotics; WBC scan can't definitively separate) · FNB 2023: ✓ **radiolabelled WBC scan** (aspiration sens 67%) — read the options/source



Periprosthetic infection — radiograph

Genes & lab markers – serum

Marker	Facts
CRP	Peaks **2–3 d**, normal by **21 d** · threshold acute (<6 wk) **<100 mg/L**, chronic **<10 mg/L** · TO sens 94%, spec 74% · first-line with ESR
ESR	Peaks **5–7 d**, normal by **90 d** · chronic **<30 mm/h** · no acute cut-off · TO sens 91%, spec 72%
IL-6	Normal **<10 pg/mL** (TO <5) · peaks **8–12 h**, normal **48–72 h** · ✓ **highest correlation with PJI** · sens **100%**, spec **95%** (TO 95/87) · ✓ **most specific** (TO, vs PCT, CRP, IL-1) · false + **RA, MS, AIDS, Paget**
Serum WBC	Useless (sens **~20%**)
D-dimer	**>860 ng/mL** · outperforms ESR/CRP, good for indolent organisms
Procalcitonin	Spec **98%**, sens **33%** · good for septicaemia
TNF-α	High spec, low sens · unstable sample
ICAM-1	✓ **Returns quickly to baseline** after clearance (✗ Wrong options: "stays elevated" (TO false statement))

Not a correct constellation (TO)

✓ **ESR, CRP, WBC count, IL-1**

Synovial fluid

Setting	WBC / PMN cut-off
Knee/hip, chronic	**>3,000 cells/µL + >80% PMN** (Orthobullets)
Knee, first 6 weeks after TKA	**>10,000** ✓ (Parvizi 2018; Bedair 2011 used 27,800)
MoM hip	**>4,350 + >85%** · **manual count** (debris falsely ↑ automated)
Hip – Schinsky criteria (TO)	**> 4,200 cells/mm³ + > 85% PMN**
Hip antibiotic spacer	**>1,166 + >64%**
TO best thresholds	Knee **1,630 + 60%** · hip **2,582 + 66%** → ✓ **2,500 cells/µL** "signifies infection" · early PJI often **>10,000 + >90%**
OB others	Chronic THA **3,966** · knee **>1,100** possible infection · knee PJI **>1,000 + >64%**
Native septic joint	**50,000**

Gram stain — **specificity > sensitivity** · high false negatives in chronic PJI → **don't rely on it** (OB)

Alpha-defensin — ✓ **intra-articular antimicrobial peptide** (✗ Wrong options: serum · cytokine) · released by ✓ **neutrophils** (✗ Wrong options: macrophages · monocytes) · **100% sens, 98% spec** · ✓ **most sensitive + specific** marker (✗ Wrong options: frozen section · IL-6 · CRP · LE) · **>5.2 mg/L** · **unaffected by antibiotics** · accurate with blood · **false + in metallosis/ALTR** · ✓ **single most accurate predictor of chronic PJI**

Leukocyte esterase strip — sens **81%**, spec **97%** · cut-off **++** · blood → centrifuge · separates MoM failure from PJI

Best combined accuracy (OB) — ✓ **laboratory-based alpha-defensin + LE strips** · lab-based > lateral-flow alpha-defensin · synovial > serum > tissue tests

Synovial CRP — **>6.9 mg/L** · **high NPV, low PPV**
✗ Wrong options: "high PPV"

MSIS 2018 (Parvizi) criteria

Major (either alone = infected) — ✓ **sinus tract communicating with prosthesis** · **same pathogen from 2 separate cultures**
✗ Wrong options: purulence alone (poly debris, MoM, gout mimic) · high WBC alone · CRP/ESR

Minor – preoperative	Points
Serum **CRP >10 mg/L** or **D-dimer >860 ng/mL**	**2**
Serum **ESR >30 mm/h**	**1**
Synovial **WBC >3,000/µL** or **LE ++**	**3**
Synovial **alpha-defensin +**	**3**
Synovial **PMN >80%**	**2**
Synovial **CRP >6.9 mg/L**	**1**
Score	**≥6 infected · 2–5 inconclusive · 0–1 not infected**

– – Active space – –

Intraoperative (if 2–5 or dry tap)	Points
Histology >5 PMN/hpf in 5 hpf (×400)	3
Purulence	3
Single positive culture	2
Combined pre + intraop	≥6 infected · 4–5 inconclusive · ≤3 not infected

Performance **98% sens, 99.5% spec** · inconclusive → molecular tests (PCR, NGS)

MSIS 2011 major criteria **or 4 of 6 minor** (ESR + CRP, synovial WBC, PMN%, purulence, one culture, histology)

Worked examples CRP 12 + ESR 45 + WBC 3,200 + PMN 78% → **6 points = infected** ✓ · WBC + alpha-defensin → **6** ✓ · WBC + PMN → 5 · D-dimer + synovial CRP → 3 · ESR + synovial CRP → 2

Major criteria (at least one of the following)	Decision
Two positive cultures of the same organism	Infected
Sinus tract with evidence of communication to the joint or visualisation of the prosthesis	

		Minor criteria	Score	Decision
Preoperative diagnosis	Serum	Elevated CRP or D-dimer	2	≥6 infected
		Elevated ESR	1	
	Synovial	Elevated synovial WBC count or LE	3	2–5 possibly infected[a]
		Positive alpha-defensin	3	
		Elevated synovial PMN (%)	2	0–1 not infected
		Elevated synovial CRP	1	

	Inconclusive pre-operative score or dry tap[a]	Score	Decision
Intraoperative diagnosis	Preoperative score	–	≥6 infected
	Positive histology	3	4–5 inconclusive[b]
	Positive purulence	3	
	Single positive culture	2	≤3 not infected

MSIS 2018 criteria

Diagnostic workup – ESR/CRP then aspiration

Painful arthroplasty

- ESR + CRP normal → infection unlikely → **observe / other causes**
- ESR or CRP elevated → **aspiration (cell count, diff, culture, crystals)**
  - On antibiotics / inconclusive → **repeat after 2 wk off, or alpha-defensin**
- Still equivocal → **intraop frozen section ± sonication of implant**

First step ✓ **ESR + CRP** · AAOS: **strong recommendation to aspirate** if ESR and/or CRP abnormal
✗ Wrong options: bone scan · MRI · CT for rotation · revision · metal ions · IV antibiotics first

Aspiration before revision elevated ESR with borderline CRP → ✓ **aspirate** · instability + ↑ ESR/CRP → ✓ **aspirate first**
✗ Wrong options: Tc scan · FDG-PET · straight to revision + frozen section

Normal ESR + CRP ✓ **observe** / no aspiration (e.g. eccentric head with normal labs)

MoM hip with fever after URI ✓ **aspirate** · MoM pain with ↑ ESR/CRP + normal ions → ✓ **arthrocentesis** · ions **Co:Cr 1:1 → MoM debris**, **2:1 → trunnionosis**
✗ Wrong options: MARS MRI · metal ions · empiric antibiotics

Equivocal aspirate + recent antibiotics ✓ **repeat aspiration** (after 1–2 weeks off) · on antibiotics now → ✓ **repeat aspiration for alpha-defensin today**

Equivocal (CRP 13, ESR 19, WBC 1,000, PMN 90%, α-defensin –, synovial CRP 4)	✓ **OR for histology** (frozen section)

Wait 2 weeks off antibiotics before repeat aspiration (Orthobullets)

Intraop frozen section	**>5 PMN/hpf × 5 hpf** · **sens 85%, spec 90–95%** · equivocal cases
Intraop cultures	same organism on repeat aspirations or **≥3 of 5** specimens · **false-positive 8%** · **tissue > swabs**
Sonication (TO)	✓ **low-frequency (40 kHz) ultrasound of explanted implant** dislodges biofilm → ↑ culture sensitivity, even after antibiotics ✗ Wrong options: high frequency · treatment of overuse injuries

Hemiarthroplasty, 18 months, rapid acetabular erosion + linear lucency → ✓ infection
✗ Wrong options: cement technique · oversized head (would extrude)

5 Management

Confirmed PJI

- Acute postop (<3–4 wk) or acute haematogenous (symptoms <3–4 wk) → **DAIR + modular exchange**
  (not if loose implant, flap needed, MRSA/fungal/mycobacteria)
- Chronic (>4 wk), fit host → **two-stage exchange (gold standard)**
  - Low-virulence, healthy host, no sinus → **one-stage (Europe)**
- Positive intraop cultures at aseptic revision (≥2/5) → **6 wk IV antibiotics**
- Unfit / refuses → **suppression · salvage: resection, fusion, AKA**

Acute infection – DAIR & liner exchange

Indications	**acute postop <3–4 wk** · **acute haematogenous** (ideally **<48–72 h** of symptoms; <3 wk) · well-fixed implant
Technique	**modular exchange** (PE, femoral head – removes fibrin nidus) · synovectomy · **6–9 L irrigation** · culture-guided IV antibiotics **2–6 weeks** (TO)
Success	**50–55%** · implants removed if reinfection
Improved success	modular exchange · **within 7 days** of symptoms · **rifampin** for staph · **fluoroquinolone** for gram-negatives (FQ-sensitive E. coli ~**79%**)
Predict failure	**MRSA, VRE, Pseudomonas**, S. epidermidis (biofilm), **fungi**, ✓ **mycobacteria** · host: **age >80**, male, COPD, CRF, cirrhosis
Absolute contraindications	**loose implants** (stem subsidence) · **poor soft tissue / flap needed** (>85% failure) · compromised cement mantle · chronic PJI (even with 12+ wk antibiotics)
Best DAIR candidate	✓ **TKA 2 yr ago, 2-day haematogenous E. coli** · ✓ **THA 3 wk ago, WBC 10,700 + 91% PMN** ✗ Wrong options: 8 mo MRSA · loose stem · flap · 6 mo symptoms · 1 yr with 2–3 mo symptoms
Head + liner exchange appropriate	✓ **infection 3 days after systemic infection** ✗ Wrong options: 4 months · 8 weeks · osteolysis · Vancouver A

– – Active space – –

Persistent wound drainage 10–14 days post-op → ✓ I&D ± PE exchange in OR · drainage at 3 wk with benign aspirate → **surgical exploration + possible PE exchange** · drainage after prior HTO incision → **urgent debridement + cultures + labs**
✗ Wrong options: oral/IV antibiotics alone · ointment · immobilizer · sutures in clinic · VAC · wait for cultures

Acute haematogenous (48 h, RA, dental work, 55K WBC) → ✓ operate now
✗ Wrong options: wait for cultures · α-defensin · IV antibiotics only · IR drain

Possible acute PJI 3 wk post-TKA (fall, diverticulitis, effusion) → ✓ aspirate first
✗ Wrong options: book DAIR directly · reassure

Acute but MRSA + BMI 50 + persistent drainage (5 wk) → ✓ two-stage (I&D <50% success; failed I&D worsens later two-stage)

FNB 2024 — bilateral TKA, infected at 3 wk, low-virulence sensitive organism → options suppression / DAIR / one-stage / two-stage — answer not given by source (DAIR fits acute <3–4 wk)

Chronic infection – two-stage exchange

Indication — ✓ **gold standard for infection >4 weeks** · medically fit · adequate bone stock · preferred for **fungal**

Stage 1 — explant · debride bone–implant interface + soft tissue · **antibiotic spacer** · **IV antibiotics 4–6 weeks** (TO: 6 wk, bactericidal titre **1:8**) · excise sinus with elliptical incision

TO notes — **IV antibiotics 6 weeks** → reimplant at **~3 months** if ESR/CRP normal + aspirate negative · **no drains with a spacer** (keep local antibiotic level) · **Prostalac (Duncan + Beauchamp)** keeps length + tension · recurrent infection → **Girdlestone** · infection rate **1–2%** (vs Charnley's early **6.8%**)

Before stage 2 — benign exam, **normal ESR/CRP/WBC**, ✓ **2-week antibiotic holiday then aspiration** · OB case: 1-month holiday
✗ Wrong options: aspirate at completion of IV · wait 6 months (spacer breaks) · spacer exchange

Reimplantation — tissue for culture + **frozen section** · acute inflammation → **re-debride, new spacer** · antibiotic cement if cementing · **cementless better in hip**

Reimplantation timing	Success
Early, <2 weeks	**35%**
Delayed, >6 weeks	**70–90%** (Orthobullets) · **80%** (OB)

Cure rates — two-stage **90–93%** (OB) · **89–100%** (TO) · one-stage **75–80%** (OB) · **75–100%** (Orthobullets) · **~89%** (TO)

Prognosis — ✓ **MRSA worse than MSSA** after two-stage (eradication + function) · **BMI ≥40** ↑ reinfection · bilateral simultaneous resection + reimplant → ✓ **successful at 2 yr, <20% revision**

Typical answers — late haematogenous after colonoscopy (6 wk) → ✓ **two-stage** · 5 wk after URI, 15K WBC, GPC → ✓ **resection + antibiotics** · MR-CoNS 8 months → ✓ **remove implants + spacer** · inflamed tissue + frozen section 20 WBC/hpf at re-revision, Gram −ve → ✓ **cultures, remove implants, spacer** · sinus tract → ✓ **labs, aspirate, two-stage** (✗ Wrong options: wait · oral antibiotics · DAIR · one-stage) · culture-negative with ↑ markers → still ✓ **two-stage** · MRSA chronic → ✓ **two-stage** (✗ Wrong options: fusion · suppression · I&D · AKA)

Failed two-stage (persistent infection) ✓ **repeat debridement + spacer** · devitalized patella (lateral release kills **superior lateral genicular**) → sequestrum → consider **patellectomy**
✗ Wrong options: fusion · AKA · suppression in healthy 62-yr-old · arthroscopic I&D

One-stage exchange

Indications **Europe**, infected THA · **no sinus**, healthy host + soft tissue, no prolonged antibiotics, no bone graft · **low-virulence, sensitive** organism · antibiotic cement

Pros/cons cheaper, single surgery, early mobility vs **↑ persistent infection** · not fungal (rarely succeeds)

Antibiotic spacers & cement

	Static spacer	Articulating spacer
Antibiotic dose	**Higher** (not premade)	Lower
Use	**Flap coverage, extensive bone loss, ligament deficiency** (gross laxity)	Default
Pros	Wound healing (no motion)	↓ reimplantation exposure time · **joint space + ROM kept** · ↓ **quadriceps shortening** · satisfaction
Reinfection + function	**Equivalent**	**Equivalent** ✓ (✗ Wrong options: "↓ infection recurrence")

Spacer pros ↓ dead space, stability, high local antibiotic · **cons** – allergy, resistant organisms, **heat-stable antibiotics only**

Cement dose (per 40 g)	Orthobullets / OB	TO
Low (prophylaxis)	**≤2 g** (commercial ALBC; OB: 0.5 g tobra/genta)	**≤2 g (<5%)** keeps strength
Treatment / retained hardware	–	**2–3.6 g**
Spacer (high)	**≥3.6 g** · practical **3 g vanc + 3.6–4 g tobra** · OB: **min 3.6 g tobra + 1 g vanc**	**4–8 g** · ✓ **max 8 g**
Max without systemic toxicity	**12.5 g tobramycin · 10.5 g vancomycin**	–
Strength compromised	–	✓ **>3.6–4 g**

Elution burst first **24 h**, rapid decline, low by **5 wk** · ✓ **↑ porosity → ↑ elution** · **hand mixing** for spacers, **vacuum** for prophylaxis · **combination (vanc + aminoglycoside) ↑ elution of both** · vancomycin added **after 30 s** of mixing → ↑ elution · extra monomer **1.5–2 ampoules** · NPWT: no difference
✗ Wrong options: ↑ strength · ↑ density

Cement chemistry (TO) **liquid antibiotics contraindicated → ✓ interfere with polymerization (soft mass)** · antibiotic voids ↓ ✓ **compressive strength** · extra powder ✓ **↑ dough time** · ✓ **polymyxin B / colistin** trapped (poor elution) · penicillins heat-labile · ✓ **fosfomycin ↑ (not ↓) companion elution**, thermostable, voids · amphotericin weakens cement most

– – Active space – –

Local antibiotics	aminoglycosides (gram-negatives, synergy vs GPC, low systemic toxicity) · **vancomycin** (GPC, excellent elution) · **thermostable** · high-dose aminoglycoside spacer → **AKI** risk
Empiric IV after cultures	**1st-generation cephalosporin** · **vancomycin** if true penicillin allergy, MRSA history, or unidentified organism · tailor to sensitivity



Articulating antibiotic spacer

Fungal infection

Features	**Candida** (most common) · risk: immunosuppression, DM, CRF, IVDU · **indolent**, no fever, ESR/CRP minimally ↑
Treatment	✓ **open I&D, two-stage exchange, antifungals 6–12 months** · **DAIR very high failure** ✓ · recurrence **20–25%** after two-stage · worse than bacterial ✗ Wrong options: suppression · DAIR + suppression · resection + antimycobacterials · single-stage
Fluconazole	✓ **oral = IV efficacy** · **heat-stable** (can go in cement) · **hepatotoxic** · **resistance rising** · no renal monitoring needed

Positive intraoperative cultures at revision (TO)

3/5 cultures CoNS after aseptic revision, afebrile, dry wound → ✓ IV antibiotics 6 weeks (Tsukayama 1)
✗ Wrong options: debridement + retention · one-stage · two-stage · observation

Soft tissue coverage – gastrocnemius flap

Large anteromedial knee defect with exposed implant → two-stage + flap at stage 1 · ✓ **medial gastrocnemius flap — medial sural artery**
✗ Wrong options: free flap first · lateral gastrocnemius (lateral wounds, CPN risk) · soleus (middle/distal third) · fasciocutaneous (4–10 cm without exposed implant)

Salvage options

Option	Indications	Outcome
Chronic suppression	Unfit, refuses, low-virulence organism sensitive to low-toxicity oral drug	**10–25%** eradication · **8–21%** complications
Resection arthroplasty	Poor bone/soft tissue · MDR recurrent · unfit · failed reimplantations · nonambulatory · **polyarticular RA, limited walking** (TO)	**TKA 50–89%** · **THA 60–100%** · short limb, poor function
Arthrodesis	Reimplantation not feasible (bone stock) · virulent recurrent · TO: young, high demand, single joint, deficient extensor, poor soft tissue	**71–95%** fusion + eradication
Amputation	Recalcitrant **TKA** infection, severe pain, soft-tissue/bone/vascular loss	**AKA**



Knee arthrodesis

Special situations

THR in tuberculous hip (TO) — **active TB no longer a contraindication** · M. tuberculosis **forms no biofilm**, rifampicin penetrates cells · **ATT ≥ 3 weeks pre-op** (preferably **2–3 months**) · total ATT **9–12 months** · **sinus → two-stage** · cemented or cementless (**uncemented if good bone**) · **ATT not added to cement** (poor elution, toxicity)

Infected pseudotumour (MoM, neutrophils on H&E, ↑ ESR/CRP) → debride, explant, spacer + ✓ **systemic antibiotics**
✗ Wrong options: radiation · chemotherapy

MoM THA with 6,000 WBC + 92% PMN (manual count), 3 months → ✓ explant + spacer
✗ Wrong options: DAIR · CoP revision (that is for metallosis without PJI) · MRI

Chronic MRSA enterococcus + global laxity + sinus → ✓ two-stage with static spacer
✗ Wrong options: dynamic spacer with laxity · flap for small sinus

– – Active space – –

X vs Y

	Acute / haematogenous PJI	Chronic PJI
Timing	**<3–4 wk** post-op or symptoms <3–4 wk	**>4 wk**
Biofilm	No	**Yes** (glycocalyx, bone–implant interface)
Organism	**S. aureus**	**S. epidermidis / CoNS**, P. acnes
Synovial WBC	**>10,000** (early)	**>3,000 + >80%**
Treatment	**DAIR + modular exchange**	**Two-stage**

	Static spacer	Articulating spacer
Best when	Flap, bone loss, **laxity**	Standard
Infection control	Equal	Equal

How it's asked

"Not associated with infection after TKA?"	→ **asymptomatic bacteriuria** (TO) ✗ Wrong options: RA · hinged prosthesis · psoriasis
"Not associated with infection after knee replacement?"	→ **age >65** (TO)
"THA infection risk – true?"	→ **allogeneic transfusion ↑ risk** ✗ Wrong options: BMI 43 safe · low albumin protective · staples better
"Order of PJI risk?"	→ **primary hip < primary knee < revision hip < revision knee**
"Highest PJI risk?"	→ **revision TKA**
"Early vs late TKA revision causes?"	→ **infection more common early than stiffness**
"Would have reduced PJI risk?"	→ **cefazolin within 1 h of incision**
"130 kg aseptic revision prophylaxis?"	→ **weight-based cefazolin + 3 weight-based doses over 24 h**
"1 g vanc + 1 g tobra per 40 g in primary TKA?"	→ **reduced infection rate**
"HbA1c 8.1% – counsel?"	→ **↑ infection risk (HbA1c >8%)**
"Dilute povidone-iodine lavage?"	→ **cost-effective infection reduction**
"Septic vs aseptic loosening, THA at 8 yr?"	→ **hip aspiration** (TO) · FNB 2023 version → **radiolabelled WBC scan**
"Sonication refers to?"	→ **low-frequency ultrasound of explanted prosthesis to aid diagnosis** (TO)
"Correct statement on PJI biomarkers?"	→ **MRI insensitive due to metal artefact** (TO)
"Equivocal aspirate after recent 14-day antibiotics?"	→ **repeat aspiration after 1 week**

"Pain 7 months after TKA, normal X-ray – next?"	→ **CRP, ESR, WBC** ✗ Wrong options: bone scan
"Lacey periosteal new bone 15 months after THA – next?"	→ **ESR, CRP, WBC**
"Instability + raised ESR/CRP after TKA?"	→ **aspirate**
"RA, loose stem, ↑ ESR – next?"	→ **hip aspiration for culture**
"Painful THA, normal ESR + CRP?"	→ **observation**
"MoM hip, 6 months pain, ↑ ESR/CRP, normal ions?"	→ **arthrocentesis**
"Equivocal labs, α-defensin negative?"	→ **OR for histology**
"Aspirate 500 WBC while on oral antibiotics?"	→ **repeat aspiration today for alpha-defensin**
"Diabetic, 10 months pain, ↑ ESR/CRP?"	→ **knee aspiration**
"MoM THA, psoas fluid, ↑ ESR/CRP, normal ions?"	→ **hip aspiration**
"Most specific marker post-THR (TO)?"	→ **IL-6**
"Not true about markers?"	→ **ICAM-1 stays elevated after clearance** (TO)
"Not a correct constellation?"	→ **ESR, CRP, WBC count, IL-1** (TO)
"Cell count signifying PJI?"	→ **2,500 cells/µL** (TO)
"Alone sufficient for PJI (MSIS)?"	→ **sinus tract communicating with prosthesis**
"Serum marker with highest correlation?"	→ **IL-6**
"Alpha-defensin tests for?"	→ **intra-articular antimicrobial peptide**
"Most sensitive and specific marker?"	→ **alpha-defensin**
"Alpha-defensin released by?"	→ **neutrophils**
"Single most accurate predictor of chronic PJI?"	→ **positive synovial alpha-defensin**
"Best combined sensitivity + specificity?"	→ **lab-based alpha-defensin + LE strips**
"WBC 12,500 five weeks after TKA?"	→ **above 10,000 cut-off – PJI likely**
"Which labs meet 2018 criteria?"	→ **CRP 12, ESR 45, WBC 3,200, PMN 78%**
"Pair that alone suggests PJI?"	→ **synovial WBC + positive alpha-defensin**
"P. acnes PJI – true?"	→ **slow onset of symptoms** (TO)
"Late (>3 mo) THA infection organism?"	→ **S. epidermidis**
"Early THA infection organism?"	→ **S. aureus**
"MRSA vs MSSA after two-stage?"	→ **MRSA poorer prognosis**
"Why antibiotics alone fail?"	→ **biofilm**

– – Active space – –

"Quorum?"	→ **critical bacterial density to establish biofilm**
"Febrile, 65K WBC, chronic – organism and step?"	→ **catalase +, coagulase + GPC; two-stage**
"IV antibiotics after DAIR?"	→ **2–6 weeks** (TO)
"Predicts DAIR failure?"	→ **mycobacteria**
"Bilateral resection + spacers + antibiotics – outcome?"	→ **successful reimplantation, <20% revision**
"Articulating vs static spacer EXCEPT?"	→ **decreased infection recurrence**
"↑ spacer porosity leads to?"	→ **↑ antibiotic elution**
"Best antibiotic cement for spacer?"	→ **≥3.6 g tobramycin + 1 g vancomycin per 40 g**
"Why liquid antibiotics are contraindicated in PMMA?"	→ **interfere with polymerization** (TO)
"Threshold where strength compromised?"	→ **>3.6–4 g per 40 g** (TO)
"Property most hurt by voids?"	→ **compressive strength** (TO)
"Dough time with extra antibiotic powder?"	→ **increases** (TO)
"Max antibiotic in 40 g cement for spacer?"	→ **8 g** (TO)
"Antibiotic avoided – poor elution?"	→ **polymyxin B** (TO)
"Not a property of fosfomycin in cement?"	→ **reduces elution of companion antibiotic** (TO)
"Persistent infection after spacer + antibiotics?"	→ **repeat debridement + new spacer**
"After 6 weeks IV, normal CRP/ESR – next?"	→ **2-week antibiotic holiday then aspiration**
"Candida PJI in transplant patient?"	→ **open I&D, two-stage, antifungals 6–12 months**
"Fluconazole – true?"	→ **oral = IV efficacy**
"DAIR for Candida?"	→ **very high failure rate**
"Anteromedial defect with exposed tibial tray – flap?"	→ **medial gastrocnemius – medial sural artery**
"3/5 intraop cultures CoNS, afebrile, dry wound?"	→ **6 weeks parenteral antibiotics** (TO)
"Infected MoM pseudotumour – surgery plus?"	→ **systemic antibiotics**
"Persistent drainage 12 days after THA?"	→ **open debridement + PE exchange + culture-guided antibiotics**
"Drainage 14 days, culture pending?"	→ **debride before cultures return**
"Streptococcus 10 days after TKA?"	→ **open I&D + PE exchange + IV antibiotics**
"5 weeks, MRSA, BMI 50, drainage?"	→ **two-stage with antibiotic spacer**
"11 weeks, MRSA PCR, dry tap, drainage?"	→ **debride, remove implants, spacer**

"Methicillin-resistant enterococcus, gross laxity?"	→ **two-stage with static spacer**
"RA patient after THA (vs OA)?"	→ **PJI risk ↑ (16,000 WBC, 88% PMN at 6 wk)**
"Active TB hip — THR?"	→ **not contraindicated; ATT ≥ 3 wk pre-op, 9–12 months total** (TO)
"Antibiotic spacer — drain?"	→ **no drain** (TO)

– – Active space – –

– – Active space – –

THA ACETABULAR SCREW FIXATION

1 Basics

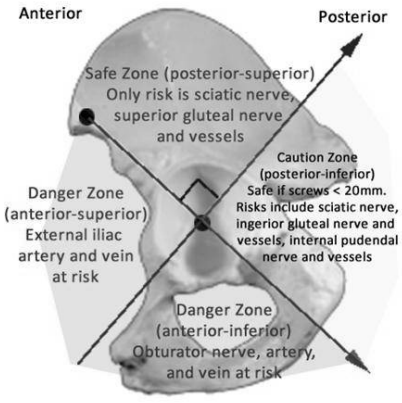
Acetabular quadrants & screw neurovascular risk

Construction line 1: ✓ **ASIS → centre of acetabulum** (→ posterior fovea) · line 2: **perpendicular** at acetabular centre

Quadrant	Zone	At risk
Posterior-superior	**Target / safe zone** – ideal for screws	**Sciatic nerve**, **superior gluteal nerve + vessels** (✓ SGA – TO)
Posterior-inferior	**Caution zone** – screws **≤20 mm**	**Sciatic nerve**, **inferior gluteal nerve + vessels**, **internal pudendal nerve + vessels**
Anterior-inferior	**Danger zone**	**Obturator nerve, artery, vein**
Anterior-superior	**"Death zone"** (Orthobullets/TO) · "danger zone" (OB)	✓ **External iliac artery + vein** (vein more medial → more at risk)

General rule posterior quadrants safe except **long posteroinferior** screws · safe areas **superior + posterior** · external iliac at risk with **anteromedial** screws

High hip centre anterosuperior + anteroinferior structures shift into the **posterosuperior quadrant** · safest = **peripheral half of posterosuperior** quadrant · elevated revision centre → **sciatic** risk with posterosuperior screws



Acetabular quadrant safe zones

6 Complications

Screw injuries

Foot drop + radicular pain after revision, screw in sciatic notch → ✓ posterior-superior quadrant (long screw, high hip centre)

Gluteus maximus lurch (trunk extends at heel strike) after transacetabular screws → ✓ **posterior-inferior quadrant → inferior gluteal nerve** injury

✗ Wrong options: posterior-superior (sciatic/SGN) · anterior-inferior (obturator) · anterior-superior (external iliac)

– – Active space – –

Greatest vascular risk → ✓ anterior-superior
✗ Wrong options: superior · posterior-superior · posterior-inferior · superolateral

Retractor placement & vascular injury

External iliac vessels — only **7 mm** from bone at **AIIS**, may lie on bone → **anterior retractor in anterosuperior quadrant** (✓) or anterosuperior screw

Common femoral artery/vein — separated from capsule only by **iliopsoas**, crosses hip near **AIIS** → anterior acetabular retractor (✓ position 2, anterosuperior) · esp. **anterolateral approach**

Inferior retractor under transverse acetabular ligament → obturator vessels/nerve

Retractor position	Injury
Anterior-superior	**External iliac / femoral vessels**
Anterior-inferior / directly inferior (under TAL)	**Obturator** vessels + nerve
Posterior-inferior	Sciatic, inferior gluteal, internal pudendal
Posterior-superior	Relatively safe – superior gluteal, sciatic

Avoiding prosthesis impingement

Proud anterior cup rim → iliopsoas impingement · evaluate with **cross-table lateral** + **anaesthetic injection of iliopsoas sheath**

Cause	Impingement type
Medialized + raised cup (↓ femoral offset)	**Bone-on-bone**
Lateralized cup	**Metal neck-on-metal cup**
Head-to-neck ratio <2:1 (skirted heads, small heads)	Prosthetic impingement

How it's asked

"Posterosuperior quadrant screw risks (TO)?"	**→ superior gluteal artery**
"Zone most risky for external iliac vessels?"	**→ anterior-superior**
"Line drawn to define quadrants?"	**→ ASIS to centre of acetabulum (A and C)**
"Foot drop, screw in sciatic notch after revision?"	**→ posterior-superior quadrant**
"Greatest risk of vascular injury?"	**→ anterior superior**
"Trunk extension at heel strike after transacetabular screws?"	**→ posterior-inferior quadrant (inferior gluteal nerve)**
"Posterior approach, bleeding from external iliac after retractor?"	**→ anterior-superior**
"Femoral artery bleeding from retractor?"	**→ anterosuperior (position 2)**
"Caution zone screw length?"	**→ ≤20 mm**

– – Active space – –

"Obturator vessels at risk?"	**→ anterior-inferior quadrant / retractor under TAL**

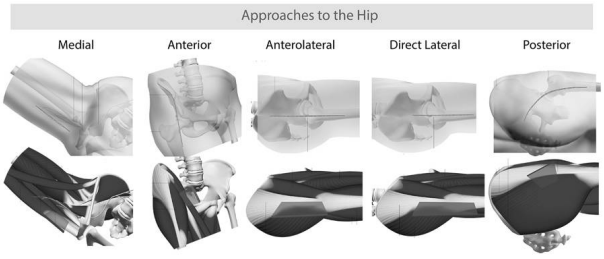
THA APPROACHES

– Recon · 20 / 65 –

1 Basics

Choice of approach

Dictated by	surgeon preference · prior incisions · obesity · dislocation risk · implant · deformity
Standard	direct anterior · anterolateral · direct lateral · posterolateral · **extensile** – trochanteric osteotomy · **MIS** variants
Original Charnley THA	✓ **anterolateral** approach, supine, **greater trochanter osteotomy**, anterior dislocation (TO)



Approaches to the hip

Intervals & nerves

Approach	Superficial interval	Deep interval	Internervous?
Direct anterior (Smith-Petersen)	**Sartorius (femoral) – TFL (superior gluteal)**	**Rectus femoris (femoral) – gluteus medius (SGN)**	✓ **Femoral / superior gluteal**
Anterolateral (Watson-Jones)	**TFL – gluteus medius** (both SGN)	–	No – intermuscular only
Direct lateral (Hardinge)	**Splits gluteus medius + vastus lateralis**	–	No
Posterolateral (Moore / Southern)	**Splits gluteus maximus** (≤7 cm from GT tip – IGN/IGA)	Short ER + capsule	No
Medial (Ludloff)	Sartorius – adductor longus	Pectineus – iliopsoas	–
Medial (Ferguson/ Hoppenfeld)	Gracilis – adductor longus	Adductor magnus – brevis (dual innervation)	Protect posterior obturator branch

Anterolateral approach (Smith-Petersen variant, TO)	"7"-shaped incision · flap of **TFL, gluteus minimus, anterior medius**
Somerville (TO)	bikini incision · DDH open reduction · TFL–sartorius
Harris (TO)	U-incision + GT osteotomy · **Gibson** – anterior border of maximus, cut medius/minimus, dislocate **anteriorly** · **modified Gibson (Marcy–Fletcher)** – cut short ERs, dislocate **posteriorly** · **Osborn / Moore** – split maximus, cut short ERs, posterior capsule

– – Active space – –

5 Management

Direct anterior – interval & LFCN injury

Interval	✓ **TFL and sartorius** superficially · ✓ **internervous plane femoral + superior gluteal** (TO) ✗ Wrong options: sartorius–maximus · minimus–rectus · rectus–sartorius · TFL–rectus
Bleeding in TFL–sartorius interval	✓ **ascending branch of LFCA** (~5 cm below hip) → ligate/coagulate ✗ Wrong options: SGA · femoral · profunda · MFCA
LFCN	**L2–L3**, under inguinal ligament, pierces fascia **~1 cm distal to ASIS** near TFL–sartorius interval · ✓ **structure at greatest risk** (✗ Wrong options: femoral nerve/artery/vein (medial) · MFCA (posterior to neck)) · ✓ **most common complication** of anterior approach (81% altered thigh sensation, no functional loss) · ✓ **common, improves over time, no functional deficit** (✗ Wrong options: uncommon · intermediate femoral cutaneous)

Fascial incision lateral, over TFL muscle belly → ✓ minimize LFCN injury
✗ Wrong options: protects femoral bundle · trajectory · PJI

Advantages	**↓ dislocation vs posterior** (~**0.5%**, regardless of capsule repair) · abductors intact · **no gait difference at 3 months** vs other approaches · unproven: less muscle damage/pain, faster recovery
Disadvantages	**steep learning curve (>100 cases)** · **wound infection ↑ in obese with large pannus** · **femoral nerve injury** from anterior retractor · femoral exposure difficult (special table) · **LFCN paraesthesia** · **↑ intraop fracture**

Direct anterior – femoral complications

Highest femoral-sided complication risk	✓ **direct anterior** (intraop **femur fracture, subsidence**, **femoral-sided revision for loosening**) · interval ✓ **femoral–superior gluteal plane** · superficial ✓ **TFL–sartorius**
Distractors	direct lateral = medius/vastus split → **Trendelenburg, HO** · posterior = maximus split → **dislocation** · anterolateral = TFL–medius → **limp**, lower dislocation

Anterolateral (Watson-Jones)

Interval	✓ **TFL (anterior) and gluteus medius (posterior)** (TO) · start midway ASIS–GT, expose SGN proximally · capsule incised along anterosuperior neck
Pros/cons	**lower dislocation** than posterior · **violates abductors → limp** · retractor through iliopsoas → **femoral nerve** risk, not instability

Direct lateral (Hardinge) – abductor injury

Detached from femur	✓ **vastus lateralis + gluteus medius (anterior ⅓–½) + gluteus minimus** (+ ligament of Bigelow) ✗ Wrong options: psoas/sartorius · piriformis/obturator internus (posterior) · maximus/TFL · minimus/rectus
Superior gluteal nerve	runs between medius and minimus **4–6 cm above GT tip** · split ≤**4–5 cm** above GT (TO: **≤5 cm** from tip, or **≥2 cm** above acetabular lip – Frndak)

Persistent limp + fatty atrophy of anterior abductors on CT → ✓ split >5 cm above GT tip (SGN injury)
✗ Wrong options: GT fracture · retractor (femoral nerve) · PT · offset

Testing suspected SGN injury after extended incision (TO) → ✓ **stand on operated limb** (Trendelenburg)
✗ Wrong options: stand on other limb · ankle DF/PF

– – Active space – –

Pros — **lower dislocation** than posterior (OB: **lowest**) · anterior + posterior access without osteotomy · **cons** – abductor violation → **limp** · **HO common**

Posterolateral (Moore / Southern)

Most common for primary + revision · abductors intact · excellent femur + acetabulum exposure · extensile

Dislocation — higher than anterior · ↓ with **capsule + short ER repair** · ✓ **Moore approach with capsulectomy = greatest dislocation risk** · overall dislocation **1–3%**
✗ Wrong options: Smith-Petersen failed capsule repair · Watson-Jones with psoas penetration · Hardinge cup 40° · anterior with 15° anteversion

Minimally invasive approach

Mini-incision (≤10 cm) — only proven advantage ✓ **better cosmesis**
✗ Wrong options: pain · transfusion · LOS · analgesia

Intraop femoral fracture during MIS — ✓ **convert to extensile approach** to visualize + reduce · bypass by **2 cortical diameters** with long stem
✗ Wrong options: limit WB alone · switch to cement · delay · autograft

SuperPATH (TO) — supercapsular percutaneously assisted · **modified posterolateral MIS** · interval ✓ **gluteus minimus–piriformis** · medius + minimus retracted anteriorly · **short ERs + ITB spared**, **no dislocation** · femoral canal prepared through **piriformis fossa** in situ · **accessory percutaneous portal for acetabulum** · ↓ LOS, blood loss, readmission
✗ Wrong options: minimus–medius (the EXCEPT answer)



SuperPATH interval

Extensile – trochanteric osteotomy

Types — standard · **trochanteric slide** · **extended trochanteric osteotomy (ETO)**

Trochanteric osteotomy (TO)	Feature
Standard	Cut below **vastus lateralis tubercle at 45°**, lifted proximally · **difficult reattachment**
Trochanteric slide	**Abductors proximally + vastus lateralis distally stay attached** → easier reduction/fixation, **less abductor lurch**
ETO	Longer distal cut → canal exposure · **stuck / broken stem removal**

Fixation (TO) — **Harris 4-wire** (two vertical + two horizontal at 90°) · **Dall-Miles cable** system

– – Active space – –

Pros	excellent acetabular exposure · component removal · difficult primary/ revision
Complications	nonunion · HO · trochanteric bursitis · abductor weakness · ✓ **fracture of osteotomy fragment = most common** · **ETO needs a diaphyseal-engaging stem**

Surgical hip dislocation (Ganz) – TO

Preserve	✓ **deep branch of MFCA** (protected by intact **obturator externus**) · **posterior approach → trochanteric flip osteotomy → anterior (Z) arthrotomy → anterior dislocation** ✓ ✗ Wrong options: posterior arthrotomy/dislocation

X vs Y

	Direct anterior	Anterolateral	Direct lateral	Posterior
Abductors	Intact	**Violated**	**Violated**	Intact
Dislocation	**Low (~0.5%)**	Low	**Lowest** (OB)	**Highest**
Signature complication	**LFCN**, femoral fracture/ loosening, obese wound	Limp	**Limp, HO, SGN**	**Dislocation**, sciatic

How it's asked

"Internervous plane of direct anterior approach?"	→ **femoral and superior gluteal** (TO)
"Bleeding between TFL and sartorius?"	→ **ascending branch of LFCA**
"Anterior approach muscle plane?"	→ **TFL and sartorius**
"Greatest potential injury in anterior approach?"	→ **LFCN**
"Most common complication of Smith-Petersen THA?"	→ **LFCN injury**
"Superficial internervous plane of DAA?"	→ **sartorius (femoral) & TFL (SGN)**
"Why fascial incision over TFL belly?"	→ **minimize LFCN injury**
"Will my leg be numb?"	→ **LFCN injury common, improves, no functional deficit**
"Highest femoral-sided complication risk?"	→ **direct anterior**
"Interval of approach with most femoral complications?"	→ **femoral–superior gluteal plane**
"Superficial interval with ↑ femoral revision for loosening?"	→ **TFL and sartorius**
"Hardinge – muscles detached besides vastus lateralis?"	→ **gluteus medius + minimus**
"Limp + fatty abductors after modified Hardinge?"	→ **medius split >5 cm above GT**
"SuperPATH – all except?"	→ **interval between gluteus minimus and medius** (TO)

"True about approach in picture (TO)?"	→ **SupraPATH (SuperPATH) approach** · explanation not provided by source ✗ Wrong options: accessory portal for canal prep · fascia-lata-only no muscle violation · reaming through piriformis fossa portal without dislocation
"Mini-incision advantage?"	→ **better cosmesis**
"Femoral fracture during MIS THA?"	→ **convert to extensile approach**
"Greatest dislocation risk?"	→ **Moore approach with capsulectomy**
"Approaches compared — true?"	→ **no difference in gait at 3 months**
"Incision 5 cm above GT, nerve injury suspected — test?"	→ **stand on operated limb** (TO)
"Original Charnley approach?"	→ **anterolateral** (TO)
"Watson-Jones plane?"	→ **TFL and gluteus medius** (TO)
"Most common ETO / trochanteric osteotomy complication?"	→ **fracture of osteotomy fragment**

– – Active space – –

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THA ASEPTIC LOOSENING

– Recon · 21 / 65 –

1 Basics

Pathophysiology

Definition	**macrophage-induced inflammatory** bone loss → implant loosening **without infection**
Sequence	**prosthesis micromotion → particulate debris → macrophage-activated osteolysis** (RANK/RANKL pathway ↑)
Debris source	conventional PE (submicron particles) · proximal femoral lysis around cemented stem → mostly **cement debris** (OB)
Epidemiology	✓ **most common cause of late THA revision**
Early (≤6 wk) cementless stem subsidence	lack of initial stability → ✓ **high BMI (e.g. 44)** or **undersized stem** ✗ Wrong options: oversized · Dorr A · tapered stem · modular neck

3 Presentation

Pain	**groin** (cup), **thigh** (stem), knee · **activity related** · **start-up pain** (first steps) = loose femoral stem
Exam	minimal pain with ROM · pain with weight bearing · mild discomfort with SLR/rotation

Genes & lab markers

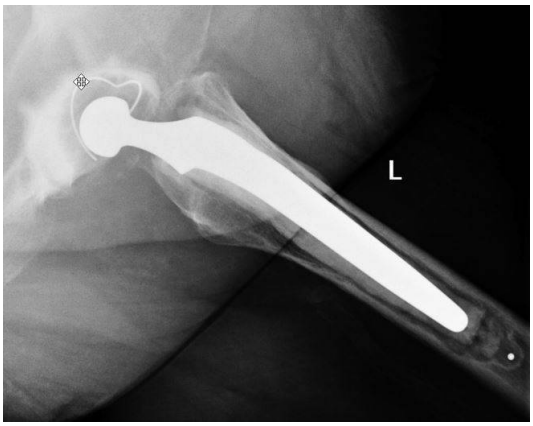
ESR + CRP normal in aseptic loosening → rule out infection first; normal labs + loosening on X-ray → **no aspiration needed**

4 Diagnosis & Classification

Key radiology findings

Views	AP + lateral · X-rays **insensitive**, **underestimate** osteolysis
General	**lucency >2 mm** at cement–bone or metal–bone interface · **progressive / continuous** radiolucent line · **component motion on stress views** · change in position on serial films · positive bone scan
Femoral component	**subsidence >1 cm** · **cement mantle fracture** · **debonding at lateral shoulder (Gruen zone 1)** · **pedestal** at stem tip · circumferential lucency · lateral cortical stress reaction
Acetabular component	migration / change in inclination · **PE liner wear → femoral head sits eccentric (superior)** in cup

– – Active space – –



Aseptic loosening of THA

Resurfacing-specific (Orthobullets) change in **peg–neck angle** · ↓ peg tip–lateral cortex distance (also head collapse) · **neck narrowing >10% / after 3 yr** (also impingement)

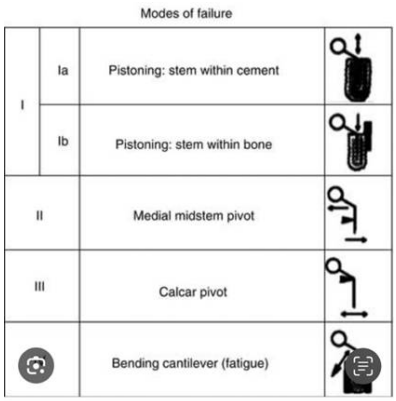
Roentgen stereophotogrammetric analysis (RSA)

Accuracy **2°, 0.5 mm**

Finding ✓ **migration >1–2 mm in the first year → ↑ risk of loosening** (varus or retroversion) · both cemented and cementless stems migrate
✗ Wrong options: cemented do not migrate · any early migration = failure · abandoned

Cemented stem failure modes (Gruen) – TO

Mode	Mechanism
I – Pistoning ✓	Stem subsides within cement (or stem + cement within bone) · **linear lucency around stem and cement** · **commonest** · poor cementing
II – Medial midstem pivot	Proximal stem medial, distal stem lateral · weak calcar support
III – Calcar pivot (distal toggle)	Distal tip toggles medial–lateral ("windshield wiper") · poor distal support
IV – Bending cantilever (fatigue)	Proximal support lost, distal fixed → **stem fracture**



Gruen failure modes of cemented stems

– – Active space – –

Cemented stem philosophies (TO)

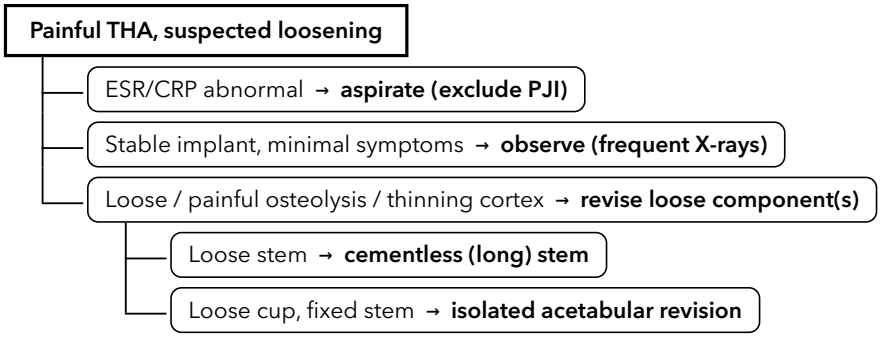
	Force-closed / taper-slip	Shape-closed / composite beam
Surface	**Polished** (Ra **<1 µm**)	**Rough** (Ra >1–2 µm), flanges, collar
Examples	**Exeter**, C-stem	**Charnley**
Subsidence	**Intended** (radial compression, no shear) · loosening if progressive **after 2 yr** or **>5 mm** total	**Not intended** · >**0.15 mm** after 2 yr → ↑ revision (**4% per 0.1 mm**)
Collar	**None** (would block subsidence) · use centralizer	Allowed (calcar load transfer)
Fracture	Slightly ↑ **periprosthetic fracture** (hoop stress)	Anatomical stems fewer early fractures

Bone quality (TO) composite beam: collar, **good in osteoporotic bone** · taper slip: proximal wedge + hoop stress → **avoid in poor bone** (fracture risk)

Earliest clinical sign (TO) **start-up pain** (groin / thigh on rising) · lucency typically appears **5–6 yr** post-op on serial films

Cement mantle **2–4 mm**, continuous · thin → fissures · **>5 mm** → micromotion, creep · **over-broaching** (stem smaller than rasp, ≥2 mm mantle) vs **line-to-line ("French paradox")** thin mantle · **retrograde gun filling** · 4th-generation: centralize stem · **Barrack–Mulroy** grades mantle quality · setting faster with heat, humidity, longer mixing, handling

5 Management



Revision indications

Revise pain from aseptic loosening · pain + osteolysis · **extensive osteolysis that would compromise future revision** · pathological / impending fracture

Observe stable implant with minimal symptoms · ↑ frequency of radiographs if wear/lucency

Untreated loose osteolytic stem highest risk ✓ **periprosthetic femoral fracture**
✗ Wrong options: infection · pseudotumour (MoM) · acetabular fracture · dislocation

Circumferential lucency + cement mantle fracture + thin cortex → ✓ revise to long-stem cementless soon
✗ Wrong options: observe · cane (use walker) · bisphosphonates (unproven, no effect on a loose stem) · long cemented stem

Loose femoral stem, normal ESR/CRP, start-up thigh pain → ✓ cementless femoral revision
✗ Wrong options: aspiration · CT · MRI · cemented revision (high early failure)

Acetabular osteolysis + loose cup, fixed stem → ✓ acetabular component revision
✗ Wrong options: head/liner exchange + grafting (loose cup) · 2-stage (not infected) · femoral · both

– – Active space – –

Typical diagnoses hemiarthroplasty 5 yr with pedestal + large lucency, normal labs → ✓ **aseptic loosening** · cement mantle fracture + lateral shoulder debonding → ✓ **femoral loosening** (✗ Wrong options: Brooker I HO · Parkinsonism) · activity thigh pain + zone 1 debonding → ✓ **loose femoral component** (✗ Wrong options: PE synovitis · psoas · asymptomatic lysis · modulus mismatch)

X vs Y

	Aseptic loosening	Septic loosening
ESR / CRP	**Normal**	↑
Radiograph	Lucency, subsidence, pedestal	+ periosteal reaction, patchy lysis, early lucency
Revision	**One-stage** of loose parts	**Two-stage**

How it's asked

"Hemi 5 yr, thigh/knee pain, pedestal, normal labs?"	→ **aseptic loosening**
"RSA finding?"	→ **migration >1–2 mm in first year ↑ loosening risk**
"Cement debonding + mantle fracture?"	→ **femoral loosening**
"Zone 1 lucency after hybrid THA, activity pain?"	→ **loose femoral component**
"Loose osteolytic stem, refuses surgery – risk?"	→ **periprosthetic femoral fracture**
"Circumferential lucency + mantle fracture at 12 yr?"	→ **revision to long-stem cementless**
"Acetabular osteolysis, loose cup, stable stem?"	→ **acetabular component revision**
"Start-up thigh pain, continuous lucency, normal labs?"	→ **femoral revision with cementless stem**
"5 mm subsidence at 6 weeks – contributing factor?"	→ **BMI 44** (undersizing)
"Linear lucency around cemented stem – failure mode?"	→ **pistoning** (TO)
"Radiographic loosening threshold?"	→ **lucency >2 mm**, subsidence >1 cm

– – Active space – –

THA DISLOCATION

– Recon · 22 / 65 –

1 Basics

Epidemiology

Incidence **1–3%** primary (OB: 0.5–10% primary, **up to 28% after revision**)
✗ Wrong options: "5–10%"

Timing **70% within first month** · 52% within 3 months · **majority within first year**, then constant lifelong rate · **late = >5 yr**

Direction **75–90% posterior**

TO notes incidence **~3%**, most within **first 3 months**

Diagnosis	Relative dislocation risk vs OA (TO)
AVN	**× 2**
DDH	**× 3**
Fracture	**× 4**
Nonunion	**× 5**
Revision	**× 11**

Mechanism

Dislocation	Position
Posterior (posterior approach)	✓ **Flexion + internal rotation + adduction** – shoe tying, low seat/toilet, bending forward
Anterior (anterior approach)	**Extension + external rotation**

Risk factors

Category	Factors
Patient	✓ **Prior hip surgery (greatest; ~2×)** · **higher comorbidity burden** (Elixhauser/ASA) ✓ · **BMI <20 or >35** · neuromuscular/spastic (**Parkinson**), cognitive impairment · drug/alcohol abuse · female (OB) / gender conflicting (Orthobullets) · age >70 · inflammatory arthritis, AVN
Spine	✓ **Prior lumbar(osacral) fusion / fixed spinopelvic alignment**
Surgical	**Posterior approach** (capsule + ER repair → near anterior rate) · component malposition (commonest cause of recurrent) · **↓ femoral offset** · ✓ **↓ head-to-neck ratio**
Implant	**MoP or MoM** bearings (vs CoP) · **cemented** components (vs uncemented)
Late	**PE wear (>2 mm)** – commonest late cause · progressive neuro decline · abductor insufficiency · loosening

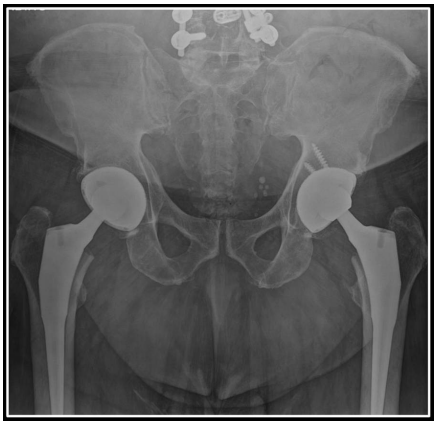
Not risk factors male sex · cup 16° anteversion · 45° abduction · high offset (tightens abductors) · direct lateral approach (lowest) · MoM large head (lower dislocation)

FNB 2024	fold increase in dislocation after prior spinal fusion (options none/2/3/4×) – answer not given by source

– – Active space – –

Spinopelvic mechanics

Normal (TO)	✓ **pelvis retroverts when sitting upright**, anteverts during sit-to-stand · lumbar lordosis **↓ sitting** ✗ Wrong options: pelvis anteverted sitting · lordosis ↑ sitting
Sitting with mobile spine	pelvis tilts posteriorly → **acetabular anteversion ↑** to accommodate femoral flexion · **sacral slope ↓**, pelvic tilt ↑
Stiff spine	**<10° change in sacral slope** standing → sitting · ✓ **lack of change (↓) in pelvic tilt** = fixed alignment → **anterior neck impingement → posterior dislocation** ✗ Wrong options: ↓ lordosis · ↓ sacral slope · ↑ anteversion · pelvic incidence (fixed parameter)
Flatback	standing anterior pelvic plane **−10°** or **PI–LL mismatch >10°** → posterior tilt → **anterior dislocation** in standing
Management	lateral **sitting + standing** spine–pelvis–hip films · ✓ **↑ cup anteversion** ± ↑ offset, **dual mobility**, navigation/robotics ✗ Wrong options: ↑ abduction (vertical cup) · retroverted stem · ↓ offset · constrained liner primary
Highest-risk history	✓ **lumbosacral fusion** ✗ Wrong options: kyphoplasty · radiation · laminectomy · ablation



Stiff spine – anterior impingement levering posterior dislocation

4 Diagnosis & Classification

Key radiology findings

Views	**AP + cross-table lateral** · **CT** for version/malposition
Malposition	inclination **>60°** · anteversion **>20°** · **retroversion**
Eccentric femoral head	**PE wear** → impending dislocation

– – Active space – –

Acetabular safe zone

Source	Abduction	Anteversion
Orthobullets	**40°**	**15°**
Lewinnek (OB)	**30–50°**	**5–25°**
OB other	30–50° · 35–45° · 40°	15–20° · 20°
Femoral stem	–	**10–15°**

Lewinnek (1978, TO) inclination **40° ± 10°** · anteversion **15° ± 10°** · radiographic anteversion = **$\sin^{-1}$ (CD / AB)** · dislocation can still occur inside the safe zone (patient + soft-tissue factors)

Navigation shows 20° abduction / 20° anteversion → ✓ increase abduction by 20°, keep anteversion (too adducted → flexion impingement, inferior dislocation)

Cup position and dislocation direction (TO)

Cup	Dislocation
Excessive anteversion	**Anterior** with **extension, adduction, ER** (✓ TO false statement: "flexion, adduction, IR")
Retroversion	**Posterior** with flexion, adduction, IR
Vertical (excess inclination)	**Superior** with adduction (± adduction contracture, inferior osteophytes)
Horizontal	**Posterior** with early flexion impingement

5 Management

THA dislocation

- First early dislocation, components well placed → **closed reduction + education**
- Recurrent (≥2) + malposition → **revise malpositioned component**
- Recurrent, late + PE wear → **revision (liner/head ± cup)**
- Recurrent, well positioned + abductor/soft-tissue deficiency → **dual mobility or constrained liner**

Wera classification of instability (2012, TO)

Type	Cause	Management (TO)
I	**Acetabular malposition**	**Cup revision**
II	**Femoral malposition**	**Stem revision**
III	**Abductor deficiency**	**Constrained liner**
IV	**Impingement**	Remove source / ↑ offset
V	**Late PE wear**	**Head + liner exchange**
VI	**Unknown cause**	Trochanteric advancement

Elevated-rim liner (TO)	↓ dislocation (**2.2% vs 3.8%**) but ↑ impingement + wear · **dual mobility** – ↑ stability, ↑ jump distance · cost: PE wear + **intraprosthetic dislocation**

First dislocation management

Closed reduction ± immobilization	**two-thirds** of early dislocations · spica, abduction brace, knee immobilizer

First dislocation at 4 weeks, components well placed → ✓ observation + education on precautions
✗ Wrong options: larger head · constrained · 3-month orthosis (only if cognitively impaired)

Component malposition – acetabular revision

Revision indication	**≥2 dislocations** with malalignment, implant failure, or PE wear · malposition = **most common cause of recurrent dislocation** → correct first
Answers	vertical cup, 3 posterior dislocations → ✓ **revise acetabular component** (✗ Wrong options: spica · modular stem · brace · constrained liner on malposition) · 10° retroverted cup → ✓ **revise entire cup** to **15–20° anteversion, ~45° abduction** (✗ Wrong options: constrained · offset · 36 mm head · GT osteotomy) · 5° retroversion + 55° abduction → ✓ **acetabular revision** (✗ Wrong options: both · head-neck/length · brace) · underanteverted vertical cup, 3 dislocations bending forward → ✓ **revise cup + larger head/ liner** (✗ Wrong options: head/liner exchange alone · PT · brace)

Revision techniques to increase stability

Technique	Mechanism
Realign components	Retroverted/vertical cup, short neck, low offset, retroverted stem
Larger head	↑ head-neck ratio → ↑ ROM to impingement + jump distance
Lateralized liner / extended offset stem	↑ offset → abductor tension
Trochanteric advancement	↑ abductor tension + compression
Elevated / lipped liner	Rotate lip toward instability
Dual mobility	Larger effective head-neck ratio
Constrained liner	Mechanical capture
Medializing cup	✓ **↓ stability** (shortens abductor arm, laxity) – the EXCEPT answer

Stem subsidence (cemented, 3 yr)	effective **neck shortening → ✓ ↓ abductor tension →** ↑ JRF + instability ✗ Wrong options: ↓ excursion distance · ↑ arc · ↓ JRF · ↑ offset
Posterior soft-tissue repair (TO)	**single myocapsular sleeve (Yadav–Ashok)** = ✓ **capsule + short ERs sutured together to greater trochanter** · ↓ IR by 50%

Late instability – PE wear & isolated liner exchange

Late instability (>5 yr)	✓ **PE wear (eccentric head)** commonest · usually needs **surgery** ✗ Wrong options: GT detachment · stem loosening · osteolysis · infection · HO

– – Active space – –

Recurrent late dislocations + eccentric wear, normal labs → ✓ revision THA for PE wear
✗ Wrong options: MRI · re-aspiration · observe · PT

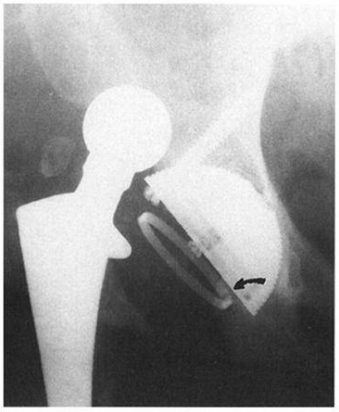
Isolated liner exchange	stable, well-aligned implants with PE wear as **sole** cause · ✓ **most common complication = dislocation / instability** ✗ Wrong options: osteolysis · shell–liner failure · PE fracture · sciatic injury · acetabular fracture

Cementing a liner into a well-fixed shell (broken locking tines) — **liner several sizes smaller** than shell (room for cement) ✓ · **score PE + smooth shell** · fully seat · avoid impingement · commonest complication still **dislocation**
✗ Wrong options: "closely match liner and shell size"

Constrained liners

Indications	✓ recurrent instability with **well-positioned components** · **abductor deficiency / unrepairable capsule** · **neuromuscular / cognitive** disease · no identifiable cause · failed larger head + trochanteric advancement · elderly low-demand
Contraindications	**component malposition** ✓ (revise first) · **PE wear alone** ✓ (EXCEPT answer) · pelvic discontinuity · infection
Limited because of	✓ **high failure rates** (fatigue, **ring/liner dissociation**) · ↑ bone–implant stress · ↓ ROM, worse function · last resort
Answers	cup 20° anteversion + 40° abduction, recurrent → ✓ **constrained** (✗ Wrong options: discontinuity · infected · 15° retroverted stem · 10° retroverted/60° cup) · bariatric weight loss, failed big head + GT advancement, aligned → ✓ **constrained** (✗ Wrong options: resection · brace · capsular ablation · bipolar) · abductors absent, components well placed → indication ✓ **incompetent abductors** · post-MoM deficient abductors, Trendelenburg → ✓ **constrained liner** (✗ Wrong options: medialize cup · offset · large ceramic head)

Broken constrained ring → ✓ ↑ dislocation risk · dislocated constrained + malposition + Parkinson → ✓ **acetabular revision with constrained implant**
✗ Wrong options: closed or open reduction



Broken constrained liner ring

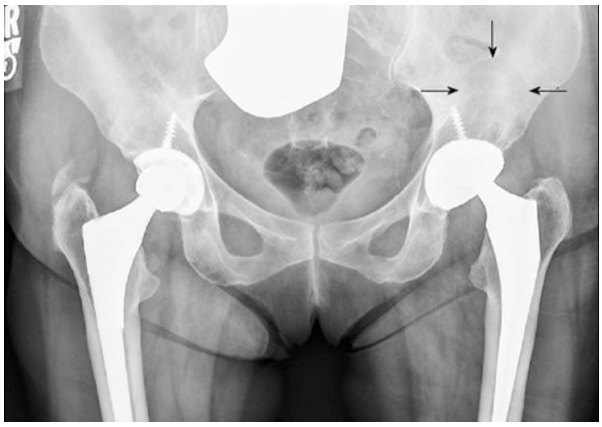
Dual mobility & abductor deficiency

Design	small inner head captive in mobile PE liner, articulating with shell (Bousquet 1974, France) · Charnley low friction + McKee–Farrar ↑ head-neck ratio · ✓ **↑ arc of motion before impingement** + ↑ jump distance ✗ Wrong options: locking ring (constrained) · ↓ jump distance · lipped liner · face-changing liner

Unconstrained — tripolar may add constraint ring · ✓ **"constrained design reduces dislocation" is NOT true** of DM (TO) · both articulations share centre · inner head dominates normal ROM, outer in high ROM

Indications (TO) — high instability risk: neuromuscular, cognitive, **ASA ≥3**, **>75 yr with prior hip surgery** · revision THA · THA after **femoral neck fracture** · tumour · ✓ **reduces instability in revision**
✗ Wrong options: jumbo cup (↑ instability) · small head · capsular debridement · ↓ abductor tension

Intraprosthetic dislocation (IPD) — unique to DM · ✓ **wear of PE liner's retentive chamfer** · head articulates on metal shell → **acute shortening, limp**, metallosis · X-ray: **eccentric head** + **"bubble sign"** (liner in soft tissue) · can follow closed reduction attempt



Dual mobility – bubble sign of intraprosthetic dislocation

Salvage

Conversion to hemiarthroplasty/ bipolar — soft-tissue deficiency · contraindicated if acetabular bone compromised · rarely used (DM) · groin pain, migration

Resection arthroplasty — options exhausted · bone + soft-tissue loss · **psychiatric secondary gain**

X vs Y

	Dual mobility	**Constrained liner**
Mechanism	↑ ROM before impingement, ↑ jump distance	**Mechanical capture** (locking ring)
ROM	Large	**Restricted**
Failure	**Intraprosthetic dislocation**	**Ring/liner dissociation**, loosening
Use	High-risk primaries, revision	Last resort – abductor/soft-tissue deficiency

	Early dislocation	**Late dislocation (>5 yr)**
Cause	Malposition, soft tissue, noncompliance	**PE wear**, neuro decline, abductor loss
Treatment	Often closed reduction	Usually **surgery**

– – Active space – –

How it's asked

"Normal pelvis in sitting (TO)?"	→ retroverted
"Broken ring (arrow) after revision — risk?"	→ dislocation
"Posterior approach — most dangerous position?"	→ flexion + internal rotation
"Higher dislocation risk?"	→ previous hip surgery
"Increases instability?"	→ reduced femoral head-to-neck ratio
"Lumbar fusion — spinopelvic change increasing risk?"	→ decreased (unchanged) pelvic tilt
"Dislocation after primary THA (2023) — true?"	→ higher comorbidity burden ↑ risk
"Stiff spine — adjust implants?"	→ increase cup anteversion
"Which history ↑ dislocation?"	→ lumbosacral fusion
"False about cup position (TO)?"	→ anteverted cup dislocates anteriorly in flexion, adduction, IR
"3 dislocations, vertical cup?"	→ revise acetabular component
"Retroverted cup 10°, 3 dislocations?"	→ revise entire acetabular component
"Unstable after reduction — EXCEPT increases stability?"	→ medializing the cup
"Navigation 20/20 — correct to safe zone?"	→ ↑ abduction 20°, keep anteversion
"Five dislocations, retroverted cup?"	→ revise cup to 15° anteversion, 45° abduction
"Underanteverted cup, 3 dislocations, good abductors?"	→ revise cup + larger head/liner
"Cup 5° retroverted, 55° abduction?"	→ acetabular component revision
"Most common complication after liner revision?"	→ dislocation
"Liner exchange + grafting — disclose?"	→ postoperative instability
"5 late dislocations, eccentric wear, normal labs?"	→ revision for PE wear
"PE exchange with standard UHMWPE + grafting — ↑ risk?"	→ dislocation
"Late instability with eccentric head?"	→ wear
"Cemented liner into shell — most common complication?"	→ dislocation
"First instability at 15 yr?"	→ polyethylene wear
"Appropriate constrained liner use?"	→ recurrent dislocation with well-positioned cup (20°/40°)
"Constrained liner EXCEPT?"	→ PE wear
"Indication shown on revision radiographs?"	→ instability
"Post-MoM, deficient abductors, 4 dislocations?"	→ constrained liner
"Failed head size + GT advancement, aligned?"	→ constrained liner
"Cemented constrained liner dissociated — EXCEPT?"	→ closely matching liner and shell size

"Dislocated constrained implant + malposition, Parkinson?"	→ **acetabular revision with constrained implant**
"What limits constrained liners?"	→ **high failure rates**
"Avoid constrained liner because?"	→ **component malposition**
"Not true about dual mobility (TO)?"	→ **constrained design reduces dislocations**
"DM hip, shortening after fall?"	→ **wear of retentive chamfer (IPD)** (TO)
"Second-stage revision — reduce instability?"	→ **dual mobility**
"Constrained liner indication (abductors absent)?"	→ **incompetent abductors**
"DM with PFR — mechanism?"	→ **↑ arc of motion before impingement**
"Eccentric head + liner in soft tissue after reduction?"	→ **intraprosthetic dislocation**
"Cemented stem subsidence — effect?"	→ **↓ abductor tension**
"First dislocation, well-placed implants?"	→ **observation + education**
"Single myocapsular sleeve repair (TO)?"	→ **capsule + ERs sutured together to GT**
"Recurrent dislocation from abductor deficiency (Wera III)?"	→ **constrained liner** (TO)
"Relative dislocation risk highest in?"	→ **revision THR (× 11)** (TO)

– – Active space – –

THA ILIOPSOAS IMPINGEMENT

– Recon · 23 / 65 –

1 Basics

Etiology

Definition	under-recognized cause of **recurrent groin pain** after THA (up to **4.3%** of painful THAs)
Causes	✓ **malpositioned acetabular component with anterior overhang** (most common) · cup **too large**, **retroverted** or **lateral** · **excessively long screws** penetrating ilium · **retained cement** · LLD (Orthobullets lists; OB: not significant)
Version trap	anterior overhang from **retroversion** · (✗ Wrong options: excessive offset → **trochanteric bursitis**) ✗ Wrong options: excessive anteversion
Anatomy	iliopsoas from lumbar transverse processes, passes **anterior to hip** → **lesser trochanter**

3 Presentation

Onset	average **20 months** after index THA ✗ Wrong options: 2 weeks
Pain	**groin**, with **active hip flexion**: **ascending stairs** (not descending), getting into/out of car, rising from chair · **pain with passive extension**
Exam	subtle: slight limp, groin tenderness, rare palpable snap · ✓ **pain with resisted seated hip flexion or active straight-leg raise** · passive ROM/SLR painless
Not iliopsoas	**start-up groin pain** → cup loosening · start-up **thigh** pain → stem loosening · constant night pain + fevers → PJI

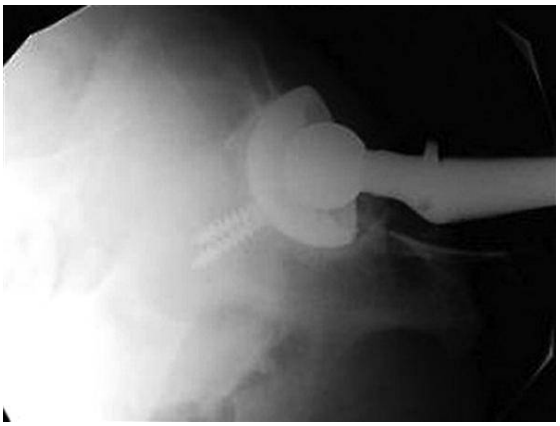
4 Diagnosis & Classification

Key radiology findings

Radiographs	AP pelvis + AP/lateral hip · ✓ **cross-table lateral** shows **anterior cup prominence** (TO next step; ✗ Wrong options: MRI · PET · bursa aspiration)
CT	cup position, **anterior overhang**, excludes other causes · **MRI** – limited by artefact (shows tendon inflammation)
Rule out	infection (ESR, CRP, aspiration) + loosening first

Diagnostic injection

Image-guided local anaesthetic ± steroid into iliopsoas sheath	✓ **most reliable confirmation** · diagnostic **and** therapeutic ✗ Wrong options: GT bursa injection · repeat aspiration · Ober test · knee films · bone scan · revision before diagnosis



Anterior cup overhang causing iliopsoas impingement

5 Management

Groin pain with resisted hip flexion after THA

- Exclude infection + loosening → **diagnostic iliopsoas sheath injection**
- Cup well positioned (normal X-ray/CT) → **arthroscopic iliopsoas tenotomy**
- Excessive anterior overhang (>8 mm) → **acetabular component revision**

Treatment – tendon release vs cup revision

Nonoperative rest, NSAIDs, stretching, CT-guided steroid injection · **rarely sufficient** (relief in only **~50%**)

Normal radiographs / well-placed cup → ✓ **iliopsoas tenotomy** (arthroscopic)
✗ Wrong options: indefinite activity modification · femoral/acetabular revision

Anterior overhang → ✓ **acetabular component revision** for permanent relief · **>8 mm prominence** → revision far better than tenotomy (Chalmers: persistent pain 50% nonop vs 24% operative)
✗ Wrong options: tenotomy alone · lipped liner (worsens) · repeat injection · both components

Arthroscopic tenotomy technique **extra-capsular** (at lesser trochanter, common after THA) or **trans-capsular** (between anterior labrum and zona orbicularis) · hip in slight flexion + ✓ **external rotation** → brings tendon toward surgeon, away from **femoral neurovascular bundle** · anterior portal ~**15 mm** from LFCN branches → shift portal lateral
✗ Wrong options: IR (for deep gluteal space/sciatic) · MFCA (posterior)

X vs Y

	Iliopsoas impingement	**Trochanteric bursitis**	**Cup loosening**
Pain	Groin, **active flexion / stairs up**	**Lateral**, side lying	Groin **start-up**
Cause	**Anterior cup overhang**, long screws	↑ offset	Osteolysis, micromotion
Test	**Sheath injection**	Bursa injection	Serial X-rays

– – Active space – –

How it's asked

"Groin pain 3 months post-THA, painful active SLR, normal labs/X-ray — next (TO)?"	→ cross-table lateral X-ray
"Most likely to determine cause of pain (stairs, passive extension)?"	→ local anaesthetic injection of iliopsoas sheath
"Resisted seated flexion pain, well-placed components — next?"	→ image-guided lidocaine injection into iliopsoas sheath
"Structure labelled 1 (groin pain with stairs up)?"	→ iliopsoas tendon
"MRI inflamed iliopsoas — cause?"	→ anterior overhang of acetabular component
"Stair-climbing groin pain, prominent cup anteriorly?"	→ iliopsoas impingement
"Offset ↓ 2 mm, length ↑ 3 mm, prominent cup — likely?"	→ iliopsoas tendonitis
"History + exam typical of psoas impingement?"	→ pain with passive extension + resisted flexion, onset ~20 months
"Injection relieved pain, normal X-rays — next?"	→ iliopsoas tendon release
"Arthroscopic tenotomy — hip position?"	→ external rotation (protect femoral neurovascular bundle)
"Recurrent pain with confirmed anterior overhang — permanent fix?"	→ acetabular component revision

– – Active space – –

THA IMPLANT FIXATION

– Recon · 24 / 65 –

1 Basics

Types of fixation

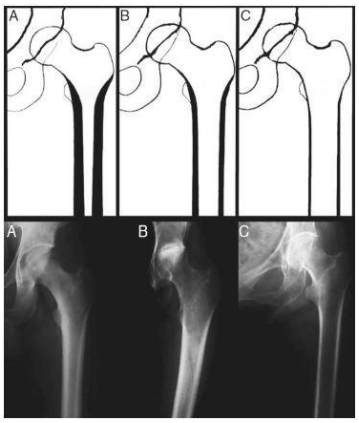
Cement	**PMMA**, acts as **grout** (interlock, no adhesion)
Biologic (cementless)	**ingrowth** into porous coating · **ongrowth** onto grit-blasted surface
Collared stem	transfers load to **calcar**, ↓ early subsidence, controls depth · cemented or cementless · **protective against early periprosthetic fracture**
History	cement: **Gluck 1891**, **Charnley 1950s** (dental cement) · "cement disease" 1980s → cementless · **AML (1983)** first microporous ingrowth stem → proximal coating (thigh pain, osteolysis)
Trend	**93% of US THAs cementless** (2012)
Templating	calibrated radiograph · goal: **limb length + femoral offset**

Dorr classification

Ratio canal diameter **10 cm below mid-lesser trochanter ÷ inner diameter at mid-lesser trochanter**

Type	Ratio	Features	Stem
A ("champagne flute")	**<0.5**	Thick cortices AP + lateral · thick posterior **fin** · young, heavy, male	**Uncemented** (flat taper)
B	**0.5–0.75**	**Posterior cortex thinning**, fin eroded, **"rat bites"**	Uncemented · flat taper may be **rotationally unstable** → metadiaphyseal/diaphyseal or cemented
C ("stovepipe")	**>0.75**	Thin, fuzzy cortices both views · elderly, thin, female	**Cemented**

TO EXCEPT (Dorr B) ✓ **"ratio <0.5"** is false (that is A)



Dorr classification

– – Active space – –

Cemented stem types (Orthobullets)

Type	Subtypes / examples
1 – Taper-slip	1A anatomic (Olympia) · **1B double taper (Exeter, CPT)** · 1C triple taper (**C-stem**, CPCS) · traditional or line-to-line (Corail, Avenir, CMK)
2 – Composite beam	2A anatomic (Lubinus SP) · 2B straight (Summit, Accolade C, Versys, Spectron) · 2C wedge (Müller)

Cementless stems A flat taper · B quadrangular taper · C fit-and-fill · D conical · E cylindrical · F calcar-guided short neck-preserving

Modularity 1 head · 2 neck · 3 subtrochanteric · 4 metaphyseal sleeve

5 Management

Operative criteria – cemented vs cementless

Component	Cemented	Cementless
Femur	**Elderly / osteopenic** · **irradiated bone** · **Dorr C (stovepipe)**	Young, active males · old with good bone · **revision** (cemented revision stems fail more)
Acetabulum	Poor bone, complex revision · **fails more** (cement poor in **shear**)	**Preferred** (porous hemispheric) – except poor bone stock or irradiation

Elderly woman, minimize fracture + durable → ✓ hybrid (cementless cup + cemented stem)
✗ Wrong options: cemented cup (early loosening) · hemi · press-fit stem (fracture) · resurfacing

Cemented vs cementless femur in elderly (registry) → ✓ lower early revision, ↓ iatrogenic + early periprosthetic fracture, ↓ early aseptic loosening · LLD no different · intraop fracture **3.5% uncemented vs 0.4% cemented** · overall **3% vs 0.23%** (female, >65)

Cemented fixation

Generations

Generation	Technique
1st	Hand mixed, **finger packed**, no restrictor/preparation
2nd	**Cement restrictor**, **cement gun**, brush + dry canal
3rd	**Vacuum mixing** (↓ porosity), **pressurization**, **pulsatile lavage**
4th (TO)	+ **stem centralization**

Optimized by ↓ porosity (vacuum) · pressurization · **mantle >2 mm** (OB ~2 mm ideal; TO **2–4 mm**, >5 mm ↑ micromotion) · **stiff stem** · centralization · **clean dry bone** · **smooth stem, rounded corners** · **no mantle defects** (stem touching cortex) · neutral position
✗ Wrong options: varus/valgus

↑ Failure ✓ **PMMA-precoated stems** (extra interface)
✗ Wrong options: collar–calcar contact · smooth corners · 2 mm mantle · stiff stem >115 GPa (stiffer improves)

Osteopenia ✓ **stronger bone–cement interface** (porosity → deeper penetration) · young male dense bone → weaker interface · pressurization ↑ penetration + fracture resistance

Barrack–Harris cement grading

Grade	Appearance
A	Complete canal fill, **"white-out"**
B	Slight radiolucency (<50% of interface)
C	✓ **Radiolucency >50%** or **incomplete mantle**
D	**Gross radiolucency** and/or **no cement around stem tip**

Bone cement properties (TO)

Mechanics — **strong in compression**, ✓ **weak in shear** (EXCEPT answer) · **E ≈ 2.2 GPa** · grout aiding load transfer

Chemistry — liquid **MMA monomer** + powder (PMMA–styrene co-polymer) · initiator **benzoyl peroxide** (powder) · accelerator **DmpT** (liquid) · stabilizer **hydroquinone** · radiopacifier **zirconium dioxide** / barium sulphate · ✓ **never 100% conversion** – **2–6% residual monomer** → 0.5% at 2–3 wk (EXCEPT answer) · glass transition **~70 °C** wet (above body temperature)

Curing — mixing → sticky → working → hardening · **heat, humidity, faster mixing ↓ working time** · pre-chilling prolongs it · antibiotics **<2 g** per pack keep strength

Calcium phosphate cement — no heat, osteoconductive · **not for load bearing** (EXCEPT answer) – weak, brittle, low cohesion in water

Biologic fixation

Parameter	Requirement
Pore size	**50–300 µm**, ideally 50–200 (Orthobullets) · **50–400 µm** (OB)
Porosity	**40–80%** (more → ingrowth but weaker, metal shearing)
Gaps	**<50 µm**
Micromotion	**<150 µm** (>150 → fibrous) · initial rigid fixation **<50 µm**
Cortical contact	Maximal
Bone	Viable – **irradiation ↑ aseptic loosening**

Press-fit — implant slightly larger than reamed (no screws needed) · **line-to-line** – same size, screws often added

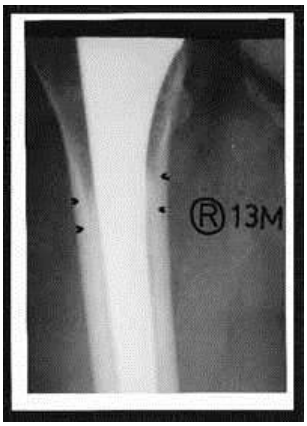
Coatings — **proximal porous** (less stress shielding) · **extensive porous** (more proximal shielding; revision) · **grit-blasted** ongrowth – always extensively coated (weaker fixation) · **hydroxyapatite** – osteoconductive adjunct · ✓ **shorter time to biologic fixation in animals**, no clinical advantage · needs **high crystallinity**, optimal (not thick) thickness, rough surface

Radiographic signs of stable fixation

Cementless stem — **spot welds** (endosteal bone to porous surface) · no lines around porous part · **proximal stress shielding** (extensively coated) · no subsidence

Cementless cup — no migration · no progressive lines · intact screws

– – Active space – –



Spot welds

6 Complications

Aseptic loosening	poor initial fixation, mechanical loss, particle osteolysis · cup → groin/buttock pain · stem → **thigh, start-up pain** · serial X-rays, bone scan · revise loose parts
Stress shielding	proximal bone loss around well-fixed stem · risk: ✓ **stiff stem (modulus mismatch; Co-Cr > Ti)** – most important · **large diameter** (biggest contributor to stiffness) · **extensive porous coating** · preoperative osteopenia · clinical significance unknown, no treatment
Intraoperative fracture	risk: **press-fit** · collar protective · **acetabular**: stable cup → add screws; unstable → remove, fix, reinsert with screws · **femoral**: stable stem → limit WB ± cerclage; unstable → remove, fix, stem bypassing by **2 cortical diameters**

X vs Y

	Cemented femur	**Cementless femur**
Patient	Elderly, osteopenic, **Dorr C**, irradiated	Young, active, Dorr A/B, **revision**
Fracture	↓	↑ (press-fit)
Early revision (elderly)	↓	↑
Failure mode	Debonding, mantle defects	Fibrous ingrowth if micromotion >150 µm

How it's asked

"Optimal pore size?"	→ **50–400 µm** (OB) · 50–300 µm (Orthobullets)
"HA-coated vs porous stems in animals?"	→ **shorter time to biologic fixation**
"Barrack–Harris – incomplete mantle, >50% lucency?"	→ **grade C** (TO)
"Bone cement EXCEPT?"	→ **strong in shear** (TO)

"Bone cement EXCEPT?"	→ **hardening = polymerization of 100% MMA** (TO)
"Calcium phosphate vs acrylic — false?"	→ **better in load-bearing areas** (TO)
"↑ cemented femoral failure?"	→ **PMMA-precoated stems**
"Osteopenia and bone–cement interface?"	→ **improved mechanical integrity**
"78-yr-old woman — minimize fracture, durable?"	→ **hybrid: cementless cup + cemented stem**
"Cemented vs cementless stem in elderly?"	→ **lower risk of early revision**
"Dorr B — EXCEPT?"	→ **cortical thickness ratio <0.5** (TO)
"Most important stress-shielding factor?"	→ **stem stiffness (modulus mismatch)**
"Micromotion threshold for bone ingrowth?"	→ **<150 µm** (gaps <50 µm)

– – Active space – –

– – Active space – –

THA LEG LENGTH DISCREPANCY

– Recon · 25 / 65 –

1 Basics

Epidemiology

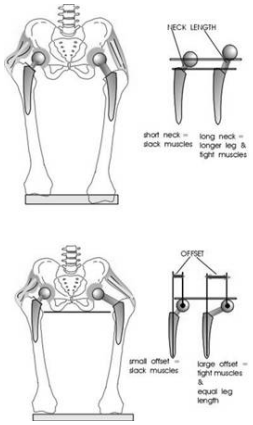
Litigation	**2nd most common cause of THA litigation** (after **nerve injury**)
Prevention	**preoperative planning/templating** ↓ incidence

Mechanism

Cause	Effect
ABDuction contracture	Involved hemipelvis **lower** → apparent **LONG** leg
ADDuction contracture	Involved hemipelvis **higher** → apparent **SHORT** leg
Weak abductors	Sensation of **long leg without true LLD** · resolves **3–6 months**
Unilateral OA	Pre-op shortening (cartilage loss, contracture) → equalized leg **feels long**

What changes length

Lengthens	Does NOT change length
Longer femoral neck	✓ **↑ femoral offset** (moves femur laterally → abductor tension, stability)
High (proximal) neck cut	**Medialized cup** (tension/JRF only)
Incomplete seating of cemented stem	Cemented vs cementless stem (if done properly)
Inferior cup (graft above to seat lower)	–
–	**Short-neck** component → **shortens**



Neck length vs offset effect on limb length

Offset restoration

Decreased offset	femur closer to pelvis → ✓ **bone-on-bone impingement** (GT on pelvis) → levering, instability · **↓ abductor moment, ↑ JRF**, abductor weakness, soft-tissue laxity ✗ Wrong options: trochanteric bursitis
Increased offset	**trochanteric pain / bursitis** · ↑ stability, ↓ impingement

3 Presentation

Symptoms	**perceived LLD** despite anatomic equality
Tolerance (TO)	**< 1 cm** unnoticed · **1–2.5 cm** may dissatisfy · **> 2.5 cm** symptomatic · lengthening **> 2.5–4 cm** → nerve injury risk · both over-lengthening and shortening ↑ contralateral hip OA risk (TO)
True length	**ASIS → medial malleolus**
Apparent length	includes contractures + pelvic obliquity · hard to measure

Perceived LLD at 6 weeks, equal lengths on X-ray, normal motion/strength → ✓ perceived LLD
✗ Wrong options: flexion/adduction contracture (would make the operated leg feel short) · medialized cup · femoral malposition

4 Diagnosis & Classification

Key radiology findings

AP pelvis	compare limb lengths (inter-teardrop / ischial line to lesser trochanter) · ↑ neck length · **offset change does not alter length**



Leg length discrepancy on AP pelvis

5 Management

LLD after THA

- Perceived / apparent (contracture, weak abductors) → **reassure, PT, wait 3–6 months**
- True LLD persisting >6 months → **shoe lift (most)**
- Significant LLD affecting life >6–12 months → **revision (rare; dislocation risk if shortening)**

Shoe lift	adequate in most · **wait 6 months** for muscle relaxation

– – Active space – –

Revision rare · significant LLD unresolved at **6–12 months** affecting quality of life · **shortening revision risks dislocation**

Intraoperative factors

Templating (TO) bilateral AP pelvis, hips **15° IR** · **inter-ischial line** + **top of lesser trochanter** each side → shortening · copy centre of rotation from normal side · cup: **inferior edge on teardrop line**, medial edge not past vertical teardrop line, ~**40°** · femur: GT + LT distances to head centre → neck cut + offset · femoral + acetabular centres at **same height** → equal length

Intra-op check (TO)	Meaning
Ranawat technique	Greater trochanter / pelvic reference for length
Shuck test	Traction → **half the head stays in the socket** = adequate tension
Drop-kick test	Soft-tissue (rectus) tension

Restore anatomic length templating · variable neck lengths · intraoperative length check · clinical + radiographic pre-op assessment · ✓ **variable-offset modular system does NOT restore length** (EXCEPT answer)

How it's asked

"Help restore limb length EXCEPT?"	→ **modular system with variable femoral offset**
"Lengthening techniques EXCEPT?"	→ **increasing femoral neck offset**
"Most likely to lengthen the leg?"	→ **very proximal neck cut** ✗ Wrong options: medialized cup · high-offset stem · short neck · cemented stem
"Equal lengths on X-ray, operated leg feels long at 6 weeks?"	→ **perceived LLD**
"Failure to restore offset (TO)?"	→ **bone-to-bone impingement**
"Abduction contracture causes?"	→ **apparent long leg**
"When to treat LLD with shoe lift?"	→ **after 6 months**
"Second most common THA litigation cause?"	→ **LLD** (first: nerve injury)
"Shuck test – adequate tension?"	→ **half the femoral head stays in the acetabulum** (TO)
"LLD after THR usually tolerated up to?"	→ **≤ 2.5 cm** (TO)

THA OTHER COMPLICATIONS

– Recon · 26 / 65 –

1 Basics

Scope

This chapter — **heterotopic ossification**, **squeaking**, **postoperative anaemia/transfusion**, stem fracture, medical complications, femoral nerve palsy, persistent pain · other THA complications have their own chapters

6 Complications

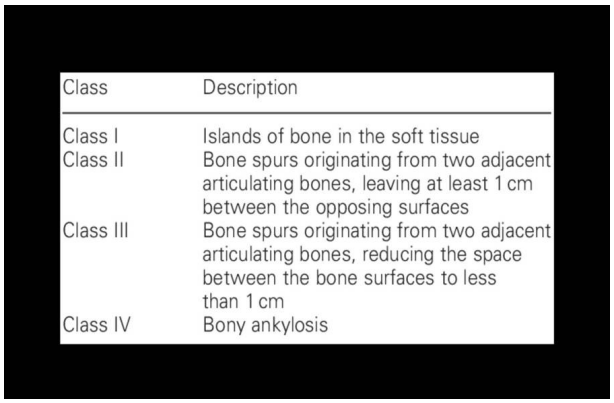
Heterotopic ossification

Frequency — common **radiographic** finding (first year), rarely functional · **~15%** clinically significant stiffness (Brooker III/IV) · more after THA than TKA · 2–5% despite prophylaxis

Biology — mesenchymal cells in muscle/fascia → osteoblasts · starts **within 16 h** of insult, matures over >1 yr into lamellar bone

Risk factors — prolonged surgery · excessive soft-tissue handling · **hypertrophic OA** · ✓ **male** · ✓ **ankylosing spondylitis** · **prior HO** (other site or **contralateral hip**) · DISH · traumatic brain injury

Brooker grade	Clinical significance
I–II	Usually asymptomatic, inconsequential (I not a pain source)
III–IV	Clinically significant stiffness, may need excision



Heterotopic ossification after THA (Brooker)

– – Active space – –

HO prophylaxis

Method	Regimen
NSAIDs	**Indomethacin** classic (other NSAIDs equivalent) · **7–21 days** (≥10 days; longer no benefit) · not 2 days · not in renal insufficiency/warfarin
Radiation	**600–800 cGy** (700–800 as effective as higher) · Orthobullets: within **24–48 h post-op** · OB: **within 24 h pre-op** or **72 h post-op** · Iorio/Healy: **≤6 h pre-op** or **≤4 days post-op** · pre-op = post-op efficacy (73% vs 76%)
Not effective	Drains · **etidronate (EHDP)** – delays mineralization only · radiation 1 week before surgery · 1,500 cGy (unnecessary)

Answers prior HO + renal insufficiency + warfarin → ✓ **800 cGy pre-op on morning of surgery** (✗ Wrong options: indomethacin 3–6 wk · 400 cGy day 2 · no treatment) · ankylosing spondylitis → ✓ **pre-op radiation 700 cGy** (✗ Wrong options: sciatic monitoring · 1,500 cGy · CPM · soft collar) · contralateral HO → ✓ **radiation 8 h post-op** (✗ Wrong options: indomethacin 2 days · drains · EHDP · RT 1 week pre-op)

Prophylaxis candidates (Purcell–Lachiewicz) bilateral hypertrophic OA · OA + lumbar **DISH** · **AS** · HO after contralateral hip surgery

HO treatment

Early Brooker III at 6 weeks, asymptomatic → ✓ observe, repeat X-ray in 6 weeks
✗ Wrong options: indomethacin (too late) · RT (too late) · excision at 10 wk (immature, ↑ recurrence)

Formed HO prophylaxis ineffective · ✓ **only surgical excision** eradicates

Excision severe loss of motion · **wait ≥6 months** for maturation + capsule · confirm maturity: stable serial films, normal bone scan uptake, **normalized serum ALP** (non-specific) · **pre-op CT** for resection planning · **perioperative RT or NSAIDs** after excision ✓ (mature grade III/IV + severe limitation → **excision + post-op irradiation**) · instability after excision common → have **constrained liners** available · protect sciatic nerve

Stiffness at 6 months with HO → ✓ first check ALP, ESR, CRP (exclude PJI, assess maturity)
✗ Wrong options: late indomethacin/RT · immediate excision · excision + constrained revision

Squeaking

Definition high-pitched audible sound with hip motion (e.g. extension from flexion on stairs)

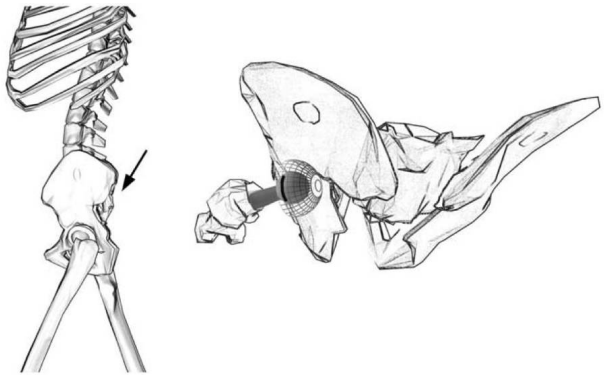
Incidence **CoC 0.5–10%** · **MoM 4–5%** · revision for squeaking **0.5%**

Causes **impingement** · **edge loading** · **component malposition** · **loss of fluid-film lubrication** · **third-body particles** · **thin flexible titanium stem** · ✓ **NOT subclinical infection**

Painless squeak, well-positioned implants → ✓ observe
✗ Wrong options: revise to PE · change cup abduction

10-yr alumina CoC results ✓ **low osteolysis, squeaking and head fracture** (not eliminated)
✗ Wrong options: same osteolysis as MoP · noisy hips ↑ osteolysis · fractures eliminated

– – Active space – –



Edge loading causing squeaking

Metal wear & ARMD

Pseudotumour risk (MoM)	✓ **edge loading** (most significant) · double heat treatment · low carbon content · **female**, **dysplasia** ✗ Wrong options: obesity · age >65

Large proximal femoral osteolysis, fixed MoM implants, normal ESR/CRP → ✓ adverse reaction to metal debris
✗ Wrong options: loosening · metastasis · PJI

Stem fracture

Mechanism	✓ **cantilever bending** fatigue: **loss of proximal (calcar) support + solid distal fixation → middle-third** fracture · modular fluted stems fracture at **modular junction** with poor proximal bone · body weight + cyclic stress ✗ Wrong options: torsion · pistoning · subsidence

Postoperative anaemia

Best predictor of transfusion	**low preoperative haemoglobin** · prevention **TXA** · transfuse at **Hb 7–8**

Medical complications

Mortality (TO)	primary THR **~1% at 90 days** · revision THR **2.5–3× higher**
Timing (TO)	early: **DVT (most common)**, dislocation, infection · late: osteolysis, aseptic loosening, periprosthetic fracture · **IL-6 > CRP > procalcitonin** as infection marker (TO)
Intraoperative death	fat + marrow embolism during femoral preparation/cementing · lung histology may show ✓ **methylmethacrylate cement in pulmonary capillaries** ✗ Wrong options: thrombus PE · infarct · granuloma · amyloid
Ileus	✓ **~1%** of TJA · **THA > TKA** · risk: **older age, male, prior abdominal surgery**

Ogilvie syndrome (acute colonic pseudo-obstruction) – elderly, male, narcotics, hip arthroplasty · distended non-tender abdomen POD 3 → ✓ **abdominal radiograph** first
✗ Wrong options: occult blood · NG lavage · US · CT

Femoral nerve palsy (TO)

Presentation	weak SLR, **cannot actively extend knee** next day

– – Active space – –

Causes	retractor **anterior to iliopsoas** · **anterior capsulotomy/capsulectomy** · femoral retraction · **extruded cement** (acetabular pressurization) · **iliacus haematoma**
NOT a cause	✓ **lengthening >3 cm → sciatic (peroneal)** palsy instead

Postoperative shortening (TO)

Shortening on POD 1	✓ **low femoral neck-shaft angle** (↓ vertical offset) ✗ Wrong options: +8 head (lengthens) · high NSA (lengthens) · undersized stem (subsides later with ambulation)

Persistent pain workup

No improvement after THA, groin → medial thigh → past knee, stairs/uphill, normal labs → ✓ image lumbar spine (stenosis) · stenosis ↑ nerve palsy risk · operative order debated (spine first ↑ hip dislocation)
✗ Wrong options: aspiration · EMG first · bone scan · CT

Osteolysis revision

Failed cup with bone loss → ✓ cementless porous-coated acetabular revision ± structural graft
✗ Wrong options: bipolar · impaction grafting both sides · cemented jumbo cup · protrusio ring (Paprosky III only)

Disparities

Black non-Hispanic + Hispanic vs white non-Hispanic	↑ **LOS**, ↑ 30-day readmission, ↑ non-home discharge · ✓ **WNH shortest LOS**

X vs Y

	HO prophylaxis	HO treatment
Timing	Peri-op (RT ±24–72 h, NSAIDs 7–21 d)	**≥6 months**, mature
Tool	Indomethacin / 600–800 cGy	**Excision** + RT/NSAID

	Femoral nerve palsy	Sciatic (peroneal) palsy
Deficit	Quadriceps (SLR, knee extension)	Foot drop, EHL
Cause	Anterior retractor, cement, iliacus haematoma	**Lengthening**, posterior approach, haematoma

How it's asked

"Symptomatic mature HO 2 yr after THA?"	**→ surgical excision + post-op irradiation**
"Prior HO, renal insufficiency, warfarin — prophylaxis?"	**→ 800 cGy pre-op morning of surgery**
"Brooker III at 6 weeks, asymptomatic?"	**→ observe, repeat X-rays in 6 weeks**
"AS patient for THA — consider?"	**→ pre-op radiation 700 cGy**
"Most effective to prevent contralateral HO?"	**→ radiation 8 h post-op**

"Stiff THA with HO at 6 months — next?"	→ **ALP, ESR, CRP**
"Squeaking contributors EXCEPT?"	→ **subclinical infection**
"Painless squeak, good position?"	→ **observe**
"10-yr alumina CoC results?"	→ **low osteolysis, squeaking, head fracture**
"Most significant pseudotumour risk with MoM?"	→ **edge loading**
"MoM, large femoral osteolysis, normal labs?"	→ **adverse reaction to metal debris**
"Stem fracture mechanism?"	→ **cantilever bending**
"Lung histology after intraop death?"	→ **methylmethacrylate cement**
"Ileus risk?"	→ **~1% of TJA**
"Distended abdomen POD 3, BM yesterday?"	→ **abdominal radiograph (Ogilvie)**
"Weak SLR next day — cannot cause it (TO)?"	→ **lengthening >3 cm**
"Shortening on POD 1 (TO)?"	→ **low femoral neck-shaft angle**
"No relief after THA, radiating thigh pain past knee?"	→ **lumbar spine imaging**
"Failed cup with bone loss at 12 yr?"	→ **cementless acetabular revision**
"Minorities vs WNH after TJA?"	→ **WNH shorter LOS**
"Best predictor of transfusion?"	→ **low preoperative haemoglobin**

– – Active space – –

THA PERIPROSTHETIC FRACTURE

– Recon · 27 / 65 –

1 Basics

Epidemiology

Type	Incidence
Intraop femur – primary uncemented	**3.5%** (OB: ~4%; 1–18% / 1.5–27.8% quoted)
Intraop femur – cemented	**0.4%**
Intraop femur – primary / revision THA	**0.1–5% / 3–21%**
Intraop acetabulum	**Cemented 0.2% · cementless 0.4%**
Postoperative femur	**0.1%** overall · **0.1–3%** primary cementless · most at **stem tip**
Overall (Abdel, 32,000 THAs)	**1.7%** · uncemented **3%** vs cemented **0.23%** · female, >65

Definitions (TO) **prosthetic** = break in the implant · **periprosthetic** = bone around implant · **interprosthetic** = between two prostheses (ipsilateral THR + TKR) · fracture **distal to stem but in the same bone** = still periprosthetic

Mechanism (TO) mostly **low-energy / trivial fall** in elderly

Prevention **pre-operative templating** · adequate exposure · care with cementless stems in poor bone (RA, osteoporosis) · low threshold to cement

Risk factors

Intraoperative femur	Postoperative femur
Female ✓ · **cementless / press-fit** · osteoporosis · revision · impaction grafting · technical error · **MIS** (controversial) · sclerotic bone (sickle cell) · metabolic bone disease, Paget	Poor bone · **cementless** stems · compromised bone stock · revision · **collarless** stems

Other risks (TO) elderly, **female**, osteoporosis, **RA**, **chronic steroids** · **mini-incision** (limited exposure) · inadequate reaming · **implant size mismatch** · **uncemented + revision THR** (poor bone stock) · uncemented > cemented

Timing pattern **cementless** stems fracture **early (<6 months)** – wedge-tapered → proximal, cylindrical fully coated → distal split · **cemented** stems fracture **late (~5 yr)** at/below tip

4 Diagnosis & Classification

Key radiology findings

Radiographs AP + lateral of hip **and whole femur** · CT for detail

Loose-stem clues **prodromal thigh/start-up pain** · **subsidence** vs earlier films · **cement-column disruption** · long spiral fracture to stem tip

Stem stability ✓ **intraoperative assessment = gold standard** · up to **20% discrepancy** pre- vs intraop · **B1 vs B2 hard to separate on X-ray**

General principles (TO)

Prosthesis	Fracture	Bone stock	Management (TO)
Stable	Undisplaced	Good	**Non-op / ORIF**
Stable	Displaced	Good	**ORIF** (plate, nail or cerclage)
Unstable	Any	Good	**Revision** (long stem ± plate)
Unstable	Any	**Poor**	**Allograft–prosthesis composite / tumour (mega) prosthesis**

Goals (TO) fracture union + restore / maintain prosthesis stability · **Vancouver = most accepted** classification for THR

Vancouver classification – postoperative

Type	Description	Treatment
AG	**Greater trochanter** · often **osteolysis**	**<2 cm** displaced → nonop, partial WB, limit abduction · **>2 cm** (TO: **>2.5 cm**) → **ORIF claw plate / cables / TBW** · address osteolysis
AL	**Lesser trochanter**	**Nonoperative** (ORIF if large medial cortex – TO)
B1	Around/just below stem, **well-fixed**	**ORIF** – locking plate + cerclage cables ± strut
B2	Around/below stem, **loose**, **good proximal bone**	✓ **Revision to long porous-coated cementless (or tapered fluted) stem** bypassing **2 cortical diameters** + fixation of fragments
B3	Loose + **poor / comminuted proximal bone**	**PFR (elderly, low demand)** · **APC / proximal femoral allograft (young)** · tapered fluted stem + struts
C	**Well below** stem	**ORIF plate**, leave arthroplasty alone

TO "Vancouver type 2" → ✓ **shaft around stem**
✗ Wrong options: GT · LT · distal

Type	Subtype	Fracture description	Treatment
Type A		Fracture in trochanteric region	
	AG	Fractures of the greater trochanter	Conservative or cable wires
	AL	Fractures of the lesser trochanter	Conservative or cable wires
Type B		Fracture around stem or just below it	
	B1	Well-fixed stem	ORIF
	B2	Loose stem with good proximal bone stock	Revision THR
	B3	Loose stem with poor-quality bone stock	Revision THR
Type C		Fracture occurring well below the tip of the stem	ORIF

ORIF: open reduction and internal fixation; THR: total hip replacement.

Vancouver classification of postoperative fractures

– – Active space – –

Vancouver classification – intraoperative

Type	Location · subtype	Treatment
A1	Proximal metaphysis · **cortical perforation**	**Bone graft** (reamings)
A2	Proximal · **nondisplaced crack** (calcar)	✓ **Cerclage before inserting stem**
A3	Proximal · **displaced unstable**	**Fully porous-coated or tapered fluted stem**
B1	Diaphyseal perforation (cement removal)	Fully porous stem bypassing **2 cortical diameters** ± strut
B2	Diaphyseal nondisplaced crack (hoop stress)	**Cerclage** if stable · fully porous stem if unstable ± strut
B3	Diaphyseal displaced	Fully porous stem bypassing ± strut
C1	Distal to tip, perforation	Morcellized graft, long stem, strut
C2	Distal nondisplaced	**Cerclage + strut**
C3	Distal displaced	**ORIF**

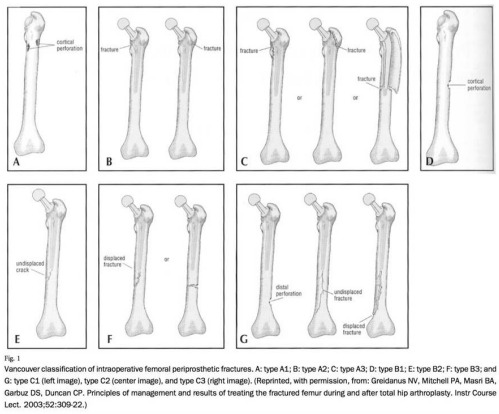


Fig. 1
Vancouver classification of intraoperative femoral periprosthetic fractures. A: type A1; B: type A2; C: type A3; D: type B1; E: type B2; F: type B3; and G: type C1 (left image), type C2 (center image), and type C3 (right image). (Reprinted, with permission, from: Greidanus NV, Mitchell PA, Masri BA, Garbuz DS, Duncan CP. Principles of management and results of treating the fractured femur during and after total hip arthroplasty. Instr Course Lect. 2003;52:309-22.)

Intraoperative Vancouver classification

5 Management

Postoperative periprosthetic femur fracture

- Trochanteric (A) → **nonop if <2 cm · ORIF if >2 cm · treat osteolysis**
- Around stem, stable (B1) or below stem (C) → **ORIF (plate + cables)**
- Around stem, loose, good bone (B2) → **long cementless revision + fixation**
- Loose + poor proximal bone (B3) → **PFR (elderly) / APC (young)**

Intraoperative femoral (calcar) fracture

Clue — **change in resistance / audible change** · stem sinks further than final broach → ✓ **calcar fracture**

Management ✓ **remove stem/broach → explore extent (intraop X-ray) → cerclage above lesser trochanter → reinsert SAME-size stem** · braided cables > wires · restores **"barrel hoop"** stability
✗ Wrong options: continue insertion + cable · keep implant and cable · undersized stem (subsides) · convert to cement without cerclage (loses hoop stress) · delayed staged reconstruction · long stem without fixing calcar

Same stem gives good outcome ✓ **female sex is a risk factor** · Berend: no revisions at 7.5 yr
✗ Wrong options: ↑ subsidence/early revision · long stem better · press-fit/MIS not risk factors

Weight bearing – conflict OB (Berend, 1349, 211416): ✓ **immediate full WB** after cerclage of minimal calcar fracture · OB (7026, 8433): **partial/protected WB 6 weeks** with 2-weekly X-rays – read the options

Extends past lesser trochanter / displaced (A3) → fully porous-coated or tapered fluted long stem · distal cracks found post-op → touch-down WB

Mechanisms **proximal**: rasping, insertion, dimension mismatch · **mid**: exposure/preparation force · **distal**: straight-stem tip impacting **femoral bow**

Intraoperative acetabular fracture

Mechanism during **cup impaction** · ✓ **most common complication of press-fit cup** (TO) · risk: **under-reaming >2 mm** (most common cause), **elliptical/monoblock** cups, osteoporosis, cementless, dysplasia, radiation

Stable cup → ✓ no change or **add screws**, protected WB **8–12 wk** (OB: flat-foot 2 wk) · nondisplaced anterior column with stable cup → ✓ **multiple screws + brief WB restriction**
✗ Wrong options: larger cup (propagates) · plating · cup-cage · 6 wk bed rest

Unstable cup → screws ± **jumbo cup** ± ORIF + revision · **posterior column displaced → ✓ remove cup, fix posterior column, antiprotrusio cage** (or cage-over-cup)
✗ Wrong options: larger cup · graft · WB protection · cemented PE

Acetabular periprosthetic – Peterson (TO)

Timing usually **intraoperative** rather than postoperative (TO)

Peterson	Features	Management (TO)
Type 1	**Stable**, undisplaced, minimal pain	**Retain cup ± trans-acetabular screws**
Type 2	**Displaced / unstable** cup, pain, instability	**Revision + column plating ± antiprotrusio cage**

Vancouver A, B1 & C – ORIF

B1 ✓ **ORIF: lateral locking plate + proximal cables** · stable stem confirmed intraop
✗ Wrong options: traction · revision · cemented stem

B1 fixation details ✓ **locking screws proximally** (bicortical where possible; unicortical alone insufficient) · **2–3 cables** between LT and stem tip · **≥3 bicortical screws / 6–8 cortices distal** · strut allograft only for bone loss/migration
✗ Wrong options: avoid cables · <3 distal screws

B1 reoperation ✓ **nonunion** most common (B2 revision + ORIF → **infection**)

Bridge plating (relative stability) ✓ **endochondral (secondary) healing**
✗ Wrong options: intramembranous (needs compression)

– – Active space – –

Ogden construct (TO)	✓ **periprosthetic femur fractures (Vancouver B)** · cables proximally + screws distally (now unicortical locking proximal, bicortical distal)
Vancouver C (TO)	**distal femoral LCP** · plate should **span to distal femoral metaphysis** to prevent re-fracture · interprosthetic → **Vancouver C** for the THR (**Su type I** for the TKR) → **long bridging plate ± bone graft**
Surgical pearls (TO)	**prepare for revision even in B1** (long stem + plates on the table) · pass cables **through muscle, not subperiosteal** · avoid periosteal stripping · stem **> 2 cortical diameters** beyond fracture
Vancouver C	✓ **plate + proximal cables + distal screws** (✗ Wrong options: retrograde nail (no canal) · conventional plate · traction) · **interprosthetic** (THA + TKA) → ✓ **lateral locking plate spanning whole femur** (✗ Wrong options: nail · revision · strut alone · total femur)

Below long revision stem, well fixed → ✓ ORIF plate + cables/wires
✗ Wrong options: strut + wires alone (no rotational control)

FNB 2024	mid-shaft fracture 3 months after THA, well-fixed stem → options nail / plating ± cerclage / traction / long-stem revision — answer not given (well-fixed stem → plating fits B1/C)

AG >2 cm → ✓ ORIF trochanteric claw plate · **AG with osteolysis + PE wear (eccentric head) → ✓ revision THA** (liner exchange, address lysis, fix GT)



Trochanteric claw plate

Vancouver B2 – long cementless revision stem

Treatment	✓ **revise to long uncemented diaphyseal-engaging stem** (extensively porous-coated cylindrical or **modular tapered fluted titanium**) bypassing fracture by **≥2 cortical diameters** + **ORIF of fragments** (cables/plate)
Why not cemented	cement **extrudes into fracture → nonunion**, mid-term loosening, poorer results in active patients · proximally coated stem (no proximal fixation — ↑ loosening/nonunion) · ORIF alone (loose stem) · strut alone when bone stock adequate · traction · PFR (bone stock adequate) · antibiotic spacer (no PJI)
Typical stems	prodromal thigh pain + subsidence · cement column disruption · loose at 2 weeks with subsidence · "easily extractable" stem at surgery
B2 ORIF alone (relative)	low demand, **ASA ≥3**, **CCI ≥5**, anatomic reconstruction possible, no prior loosening/pain (González-Martín) · revision favoured: high demand, ASA <3, comminution / ≥2 zones, prior pain

Loose cemented stem eroding lateral cortex (impending fracture) → ✓ cementless revision
✗ Wrong options: cemented ± strut · strut + cables · PFR

Fracture around antibiotic spacer, infection cleared (WBC 500, α-defensin −) → ✓ reimplant uncemented diaphyseal-engaging stem + cables

– – Active space – –

✗ Wrong options: new spacer · traction + suppression · ORIF

Normal ABI (0.9–1.3) with sluggish refill + B2 → ✓ revision + ORIF, serial exams

✗ Wrong options: vascular consult/arteriogram · US · traction · exploration

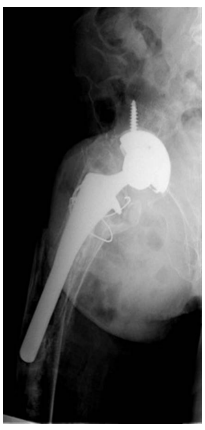


Vancouver B2 fracture

Vancouver B3 – proximal femoral replacement

Options	**PFR / megaprosthesis / tumour prosthesis** (elderly, low demand, dementia, can't restrict WB → immediate WB) · **APC / proximal femoral allograft / impaction grafting** (young, restore bone) · diaphyseal cementless or tapered fluted stem + struts
Not an option (TO)	✓ **long porous-coated stem with plate + cable** construct alone
Typical answers	18 yr post-THA, loose cemented, poor bone → ✓ **PFR** (✗ Wrong options: NWB · ORIF · long cemented · fully porous + ORIF) · 88-yr-old with dementia → ✓ **PFR** (✗ Wrong options: proximally coated · ORIF · impaction grafting · strut) · Paget + coagulopathy → ✓ **PFR** (fast) · 91-yr-old minimally ambulatory with start-up pain → ✓ **PFR** · active 83-yr-old → ✓ **PFR** (✗ Wrong options: metaphyseal press-fit stem) · loose + progressive bone loss → **cementless tapered fluted modular stem + proximal strut**

Loose stem + loose migrated cemented cup + poor bone (B3) → ✓ revise both components



Vancouver B3

– – Active space – –

Fully porous-coated stem fracture

Biggest risk

✓ **inadequate proximal bone stock** → distal ingrowth + no proximal support → **cantilever** fatigue mid-stem · also **stem <13.5 mm**, weight **>88 kg**, **young age (<60)**, ETO nonunion · revise via **ETO**, address proximal bone (graft/PFR)

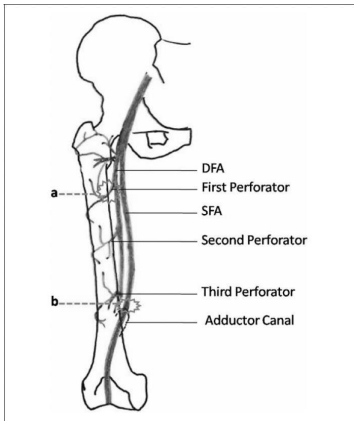
✗ Wrong options: older age · offset · exposure

6 Complications

Associated neurovascular injury

Loose stem, comminuted fracture, absent pulses, arteriogram occlusion → ✓ traction → vascular shunt → revision THA → definitive vascular repair

Cerclage cabling of proximal femur – bleeding → ✓ deep femoral (profunda) artery / perforators (close to diaphysis, tethered by perforators) · **SFA** at risk **distally** (adductor canal) · **midshaft safest** · late pseudoaneurysm (swelling, falling Hct)



Femoral arteries relative to cerclage zones

X vs Y

	B1	B2	B3
Stem	**Fixed**	**Loose**	**Loose**
Proximal bone	Good	**Good**	**Poor / comminuted**
Treatment	**ORIF**	**Long cementless stem + ORIF**	**PFR / APC**
Common reoperation	**Nonunion**	Infection	–

How it's asked

"Nondisplaced calcar fracture inserting cementless stem – next?"	**→ remove stem, cerclage, reinsert stem**
"1 cm calcar crack after final broach?"	**→ cerclage, insert standard press-fit stem, WBAT**
"Calcar cabled, same stem reinserted – true?"	**→ female sex is a risk factor**
"Cerclage for minimal calcar fracture – activity?"	**→ immediate full WB**

"Nondisplaced crack above LT with tapered wedge stem?"	→ cable above LT + partial WB 6 weeks
"Small anteromedial neck crack at final insertion?"	→ remove implant, cable, reinsert
"Displaced calcar fracture during insertion?"	→ intraoperative exploration of extent
"Implant sinks further than final broach?"	→ calcar fracture — remove, cable, insert same implant
"Sickle cell, non-displaced metaphyseal fracture?"	→ remove implant, cerclage, same implant
"Most common complication of press-fit cup (TO)?"	→ acetabular fracture
"Small posterior wall fracture, stable shell?"	→ no change in treatment
"Displaced posterior column fracture during impaction?"	→ remove cup, fix column, antiprotrusio cage
"Nondisplaced anterior column fracture, stable cup?"	→ multiple screws + brief WB restriction
"Ogden construct used for (TO)?"	→ periprosthetic femur fractures
"Fall 4 months post-THA, stable stem, no prodrome?"	→ ORIF
"Stable stems found intraop?"	→ ORIF with locking plates + cables
"Stable stem — management?"	→ ORIF plate + proximal cables
"Interprosthetic fracture?"	→ plate + screws + cerclage cables
"Which fracture for locking plate + cerclage?"	→ Vancouver C
"Fracture with well-fixed stem — classify?"	→ B1
"Below cemented stem, mantle intact?"	→ plate with proximal cables + distal screws
"Below well-fixed long revision stem?"	→ ORIF plate, screws, cables
"High-speed MVA, well-fixed cementless stem?"	→ ORIF
"Stable THA + TKA, femur fracture between?"	→ Vancouver C; lateral locking plate
"Displaced GT fracture?"	→ AG; ORIF with claw plate
"B1 ORIF — most important for fixation?"	→ locking screws proximally
"B1, stem fixed intraop — classify and treat?"	→ B1; ORIF
"B1 ORIF — commonest reoperation?"	→ nonunion
"Bridge plate — healing mode in 6 weeks?"	→ endochondral ossification
"Loose stem, fracture around stem (TO)?"	→ B2; long-stem revision + plate
"82/F, B2 (TO)?"	→ revision with uncemented long stem
"B2 — management (TO)?"	→ revision with long stem
"Loose stem, sufficient bone?"	→ long cementless stem

– – Active space – –

"Loose stem at surgery, cup fixed?"	→ uncemented femoral revision bypassing by two cortices
"Loose stem day 1 home?"	→ long diaphyseal press-fit stem
"Femoral component easily extractable?"	→ B2; long porous-coated cementless stem + fixation
"Which fracture needs long uncemented stem?"	→ the B2 radiograph
"Loose stem 3 yr post-THA?"	→ extensively porous-coated stem
"Cement column disruption?"	→ cementless revision + adjuvant fixation
"GT fracture with osteolysis + eccentric head?"	→ revision THA
"Loose stem 53-yr-old?"	→ long porous-coated cylindrical stem
"Thigh pain for 3 yr, subsidence, fall?"	→ cementless stem bypassing ≥2 cortical diameters
"Loose stem + bone loss, prior thigh pain?"	→ cementless long tapered fluted modular stem + proximal strut
"Loose cemented stem + loose cemented cup?"	→ revise acetabular and femoral components
"10 weeks post-op, loose stem?"	→ revision to long fluted tapered uncemented stem
"Fracture around spacer, infection cleared?"	→ reimplant uncemented diaphyseal-engaging stem + cables
"B2 with normal labs?"	→ revision of femoral component + ORIF
"Loose stem — femoral component?"	→ uncemented long stem + ORIF
"Which needs revision + ORIF?"	→ Vancouver B2 (active 67-yr-old)
"Loose cemented stem eroding lateral cortex?"	→ cementless femoral revision
"Which radiograph — ORIF + diaphyseal stem?"	→ B2
"B2 with sluggish refill, ABI 1.1?"	→ revision + ORIF
"Golfer, fall — revision + ORIF for?"	→ B2
"2 weeks post-op, subsidence?"	→ B2; long cementless stem + fixation
"Which radiograph for long porous stem + fixation?"	→ B2
"B3 — not an option (TO)?"	→ long porous-coated stem with plate + cable
"B3, 18 yr post-THA?"	→ proximal femoral replacement
"88-yr-old with dementia, B3?"	→ PFR megaprosthesis
"Ambulatory 76-yr-old, loose cemented, poor bone?"	→ PFR
"Paget's + coagulopathy?"	→ PFR
"85-yr-old, THA 25 yr ago — classify and treat?"	→ B3; PFR

"91-yr-old minimally ambulatory?"	→ **PFR**
"Active 83-yr-old with B3?"	→ **PFR**
"Fractured fully porous-coated stem — biggest risk?"	→ **inadequate proximal bone stock**
"Absent pulses, loose comminuted fracture?"	→ **traction, shunt, revision, vascular repair**
"Bleeding while cabling proximal butterfly?"	→ **profunda femoris (structure D)**
"Vancouver type 2 involves (TO)?"	→ **shaft around stem**
"Interprosthetic femur fracture between THR and TKR — classify for the hip?"	→ **Vancouver C; long bridging plate** (TO)
"Loose prosthesis with poor bone stock — option?"	→ **allograft–prosthesis composite / megaprosthesis** (TO)

– – Active space – –

THA POSTOPERATIVE INPATIENT MANAGEMENT

– Recon · 28 / 65 –

1 Basics

Team & phases

Team	surgeon ± hospitalist · PT · OT · case manager · nursing · patient + family
Phases	**pre-op teaching → inpatient acute care** → inpatient extended care (rehab/SNF) → outpatient/home

5 Management

Pain management

Phase	Measures
Pre-op	**NSAIDs + acetaminophen** · gabapentin/opioids (weaker data)
Intra-op	**Regional (spinal/epidural) preferred over GA** · **periarticular multimodal injection**
Post-op	**Multimodal oral therapy = gold standard**

Liposomal bupivacaine (FDA 2011)	slow release, longer action, no systemic toxicity · ✓ **comparable to peripheral nerve blocks for opioid use** · not clinically superior to standard bupivacaine ✗ Wrong options: same cost (expensive) · earlier ambulation · superior after 48 h (not even at 72 h) · shorter LOS vs fascia iliaca block

Medical management

Problem	Key facts
Hypertension	>140/90 · check reversible causes (missed meds, pain, anxiety, constipation) · oral agents preferred
Hypotension	SBP <90 or drop ≥40 · ✓ **commonest cause: intravascular volume depletion** · hold antihypertensives, IV fluids ± CBC/BMP, ECG, troponin, CXR, CTPA
Hypoxia	↓ cardiac output + O_2 uptake · **incentive spirometry for all** · consider ischaemia, COPD, CHF, **PE** · keep SpO_2 >90% · distress → ICU
↓ Urine output	Hypovolaemia, **urinary retention**, **AKI** · retention → straight cath, ↓ anticholinergics/ opioids, **tamsulosin** · AKI → fix cause, euvolaemia, stop nephrotoxins
Altered mental status	Find cause · **reorient, normalize sleep–wake cycle**
Chest pain	Troponin + ECG if cardiac · aspirin + cardiology (ischaemia) · NSAIDs (pleuritic/wall) · PPI (dyspepsia)
PONV	Opioids, anaesthetics · antiemetics, fluids, electrolytes
Ileus	NG tube may be needed
Fever	**POD 1–2 cytokine fever** – observe if well · **after POD 3** → infectious (UTI, cellulitis, pneumonia, C. diff) or non-infectious (PE, withdrawal, transfusion/drug reaction) · antibiotics only for an identified source

Postoperative ileus & Ogilvie syndrome

Ogilvie — **acute colonic pseudo-obstruction** without mechanical cause · **laxatives may worsen** it

Most severe complication — ✓ **bowel ischaemia** (→ colectomy/laparotomy)
✗ Wrong options: SMA syndrome · renal failure · cholangitis · appendicitis

Treatment — stop opioids · **NG tube** · IV fluids, electrolytes · **neostigmine** · failure → **colonoscopic decompression**



Ogilvie colonic pseudo-obstruction

Physical & occupational therapy

Start day of surgery — ↓ LOS, ↓ pain, ↑ function

Exercises — ankle pumps, knee bends, glute sets, quad sets, hip abduction, SLR → edge of bed → chair → walker → gait training

– – Active space – –

Stairs **"up with the good, down with the bad"** (up: non-operated leg first · down: operated leg first)

Walker in immediate post-op period · OT: **raised toilet seat**, shower seat, raised chair

Progression sitting → walker gait, out of bed → transfers, normal gait → independence

Discharge criteria (home)

Criteria **independent ambulation with aid 50–100 ft** · independent **transfers** · independent **ADLs** · **2 stairs with supervision** · **follows dislocation precautions** · adequate **home help**

Extended care

Earlier discharge to rehab → improved outcomes

	SNF / subacute rehab	Inpatient rehab
Who	Fail home criteria, don't qualify for IRF	Can do **3 h therapy/day** + (**>85 yr**, **BMI >50**, or **bilateral TJA**)
Cost	More cost-effective	Intensive, rarely used after TJA

How it's asked

"Liposomal bupivacaine periarticular injection — true?"	**→ comparable to nerve blocks for opioid consumption**
"POD 4, distension, no BM 7 days — most severe complication?"	**→ bowel ischaemia**
"Most common cause of post-op hypotension?"	**→ volume depletion**
"Fever POD 1, patient well?"	**→ observe (cytokine fever)**
"Stairs after THA?"	**→ up with the good, down with the bad**
"Inpatient rehab eligibility?"	**→ 3 h/day therapy + age >85, BMI >50 or bilateral TJA**

– – Active space – –

THA PROSTHESIS DESIGN

– Recon · 29 / 65 –

1 Basics

Determinants

THA biomechanics depend on	prosthesis design · **bearing surface + lubrication** · fixation
Components	femoral (cemented, press-fit: tapered, extensively coated, modular) · acetabular (cemented PE or metal; press-fit metal; **modular dual mobility** – ↓ revision dislocation, risk of **liner malseating**) · bearings (PE, metal, ceramic)

History

Year	Milestone
1891	**Gluck** – ivory femoral head
1940	**Austin Moore** – first metallic hip (bolted proximal femoral replacement)
1952	**Austin Moore prosthesis** – large uncemented stem, **fenestrations for self-locking** (impetus for biologic fixation), no PE ✓
1960s	**Charnley low-friction arthroplasty** – **small head** (↓ wear), metal stem, **PE cup**, **acrylic cement** · Teflon cup first · mentor Sir Harry Platt · manufacturer Thackray
Other	**San Baw** – ivory hips for neck nonunion
1821 (TO)	**Anthony White** – first **excision** arthroplasty (fibrous tissue fills defect)
1840s (TO)	French surgeons – **interposition** (fascia) → faster rehab, unstable
1891 (TO)	Gluck – TO lecture says **glass**; ivory (above) – read the options
1920s (TO)	**Smith-Petersen mould arthroplasty** – metal cup, temporary resurfacing
1953 (TO)	**Thompson** – **collar, no fenestrations** → load to calcar
1974 (TO)	**Bipolar** prosthesis (Bateman) – two interfaces → ↓ acetabular erosion

Arthroplasty types (TO)	**excision** · **interposition** (fascia lata) · **replacement** · hemiarthroplasty = femoral head only · THR = acetabular + femoral
Charnley principles (TO)	**small 22 mm head** (↓ friction torque) · **thick PE socket** · **medialize cup + lateralize GT** (↑ abductor arm → ↓ JRF) · **PMMA** both components · failures: **dislocation** (small head), **acetabular perforation** (over-medialization), **trochanteric osteotomy nonunion** · modern principle: **recreate native anatomy** (offset, version, inclination, length), no trochanteric transfer
Charnley head	✓ **22.225 mm (7/8 inch)** (TO)

– – Active space – –

Charnley all-PE cemented cup

✓ **20–30% failure at 35 yr** (88% intact at 20 yr, 78% at 35 yr; 77.3% at 21 yr)



Austin Moore prosthesis

5 Management

Press-fit femoral stems

Principle **biologic fixation** · **hoop stresses** give initial stability

Type	Examples
Tapered (proximally coated)	Tri-Lock, M/L Taper
Extensively coated (diaphyseal)	**AML**, VerSys Full Coat
Modular (mix-and-match body + stem)	**S-ROM**, ZMR

Cementless stem classification (TO, Khanduja) **1** single wedge (proximal, flat AP/tapered ML, broach only) · **2** double wedge (proximal, ML + AP) · **3** double-wedged, fixation at **metaphyseal–diaphyseal junction** (3a/b/c) · **4 extensively coated**, full-length fixation · **5 modular** · **6 anatomic** (posterior metaphyseal bow) · ✓ **not distally fixed = type 2** (types 3b, 3c, 4 are)

Khanduja – TO lecture version 1 single wedge (Dorr B/C) · 2 dual wedge (**Dorr A**) · 3 round / rectangular (revision, diaphyseal) · 4 fully coated (revision, dysplasia) · 5 highly modular (**S-ROM**, DDH) · 6 anatomic · type 3 described as metaphyseal–diaphyseal double wedge above – read the options

Complications **intraop fracture** (lateral approach + under-reaming) · **loosening** in **irradiated bone** · **junctional (taper) corrosion** in modular necks → metallosis, pseudotumour

General Category	Type	Geometry	Description	Location of Fixation
Straight stems				
Tapered proximal fixation	1	Single wedge	Narrows medially-laterally. Proximally coated. Flat stem, thin in anterior-posterior plane	Metaphyseal
Tapered proximal fixation	2	Double wedge, metaphyseal filling	Narrows distally in both medial-lateral and anterior-posterior planes. Wider than Type 1. Fills metaphyseal region	Metaphyseal
Tapered proximal fixation	3A	Tapered, round	Rounded tapered conical stem with porous coating at proximal two-thirds	Metaphyseal-diaphyseal junction
Tapered distal fixation	3B	Tapered, splined	Conical taper with longitudinal raised splines	Metaphyseal-diaphyseal junction and proximal diaphyseal
Tapered distal fixation	3C	Tapered, rectangular	Rectangular cross section with four-point rotational support in metaphyseal-diaphyseal region	Metaphyseal-diaphyseal junction and proximal diaphyseal
Distally fixed	4	Cylindrical, fully coated	Extensive porous coating. Proximal collar to enhance proximal bone oading and axial stability	Primarily diaphyseal
Modular	5		Metaphyseal and diaphyseal components prepared independently	Metaphyseal and diaphyseal
Curved, anatomic stem	6		Proximal portion is wide in both lateral and posterior planes. Posterior bow in metaphysis, anterior bow in diaphysis	Metaphyseal

Cementless stem types

Cemented femoral stems

Indications	**irradiated bone** · **Dorr C** (poor proximal stock) · elderly · grout; limited remodelling
Material	**Co-Cr or stainless steel** (stiff → ↓ cement stress) · not **cemented titanium** (less stiff → micromotion, debonding, **crevice corrosion** — no longer recommended)
Stem breakage	smaller stems, poor tolerance of **cantilever bending**
Desired features (TO)	**high-modulus superalloy (Co-Cr)** · broad rounded edges · collar (depth control, ↓ calcar resorption) · **macrotexturing** (grooves) · **polished surface** (less debris if debonded; Exeter matte > polished loosening) · **no PMMA precoat** · **centralizer** · ✓ **NOT microtexturing / matte finish** (that is for uncemented ingrowth)
Gruen failure modes	**I pistoning** (linear lucency around stem + cement, commonest) · II medial midstem pivot · III calcar pivot · IV bending cantilever
Force-closed vs shape-closed	taper-slip polished collarless (**Exeter, C-stem**; subsidence intended, >5 mm or progressive after 2 yr = loose) vs composite beam rough ± collar (**Charnley**; subsidence >0.15 mm after 2 yr → ↑ revision) · mantle **2–4 mm** · line-to-line = "French paradox"

Elderly Dorr C femur

82-yr-old walker, Dorr C → ✓ cemented femoral stem
✗ Wrong options: cemented cup (shear loosening) · diaphyseal stem (save for revision) · 4.5 g antibiotic for prophylaxis (≤1 g; 4.5 g for spacers) · smallest stem for >1 cm mantle

Dorr classification (TO)

Classifies ✓ **proximal femur**
✗ Wrong options: femoral stem · acetabulum

Type	Ratio	Shape	Stem
A	**<0.5**	**"Champagne flute"**, thick cortices, young men	Uncemented (either works)
B	**0.5–0.75**	Medial + posterior cortical loss, wider canal	Uncemented
C	**>0.75**	**"Stovepipe"**, thin cortices, elderly postmenopausal women	✓ **Cemented** (most suited)

– – Active space – –

Bearing surfaces

Bearing	Benefits	Disadvantages
Metal-on-PE	**Longest record**, **cheapest**, most modular	**↑ wear + osteolysis** · small head → impingement
Metal-on-metal	↓ linear wear · particles **smaller + more numerous** (smaller total volume) · large head → ↑ ROM	Cost · **ions 5–10× normal** (serum + urine; peak **12–24 months** run-in) · **pseudotumour** · **type IV hypersensitivity** · no proven cancer link
Ceramic-on-ceramic	✓ **Best wear** (alumina 4000× < CoCr-on-PE) · **lowest friction** · **inert**	Cost · **brittle – fracture** · **squeaking** · **stripe wear** · fewer neck lengths · 28 mm heads only zirconia
Ceramic-on-PE	**Standard of care** · ↓ PE wear vs MoP	**Zirconia phase transformation** (tetragonal → monoclinic: >8 yr, pressure, heat – 99 °C vs alumina 50 °C)
Titanium-on-PE	–	**High wear** – not recommended

Gold-standard bearing — **ceramic-on-PE = standard of care** (current chapter) · **metal-on-cross-linked PE = gold standard** (TO lecture) – read the options · TO: CoC least wear, brittle, squeaking, **malposition-intolerant** · MoM obsolete (ALVAL, ions)

Modular options (TO) — neck length **25–50 mm** · neck-shaft **127–132°** · head **28–32 mm ideal** (jump distance, low dislocation) · **> 36 mm → thinner PE → early wear** (TO) · PE thickness **≥ 5 mm** (TO, this lecture) · **≥ 6 mm** (TO osteolysis lecture) – read the options

Morse taper (TO) — **12/14 taper = lowest failure** · trunnionosis = **3–5% of THR revisions**

Cups (TO) — **cementless press-fit cup = gold standard** · avoid screws if possible (↑ osteolysis via effective joint space) · cemented cup rarely used · **hybrid hip (cemented stem + uncemented cup) = commonest combination** (TO)

Stiffness order (TO) — **PE (lowest) < bone < titanium < Co-Cr < ceramic (highest)** · **Ti for cementless**, **Co-Cr for cemented** · distal tapering ↓ stiffness · proximal wedging → **hoop stresses** → stability

Fixation choice (TO) — PMMA = **grout, not adhesive** · cemented → elderly, short life expectancy, poor bone · cementless → young, good bone · ingrowth (porous, tantalum) vs **ongrowth** (grit-blast, plasma spray)

Liners (TO) — **dual mobility** = two articulations, ideal for **elderly / weak abductors** · **constrained liner outdated** (high mechanical failure)

MoM contraindications — **pregnancy / child-bearing women** (ions cross placenta) · **renal disease** (renal excretion) · metal hypersensitivity

Wear order (TO) — **CoC (0.5–2.5 µm/yr) < MoM (2.5–5 µm/yr) < MoP** · ✓ **CoC lowest wear of any bearing**

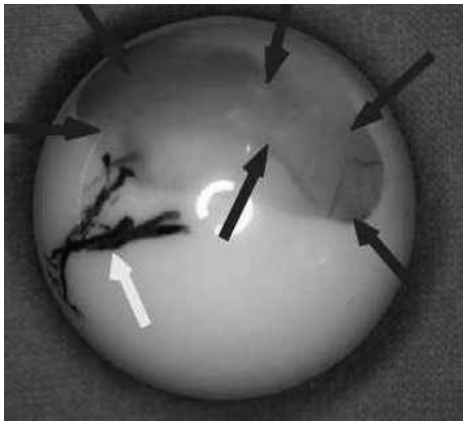
Ceramic properties

Ceramic — ✓ **non-ductile, high wettability, highest modulus, low fracture toughness** · inert, biocompatible · alumina + zirconia composites

Modulus order — ceramic > Co-Cr > stainless steel > titanium (OB) · TO: polymers < cement < cortical bone < **Ti** < SS < Co-Cr < ceramics

MoM advantage over CoC — ✓ **↑ fracture toughness**
✗ Wrong options: wettability, hardness, roughness, friction

CoC vs metal on HXLPE	✓ **lower volumetric wear** ✗ Wrong options: jump distance/ROM (head size) · cost · stripe wear (unique to CoC)
CoC complications (TO)	**stripe wear** (head–rim line contact during subluxation, crescent line), **squeaking** (0.3–24.6%; onset 14–40 months; fundamental 400–7,500 Hz; friction from lost lubrication), **fracture**, impingement · ✓ **NOT high wear rate**
Ceramic fracture risk	**28 mm** > 32/36 mm heads · **short neck** · canted liner · malposition
Ceramic generations	1st pure zirconia (roughening, fracture) · 2nd CaO/MgO (↓ grain) · 3rd **hot isostatic pressing** (ultrapure alumina, zirconia, yttrium) · 4th **strontium oxide platelets** deflect cracks
After ceramic head fracture	revise with **ceramic** head, not metal · metal head → retained ceramic fragments cause ✓ **massive third-body (abrasive) wear** ✗ Wrong options: infection · pseudotumour · metallosis · psoas



Ceramic stripe wear

Head size & wear

Large head	✓ **↑ ROM** before impingement · ↑ head-neck ratio · **↑ jump distance (~½ head diameter)** · more stable · **↓ ceramic fracture** · **↓ neck-to-rim impingement** · ✓ **↑ volumetric wear** ✗ Wrong options: "decreases ROM" · "↑ neck-rim impingement" (TO EXCEPT answers)
Volumetric wear	**$V = \pi r^2 w$** · ✓ **head size (radius²) most important** · ✓ relates to **total contact area** (TO) · main determinant of particle number
22 mm vs 32 mm	22 mm: ✓ **higher linear wear (creep)**, ↓ volumetric wear, ↑ dislocation, ↓ ROM · 32 mm: ↑ volumetric wear, ↓ linear · **28 mm compromise**
Linear wear	penetration of head into cup · **>0.1 mm/yr → osteolysis** · conventional UHMWPE **0.1–0.2 mm/yr** · **RSA most accurate** wear measure
Wear modes (TO)	**adhesive** · **abrasive** ("cheese grater"; two- and three-body) · **third-body** (cement, bone, metal, PE, HA) · **surface fatigue/delamination** (TKA) · **fretting** (THA backside shell–liner) · corrosive · hip PE wear mainly **adhesive/abrasive**; TKA **delamination**
Wear stages	run-in → steady state → rapid failure

MoM tribology

Lubrication (TO)	✓ **elastohydrodynamic** in MoM · **boundary** in MoP · mixed lubrication (OB)

– – Active space – –

Diametral clearance most important wear variable · ideal **100–200 µm** · too small → equatorial contact, seizing · too large → lost fluid film, rapid wear · ✓ TO false statement: "large clearance = better fluid film" · OB: ideal **radial mismatch ~50 µm**

High carbon (>0.20%) Co-Cr → ↓ wear · **double heat treatment** + **low carbon** → ↑ wear, pseudotumour

Larger MoM bearings wear similarly
✓ **greater sliding speed draws in more fluid**
✗ Wrong options: boundary lubrication · flexibility · carbides · smaller clearance

Not affecting MoM wear
✓ **head-to-neck ratio**
✗ Wrong options: radial mismatch, sphericity, surface finish, carbon content

ALVAL delayed-type hypersensitivity, ~**0.3%** of MoM

MoM systemic effects

True ionically charged particles · ↑ **Co + Cr in serum + urine** · **higher frictional torque than CoC** · **no activity–ion correlation** · ions **renally excreted** (not liver) · cross **placenta** · ✓ **pseudotumours** · ↑ registry revision rates

Not true / not shown ✓ **higher cancer risk** (EXCEPT answer) · sarcoma, carcinomas · ↑ dislocation · nephrotoxicity

Observed more often ✓ **leukocyte chromosomal aberrations** (translocations, aneuploidy) – significance unknown · haematopoietic malignancy data mixed

Hypersensitivity (type IV) T-cell mediated · **nickel > cobalt, chromium** · most cells are macrophages (lymphocytes **5%**) · cytokines → macrophage activation → MHC II, IL-2 → more T-cells

Taper corrosion & modularity

Taper (trunnion / neck–stem) corrosion in MoP
✓ **pseudotumour** (ALTR) · fretting + crevice corrosion (mechanically assisted) · affects **>⅓ of modular devices**
✗ Wrong options: PE wear · dislocation · squeaking · sepsis

Modular-neck ALTR findings X-ray usually normal; **medial calcar erosion** commonest · MRI **thickened capsule + fluid** · ↑ **cobalt with near-normal chromium** (Cr precipitates as orthophosphate) · black bore of female taper

Treatment
✓ **single-stage revision of liner, head, neck + stem to a MONOBLOCK stem**
✗ Wrong options: revise stable cup · two-stage · retain stem if female taper corroded · modular revision stem

Impingement ↓ ROM ✓ **skirted modular heads** · avoid skirts, maximize head-neck ratio, chamfered liners, **trapezoidal** (not circular) neck · cup 45–55° with 10–20° combined anteversion (wear/instability if >50°)

Malseated modular dual-mobility liner (Co-Cr inelastic, hard bone deforms shell; "4-quadrant test") → ✓ **↑ fretting corrosion**
✗ Wrong options: dislocation · PJI · fracture · ↓ ROM

	Cause	Location	Solution
Galvanic	Interface of two dissimilar metals creates electrical potential	Neck taper junction	Avoid dissimilar metals or form passivation layer
Fretting	Micromotion at the interface of two metals disrupts the passivation layer	Interface of two similar metals; occurs secondary to micromotion	Avoid micromotion
Crevice*	Microscopic cracks create microenvironmental that disrupts passivation layer	Interstices/cracks at modular component junctions	Increase machining tolerances; polish surface
Pitting*	Small surface defect in passivation layer	Depressions at modular component junctions or articular components	Increase machining tolerances; polish surface
Intergranular	Corrosion at the interface of grain boundaries where carbides are present	Diffuse	Metal alloys (molybdenum) and low carbon (0.2%)

* Crevice and pitting corrosion functions as a localized form of galvanic corrosion as a consequence of dissimilar charged ions from the metal surface and aqueous microenvironment.

Modes of taper corrosion

Implant choice in nickel allergy

Nickel in **Co-Cr and stainless steel**, not titanium or ceramic

Answer ✓ **press-fit titanium stem + alumina ceramic head + cementless titanium cup** · TKA: **oxidized zirconium femur + titanium tibia**
✗ Wrong options: cemented Ti (less stiff) · Co-Cr · titanium head (abrasive wear)

Stem design, stress shielding & fracture risk

Stress shielding proximal bone loss around a well-fixed stiff stem (**Hooke's law** – stiffer "spring" carries load) · **Wolff's law** bone loss

↑ Stress shielding	↓ Stress shielding
Large diameter (stiffness ∝ **r⁴**; ≥16 mm; OB >18 mm) ← most important	Small diameter
Co-Cr (10× femur; alumina 19×)	✓ **Titanium** (lower modulus)
Solid, round	Hollow, slotted, fluted, tapered, short
✓ **Cylindrical distal-fitting / extensively coated / distally fixed**	**Tapered, proximally fixed**
Cementless	✓ **Cemented** stems (TO true statement)
Osteoporosis	–

Fully coated Co-Cr stem → X-ray ✓ **proximal femur / trochanteric osteopenia** · late GT insufficiency fracture
✗ Wrong options: distal · whole femur · acetabulum

Grade III stress shielding with 16.5 mm stem – EXCEPT → ✓ titanium stem
✗ Wrong options: round · large diameter · solid

Material to ↓ shielding → ✓ titanium (highest fatigue strength among alloys) · flexible, short, hollow stems

Osteolysis location extensively coated → **proximal** · patch-coated → **distal** (joint fluid reaches diaphysis) · HA fully coated → ↑ cortical shielding vs proximally porous

FNB 2024 stem diameter threshold for proximal resorption (11 / 12.5 / 13.5 / 14 mm) and "proximal stress shielding occurs in…" – answers not given by source (diaphyseal-fitting > tapered fits the rule above)

Titanium vs Co-Cr (TO) Ti: ✓ **lower tensile strength + fatigue resistance (than Co-Cr)** · **lower modulus**, less stiff · **more biocompatible**, better osseointegration · Ti porous coat restricted to bulky proximal stem (notch sensitivity)
✗ Wrong options: higher modulus · stiffer · less biocompatible

Intraop femur fracture – highest risk ✓ **final insertion of uncemented stem via lateral approach in female >65** (medius directs broach medial/anterior → calcar stress)

– – Active space – –

Cementless ingrowth – pore size & osseointegration

Parameter	Values (source)
Pore size	**50–400 µm** (OB, TO) ✓ "<500 µm" · **100–400 µm** (TO) · 100–150 / 50–150 µm (TO list) · **50–300 µm**, ideally 50–200 (Orthobullets)
Porosity	**40–50%** (TO) · 40–80% (Orthobullets)
Gap	✓ **<50 µm** (TO false: "<100 µm")
Micromotion	Minimal (<150 µm)
Ongrowth roughness	**3–5 µm**, grit blast / plasma spray
Pore depth	↑ depth → ↑ fixation strength

Osseointegration timeline starts **4–12 weeks**, continues ✓ **up to 3 years**

Stable (bony ingrowth)	Stable fibrous	Unstable
No subsidence, minimal lines, **spot welds**, distal cortical hypertrophy, proximal stress shielding	No progressive migration · **parallel radiopaque lines ~1 mm** from stem · no cortical hypertrophy	**Progressive subsidence** · **divergent** lines · pedestal / hypertrophy under collar + tip

Acetabular osseointegration signs no lucent lines · **superolateral buttress** · medial stress shielding · **radial trabeculae** · **inferomedial buttress**

Impaction bone grafting & conversion after failed fixation (TO)

Conversion after failed neck fixation (screw holes) → ✓ long uncemented stem + prophylactic cerclage, bypass last hole by **2 cortical diameters**
✗ Wrong options: cemented (leaks through holes)

Impaction grafting (femur) Hastings & Parker 1975, Nijmegen/Exeter · **1–2 mm cancellous allograft chips** impacted → **cemented** stem in neo-canal · loads transmitted as **compression on graft** · **some subsidence expected** · indication: **Paprosky 3B in young**, type 4 · complications: dislocation **4.1%**, fracture **5.4%** · ✓ **incorrect: "stem must be tapered fluted, collarless, flexible titanium"** (no single required stem; polished collarless or collared used)

X vs Y

	MoM	CoC	MoP / CoP
Wear	Low linear, tiny numerous particles	**Lowest**	Highest (MoP)
Unique problems	**Ions, ALVAL, pseudotumour**	**Fracture, stripe wear, squeaking**	Osteolysis
Toughness	**High**	**Low (brittle)**	–
Lubrication	Elastohydrodynamic	Fluid film (wettable)	Boundary

	Titanium	Cobalt-chrome
Modulus / stiffness	**Lower**	Higher
Stress shielding	**Less**	More
Cemented use	No (crevice corrosion)	Yes
Nickel	**None**	Present

How it's asked

"Not associated with CoC (TO)?"	**→ high wear rate**
"Large head — NOT true (TO)?"	**→ decreases ROM**
"Not true about CoC (TO)?"	**→ high wear rate**
"Not true about MoM (TO)?"	**→ large diametral clearance → better fluid film**
"Predominant lubrication in MoM (TO)?"	**→ elastohydrodynamic**
"Large head NOT associated with (TO)?"	**→ increased neck-to-rim impingement**
"True about bearing wear (TO)?"	**→ CoC lowest wear of any combination**
"22 mm vs 32 mm head (TO)?"	**→ higher linear wear**
"Volumetric wear depends on (TO)?"	**→ total contact area (head size)**
"Charnley all-PE cup failure at 35 yr?"	**→ 20–30%**
"Groin pain after ceramic head fracture revised to metal head?"	**→ massive third-body wear**
"Benefit of CoC vs metal-on-HXLPE?"	**→ lower volumetric wear**
"MoM vs MoP particles?"	**→ smaller and more numerous**
"MoM wear not affected by?"	**→ head-to-neck ratio**
"Advantage of MoM over CoC?"	**→ increased fracture toughness**
"Parkinson + mild renal insufficiency — bearing?"	**→ large metal head on HXLPE**
"Larger MoM wears similarly — why?"	**→ greater sliding speed**
"Non-ductile, wettable, high modulus, low toughness?"	**→ ceramic**
"MoM EXCEPT?"	**→ higher cancer risk**
"MoM patient worried — accurate?"	**→ no correlation between activity and ion levels**
"Higher frequency with MoM?"	**→ leukocyte chromosomal aberrations**
"Taper corrosion in MoP?"	**→ pseudotumour**
"Modular neck corrosion, stable components?"	**→ revise liner, head, neck + stem to monoblock**
"Decreases ROM to impingement?"	**→ skirted modular head**

– – Active space – –

"Malseated modular dual-mobility liner?"	→ ↑ fretting corrosion
"Severe nickel allergy?"	→ press-fit Ti stem, alumina head, cementless Ti cup
"Fully coated Co-Cr stem — X-ray change (TO)?"	→ trochanteric / proximal femoral osteopenia
"Stress shielding EXCEPT (TO)?"	→ titanium stem
"More proximal stress shielding (TO)?"	→ cylindrical distal-fitting vs tapered
"Not distally fixed (TO)?"	→ type 2
"True about stress shielding (TO)?"	→ cemented less than uncemented
"Material to ↓ stress shielding (TO)?"	→ titanium
"↑ stress shielding?"	→ distally fixed components
"Highest intraop fracture risk?"	→ final insertion, uncemented stem, lateral approach, female >65
"Elderly Dorr C — PMMA use?"	→ cement the femoral stem
"Charnley head size (TO)?"	→ 22.225 mm
"Austin Moore designer?"	→ Austin T. Moore
"MoM vs MoP outcome?"	→ pseudotumours
"Converting failed neck screws to THR (TO)?"	→ long uncemented stem + cerclage
"Impaction grafting — incorrect (TO)?"	→ stem must be tapered fluted flexible titanium
"Not desired in cemented stem (TO)?"	→ microtexturing / matte finish
"Linear lucency around cemented stem (TO)?"	→ pistoning
"Ti vs Co-Cr (TO)?"	→ Ti lower tensile strength + fatigue resistance
"Dorr classification is for (TO)?"	→ proximal femur
"Femur most suited for cementing (TO)?"	→ Dorr C
"Ideal pore size (TO)?"	→ <500 µm (50–400)
"Osseointegration continues until (TO)?"	→ up to 3 years
"Not true for ingrowth (TO)?"	→ gaps <100 µm
"True about cementless stem (TO)?"	→ optimal pore size <400 µm
"Ideal trunnion taper?"	→ 12/14 (TO)
"Commonest current fixation combination?"	→ hybrid — cemented stem + uncemented cup (TO)
"Charnley's principle to reduce JRF?"	→ medialize cup + lateralize greater trochanter (TO)

THA PSEUDOTUMOR (METAL ON METAL REACTIONS)

– Recon · 30 / 65 –

1 Basics

Definition & epidemiology

Pseudotumour = ALVAL (aseptic lymphocyte-dominant vasculitis-associated lesion) – **mass-forming tissue reaction** to MoM wear · (semi-)solid or cystic periprosthetic mass **≥2 cm** not due to infection, tumour, bursa or scar (TO)

Incidence **10–15%** of MoM THAs · resurfacing **0–39%** (OB) · ALVAL ~0.3% (TO) · **45–50% of MoM revisions** are for pseudotumour/ALTR · **~40% asymptomatic** · revision for symptomatic pseudotumour only **1.7–5.6%**
✗ Wrong options: majority

Seen with ✓ **MoM THA / resurfacing** (TO) · also **MoP with head–neck (trunnion) corrosion** → ALTR

Risk factors

Risk factors	Not risk factors
↑ **Cobalt + chromium** (Co **>5 µg/L** – TO)	✓ **BMI >40** (TO EXCEPT)
Female	Metal allergy (TO)
High cup inclination >55° / ↑ anteversion → **edge loading**	Obesity
Age <40 (TO)	–
Taper/trunnion corrosion even with good position	–
Large metal heads (some studies), low diametral clearance	–

Pathophysiology

Two mechanisms **hypersensitivity to metal ions** · **local high wear debris** · ✓ pseudotumour associated with **↑ linear wear** of both components

Hypersensitivity ✓ **type IV delayed (cell-mediated)** · metal ions form **haptens** with native proteins · ✓ **lymphocyte-mediated** in metal-allergic patients (jewellery reactions) (✗ Wrong options: macrophage-mediated · large/small particle wear) · ALVAL due to ✓ **hypersensitivity to metals** (✗ Wrong options: abrasion of shell · screws · stem corrosion)

Cells – conflict Orthobullets/OB: macrophages + lymphocytes in all, **lymphocytes predominant** · high-wear cases → more macrophages, smaller lymphoid aggregates · TO: metal debris **cytotoxic to macrophages → macrophage depletion** → lymphocyte predominance · OB (DTH): effector = T-cell but **most participating cells are macrophages (lymphocytes ~5%)**
✗ Wrong options: macrophage predominant

– – Active space – –

Histology	**dense perivascular lymphocytic infiltrate** (hallmark), T **and** B lymphocytes, lymphoid follicles/aggregates, histiocytes, giant cells, necrosis · ✓ dark areas = **T + B lymphocyte infiltrates** (✗ Wrong options: particle-laden macrophages (typical of MoP)) · MoP trunnion ALTR → ✓ **predominance of perivascular lymphocytes** (✗ Wrong options: neutrophils (PJI) · lipid-laden histiocytes/giant cells (PVNS))
ALVAL true (TO)	B lymphocytes · macrophage depletion · lymphoid aggregates · ✓ **false: "diagnosis is not made by macroscopic appearance"** (solid mass visible) · definitive diagnosis histological

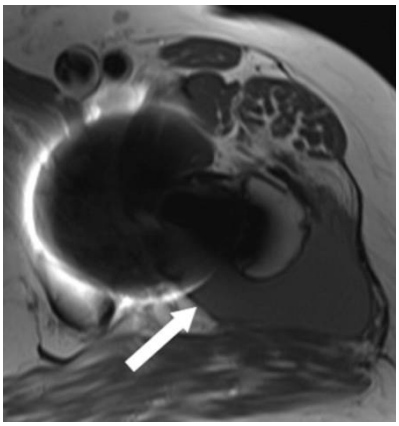
3 Presentation

History	gradual or sudden groin pain · may be **asymptomatic** · mass causing discomfort
Exam	periprosthetic soft-tissue mass · **Trendelenburg gait** (abductor destruction) · groin pain with **flexion, IR, adduction** or rising from chair · NV usually normal

4 Diagnosis & Classification

Key radiology findings

Radiographs	AP pelvis + AP/lateral hip · exclude fracture · **calcar erosion**
MARS MRI	✓ **test that best correlates with prognosis** (TO) (✗ Wrong options: ESR/CRP/WBC · metal ions (don't correlate) · CT (bone loss only)) · ✓ **best test for painful MoM** (✗ Wrong options: CT · bone scan · ion levels alone) · findings: periarticular **fluid collection / solid mass**, synovial/capsular hypertrophy, **tendon avulsion**, bone oedema · **T1 ≈ bladder (transudate)** · **T2 hyperintense** (hypointense foci = necrosis/metal) · MAVRIC, SEMAC newer
CT	bony destruction, need for structural graft/augments



MARS MRI pseudotumour

Genes & lab markers

Metal ions (Co, Cr)	peak **12–24 months** (run-in) then fall · **>7 ppb (µg/L)** → MARS MRI · ions renally excreted · **activity does not change levels** · Co:Cr ≈ **1:1** = MoM bearing, **high Co:Cr (2:1)** = trunnionosis ✗ Wrong options: pseudotumour

Aspiration — distinguishes **metallosis** from PJI · **"dishwater" fluid** (grey, watery, hazy) · ✓ **manual cell count** (fibrinous debris falsely ↑ automated) · MoM PJI cut-off **>4,350 WBC** · ARMD → **lymphocytes >40%**; PJI → **PMN >70%** (e.g. WBC >3,000 with low PMN, high lymphocytes = ✓ **pseudotumour hypersensitivity**; ✗ Wrong options: bacterial PJI · large-particle PE disease)

PJI vs ALTR — ALTR raises ESR, CRP, synovial WBC and gives **false-positive α-defensin** → ✓ **thorough work-up: cultures, manual WBC, %PMN (80% cut-off)** · α-defensin remains valid in RA
✗ Wrong options: α-defensin alone diagnoses PJI · ESR/CRP confirm · ALTR excludes PJI

5 Management

MoM THA follow-up

- Asymptomatic, ions <7 ppb, normal films → **observe (serial X-ray + ions)**
- Ions >7 ppb (even asymptomatic) → **MARS MRI**
- Symptomatic + normal ions + normal MRI → **close follow-up 3–6 months**
- Pain/↑ ions + pseudotumour (or progressive osteolysis) → **revise to CoP**
  - Abductor loss → **constrained liner ± gluteus maximus transfer**

Observation

Indication — well-functioning, **low ions**, no pseudotumour · serial radiographs (wear, loosening)

Close follow-up allowed — ✓ **symptomatic, Co 6 / Cr 5, normal ESR/CRP, MARS MRI normal**
✗ Wrong options: asymptomatic with ions >7 (needs MRI) · symptomatic with fluid (revise) · asymptomatic + fluid + progressive osteolysis (revise)

Revision indications

Revise — ✓ **MARS MRI with periarticular fluid + abductor destruction** · painful MoM + ↑ ions + pseudotumour · delay → more soft-tissue damage
✗ Wrong options: CRP 6 · HO · aspirate 900 WBC · cobalt 5 ppb alone

Technique — revise to **ceramic-on-polyethylene** · single- or two-stage (if infection concern) · lateral or anterior approach · **bimodular (modular-neck) stem + monoblock cup** → ✓ **revise both components to CoP + I&D**
✗ Wrong options: head/liner exchange (monoblock cup) · larger Co-Cr head · I&D alone · observe

Abductor deficiency — **Trendelenburg gait** → ✓ **gluteus maximus transfer** (extensor → abductor) ± **constrained liner**
✗ Wrong options: flexion contracture · medial migration (cup-cage) · psoas · ER weakness

Dislocation after head/liner exchange for pseudotumour → ✓ **abductor insufficiency**
✗ Wrong options: offset · larger head · vertical cup · impingement

Infected pseudotumour — neutrophils on histology + ↑ ESR/CRP → debride, explant, spacer, **systemic antibiotics**

– – Active space – –

6 Complications

Revision outcome	bone loss, soft-tissue destruction, abductor deficiency → harder revision, worse function · **↑ dislocation + infection**
After revision	instability · deep infection · aseptic loosening · persistent psoas tendinitis · persistent metal reaction · **14–20% re-operated within 5 yr**
Femoral nerve palsy	rare (case reports) · decompress + revise to CoP

X vs Y

	ALTR / pseudotumour	PJI
Synovial cells	**Lymphocytes >40%**	**PMN >70–80%**
α-defensin	**False positive** possible	Positive
Fluid	"Dishwater" grey	Purulent
Count	**Manual**	Automated fine

	MoM bearing wear	Trunnionosis
Co:Cr	~1:1	**Co ≫ Cr (2:1)**
Bearing	MoM	MoP or MoM with modular taper

How it's asked

"Painful MoM – test best correlating with prognosis (TO)?"	**→ MARS MRI**
"Pseudotumour risk factors EXCEPT (TO)?"	**→ BMI >40**
"Masses around resurfacing on MRI (TO)?"	**→ pseudotumour formation**
"38-yr-old woman, resurfacing, normal labs, mass on MRI?"	**→ pseudotumour**
"MoM aspirate WBC high but lymphocyte-predominant?"	**→ pseudotumour hypersensitivity response**
"Groin pain after resurfacing, X-ray normal – best test?"	**→ MARS MRI**
"Which MoM patient can just be followed?"	**→ symptomatic, ions <7 ppb, normal MARS MRI**
"Finding that mandates revision?"	**→ MARS MRI fluid + abductor destruction**
"ALTR + ↑ ESR/CRP + α-defensin + – true?"	**→ needs cultures, manual WBC, %PMN**
"ALVAL seen with (TO)?"	**→ metal-on-metal THR**
"ALVAL due to (TO)?"	**→ hypersensitivity to metals**
"ALVAL – all true EXCEPT (TO)?"	**→ diagnosis not made by macroscopic appearance**

"Dark areas in MoM pseudocapsule histology?"	→ T- and B-lymphocyte infiltrates
"MoP with ↑ ions, ALTR – histology?"	→ perivascular lymphocyte predominance
"MoM + jewellery reactions – mechanism?"	→ lymphocyte-mediated hypersensitivity
"MoM property?"	→ ions excreted mainly by kidney
"Pseudotumour on MARS – true?"	→ associated with ↑ linear wear
"Bimodular stem, monoblock cup, pseudotumour, ↑ ions?"	→ revise both components to CoP + I&D
"Dislocation 3 weeks after head/liner exchange for pseudotumour?"	→ abductor insufficiency
"Gluteus maximus transfer planned – why?"	→ Trendelenburg gait (abductor deficiency)
"Ion level prompting MRI?"	→ >7 ppb

– – Active space – –

THA REHABILITATION

– Recon · 31 / 65 –

1 Basics

Phases

Team	surgeon, PT, OT, case manager, nursing, patient + family
Phases	**pre-op teaching → inpatient acute → inpatient extended (rehab/SNF) → outpatient**

5 Management

Preoperative teaching

Pre-op PT	✓ **not shown to improve post-op outcomes**
Hip precautions	most useful if taught **before** surgery · hips dislocate in the **direction of the approach** (if implants well positioned)

Approach	Avoid
Posterolateral	**Flexion >90°**, **extreme IR**, **adduction past midline**
Anterolateral	**Extension**, **extreme ER**, adduction past midline
Direct anterior (Smith-Petersen)	**Bridging**, **extension**, **extreme ER**, adduction past midline

Told to avoid extension, adduction, ER → ✓ Smith-Petersen (anterior)
✗ Wrong options: Southern/Kocher-Langenbeck (posterior) · Stoppa · Pfannenstiel



Posterior hip precautions

Inpatient acute care

Pain	pre-op NSAIDs (± opioids) · **regional anaesthesia preferred over GA** · periarticular multimodal injection · **multimodal oral therapy = gold standard**
PT progression	sit upright → walker gait, out of bed → transfers, normal gait → independence

Discharge home	independent ambulation with aid · independent transfers · independent ADLs · stairs with supervision · home help

– – Active space – –

Extended care

Earlier discharge to rehab → improved outcomes · same discharge criteria

Outpatient – return to activity

Sport	**low-impact** preferred · **golf handicap barely changes after THA** (worsens after TKA) · **high-impact ↑ revision in patients <55**
Driving	**right THA 3–4 weeks** · **left THA <3–4 weeks** · ✓ **reaction time back to pre-op at 4–6 weeks** (OB answer: **4 weeks**) · influenced by side, lengthening, rehab, narcotics ✗ Wrong options: 1–2 weeks · 2–3 months
Work	**within a month** if not manual

How it's asked

"When does driving reaction time return after THA?"	**→ 4 weeks (4–6 weeks)**
"Avoid extension, adduction, ER — approach?"	**→ Smith-Petersen**
"Posterior approach precautions?"	**→ no flexion >90°, no IR, no adduction past midline**
"Does pre-op PT improve outcome?"	**→ no**
"High-impact sport after THA in <55 yr?"	**→ ↑ revision**

– – Active space – –

THA REVISION

– Recon · 32 / 65 –

1 Basics

Epidemiology

Volume	US revisions projected **>100% ↑ by 2030** · ~equal sex · age **65–70**
Causes	overall + after 2 yr: **mechanical failure** (aseptic loosening) · **early (<2 yr): infection, periprosthetic fracture**
Indications	wound problems · periprosthetic/implant fracture · instability · **PJI** · ALTR (MoM) · trunnionosis · osteolysis · aseptic loosening · LLD
Charnley LFA (1962, 22 mm head, cemented)	**11.8% revised at 30 yr** · ✓ **acetabular failure most common** reason (> deep infection > fracture; acetabular > femoral) · acetabular exchange = commonest Charnley revision

Categories

Category	Examples
No component change	Wound closure · **psoas release** · **HO excision** · ORIF fracture with stable implants
Modular only	**Head ± liner exchange** · titanium sleeve
Non-modular	**Cup** or **stem** exchange · both · spacer · Girdlestone

Sciatic nerve risk (TO)

Nerve palsy rates	primary OA **0.5%** · DDH **2.3%** · ✓ **revision 3.5% (highest)** · risk factors: DDH, post-traumatic, posterior approach, cementless, **revision**, female, lengthening

3 Presentation

Symptom	Suggests
Groin pain	Acetabular problem
Thigh pain	Femoral stem
Start-up pain	Component loosening
Night pain	Infection
No improvement after surgery	Wrong diagnosis

Stem subsidence	stems subside into **retroversion** → ✓ **limb externally rotated** ✗ Wrong options: IR · flexion contracture
Resisted hip flexion pain	psoas impingement

4 Diagnosis & Classification

Key radiology findings

Views — AP pelvis · orthogonal hip views · **full-length femur** · prior/immediate post-op films · **Judet views** (columns, discontinuity)

CT — ✓ **osteolysis extent** (X-rays **underestimate**) · pre-op planning, discontinuity · **CT angiogram for Paprosky IIIB ("up and in")** – iliac vessel relation ✓

Labs — ESR, CRP, CBC · Co/Cr (trunnionosis, pseudotumour) · aspirate if labs suggest infection · ✓ **↑ ESR + CRP precludes single-stage revision without further work-up**
✗ Wrong options: ↑ cobalt · metallosis · high Co:Cr · pseudotumour

Lysis around stable cup, normal labs → ✓ CT (Judet) to plan · failed cup with bone loss → ✓ **CT** to plan and detect discontinuity
✗ Wrong options: bone scan · aspiration · ions · dynamic fluoroscopy

AAOS acetabular bone loss

Type	Defect
I	**Segmental** – rim or medial wall
II	**Cavitary** – volumetric loss
III	**Combined**
IV	**Pelvic discontinuity**
V	✓ **Arthrodesis** (TO)

Paprosky acetabular bone loss

Type	Description	Typical reconstruction
I	Minimal deformity, **intact rim**	Hemispherical cup + screws
IIA	Superior (± medial) lysis, **intact superior rim**	Cup + screws
IIB	**Absent superior rim**, superolateral migration (<3 cm)	Cup + screws ± jumbo/augment
IIC	**Localized medial wall** destruction	Cup / jumbo + screws + morsellized graft
IIIA	Rim loss **10 → 2 o'clock**, **"up and out"** (>3 cm superolateral)	**Porous cup + metal augments** + graft
IIIB	Rim loss **9 → 5 o'clock**, **"up and in"** (superomedial), **Köhler line broken**, discontinuity	**Cup-cage**, **custom triflange**, structural graft

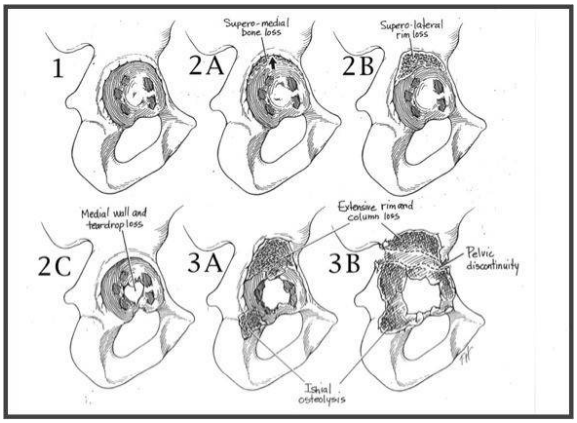
Type II vs III (TO) — hip centre migration **<3 cm vs >3 cm** · II has supportive columns (≥50% host contact), III lacks them

Superior migration cut-off — **< 3 cm** type II / **> 3 cm** IIIA (Orthobullets) · **< 2 cm** II / **> 2 cm** III (TO notes) – read the options

– – Active space – –

Cup-cage sequence (TO) medial → lateral: **graft → cage → cement → cup → head** · cage purchase on **ilium + ischium**

Defect (TO)	Management
Cavitary	**Reverse reaming** + morselized head autograft
Segmental	Structural graft / augment (posterosuperior column)
Combined	Antiprotrusio cage
Pelvic discontinuity	**Cup-cage**



Paprosky acetabular classification

AAOS femoral bone loss

Type	Defect
I	Segmental (shell)
II	Cavitary (endosteal, intact cortex)
III	Combined
IV	**Malalignment**
V	**Stenosis** (canal obliterated)
VI	**Discontinuity** (fracture/nonunion)

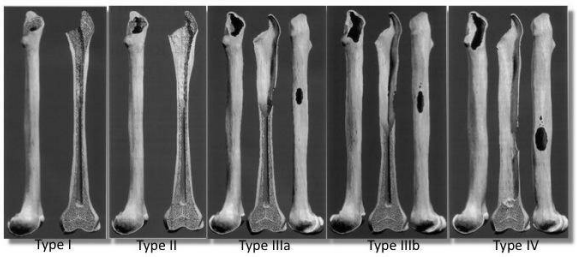
Paprosky femoral bone loss

Type	Description	Treatment
I	Minimal metaphyseal loss	**Primary components**
II	Extensive metaphyseal, **intact diaphysis**	**Extensively porous-coated long stem**
IIIA	Metadiaphyseal loss, **≥4 cm intact isthmus**	**Extensively porous-coated (cylindrical) stem** · modular tapered
IIIB	Metadiaphyseal loss, **<4 cm isthmus**	✓ **Modular fluted tapered stem** · extensively coated (if canal **<19 mm**) · **impaction grafting** · APC
IV	**Non-supportive diaphysis** (no isthmus)	**Impaction grafting**, **APC**, **EPR / PFR**, long cemented

TO (FNB 2014) <4 cm diaphysis → ✓ **IIIB: extensively porous-coated tapered stem + impaction grafting**
✗ Wrong options: IIIA · II · IV/APC

Stem logic (TO) **fluted → rotational stability** · **tapered → distal fit, prevents subsidence** · Paprosky IIIB = isthmus breached → **distally tapered fluted stem** · IV → APC / tumour prosthesis

Difficult THR principles (TO) multiple implants + graft ready · posterior approach (extensile) · **explore + protect sciatic nerve** · trochanteric osteotomy for acetabular exposure · **delay WB 2–3 months** (uncemented osseointegration)



Paprosky femoral classification

5 Management

Revision THA planning

- Exclude infection (ESR/CRP → aspirate) → **if PJI, staged**
- Well-fixed, well-positioned cup + wear/lysis → **head + liner exchange ± graft**
- Acetabulum: ≥50% host support → **porous cup/jumbo + screws ± augments**
  - Discontinuity / <50% → **cup-cage, triflange, column plating**
- Femur: Paprosky I → primary · II/IIIA → extensively coated · IIIB → tapered fluted · IV → **PFR/APC**

Acetabular reconstruction

Porous hemispherical / jumbo cup + screws needs **≥50% host bone** · cup **inferior + medial** (↓ JRF) · cavitary defects → particulate graft · ✓ **most reliable long-term** for Paprosky II · jumbo may compromise posterior column, dislocation
✗ Wrong options: cemented all-PE (high failure) · cage · bilobed · bipolar in acetabulum (pain, erosion)

Paprosky IIC (medial wall), ≥50% stock, no discontinuity, normal centre → ✓ jumbo cup + screws + morsellized graft (most stable + cost-effective)
✗ Wrong options: cage · triflange · spacer · augments

Paprosky IIB → ✓ cementless hemispherical cup + screws

Paprosky IIIA (LLD 3.5 cm, Trendelenburg) → ✓ porous cup + metal augments + graft + screws (unitize augment–cup with cement)
✗ Wrong options: particulate graft alone (IIA/IIB) · triflange (IIIB) · cemented cup (shear failure) · 2-stage

Augments Paprosky **IIB–C, IIIA–B** · survivorship **94–100% (2 yr), 92–100% (5 yr), 92% (10 yr)**

– – Active space – –

Reconstruction / antiprotrusio cage + structural graft

rim **<⅔**, <50% stock · PE cemented into cage · **allograft failure commonest complication** · without cage **40–60% failure** · major bone loss + medial wall penetration → ✓ **reinforcement cage + graft**
✗ Wrong options: jumbo

Non-contained bicolumn defects >50% of weight-bearing area → ✓ ilioischial reconstruction ring + structural corticocancellous graft (restores stock)
✗ Wrong options: morsellized graft + cup · bilobed · roof ring

Impaction grafting (acetabulum)

contained cavitary defects + cemented PE · not for discontinuity

Cemented cup

cement liner into a well-fixed shell · **revision cemented cups fail** (shear)

Pelvic discontinuity & triflange

Definition

separation of **superior ilium from inferior ischiopubic** segment · AAOS IV / Paprosky IIIB · acute pain, cannot bear weight (e.g. RA) → ✓ **pelvic discontinuity**

Principles

reconnect ilium + ischium · bleeding bone contact with ingrowth surfaces · graft defects · stable construct until ingrowth

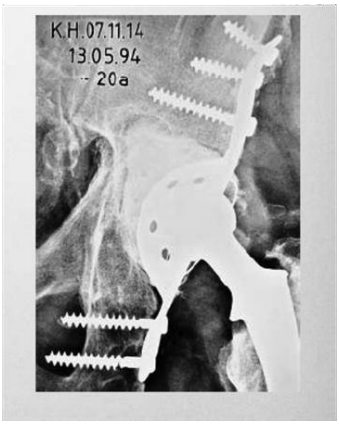
Option	Indication
Column plating + hemispherical cup	**Acute** discontinuity, minimal bone loss, healing potential
Cup-cage	Chronic discontinuity, **Gross IV/V** segmental defects · ✓ displaced transverse periprosthetic acetabular fracture with medial migration · ✓ Paprosky IIIB + structural graft
Custom triflange (CTAC)	Massive bone loss + **chronic discontinuity** · **irradiated** bone · iliac → pubic → ischial flanges, screws ± posterior column plate · ✓ **most common complication: dislocation**
Distraction	Discontinuity with shell ± augments
Bulk allograft + cemented cup	**~50% failure** (resorption, infection)
Independent cage alone	**50–60% failure**

Triflange chosen because of

✓ **pelvic discontinuity** (✗ Wrong options: dislocations · IPD · loosening · Paprosky 2B) · triflange needs posterior/extensile exposure (✗ Wrong options: anterior approach for ischial/iliac flanges) · not a bilobed cup

Paprosky IIIB with discontinuity → ✓ custom triflange or **cup-cage** (✗ Wrong options: cemented cup · multihole cup) · chronic discontinuity after infection cleared (6 months, normal labs) → ✓ **triflange + femoral revision** (✗ Wrong options: new spacer · bulk allograft · cage alone · Girdlestone)

– – Active space – –



Antiprotrusio cage

Osteolysis & isolated liner exchange

Retain shell + liner exchange + graft lysis	✓ **well-designed, well-fixed modular cup** with good locking mechanism and record · lysis via screw holes/dome hole; loosening late ✗ Wrong options: non-modular · instability · radiolucency · migration
Indications	eccentric PE wear with stable components · acute infection · trunnionosis
Typical answers	eccentric head + proximal femoral lysis, fixed components → ✓ **synovectomy + liner exchange** (✗ Wrong options: stem/cup revision) · massive cup lysis, ceramic-on-conventional PE, fixed cup → ✓ **liner exchange + grafting + switch head to METAL** (✗ Wrong options: new ceramic head on used taper (fracture; metal sleeve if ceramic wanted)) · lysis around one screw → ✓ **liner exchange + remove screw + debride ± graft** (Beaulé: no lesion recurrence) · retroacetabular lysis with clicking, normal labs → ✓ **head + liner exchange + retroacetabular graft**
Osteolytic lesions	may shrink/heal without grafting once wear source removed · grafting optional/controversial
Complication	✓ **dislocation** (11–25%) · use larger head · damaged locking mechanism → cement liner into shell (2–4 mm mantle)

Ischial lysis with dissociated dome-hole eliminator → ✓ wear-induced osteolysis
✗ Wrong options: chondrosarcoma · metastasis · corrosion

Femoral revision

Primary components	Paprosky I (no unexpected bone loss)

Extensively porous-coated long stem (or porous/grit combo) or **modular tapered** – Paprosky **II, IIIA** (IIIB modular fluted tapered) · **95% survival at 10 yr** · bypass most distal defect by **2 cortical diameters** · cavitary → particulate graft · cortical defects → struts/plates · ✓ superior to cemented revision

– – Active space – –

Typical answers	loose hemi with proximal loss → ✓ **cementless diaphyseal press-fit stem** (✗ Wrong options: bone scan · tumour prosthesis · metaphyseal cement) · IIIA with fracture → ✓ **fully coated cementless + cables + graft** (✗ Wrong options: ORIF · modular · total femur) · start-up pain + subsidence, no defects → ✓ **uncemented long fully porous stem** (✗ Wrong options: cemented · APC · PFR · spacer) · deficient proximal femur with >4 cm isthmus → ✓ **diaphyseal-engaging extensively coated** (✗ Wrong options: standard cemented · proximally coated · PFR · APC) · <4 cm isthmus (IIIB) → ✓ **modular fluted tapered stem** (✗ Wrong options: struts alone · impaction + cementless (impaction needs **cemented** stem) · metaphyseal tapered) · early loosening at 3 months → ✓ **non-modular tapered fluted Ti stem**
Failed ingrowth signs	reactive line around porous part · **hypertrophy under collar** · **pedestal** at tip (no stress shielding) → ✓ **revise femoral component**

Symptomatic loose osteolytic stem → ✓ **protected WB + urgent (not emergent) revision** when cleared
✗ Wrong options: reassure · alendronate

Impaction grafting (femur)	ectatic canal, thin cortices · **Paprosky IIIB, IV** · morsellized fresh-frozen allograft + **smooth tapered cemented stem** · **subsidence** commonest complication
APC	Paprosky **IIIB, IV** · allograft canal slightly larger than host · cement stem into allograft, **press-fit 4–6 cm into host**
EPR / PFR	massive loss, **non-supportive diaphysis** · elderly/sedentary · frozen section of osteotomy bone to exclude tumour · distal fixation cemented / compressive osseointegration / press-fit
Cemented revision stems	irradiated bone, severe osteopenia, Dorr C · **high failure** · remove old cement, neocortex, sclerotic bone

Proximal femoral replacement

Indications	✓ **<4 cm diaphyseal cortex** · extensive metadiaphyseal loss with <4 cm isthmus · **Paprosky IV** · **failed multiple osteosyntheses of proximal femoral nonunion** · elderly/sedentary · ✓ **EXCEPT: age <50**
Pre-op film indicating PFR	✓ **Paprosky IV** (no isthmus) (II → extensively coated, IIIA → diaphyseal/ modular tapered)

Extended trochanteric osteotomy

Use	remove well-fixed stem / **difficult cement mantle** (✓ ETO when only half the cement can be removed) · **12–14 cm** including vastus ridge · enables diaphyseal-engaging stem ✗ Wrong options: short metaphyseal stem · impaction on loose mantle · cement-in-cement on loose mantle · spacer
Biomechanics	✓ **reduces torque-to-failure to <50% of intact femur** → guide rehab restrictions
Proximal migration	✓ **plate + cables** ✗ Wrong options: abduct >10–15° when fixing · <10 cm osteotomy (↑ migration) · distal prophylactic cable (prevents propagation only) · strut (junctional failure)
Complications	nonunion, intraop fracture, proximal migration, infection, subsidence

Fully coated stem fracture after ETO	✓ **cantilever bending after distal ingrowth** ✗ Wrong options: poor distal fixation · anteversion · inadequate bypass · excessive diameter (thin stems fail)

Abductor deficiency & instability after revision

Gluteus maximus transfer (anterior part + TFL to GT) for irreparable abductors ✓ (label A) · Hardinge most associated with post-op Trendelenburg

Hemi in demented 91-yr-old dislocating → contributor ✓ **dementia** → convert to THA + constrained liner

MoM resurfacing with abducted cup, clunking → ✓ revise to conventional THA
✗ Wrong options: cup-only revision (leaves MoM)

Loose stem + well-fixed well-positioned cup, 10 dislocations, unknown implant, high-risk host → ✓ revise stem + cement constrained liner into existing shell (shell fixed, positioned, locking broken/unknown, room for 2–4 mm mantle)
✗ Wrong options: retrovert stem · remove all · Girdlestone

Conversion of hip arthrodesis

Indications	**low back + knee pain** · ✓ **gluteus medius function predicts ambulatory outcome** (pre-op **EMG**) ✗ Wrong options: >95% survival at 20 yr (97% at 8 yr; others <70% at 10 yr) · time limit (success 30 yr after fusion) · complications = primary (higher) · nerve palsy = primary (higher)
Technique	GT osteotomy · neck osteotomy + reaming under fluoroscopy · revision cup/stem ± **dual mobility or constrained** · glut max transfer if abductors deficient · improves knee + back pain

THA in ankylosing spondylitis (TO)

Features	↑ blood loss, ↑ infection (steroids), intraop fracture, dislocation, **HO** · **stuck-sitting, posterior pelvic tilt, sacral slope change <10° → ↓ cup anteversion** (and inclination) to prevent posterior impingement + **anterior dislocation** · dual mobility · release tight capsule/soft tissue · cement if Dorr C · ✓ **false: "difficult reaming"** (canal wide, cortex thin — perforation/fracture risk) · hip before knee if equal · spine first if both operative

6 Complications

Revision vs primary	lower survival · ↑ **dislocation** (even simple), infection, nerve palsy, fracture, abductor deficiency, DVT, LLD

– – Active space – –

X vs Y

	Paprosky IIIA (acetabulum)	Paprosky IIIB (acetabulum)
Direction	**Up and out**	**Up and in**
Rim loss	10–2 o'clock	9–5 o'clock
Köhler line	Intact	**Broken**
Discontinuity	Uncommon	**Common** – CT angiogram
Reconstruction	Cup + **augments**	**Cup-cage / triflange**

	Paprosky IIIA femur	Paprosky IIIB femur
Isthmus	**≥4 cm**	**<4 cm**
Stem	**Extensively porous-coated cylindrical**	**Modular fluted tapered** (or coated if canal <19 mm)

How it's asked

"AAOS acetabular type 5 (TO)?"	**→ arthrodesis**
"Bicolumn non-contained loss >50% – restore stock?"	**→ ilioischial reconstruction ring + structural graft**
"Fall 2 yr post THA, discontinuity on X-ray – classify/ treat?"	**→ AAOS IV; antiprotrusio cage + screws + posterior column plate**
"Custom 3D implant – true?"	**→ most common complication is dislocation**
"Reconstruction cage indicated for?"	**→ significant bone loss with pelvic discontinuity (3B)**
"Pain, shortening, medial wall penetration?"	**→ reinforcement cage + bone graft**
"Loose cemented all-PE cup after 19 yr (Paprosky 2)?"	**→ cementless porous cup + screws**
"Transverse periprosthetic acetabular fracture, medial migration?"	**→ cup-cage construct**
"Chronic discontinuity, normal labs?"	**→ cup-cage ± morsellized graft**
"Superomedial migration, discontinuity?"	**→ Paprosky 3B; custom triflange**
"Superior rim loss, superolateral migration?"	**→ IIB; cementless cup + screws**
"3.5 cm LLD, superolateral migration?"	**→ IIIA; metal augments + porous cup + graft + screws**
"Köhler line broken, superomedial migration?"	**→ IIIB; cup-cage + structural allograft**
"Proximal + posterior cup migration with discontinuity?"	**→ triflange reconstruction**
"Medial wall erosion, ≥50% stock, no discontinuity – cost-effective?"	**→ jumbo cup + screws + morsellized graft**
"RA, acute severe pain 8 yr post THA?"	**→ pelvic discontinuity**
"Why triflange?"	**→ pelvic discontinuity**

– – Active space – –

"Which defect needs CT angiogram?"	→ **Paprosky IIIB**
"Chronic discontinuity after cleared PJI?"	→ **revise stem + custom triflange**
"Leaving shell, exchanging liner, grafting — optimal when?"	→ **well-designed well-fixed modular cup**
"Ischial lysis + migrated dome-hole eliminator?"	→ **wear-induced osteolysis**
"Eccentric head, fixed components 17 yr?"	→ **synovectomy + liner exchange**
"Massive cup lysis, ceramic-on-conventional PE?"	→ **liner exchange + grafting + metal head**
"Lysis around one screw, eccentric head?"	→ **liner exchange, remove screw, debride ± graft**
"Liner + head revision — increased risk?"	→ **dislocation**
"Isolated PE exchange outcome?"	→ **↑ dislocation vs primary**
"Lysis + stable implants — next study?"	→ **CT scan**
"Retroacetabular lysis + clicking, normal labs?"	→ **head + liner exchange + retroacetabular graft**
"Failed revised cup + bone loss, normal labs — guide treatment?"	→ **CT scan**
"<4 cm diaphysis (TO)?"	→ **Paprosky IIIB; extensively coated tapered stem + impaction grafting**
"Loose hemi, proximal bone loss?"	→ **cementless diaphyseal press-fit stem**
"Loose stem, IIIA, periprosthetic fracture?"	→ **fully coated cementless stem + cables + graft**
"Start-up pain + subsidence, no defects?"	→ **uncemented long fully porous-coated stem**
"Reactive line, collar hypertrophy, pedestal?"	→ **revise femoral component**
"Loose cemented stem, deficient proximal femur, >4 cm isthmus?"	→ **diaphyseal-engaging extensively coated stem**
"Loose osteolytic stem, CAD, symptomatic?"	→ **protected WB + urgent revision when cleared**
"<4 cm isthmus distal fixation?"	→ **modular fluted tapered stem**
"Subsided stem — exam?"	→ **external rotation**
"Early loosening 3 months post-op?"	→ **non-modular tapered fluted titanium stem**
"ETO — rehab consideration?"	→ **torque-to-failure <50% of intact femur**
"Half the cement can't be removed?"	→ **extended trochanteric osteotomy**
"Prevent ETO fragment migration?"	→ **plate + cables**
"Fractured fully coated stem after ETO?"	→ **cantilever bending after distal ingrowth**

– – Active space – –

"PFR indications EXCEPT?"	**→ age <50**
"Pre-op film for PFR?"	**→ Paprosky IV (Figure A)**
"Dislocating hemi in demented 91-yr-old — contributor?"	**→ dementia**
"Clunking MoM resurfacing with abducted cup?"	**→ revision THA**
"Muscle transferred for abductor deficiency?"	**→ gluteus maximus (A)**
"PFR follow-up — which pre-op film?"	**→ Paprosky IV (Figure E)**
"Loose stem, fixed cup, 10 dislocations, unknown implant?"	**→ revise stem + cement constrained liner into shell**
"Arthrodesis conversion — true?"	**→ gluteus medius function predicts ambulation**
"Charnley 30-yr data?"	**→ acetabular failure more common than deep infection**
"Precludes single-stage revision?"	**→ elevated ESR + CRP**
"THR in AS — false (TO)?"	**→ difficult reaming**
"Highest sciatic nerve risk in THR (TO)?"	**→ revision surgery**

THA SCIATIC NERVE PALSY

– Recon · 33 / 65 –

1 Basics

Epidemiology

Incidence **0–3%** (OB ~**1%**; TO 0.6–3.7%, revision **7.6%**) · by diagnosis (TO): **primary OA 0.5%**, **DDH 2.3%** (5.8% sciatic), **revision 3.5%** (3–8%)

Litigation ✓ **nerve injury = most common cause of THA malpractice litigation (38%)** · LLD second (26%) · dislocation not reported · TO: suits arise mainly from **failure to diagnose/manage** the palsy

Nerves involved

Sciatic **~80%** (TO >90%) of THA nerve injuries · ✓ **peroneal division almost always** · peroneal: lateral, superficial, tightly packed fascicles, less connective tissue, tethered
✗ Wrong options: tibial (isolated only ~1%)

Anatomy sciatic nerve **closest to acetabulum at the ischium** · care with **posterior acetabular retractor when hip flexed**

Approach (TO)	Nerve at risk
Anterior	**LFCN** (15–81%) > femoral > sciatic
Anterolateral / direct lateral	**Superior gluteal**
Posterior	**Sciatic (peroneal)**

Other nerves (TO) **femoral** (2nd commonest, 2.3% of injuries, ~0.8% of THAs) – anterior retractors, hyperextension, **iliacus haematoma** · **obturator** (rare) – **cement extrusion** through medial wall, anteroinferior screws → **groin pain, medial thigh paraesthesia, adductor weakness** · **superior gluteal** – medius split **>5 cm** above GT → Trendelenburg, often recovers

Causes

Mechanisms **direct trauma** (retractor, laceration) · **stretch** (lengthening, lateralization, dislocation manoeuvres) · **compression – haematoma**, tight dressings · **cement heat** · **unrecognized lumbar lateral recess stenosis** ("double crush") · positioning (CPN in lateral position) · **unknown ~40%** (TO ≥50%)

– – Active space – –

Risk factors

✓ Risk factors	Not risk factors
DDH ✓ (congenitally short neurovascular structures → over-lengthening)	**Obesity / BMI**
Revision ✓ (highest)	**Rheumatoid arthritis**
Female	**Osteonecrosis / AVN**
Limb lengthening	Male sex
Post-traumatic arthritis	Anterolateral approach
Posterior approach	Posterosuperior quadrant screws (safe zone)
Cementless femoral fixation	DMARDs, chronic analgesics, low pre-albumin
Surgeon rating case as difficult ✓ (Eggli – only correlating factor)	Coxa vara (no exploration needed)
Pre-existing lumbar stenosis / spine surgery	–
Operative time ✓ · younger age · smoking · uncontrolled DM · VTE history · experienced surgeon, later case	–

Lengthening thresholds Orthobullets **>4 cm** · OB keep lengthening **<3 cm** · TO **>3.8 cm** peroneal at risk, **>4 cm** complete sciatic palsy

TO lecture notes **sciatic = commonest nerve** (esp. posterior approach) · highest risk: **revision, DDH, lengthening > 2.5–4 cm** · ↑ risk with **cementless** fixation (snug fit, over-lengthening) · exploration **only in revision** (scarred bed)

Sciatic exploration not routine · advisable with distorted anatomy: ER deformity, **protrusio**, short head/neck, **revision**, **DDH** · ✓ **not needed in coxa vara** (TO)

3 Presentation

Symptoms numbness, paraesthesia in sciatic distribution · **foot drop** (no dorsiflexion, no EHL, weak eversion) · "slapping gait"

Prognosis by deficit **worst: complete motor + sensory**
✗ Wrong options: motor with intact sensation

4 Diagnosis & Classification

Key radiology findings

CT / ultrasound suspected **haematoma**

EMG later, to confirm level + guide prognosis · MRI of hip limited by artefact
✗ Wrong options: as immediate step

Foot drop with back pain, stable Hb, no swelling (POD 2) → ✓ lumbar lateral recess stenosis (double crush) → lumbar MRI
✗ Wrong options: positioning · haematoma · infection · VTE

Foot drop after therapeutic enoxaparin for PE + transfusions → ✓ haematoma compressing sciatic nerve
✗ Wrong options: mechanical · stenosis · embolic infarct · anaemia

5 Management

New sciatic palsy after THA

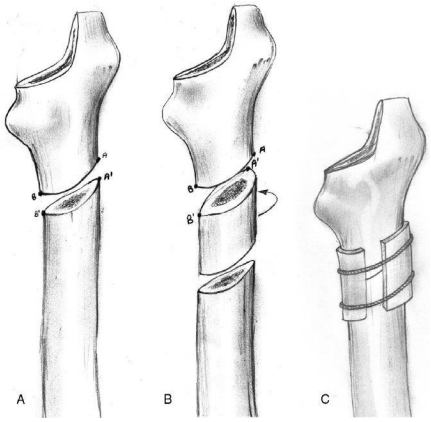
- Intraop (wake-up test / neuromonitoring) → **extend hip, ↓ lengthening (subtrochanteric shortening)**
- Immediate post-op → **hip extension + knee flexion; loosen dressings**
  - Haematoma (drop in Hb, drainage, anticoagulated) → **reverse + urgent evacuation**
- Persistent foot drop → **AFO; tendon transfer (PTT) if AFO fails**

Prevention – femoral shortening & nerve monitoring

DDH pre-op planning · lengthening **<3 cm** · ✓ **subtrochanteric (femoral) shortening osteotomy** · **downsize components** · **intraop wake-up test** · **neurophysiologic monitoring**
✗ Wrong options: anterior approach · modular components · larger head · acetabular osteotomy

Wake-up test shows no dorsiflexion with equal lengths in dysplastic hip → ✓ shortening subtrochanteric osteotomy
✗ Wrong options: EMG · continue + observe · spacer · neurology

Intraop MEP/EMG alert → ✓ extend the hip
✗ Wrong options: traction · flex hip (impinges on retractor) · extend knee · irrigation



Subtrochanteric shortening osteotomy

Postoperative haematoma & other causes

Immediate palsy ✓ **hip extension + knee flexion** (slackens nerve) · then search for reversible cause (axial imaging for haematoma)
✗ Wrong options: hip flexion + knee extension · flat bed NWB · EMG first · knee X-ray

Haematoma (warfarin INR 6, Hct drop, saturated dressing, complete palsy) → ✓ **reverse INR + urgent surgical evacuation (I&D)**
✗ Wrong options: VAC + monitor · transfuse + observe · switch to enoxaparin · IVC filter + exploration

Prognosis & management of foot drop

Recovery ✓ only **35–40% regain full strength after complete palsy → 60–65% never fully recover** (OB: 30–40%; TO: best series ~50%) · maximal recovery takes **~2 years**

– – Active space – –

Persistent foot drop ✓ **AFO first line** (restricts plantarflexion in swing) · continue AFO at 4 weeks–3 months
✗ Wrong options: spine MRI · ankle fusion · revision THA · sural grafting · early tendon transfer · neurology first

AFO not tolerated, chronic → ✓ posterior tibial tendon transfer through interosseous membrane to lateral cuneiform (donor needs **≥4/5**, loses 1 grade)
✗ Wrong options: ankle fusion · split TA transfer (TA 0/5) · peroneus longus (weak) · FHL + TAL

Retractor injury

DAA, foot drop + dorsal foot numbness after retractor slip → ✓ posterior acetabular retractor (4) · deep trochanteric retractor also risks sciatic · position 1 → superior gluteal · 2 → **femoral** (≈2 cm from anterior lip) · 3 → **obturator**

Obturator nerve injury (TO)

Persistent groin pain, no foot drop, no plantarflexion weakness, no Trendelenburg → ✓ obturator nerve
✗ Wrong options: femoral (anterolateral thigh) · sciatic (foot drop) · superior gluteal (Trendelenburg)

X vs Y

Nerve	Deficit after THA	Typical cause
Sciatic (peroneal)	**Foot drop**, dorsal foot numbness	Lengthening, posterior retractor, haematoma
Femoral	Quadriceps weakness, anterior thigh numbness	Anterior retractor, iliacus haematoma, cement
Obturator	**Groin pain**, adductor weakness	**Cement extrusion**, anteroinferior screw
Superior gluteal	**Trendelenburg**	Medius split >5 cm
LFCN	Lateral thigh numbness	Direct anterior approach

How it's asked

"Sciatic exploration not indicated (TO)?"	**→ THR in coxa vara**
"Highest sciatic risk (TO)?"	**→ revision surgery**
"Motor palsy risk factors EXCEPT?"	**→ obesity**
"Sciatic palsy risk factors EXCEPT?"	**→ rheumatoid arthritis**
"Associated with palsy EXCEPT?"	**→ AVN of femoral head**
"↑ sciatic injury in primary THA?"	**→ DDH**
"Most commonly injured nerve?"	**→ peroneal branch of sciatic**
"Risk factor for peripheral nerve injury?"	**→ operative time**
"Wake-up test – no dorsiflexion?"	**→ shortening subtrochanteric osteotomy**
"Neuromonitoring alert – reduce tension?"	**→ extend the hip**

"Foot drop after DDH THA — prevent?"	→ femoral shortening osteotomy
"INR 6, Hct drop, complete sciatic palsy?"	→ reverse INR + urgent evacuation
"Foot drop POD 2, no drainage, normal anticoagulation?"	→ hip extension + knee flexion
"Back pain, foot drop, stable Hb?"	→ lumbar lateral recess stenosis
"Enoxaparin for PE, transfusion, new foot drop?"	→ haematoma
"Complete peroneal palsy — % never recovering full strength?"	→ 60–65%
"Foot drop 2 yr, AFO intolerant?"	→ PTT transfer to lateral cuneiform
"Slapping gait at 3 months?"	→ continue AFO
"Slapping gait at 4 weeks?"	→ AFO
"Nerve injury after THA — true?"	→ most common cause of malpractice litigation
"DAA retractor slip → foot drop?"	→ posterior acetabular retractor
"Groin pain, no foot drop/Trendelenburg (TO)?"	→ obturator nerve

– – Active space – –

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THA STABILITY TECHNIQUES

– Recon · 34 / 65 –

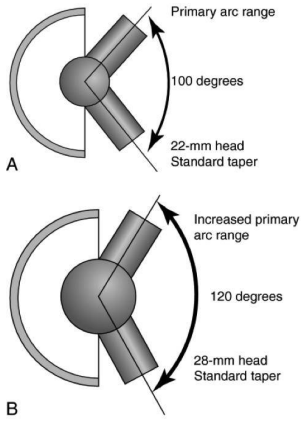
1 Basics

Four determinants of stability

1	**component design** · **2** – **component position** · **3** – **soft-tissue tensioning** · **4** – **soft-tissue function**

Component design – head size & head-neck ratio

Head-neck ratio	head diameter ÷ neck diameter · ✓ **larger ratio → greater arc of motion before impingement** (primary arc) · ✓ **larger head = most closely linked to ↑ ROM** ✗ Wrong options: approach · extended offset · anteversion >18° · abduction <35°
Skirts	extend neck length but ✓ **↓ head-neck ratio → ↑ impingement + dislocation** ✗ Wrong options: large ratio · 15° stem/cup anteversion · 50° abduction are safe
Jump distance	translation before dislocation (~½ head diameter) · larger heads seat deeper → ↑ jump distance → ↑ stability
36 vs 28 mm head in revision	✓ **delays neck–socket impingement** ✗ Wrong options: compensates abductor deficiency · ↓ volumetric wear (↑) · ↓ trunnion stress (↑) · compensates vertical cup
Large head will NOT compensate for	**abductor deficiency** or **vertical cup** · ✓ **high inclination negates the benefit of a larger head** (dislocates with minimal translation)
Primary arc ↑ by	✓ **larger head** ✗ Wrong options: collar · hooded liner · offset (affects JRF not arc) · more anteversion



Head-neck ratio and impingement-free arc

Neck-shaft angle

Change vs native	Leg length	Offset
↑ NSA (valgus)	↑	↓
↓ NSA (varus)	↓	↑

Longer leg + less offset post-op → ✓ increased prosthetic neck-shaft angle
✗ Wrong options: medialized cup (↓ offset only) · standard offset stem (offset only) · longer neck (↑ both) · constrained liner

Trendelenburg + 3 dislocations + already long leg → ✓ more varus (high-offset) stem – ↑ offset without length
✗ Wrong options: skirt · longer neck · valgus stem · medialized cup

Acetabular liners

Elevated (lipped) rim placed posteriorly ↑ stability · ↓ revision for instability

Lateralized (offset) liner +4–10 mm acetabular offset (+ small lengthening) · ↑ tension · ✓ ↑ **acetabular component loosening** (eccentric load, torsion at liner–shell + bone–implant interfaces, ↑ JRF + wear) – only liner type linked to early loosening

4 Diagnosis & Classification

Component position

Component	Target
Cup anteversion	**5–25°** (Lewinnek 15 ± 10)
Cup abduction (inclination)	**30–50°** (Lewinnek 40 ± 10)
Stem anteversion	**10–15°** (hard to adjust in cementless)
Combined anteversion	**37°** (stiff spine: 35–40°)

Approach adjustments **posterior → more anteversion · anterior → less anteversion**

Malposition	Result
Excessive retroversion	**Posterior** dislocation
Excessive anteversion	**Anterior** dislocation
Excessive abduction (vertical, high theta)	**Posterosuperior** dislocation · **eccentric PE wear**, late instability
Excessive adduction (horizontal)	**Flexion impingement**, **inferior** dislocation

Cup medialization ↓ body-weight moment arm → ↓ JRF ✓ (needs ↑ femoral offset to keep abductor tension) · not possible with resurfacing

AIIS hypertrophy impingement + instability in straight flexion → resect · ✓ **direct head of rectus femoris** arises from AIIS (reflected head – groove above acetabular brim; sartorius – ASIS)

– – Active space – –

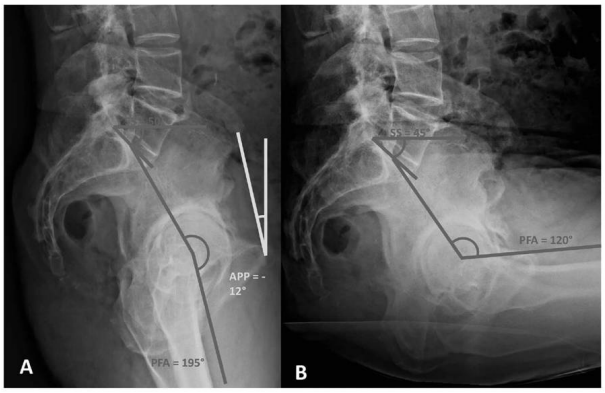
Spinopelvic mobility & cup position

Normal standing → sitting pelvis tilts posteriorly **20–35°** (15–20°) · femur flexes **55–70°** · sacral slope change **10–30°** (11–29°) · pelvic-femoral angle change **51–69°** · sitting anteinclination **41–63°** · standing PFA 165–195° · sitting CSI 162–198°

Stiff spine ✓ **Δ sacral slope <10° → hip hypermobility** → ↑ dislocation · every 1° ↓ spinopelvic motion → 0.9° ↑ femoral motion

Phenotype	Pelvis	Impingement → dislocation
Stuck sitting (APP tilted back 10–13°)	Fixed posterior tilt	**Posterior impingement → ANTERIOR dislocation** ✓ (even via posterior approach, if combined anteversion excessive)
Stuck standing (APP neutral)	Fails to tilt back when sitting	**Anterior impingement → POSTERIOR dislocation**

Stiff-spine targets Lum: inclination **45–50°**, anteversion **20–25°**, combined **35–40°** · McKnight (stuck sitting): **less inclination (35–40°) + less anteversion (15–20°)** · large head / **dual mobility**, restore offset (high-offset stem), navigation/robotics



Stuck-sitting pelvis — anterior instability

5 Management

Femoral offset & abductor tension

Femoral offset ✓ **perpendicular distance from head centre to femoral shaft axis** (line 4) · **horizontal offset (label B)** most affects JRF (neck length also ↑ offset but lengthens leg; vertical height = length only)

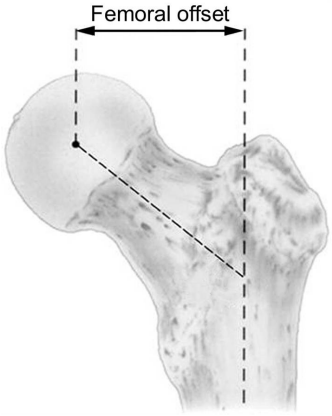
↑ Offset	↓ Offset
↓ impingement · ↓ **JRF** · ↑ soft-tissue tension **without ↑ length** · ↑ stability · ↓ wear, ↓ cup strain, ↓ limp	**Instability** · **abductor weakness** · **gluteus medius lurch / Trendelenburg** · ✓ **↑ osseous (bony) impingement** · ↑ JRF
Risk: lateral prominence → **trochanteric bursitis**	↓ bursitis risk

High-offset stem ✓ **↑ soft-tissue tension** (✗ Wrong options: ↑ JRF · ↑ length · ↑ cup loosening (that is lateralized liner) · ↑ dislocation) · ✓ **improves stability by ↑ abductor tension** (✗ Wrong options: shorter neck · low offset · larger head · medialized cup)

Exacerbates Trendelenburg
✓ **decreasing femoral offset**
✗ Wrong options: extended offset neck · bigger head · longer neck · inferior cup

↓ JRF
✓ **medialize cup** or ↑ offset
✗ Wrong options: ↑ NSA · ↓ offset · ↑ femoral anteversion · head-neck ratio (stability only)

Ways to ↑ offset
longer neck · **↓ NSA** · medialize neck + lengthen · **trochanteric advancement** · lateralized liner



Femoral offset measurement

Soft-tissue function

System	Conditions impairing stability
Central nervous	Stroke, cerebellar disease, **dementia**, MS, **Parkinson**, myelopathy, delirium, **alcoholism**
Peripheral nervous	**Spinal stenosis** (medius = **L5**), neuropathy, radiculopathy, paresis
Local tissue	Trauma, myoligamentous disruption, deconditioning, ageing, irradiation, osteolysis, collagen disease, myopathy, malignancy, infection

Abductor integrity
gluteus medius main abductor · chronic insufficiency → **gluteus maximus ± TFL transfer**

Constrained & lipped liners

Constrained indications
absent abductors · recurrent dislocation with **well-positioned** implants · failed unconstrained solutions · lower threshold in elderly, low-demand, neurologically impaired

Contraindication
✓ **poorly positioned cup** · young patient with large ROM (relative) · risks: loosening, ↓ ROM, mechanism failure
✗ Wrong options: well-positioned cup in elderly · absent abductors · cage · central neurologic decline

Recurrent instability revision

Vertical (high theta) cup with recurrent dislocation → ✓ uncemented acetabular revision
✗ Wrong options: lipped liner · bigger head/thinner liner · cemented cup (↑ revision) · constrained (early failure on malposition)

Posterior capsular insufficiency + irreparable medius tear (trunnionosis ALTR) → ✓ split gluteus maximus transfer (one limb to anterior capsule / minimus, one to GT / medius; ± TFL – Whiteside)

– – Active space – –

✗ Wrong options: constrained liner alone · PFR + direct repair · medialize cup (worsens) · high offset + dual mobility alone

Dislocation rates primary **0.5–6%** · revision **2–20%** · patient risks: Parkinson, dementia, spasticity, alcoholism, previous surgery, ON, obesity, hip fracture

X vs Y

	Larger head	↑ Offset	Cup medialization
Main effect	↑ arc + jump distance	↑ abductor tension, ↓ JRF	↓ body-weight arm, ↓ JRF
Leg length	–	**No change**	–
Fixes abductor deficiency?	**No**	Partly	**No (worsens tension)**

	Stuck sitting	Stuck standing
Dislocation	**Anterior**	**Posterior**
Cup	Less anteversion + inclination	More anteversion

How it's asked

"Increases ROM before impingement?"	**→ larger femoral head-to-neck ratio**
"CoC impingement – increase primary arc?"	**→ larger femoral head**
"Most likely to ↑ dislocation?"	**→ skirted femoral head**
"Impingement in straight flexion from labelled structure – muscle origin?"	**→ direct head of rectus femoris (AIIS)**
"36 mm vs 28 mm head in revision?"	**→ delays neck–socket impingement**
"High cup inclination negates benefit of?"	**→ larger femoral head**
"Template with reduced offset leads to?"	**→ ↑ osseous impingement**
"Most closely related to ↑ post-op ROM?"	**→ larger femoral head**
"Worsens Trendelenburg?"	**→ decreasing femoral offset**
"↑ abductor tension for stability?"	**→ high-offset femoral component**
"Line representing femoral offset?"	**→ head centre to femoral shaft axis (line 4)**
"Adjust to ↓ JRF most?"	**→ horizontal offset (B)**
"High-offset stem leads to?"	**→ ↑ soft-tissue tension**
"↓ JRF during THA?"	**→ medialize the cup**
"Trendelenburg, 3 dislocations, long leg – implant change?"	**→ more varus femoral stem**
"Longer leg + less offset post-op – cause?"	**→ ↑ prosthetic neck-shaft angle**

"Stuck sitting — risk if ignored (posterolateral approach)?"	→ **anterior dislocation**
"Parameter indicating hypermobile hip?"	→ **Δ sacral slope <10°**
"Contraindication to constrained liner?"	→ **poorly positioned cup**
"Liner linked to early cup loosening?"	→ **lateralized (offset) liner**
"Recurrent dislocation, vertical cup?"	→ **uncemented acetabular revision**
"Capsular insufficiency + irreparable medius tear?"	→ **gluteus maximus split transfer**

– – Active space – –

THA TEMPLATING

– Recon · 35 / 65 –

1 Basics

Definition	anticipating **implant size + position** pre-op · reproduces hip biomechanics · minimizes **LLD**
Accuracy	**52–98%** within ± one size · improves with experience
Order	radiographs → patient data on template → landmarks → **LLD** → ✓ **acetabulum first** (sets new centre of rotation) → femur
Tip	aim for sizes in the **middle range** of the system · otherwise consider a different system

4 Diagnosis & Classification

Key radiology findings

Film	Requirement
AP pelvis	Centred over **pubic symphysis**
AP hip	**10–15° internal rotation** (Orthobullets) · **15–20° IR** (TO) – matches native anteversion · neck parallel to cassette
Frog lateral hip	–
Magnification	**20%** standard (tube 1 m, film 5 cm below table) · ↑ in obese, ↓ in slim · use magnification markers

Position of legs for templating	✓ **15° IR** (TO) ✗ Wrong options: 30° IR · 15° ER · 30° ER
External rotation on film	**falsely ↓ offset** · **valgus-appearing neck** · **falsely ↓ canal diameter**
Secondary checks	**pelvic obliquity** (spinal deformity → LLD issues) · **acetabular retroversion** (harder cup positioning)
Pelvic sagittal tilt (TO)	sacrococcygeal joint to top of symphysis · neutral **32 mm women (8–50)** · **47 mm men (15–72)** · ↑ when pelvis tilts forward · standing film with symphysis in line with sacrum (TO)

Radiographic landmarks

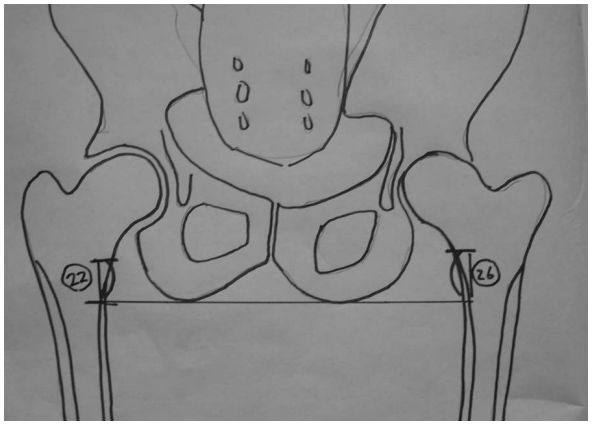
Side	Landmarks
Femur	Medullary canal · GT · LT · **saddle point** – most distal junction of superior neck and GT
Acetabulum	Roof · **teardrop** – superimposed distal medial wall + tips of anterior/posterior horns · lies behind **TAL**, at superior border of obturator foramen (TO)
Pelvis	**Ischial tuberosities** (LLD)



Teardrop landmark

Limb length discrepancy

Step 1	horizontal line across **ischial tuberosities** (extend past medial femoral cortices) · alternative **inter-teardrop line** (may be **more accurate**)
Step 2	mark **proximal-most point of both lesser trochanters**
Step 3	LLD = difference in distance from the reference line to each LT



LLD — ischial line to lesser trochanters

5 Management

Template the acetabulum

Abduction	roughly **40°**
Medial border	approximates **ilioischial line**, close to **teardrop** · medial cup → ✓ ↓ **JRF** + ↓ **abductor force** to keep pelvis level ✗ Wrong options: ↑ abductor force · ↑ leg length (set by cup + stem together) · ↑ / no change in JRF
Inferior border	at **inferior teardrop line**
Then	mark **acetabular centre of rotation (COR)**
Acetabular offset (TO)	acetabular COR to vertical line through teardrop · also: centre of femoral head to **true floor of acetabulum**

Template the femur

Size	fills **medullary canal**

– – Active space – –

Insertion depth set to equalize limb length

Neck cut **posterior approach → lesser trochanter** reference · **anterior approach → saddle point**

Femoral offset ✓ **femoral head COR to line bisecting long axis of femur** (TO) · ↑ offset → ↑ stability, ↓ JRF, ↓ impingement · ↓ offset → instability, abductor weakness, **gluteus medius lurch**

Restore offset by stem with more/less offset · different **neck-shaft angle** · change **neck length**

Templating effects on leg length & offset

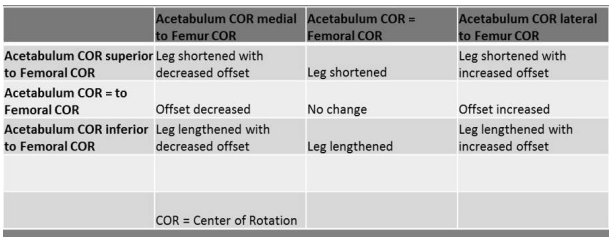
Rule acetabular COR is **fixed** · femoral COR moves to meet it intra-op

Femoral COR relative to cup COR	Effect
Superior	**Lengthens** limb
Inferior	✓ **Shortens** limb (directly inferior → shortens, **no offset change**)
Medial	**↑ offset**
Lateral	**↓ offset**
Overlapping	No change in length or offset

Cup COR vs femoral COR (TO table)	Cup medial to femur	Equal	Cup lateral to femur
Cup superior	Short + ↓ offset	Shortened	Short + ↑ offset
Equal	**↓ offset**	No change	↑ offset
Cup inferior	Long + ↓ offset	Lengthened	Long + ↑ offset

EXCEPT trap (TO) "femoral COR **inferior** to cup COR **lengthens**" = ✓ **false** · true statements: AP in 15° IR at 20% magnification · inferior → shortens · cup COR medial to femoral COR → ↓ offset

↑ offset with same leg length ✓ femoral COR **medial** to cup COR at the **same level** (Figure E)



Template positions – length and offset

Large femoral heads

Associated with ✓ **acceptable wear with modern bearings** · ↓ dislocation · ↑ ROM before impingement · available in all bearing materials
✗ Wrong options: ↑ dislocation · ↓ ROM · fewer materials · ↑ impingement

Acetabular anteversion measurement (TO)

Film	Method
AP radiograph	**Liaw** ($\sin^{-1}\tan\beta$) · **Lewinnek** (arcsin D1/D2 — short/long axis of ellipse) · **Widmer** (short axis / total length) · **Hassan** · **Ackland**
Cross-table lateral	✓ **Woo and Morrey** — measured directly · distinguishes anteversion vs retroversion

NOT an AP method ✓ **Woo and Morrey** (TO)

X vs Y

	Femoral offset	Acetabular offset
Definition	Head COR → femoral shaft axis	Acetabular COR → teardrop vertical (or true floor)
↓ when	Leg externally rotated on film	Cup medialized

	Posterior approach	Anterior approach
Neck-cut reference	**Lesser trochanter**	**Saddle point**

How it's asked

"Templating truths — EXCEPT?"	→ **femoral COR inferior to cup COR lengthens the limb** (TO)
"Femoral offset definition?"	→ **femoral head COR to line bisecting femoral long axis** (TO)
"Leg position while templating?"	→ **15° internal rotation** (TO)
"Template that ↑ offset, same leg length?"	→ **Figure E — femoral COR medial, same level**
"Femoral COR directly inferior to cup COR?"	→ **decreases limb length**
"Femoral COR medial to cup COR at same level?"	→ **length same, offset increased**
"Cup moved medially (A → B)?"	→ **decreases joint reactive force**
"Large heads associated with?"	→ **acceptable wear with modern bearings**
"Not an AP-radiograph anteversion method?"	→ **Woo and Morrey** (TO)

– – Active space – –

THA TRUNNIONOSIS

– Recon · 36 / 65 –

1 Basics

Definition — **mechanically assisted crevice corrosion (MACC)** · metal ion release from corrosion at the **femoral head–neck Morse taper** → hip pain + **adverse local tissue reaction (ALTR)**

Occurs with any bearing — ✓ **metal-on-polyethylene** as well as MoM · relatively rare · lucency between head and cup = PE liner → not MoM
✗ Wrong options: only in MoM

Modularity — lets surgeon tune length + offset · extra junction = corrosion site · **dual-modular (modular neck) stems** → two junctions (head–neck + neck–stem) → **fretting corrosion** most common failure → revise to **monoblock stem**

Risk factors

Factor	↑ Corrosion	↓ Corrosion
Head size	✓ **Larger CoCr heads** (↑ torque on trunnion)	Smaller heads
Trunnion geometry	✓ **Shorter, thinner** taper (flexible, ↓ interference fit, ↑ micromotion, edge loading)	**Longer, wider, more rigid** taper
Materials	✓ **Mixed alloy** taper/bore (CoCr head on Ti stem)	Similar alloy couple · **ceramic head**
Head length	✓ **Long (high-length) heads**	Low-length heads
Stem	**High-offset** stems (↑ fretting) · low-rigidity stems	–
Assembly	**Impaction <4,000 N** · **wet taper** · impaction off-axis	**4,000 N**, dry, in line with taper
Patient	**High BMI**, ↑ cyclical loading	–
Bearing	MoM (ions contribute to taper wear)	–

Cemented vs cementless fixation — unrelated to trunnionosis

2 Mechanism

– – Active space – –

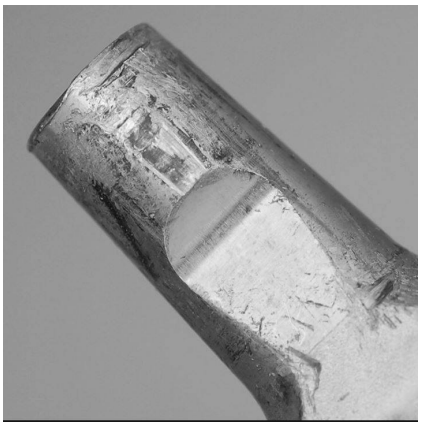
Mechanism – fretting & crevice corrosion

Type	Mechanism
Galvanic	Electrochemical electron transfer between **dissimilar metals** in conductive medium (CoCr head on Ti stem)
Fretting ✓	**Micromotion** between two materials → wear + ion release · disrupts **passivation (oxide) layer** · ✓ **micromotion at head–neck junction → fretting** · commonest in modular necks
Crevice / pitting	Physicochemical metal–environment interaction · low-O_2 area anodic, O_2-rich cathodic · in fatigue cracks · **316L stainless steel most prone**, **titanium least**
MACC	Fretting + crevice combined → passivation layer breaks, O_2-rich fluid enters crevices, acidic layer, ↑ ions → catastrophic failure

Self-passivation Ti and CoCr form adherent oxide coat → **protective**
✗ Wrong options: cause of symptoms

Adhesive wear PE adheres to bearing → osteolysis
✗ Wrong options: a corrosion type in trunnionosis

Tissue response **ALVAL / pseudotumour** = **lymphocyte**-mediated (type IV) · same as MoM · **macrophages** → PE debris osteolysis · neutrophils → infection



Trunnion wear

3 Presentation

Symptoms **pain + limp** commonest · delayed groin/buttock/thigh pain · "clunking"

Exam palpable pseudocyst if ALTR large · **abductor weakness / Trendelenburg** (attritional abductor destruction by ARMD) – not primary avulsion

Catastrophic failure sudden pop, inability to walk → **head–neck dissociation** / stem fracture at trunnion · not dislocation from failed precautions

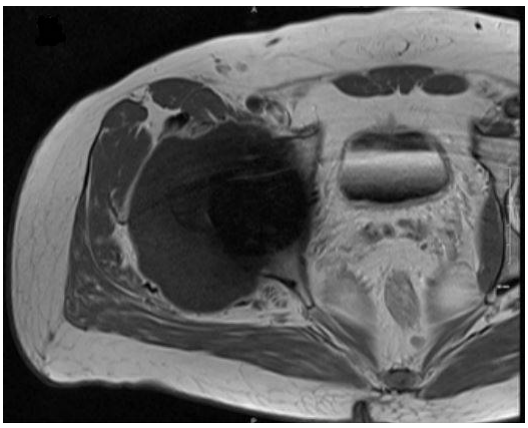
4 Diagnosis & Classification

Key radiology findings

Radiographs AP + lateral · ✓ **may be normal** · osteolysis (calcar, GT) late · **Judet / CT** if advanced osteolysis

– – Active space – –

MARS MRI — **periarticular pseudotumour**, fluid collection, capsular thickening, soft-tissue destruction · **US** – fluid collection adjacent to joint



Pseudotumour on MARS MRI

Genes & lab markers

Rule out infection first — ESR, CRP, WBC · aspiration with ✓ **manual cell count** (metal debris **falsely ↑ automated count**)
✗ Wrong options: automated count confirms

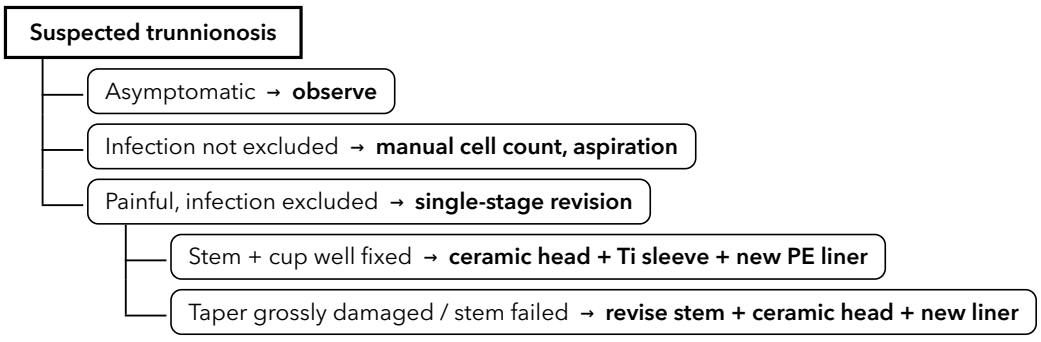
Metal ions — ✓ **cobalt ↑ much more than chromium** · no pathognomonic level

	Co : Cr ratio
Trunnionosis	**≥2:1** (Orthobullets) · **~5:1** / Co **4–5×** Cr (OB explanations)
MoM bearing wear	**~1:1** (1–2:1)

Cobalt threshold for MACC work-up — **>1 ng/mL (ppb)** (Orthobullets; OB: MoP with ions >1.0 µg/L) · **≥1.6 ng/mL** suggested (OB) – read the options

Elevated serum cobalt — ✓ expected with CoCr head + fluid collection
✗ Wrong options: pedestal (loosening) · macrophages (PE) · HO · LLD

5 Management



Revision – ceramic head with titanium sleeve

Indication — painful THA + risk factors + **Co elevated > Cr**, no infection

Technique — **frozen sections + cultures** · **extensive synovectomy / debridement** (remove metal debris, devitalised tissue → prevents recurrence) · inspect + **clean taper** · retain vs revise stem

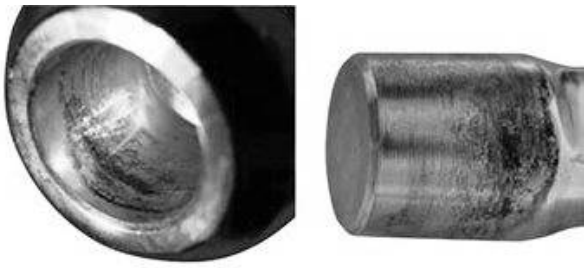
Head ✓ **ceramic head with titanium sleeve** · **sleeve prevents ceramic fracture** on damaged taper
✗ Wrong options: CoCr → CoCr (recurrent ALTR) · CoCr + sleeve

Liner ✓ **always exchange** (embedded metal debris)

Lowest recurrence ✓ **ceramic-on-polyethylene**

Prevention ✓ **ceramic femoral head** eliminates CoCr–Ti interaction
✗ Wrong options: larger head · cemented fixation · diaphyseal stem

Scenario	Answer
Large CoCr head, black fluid, Co 12/Cr 2, manual WBC 1,500	✓ **ceramic head + Ti sleeve + new PE** (✗ Wrong options: revise both components · spacer · bursa injection)
Intra-op corroded trunnion, well-fixed parts	✓ **ceramic head + Ti sleeve** (✗ Wrong options: I&D only (recurrence) · 2-stage · IV antibiotics · cup revision)
Mild trunnion damage, 44 mm CoCr head	✓ **clean trunnion + ceramic head + Ti sleeve + liner exchange** (✗ Wrong options: remove all components)
Catastrophic stem/trunnion failure, cup fixed	✓ **revise femoral component + ceramic head + new liner** (✗ Wrong options: observe · ions first · sleeve only · revise cup)
Modular-neck stem with ALTR	**Revise entire femoral component to monoblock**



Corroded trunnion at revision

Trunnionosis vs other causes

	Trunnionosis	MoM wear	Aseptic loosening	PJI
Ions	**Co** ≫ **Cr**	Co ≈ Cr	Normal	Normal
Cells	**Lymphocytes**	Lymphocytes	**Macrophages**	**Neutrophils**
Imaging	Pseudotumour, may be normal X-ray	Pseudotumour	Subsidence, lucency, start-up pain	Variable
Revision	**1-stage** head + liner	Bearing revision	Loose component	**2-stage**

How it's asked

"Head–neck dissociation at 5 yr, infection excluded — true statement?"	**→ micromotion and/or electrochemical transfer of metals**
"↑ taper corrosion associated with?"	**→ mixed alloy taper and bore**
"Sudden pop, Co 15 / Cr 3, PE liner — cause?"	**→ mechanically assisted crevice corrosion**

– – Active space – –

"CoCr head, Trendelenburg, MARS fluid, Co ≫ Cr — root cause?"	→ corrosion at head–neck Morse taper
"Modular-neck stem, Co 6.1, Cr 2.3, normal ESR — aetiology?"	→ micromotion at head–neck or stem–neck junction
"Micromotion at head–neck — which corrosion?"	→ fretting corrosion
"Retrieved corroded stem — design feature contributing most?"	→ shorter trunnion mating surface
"Why switch to ceramic heads — true about trunnionosis?"	→ elevated serum Co:Cr ratio
"44 mm CoCr on UHMWPE, fluid collection — likely present?"	→ elevated serum cobalt
"Cell mediating ALVAL in trunnionosis?"	→ lymphocyte
"Black fluid, manual WBC 1,500, Co 12 / Cr 2 — next step?"	→ ceramic head + titanium sleeve + new PE liner
"Corroded trunnion, well-fixed parts — best option?"	→ ceramic femoral head with titanium sleeve
"Calcar osteolysis, Co 4.2 — what would have reduced risk?"	→ ceramic femoral head
"Mild trunnion damage at revision — perform?"	→ clean trunnion + ceramic head + Ti sleeve + liner exchange
"Catastrophic femoral failure, cup well fixed?"	→ revise femoral component + ceramic head + new liner

THA VASCULAR INJURY & BLEEDING

– Recon · 37 / 65 –

1 Basics

Incidence vascular injury **0.1–0.2%** · rare but devastating

Risk factors **acetabular screws** · prior vascular bypass / dysvascular limb · **revision** · **intrapelvic cup** and femoral cerclage wiring

Mechanisms screw puncture/laceration · **errant retractor** · compression by **anti-protrusio cages** · direct laceration during approach

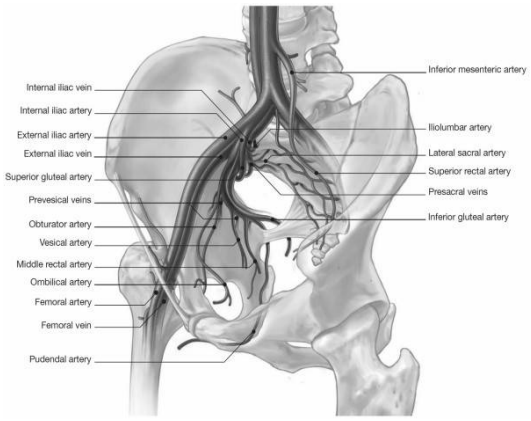
Vessels at risk

Vessel	Location / threat
Obturator a. + v.	Distal to **transverse acetabular ligament (TAL)** · along quadrilateral surface, exits at superolateral obturator foramen · **inferior retractor under TAL** · **anterior-inferior screw**
External iliac a. + v.	**Anterior-superior quadrant** ("zone of death") · ~**7 mm** from bone at ASIS (may lie directly on bone)
Common femoral a. + v. (+ nerve)	Superficial to iliopsoas · only **anterior capsule + iliopsoas** separate from hip · **anterior rim retractor**
Medial femoral circumflex a.	Under **quadratus femoris** / gluteus maximus tendon · posterior approach
Ascending branch of LFCA	Between **TFL and sartorius** · **direct anterior approach**
Superior gluteal a.	Posterior-superior quadrant (safe zone, still possible)
Inferior gluteal + internal pudendal	Posterior-inferior quadrant

Acetabular quadrants

Quadrant	At risk	Zone
Posterior-superior	Superior gluteal vessels, sciatic n.	**Safe** (best bone)
Posterior-inferior	Inferior gluteal + internal pudendal vessels, sciatic n.	Safe-ish
Anterior-inferior	**Obturator** vessels + nerve	Danger
Anterior-superior	**External iliac** vessels	**Danger ("death")**

– – Active space – –



Obturator artery and TAL

Obturator artery injury

Retractor deep to TAL → sudden massive bleed → ✓ obturator artery (TO)
✗ Wrong options: external iliac a./v. · MFCA

Greatest risk to obturator a. → ✓ inferior retractor under TAL (into obturator foramen)
✗ Wrong options: posterior wall retractor (sciatic n.) · posterior-superior screw (sciatic n. + superior gluteal a., safe zone) · anterior-superior screw (external iliac) · anterior wall retractor (external iliac/ femoral)

Uncontrollable obturator bleed	**ligate internal iliac** artery reported

Sources of hematoma (TO)

Most common source	✓ **branches of obturator vessels** (near ligamentum teres, TAL, inferior acetabular osteophytes) (TO)
High-risk zones (TO)	**inferior edge of TAL → obturator vessels** · **gluteus maximus insertion → profunda femoris branches** · **medial wall penetration → iliac vessels** · avoid deep reaming / over-medialization
Drain (TO)	only if a major vessel was injured · **keep clamped 6–8 h** → tamponade
Others	**first perforating branch of profunda femoris** (deep to gluteus maximus insertion) · femoral vessel branches near anterior capsule · superior + inferior gluteal branches (TO)

Anterior retractor – femoral neurovascular bundle (TO)

Hohmann/cobra on anterior acetabular rim	between rim and **psoas tendon** · over psoas → ✓ **femoral artery + nerve** at risk (TO) · risk ↑ the **more inferior** the retractor · safest near **ASIS level** (TO) ✗ Wrong options: psoas tendon · external iliac · obturator

Nerve injury link (TO)

Sciatic	**80%** of THA nerve injuries (femoral **20%**) · **compression** commonest mechanism · closest to acetabulum at **ischium** · **peroneal division** most affected · errant retractor commonest intra-op cause (TO)
Nerve injury risk factors (TO)	**female** · post-traumatic arthritis · **revision** · **DDH** · **lengthening >3 cm**

3 Presentation

– – Active space – –

Signs hemodynamic instability · **pulsatile intra-op bleeding** · **abdominal distension** · loss of limb pulses (palpation/Doppler)

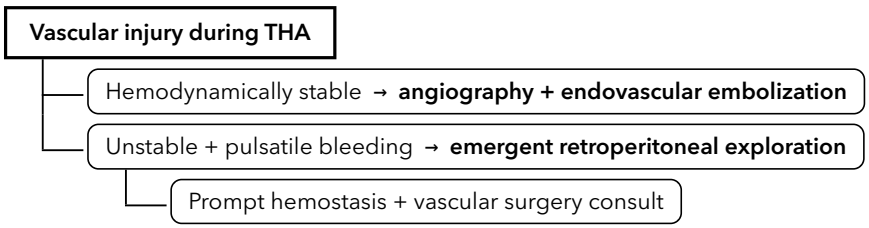
4 Diagnosis & Classification

Key radiology findings

Radiograph **extra-osseous acetabular screw**

Angiography stable patient with high suspicion of uncontrolled bleed · delineates vessel · allows **immediate embolization**

5 Management



Outcome with treatment mortality **9%** · amputation **12%** · permanent ischaemic disability **17%**

Transfusion predictors

Best predictor of transfusion ✓ **low pre-op hemoglobin**
✗ Wrong options: shorter height · lower weight · previous contralateral THA · female gender

Other risks **RA** · advanced age · longer operative time · no clear link with BMI, gender, prophylactic anticoagulation

Anemia symptoms syncope, dyspnea, chest pain · ↓ SpO_2, tachycardia, tachypnea, delayed capillary refill, pallor · check CBC, coagulation, iron

Prevention pre-op **erythropoietin** · topical/IV **tranexamic acid** · hypotensive regional anaesthesia

Transfusion criteria

Patient	Transfuse if
No ischaemic cardiovascular disease	**Hb <7** + symptomatic anemia
Ischaemic CVD (MI, CHF)	**Hb <8** + symptomatic anemia

Transfusion complications ✓ **↑ PJI** · transfusion reactions

– – Active space – –

X vs Y

Retractor / screw	Structure at risk
Inferior retractor under TAL	**Obturator a.**
Anterior rim retractor over psoas	**Femoral a. + n.**
Posterior wall retractor	**Sciatic n.**
Anterior-superior screw	**External iliac a./v.**
Anterior-inferior screw	**Obturator** a./v./n.
Posterior-superior screw	Sciatic n., superior gluteal a. (safe zone)

How it's asked

"Best predictor of post-op transfusion?"	→ **lower preoperative hemoglobin**
"Massive bleed after retractor deep to TAL?"	→ **obturator artery** (TO)
"Most common source of hematoma in THR?"	→ **branches of obturator vessels** (TO)
"Manoeuvre placing obturator artery at greatest risk?"	→ **inferior retractor under the TAL**
"Hohmann on anterior acetabular rim — structure at risk?"	→ **femoral artery and nerve** (TO)

TKA APPROACHES

– Recon · 38 / 65 –

1 Basics

Approach dictated by	surgeon preference · **prior incisions** · degree of deformity · **patella baja** · obesity
Simple primary	medial parapatellar · midvastus · subvastus · minimally invasive
Complex primary / revision	medial parapatellar · **quadriceps snip** · **V-Y turndown** · **tibial tubercle osteotomy (TTO)**
Outcome	✓ **no midterm/long-term difference** in function between medial parapatellar, midvastus, subvastus · similar **blood loss** · best long-term practice → ✓ **approach the surgeon is most comfortable with** · subvastus had higher mean ROM at 6 mo only (meta-analysis, 60 RCTs) ✗ Wrong options: midvastus · mini-medial parapatellar · subvastus · standard MPP

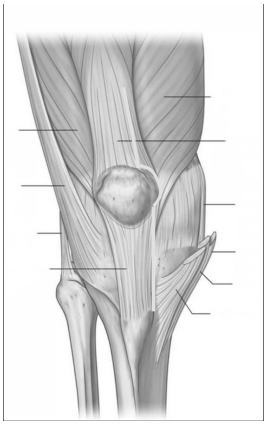
Prior incisions

Skin blood supply	mainly from **medial side** (saphenous a. + medial genicular perforators) → **lateral flap** at greatest necrosis risk
Multiple longitudinal incisions	✓ use the **most lateral** (TO) (or most recently healed) ✗ Wrong options: medial · transverse · new midline
Dual-plated plateau fracture with medial + lateral scars	✓ **lateral one** (TO) · preserve medial flap (perforators) · avoid excess lateral subcutaneous dissection (TO)
Single previous scar	reuse if suitable · **transverse scar** → cross at **right angles** (perpendicular), avoid acute angles
Skin bridge	**≥5 cm** if old scar cannot be used (TO) · exact length controversial (Orthobullets)
Old transverse incision years earlier	standard anterior longitudinal incision usually fine (TO)
Post-traumatic (plateau ORIF) TKA	risk of TKA **5× controls (7.3% at 10 yr)** · remove only hardware that interferes with prosthesis · stiff knees → snip, V-Y, TTO, lateral release · bone loss → cement, graft, metal wedges · computer assistance helpful (TO)

5 Management

Primary approaches

Approach	Technique	Advantages	Disadvantages / relative CI
Medial parapatellar	Straight midline incision, arthrotomy medial to patella	Familiar · **excellent exposure** · easy patellar eversion	Medial capsular repair failure · lateral patellar subluxation · indirect lateral retinaculum access · **patellar circulation risk if lateral release added**
Lateral parapatellar	Lateral arthrotomy	**Fixed valgus** · preserves patellar blood supply · prevents lateral subluxation · direct lateral access	Technically demanding · **difficult medial patellar eversion** · may need TTO
Midvastus	Through **VMO muscle belly** · spares VMO insertion on quad tendon	Possibly faster rehab · better patellar tracking	Less extensile · hard in obese, flexion contracture · ✓ **partial VMO denervation** · CI: ROM <80°, obese, hypertrophic OA, previous HTO
Subvastus	**Vastus medialis belly lifted off intermuscular septum** (incision not through muscle)	**Patellar vascularity preserved** · extensor mechanism intact · minimal lateral release · accelerated rehab	✓ **Least extensile** · CI: **revision**, large quads, previous HTO, obese, previous parapatellar arthrotomy
Minimally invasive / quad-sparing	Special instruments	–	Technically demanding · ↑ **component malposition** · longer tourniquet time



Medial parapatellar, quad-sparing, subvastus, midvastus

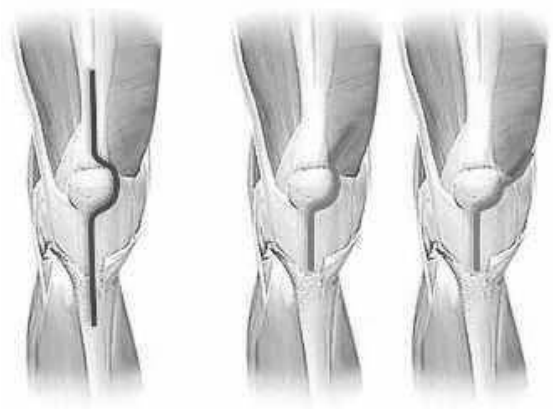
Exposure tip (TO) always raise the **medial periosteal sleeve** + remove **patellar fat pad** · subvastus / midvastus not for **obese or revision** · **rectus snip** for stiff / revision knees

Ankylosed knee (TO) needs **intact extensor mechanism** (else mesh / allograft) · **flexion > 90° hard to achieve**, high complication rate, poor prognosis

Vastus approaches & denervation

Greatest quadriceps denervation potential	✓ **midvastus** (abnormal EMG · blunt interval dissection recovers better than sharp) ✗ Wrong options: subvastus · quad sparing · MPP · TTO
Subvastus EXCEPT (TO)	✓ **"spares quad tendon, carried through belly of VMO"** = false (that is **midvastus**) · true: VM belly lifted off septum · accelerated rehab · not for revision (TO)

Picture with incision not through muscle substance → ✓ subvastus (TO)



Subvastus — incision below VM, not through muscle

Minimally invasive & quadriceps-sparing approaches

Outcomes	✓ **no clinically significant improvement in PROs**, gait or quad strength ✗ Wrong options: quickest rehab · normal gait sooner · shorter tourniquet / less pain (longer tourniquet) · no malalignment risk
Higher component malposition vs MPP	✓ **quadriceps sparing** ✗ Wrong options: lateral parapatellar · midvastus · snip · V-Y
Mini-subvastus	preserves all four quadriceps attachments to patella
Convert to standard parapatellar when	✓ **jig alignment difficult / incision too small** · **patellar tendon starts peeling off tubercle** ✗ Wrong options: subvastus · TTO · two-incision · ligament release

Lateral parapatellar approach for valgus knee (TO)

Approach of choice	**fixed valgus** · lateral release integral to approach · preserves **medial** patellar blood supply · medial displacement of extensor mechanism internally rotates tibia → brings **posterolateral corner** into view · optimises PF tracking · tubercle can be medialised (TO)
Medial arthrotomy + lateral release	significant impairment of extensor mechanism blood supply (TO)
EXCEPT (TO)	✓ **"more patellar necrosis than medial arthrotomy"** = false · true: less maltracking · difficult exposure (patellar eversion, may need TTO) · **fixed varus = contraindication** (TO)

– – Active space – –

Extensile exposure in revision

First steps extend medial release posteriorly (tibial ER → lateralises extensor mechanism) · ✓ **scar debridement recreating medial + lateral gutters + remove polyethylene** · scar under extensor mechanism commonest cause of failure to evert patella · need **~110° flexion** for safe implant removal

Stepwise debridement → **quad snip** → **TTO (preferred)** or quad turndown → collateral release + hinge = **last bail-out**

Lateral release alone for lateral patellar subluxation after implants · does not improve intra-articular exposure

	Quadriceps snip	V-Y turndown	Tibial tubercle osteotomy
Technique	Oblique cut from **apex of quad tendon at 45° into vastus lateralis**	MPP arthrotomy + diverging cut down **vastus lateralis tendon** toward lateral retinaculum (inverted V, closed as Y)	**6–10 cm** fragment cut **medial to lateral** (lateral bony hinge) · screws/ wires
Exposure	Least extensile of three	**Excellent** (ankylosed knee) · lengthens quad tendon	**Excellent** · best for **well-fixed cemented tibial stem / cone**
Post-op	✓ **No change** – **WBAT + ROM as tolerated**, same as MPP (TO)	**Immobilize in extension** until extensor heals	ROM limited **4–6 wk** · some limit WB
Problems	Minimal long-term · early quad weakness, no extensor lag	✓ **Extensor lag** · quad weakness · **patellar devascularisation** (inferior geniculate) → necrosis/fracture	Tubercle avulsion fracture · **non-union (low – union ~100%)** · wound problems

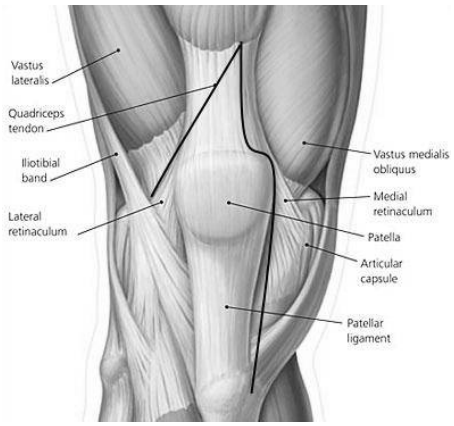
Snip + proximal lateral release joined = **quadriceps turndown** → extensor lag, patellar blood supply loss, worse function

TTO needed for ✓ **well-fixed tibial cone + cemented stem**
✗ Wrong options: maltracking · metaphyseal bone loss (cones/stems) · extensor lag (extensor reconstruction) · elevated joint line (distal femoral augments)

V-Y vs TTO ✓ **V-Y → more extensor lag** · V-Y → higher patellar fracture risk · PROMs similar · TTO low non-union · V-Y starts ROM later

Patella baja + exposure difficulty – which needs rehab change? → ✓ **V-Y turndown**
✗ Wrong options: MPP · lateral parapatellar · midvastus · quad snip (all standard rehab)

Most difficult TKR (TO, FNB 2024) options Insall–Salvati <0.7, Caton–Deschamps <0.7, bipartite patella · answer **not provided by the source** · patella baja complicates exposure (Orthobullets)



V-Y turndown

Bilateral TKA

Term	Definition
Simultaneous	Two surgeons, both knees at the same time
Sequential	One surgeon, one knee then the other, **one anaesthetic**
Staged	Separate anaesthetics · **3 days to 1 year** apart

Antibiotic-loaded bone cement ↓ deep infection in **revision** · primary use controversial (theoretical ↓ strength) · not shown to ↑ aseptic loosening

X vs Y

	Midvastus	Subvastus
Plane	**Through** VMO belly	**Under** VM, off intermuscular septum
Risk	**Denervation**	**Least extensile**
Revision	Avoid if previous HTO, ROM <80°	**Contraindicated**

	Medial parapatellar	Lateral parapatellar
Use	Default, all knees	**Fixed valgus**
Patellar eversion	Easy	**Difficult** (may need TTO)
Patellar blood supply	Risked if lateral release added	**Preserved**

How it's asked

"Quad-sparing MIS, jig alignment difficult – next step?"	**→ convert to standard parapatellar arthrotomy**
"Approach with higher component malposition?"	**→ quadriceps sparing**
"True about MIS quad-sparing approaches?"	**→ no clinically significant improvement in PROs**
"Approaches A, B, C (MPP, midvastus, subvastus) – accurate?"	**→ similar blood loss and midterm function**

– – Active space – –

"Best long-term practice for a new surgeon?"	→ **the approach the surgeon is most comfortable with**
"Subvastus – EXCEPT?"	→ **spares quad tendon, carried through belly of VMO** (TO)
"Picture of approach (incision not through muscle)?"	→ **subvastus** (TO)
"Greatest quadriceps denervation potential?"	→ **midvastus**
"Rehab after Q-snip?"	→ **same as midline parapatellar** (TO)
"Quad snip in revision – therapy regimen?"	→ **no restriction in ROM or weight bearing**
"TTO likely needed because?"	→ **well-fixed tibial cone and cemented stem**
"Quad snip in revision for aseptic loosening – instructions?"	→ **WBAT, ROM as tolerated**
"Stage-2 revision, stiff, cannot evert patella – next exposure step?"	→ **scar debridement recreating gutters + remove polyethylene**
"V-Y turndown (A) vs TTO (B)?"	→ **A more likely to have extensor lag**
"Two longitudinal + one transverse scar – which incision?"	→ **lateral** (TO)
"Prior dual plating, medial + lateral scars – incision?"	→ **lateral one** (TO)
"Patella baja, exposure needed – which technique changes rehab?"	→ **V-Y turndown (Figure F)**
"Lateral parapatellar for valgus – EXCEPT?"	→ **more patellar necrosis than medial arthrotomy** (TO)

TKA ASEPTIC LOOSENING

– Recon · 39 / 65 –

1 Basics

Definition — **macrophage-induced inflammatory** bone loss → implant loosening **without infection**

Epidemiology — ✓ **most common mode of long-term (>2 yr) TKA failure** · HXLPE has ↓ wear failures but loosening persists

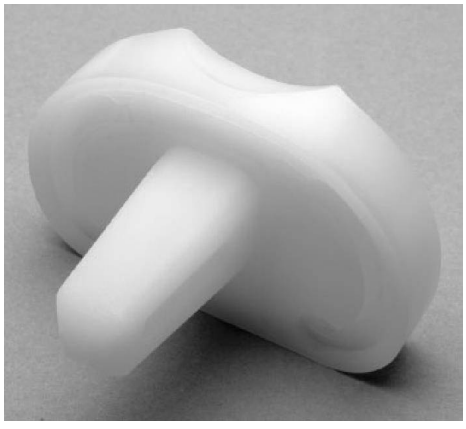
Steps — **particulate debris formation** → **macrophage-activated osteolysis** (RANKL → osteoclasts) → **prosthesis micromotion** → **debris dissemination**

Osteolysis & prevention

Factors affecting PE wear rate — **sterilization method** (gamma in air ↑) · manufacturing (conventional vs cross-linked) · **shelf life** (prolonged ↑) · third-body debris · **backside wear** (modular insert on metal tray) · femoral counterface roughness · **alignment + stability** · **activity level**

Malalignment — asymmetric loading → early loosening · ✓ **varus more than valgus**

↓ Osteolysis	↑ Osteolysis
✓ **All-polyethylene tibial component** (no backside wear)	Metal tray + **modular PE insert** (backside wear)
HXLPE	Avoiding HXLPE
Congruent PE	**↓ congruency** → sliding → adhesive wear
Less constraint	**↑ constraint** (VVC) → ↑ stress at articulation



All-polyethylene tibial component

Early cement debonding

Weak link — ✓ **implant–cement interface** (more common than bone–cement)

↑ Risk — ✓ **ROM + ligament stability testing before cement fully cures** (↓ tibial tray pull-out strength) → tibial tray loosening with **varus collapse**

– – Active space – –

Prevention (✗ Wrong options: as risk factors) **fresh gloves** before handling implants (lipid) · **suction + dry bone surface** · cement on **both implant undersurface and bone** · **roughened** tray undersurface (✗ Wrong options: smooth polished) · let cement fully cure

3 Presentation

Symptoms painless early · later pain localised around loose component · **weight-bearing / activity related**

Exam minimal pain with ROM · ↑ pain with weight bearing · effusion possible

4 Diagnosis & Classification

Key radiology findings

View	Use
AP	**Tibial osteolysis** readily seen · **femoral osteolysis hard to see** (posterior condyles hidden by component)
Lateral	Routine
Oblique	✓ **Best for femoral osteolysis**
CT / MRI	Size large lesions for pre-op planning

Findings radiolucency with **sclerotic border**, especially **>2 mm** · implant position change (**varus/valgus tibial subsidence**) · **progressive widening** of cement–bone / bone–prosthesis interface · **cement cracking / fragmentation**



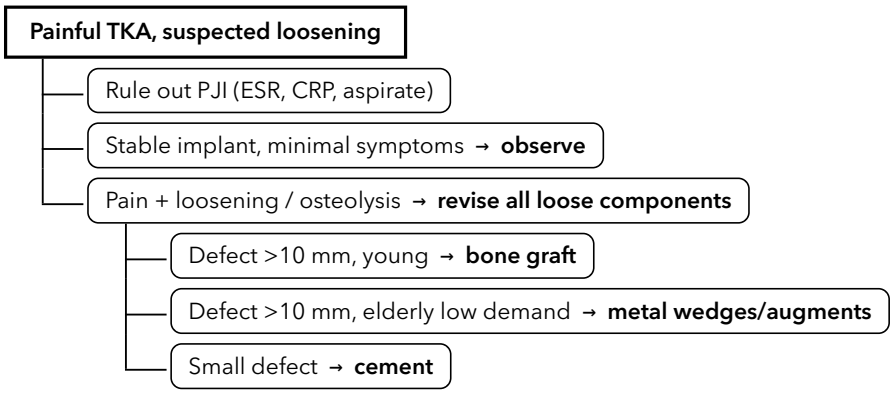
Femoral and tibial osteolysis

Genes & lab markers

ESR + CRP normal · always exclude **PJI** first

Osteolysis vs others effusion, near-normal ESR, normal CRP, lytic lesions, no loosening → ✓ **osteolysis**
✗ Wrong options: infection · aseptic loosening (no loosening signs) · PE dislodgement (locking mechanism intact) · crystal (doesn't explain lysis)

5 Management



Revision indications

Revise pain due to aseptic loosening · pain + osteolysis · **extensive osteolysis that would compromise future revision**

Observe stable implant, minimal symptoms

Defect	Fill
>10 mm, younger	**Bone graft** (preserves bone stock)
>10 mm, elderly / low activity	**Metal wedges / augments**
Small	**Bone cement** (heat → thermal necrosis → can cause loosening)

X vs Y

	Osteolysis	Aseptic loosening	PJI
Implant	Fixed	**Loose / subsided**	Variable
Labs	Normal	Normal	↑ ESR, CRP
Treatment	Observe or liner exchange + graft	**Revise loose parts**	Two-stage / DAIR

How it's asked

"7 yr TKA, effusion, CRP normal, lytic lesions, no loosening?"	→ **osteolysis**
"Reduces osteolysis risk?"	→ **all-polyethylene tibial component**
"Tibial tray varus collapse at 3 yr – technical risk?"	→ **ROM/stability testing before cement fully cures**
"Best view for femoral osteolysis?"	→ **oblique** (AP misses posterior condyles)
"Most common weak link in cemented TKA?"	→ **implant–cement interface**

– – Active space – –

TKA AXIAL ALIGNMENT

– Recon · 40 / 65 –

1 Basics

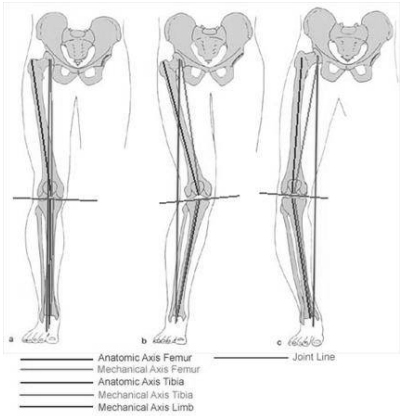
Relevance pre-op planning · component insertion · ligament balancing · prosthesis selection

Normal anatomy

Measure	Value
Distal femur (anatomic axis vs joint line)	**~9° valgus**
Femoral anatomic vs mechanical axis	**5–7° valgus**
Proximal tibia (anatomic axis vs joint line)	**2–3° varus** (tibial joint line ~3° varus to mechanical axis)
Q angle	**11 ± 7°**

Mechanical axis

Limb mechanical axis ✓ **vertical line from femoral head centre through centre of knee to centre of ankle** · re-established at TKA
✗ Wrong options: 5–7° valgus between shafts (anatomic axis) · 3° varus between shafts (varus knee) · passes 1.5 cm lateral (valgus knee) · 1.5 cm medial (varus knee)



Mechanical vs anatomic axes

Technical goals

Goals restore **mechanical alignment (0°)** · restore **joint line** (lets preserved ligaments e.g. PCL function) · **balanced flexion + extension gaps** · maintain **normal Q angle**

4 Diagnosis & Classification

Key radiology findings

View	Purpose
Standing AP + lateral knee	Joint-space narrowing · collateral insufficiency (**lateral gapping in varus**, **medial gapping in valgus**) · femorotibial subluxation · bone defects
Standing full-length AP + lateral	Accurate **valgus cut angle** when **femoral/tibial deformity** or **very tall / short** stature
Flexion + extension laterals	–
Sunrise	Patellofemoral

5 Management

Femoral alignment

Anatomic axis (AAF) bisects femoral canal · sets **IM guide entry point** · IM guide runs down AAF

Mechanical axis (MAF) femoral head centre → point where AAF meets **intercondylar notch** · neutral → even medial/lateral condylar load

Valgus cut angle (VCA) AAF–MAF difference (**~5–7°**) · distal cut perpendicular to MAF · jig usually **6°** from AAF · **tall → VCA <5°** · **short → VCA >7°** · measure on full-length standing AP

Tibial alignment

Anatomic axis (AAT) bisects canal · IM/EM guide parallel to it

Mechanical axis centre of proximal tibia → **centre of talus** · tibial cut **perpendicular to mechanical axis**

Usually AAT = mechanical axis → **IM jig** fine · **tibial deformity** (axes differ) → ✓ **extramedullary guide**

Patellofemoral alignment

Abnormal patellar tracking ✓ **most common complication of TKA** (not most serious)

Most important variable preserve **normal Q angle (11 ± 7°)** · Q = ASIS–patella centre vs patella centre–tibial tuberosity

↑ Q angle errors **IR of femoral component** · **medialised femoral component** · **IR of tibial component** · **lateral placement of patellar button** → lateral subluxation, pain, wear, dislocation

– – Active space – –

Joint line preservation

Goal implant thickness = bone + cartilage removed · fill defects so joint line not jeopardised

	Elevated joint line (>8 mm → motion problems)	Lowered joint line
Effects	**Mid-flexion instability** · PF tracking problems · **pseudo patella baja**	**Lack of full extension** · **flexion instability**

Valgus knee never elevate joint line until balanced to full extension

Deformity & risk (peroneal palsy)

Valgus vs varus TKA valgus ↑ ✓ **peroneal nerve palsy** (lateral lengthening → traction) · also instability, **recurrent valgus**, wound breakdown, patellar maltracking · lateral contracture + medial laxity · lateral condyle / plateau deficiency → hard to find epicondylar axis → **femoral IR** → maltracking
✗ Wrong options: recurrent varus · ↑ revision · PJI · longer stay

67 yo, old ACL tear, varus with medial tibial deficiency + subluxation → ✓ TKA
✗ Wrong options: HTO (older) · UKA (ACL deficient) · interpositional (0/12 excellent) · closing wedge

UKA criteria limited unicompartmental disease · fixed **≤10° varus or ≤5° valgus** · **intact ACL** · no ML subluxation

Extra-articular deformity (McPherson rule) (TO)

Principle the **closer** a coronal extra-articular deformity is to the knee, the **greater** knee malalignment (TO)

McPherson rule ✓ deformity within **distal ¼ femur or proximal ¼ tibia** and **≥20° → concomitant osteotomy + TKA** (TO) · closing wedge osteotomy + **long diaphyseal-fit stem** (no extra fixation) · bone cuts alone → excessive resection threatening collaterals or incompetent ligaments (TO)

Traps (TO) ✗ Wrong options: farther deformity → greater malalignment · farther deformity needs bigger intra-articular cut · deformity within distal/proximal ¼ easier to correct with cuts

Robotic-assisted TKA

Greatest positioning precision ✓ **robotic assistance** (↓ outliers)
✗ Wrong options: spinal anaesthesia · TXA · IM / EM guides

RA-TKA literature ✓ **better implant position + limb alignment, no difference in long-term outcomes** · no consistent difference in complications, revision, survivorship · surgical time normal after **15–20 case** learning curve · dissatisfaction after TKA **5–20%**
✗ Wrong options: longer time for experienced · ↑ longevity

Patient-specific instrumentation (TO)

PSI 3D model from imaging → custom cutting jigs → sizing, resection, rotation, slope · ↓ steps → **less operative time** (TO)

Uses (TO) **extra-articular deformity** · **retained femoral hardware** (IM guides impossible)

EXCEPT (TO) ✓ **"fewer outliers than conventional TKR"** = false · PSI equal or **less accurate**, more outliers · uses **measured resection** for rotation, ignores soft tissue → femoral rotation errors (TO)

X vs Y

	Robotic / navigation	PSI
Outliers	↓	Same or ↑
Long-term outcome	No difference	–
Use without IM canal	✓	✓

	IM tibial guide	EM tibial guide
When	Anatomic = mechanical axis	**Tibial deformity**

How it's asked

"Mechanical axis of a normal limb?"	**→ vertical line femoral head → centre of knee → centre of ankle**
"67 yo, old ACL tear, varus with subluxation?"	**→ total knee arthroplasty**
"Valgus knee vs varus knee – ↑ risk of?"	**→ peroneal nerve palsy**
"Greatest implant positioning precision?"	**→ robotic assistance**
"RA-TKA vs conventional – true?"	**→ better positioning/alignment, no difference in long-term outcomes**
"McPherson rule?"	**→ deformity in distal ¼ femur / proximal ¼ tibia ≥20° → concomitant osteotomy + TKR** (TO)
"Patient-specific TKR – EXCEPT?"	**→ fewer outliers than conventional TKR** (TO)
"Most common complication of TKA?"	**→ abnormal patellar tracking**

– – Active space – –

TKA CORONAL PLANE BALANCING

– Recon · 41 / 65 –

1 Basics

Principle **concave side = tight → release · convex side = stretched → tighten** (fill gap with thicker poly) · always test in **flexion and extension**

Epidemiology **>85%** of TKA patients neutral or varus · valgus = more complex

Kinematic alignment implants in patient's constitutional varus/valgus · outcomes ≈ neutral mechanical alignment



Coronal balancing – concave vs convex

Posterior oblique ligament function

Structure	Primary role	Secondary
POL	✓ **Resists internal rotation**	Valgus, external rotation
Superficial MCL	**Valgus**	Valgus + IR combined
Deep MCL	Released in standard approach	–
LCL	**Varus**	–
LCL + posterolateral release	→ **external rotation laxity**	–

✗ Wrong options: POL resists flexion

Medial release sequence (varus knee)

Step	Orthobullets	TO (FNB 2022)
1	**Deep MCL** release to mid-coronal plane of tibia	Deep MCL + posteromedial capsule from tibia + osteophytes
2	**Medial osteophytes**	**POL**
3	**Posteromedial corner (POL)**	**Semimembranosus** (+ transect PCL if preserved)
4	**Medial tibial reduction osteotomy**	**Pie-crust superficial MCL** + release **pes anserinus**
5	**PCL release / substitution** if imbalance persists	Strip periosteum + sMCL off tibia distally **4–5 cm**
6	**Semimembranosus** (esp. with flexion contracture)	–
7	**Pie-crust superficial MCL** (**18-gauge needle**)	–
8	Complete **sMCL / pes** release – rarely needed · destabilises medial flexion gap → **constrained prosthesis**	–

Differential sMCL release — **posterior (oblique) portion tight in extension** · **anterior portion tight in flexion**

Extension vs flexion space (TO) — posterior structures (**POL → posterior capsule → semimembranosus**) affect ✓ **extension** space · **anterior half of sMCL + pes anserinus** affect **flexion** space (TO)

Tight medial in extension, flexion balanced (FFD 15°) → ✓ **release POL** (TO)
✗ Wrong options: sMCL · pes · PCL

Pie-crusting — ✓ **soft-tissue balancing in TKR** (TO) · multiple stabs with blade/needle **parallel to joint line** on contracted side · titrated lengthening of varus or valgus sleeve
✗ Wrong options: syndesmosis · deltoid · bone cuts

Lateral side (varus knee) — fill gap to tension stretched lateral ligaments · **poly >15 mm needed → constrained** (avoid joint-line elevation)

Blount's varus 25° with thrust → ✓ **constrained implants may be needed** (posteromedial bone loss + extensive posteromedial releases)
✗ Wrong options: peroneal palsy common (that is valgus) · arthrodesis · lateral reefing (attenuates) · TKA then osteotomy

– – Active space – –

Valgus knee (lateral release, popliteus, peroneal nerve)

Valgus pathology ✓ **lateral femoral condyle hypoplasia** · tight lateral retinaculum, LCL, posterolateral corner, ITB · **attenuated MCL** · **lateral patellar tracking** · **tibial external rotation** relative to femur
✗ Wrong options: tibial IR · medial tracking · tight MCL

Order	Lateral release (Orthobullets)	Lateral release (TO)
1	**Osteophytes**	Lateral osteophytes + posterolateral capsule
2	**Posterolateral capsule**	**PCL always released**
3	**ITB** if **tight in extension** (pie-crust or off Gerdy's)	Pie-crust ITB (extension) at joint line or at Gerdy's
4	**Popliteus** if **tight in flexion** (anterior part of insertion)	Strip LCL + popliteus (flexion) off LFC
5	**LCL** (some release first if tight in flexion + extension)	Lateral gastrocnemius + posterolateral capsule off femur

Tight	Release
Extension only	**Iliotibial band**
Flexion only	✓ **Popliteus** (anterior portion) → ↑ **lateral flexion space** (tibia max IR in flexion → popliteus tightest)
Flexion + extension	**LCL**
LCL + popliteus both released	Flexion gap unstable → **constrained prosthesis**

Balanced in extension, lateral tight in flexion → ✓ release anterior popliteus
✗ Wrong options: pie-crust ITB (extension) · PCL recession (medial flexion) · LCL (instability → constraint) · semimembranosus (medial extension)

Medial side (valgus) fill until MCL taut · poly **>15 mm** (OB: >14 mm) → **constrained** · CCK / CPS for longstanding deformity

Femoral rotation trap hypoplastic LFC + **posterior referencing → ✓ femoral component IR →** PF maltracking + asymmetric flexion gap · check against **epicondylar + AP (Whiteside) axes**
✗ Wrong options: angulation (IM guide controls) · exposure (easier in valgus) · flexion · tibial rotation

Severe valgus (≈45°) – IM femoral entry point (TO, FNB 2024) options more lateral / medial / superior / standard · answer not provided for this item · another TO explanation (FNB 2023): entry point in a valgus knee is usually ✓ **more medial** than standard (TO)

Valgus – TO lecture notes distal femoral cut **3° valgus (not 6°)** in a valgus knee · **valgus > 20° → constrained prosthesis** (TO)

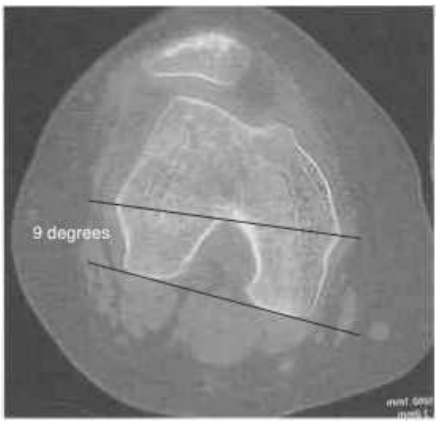
Varus release (TO notes) capsule off tibia → strip **sMCL → pes anserinus →** periosteum **4–5 cm** distally

Femoral component rotation

Posterior condylar axis normally **3° IR** relative to TEA / tibial joint line → cutting block has **3° ER** built in

Valgus with hypoplastic LFC

posterior condylar angle **>3°** (e.g. **9°**) → standard 3° guide leaves **6° IR** → ✓ **add 6° more ER**
✗ Wrong options: 0° guide → 9° IR · upsize · IR 6° vs TEA · rotate tibia



Posterior condylar angle vs TEA

Peroneal nerve palsy

Highest risk correction of **valgus + flexion contracture**

Risk factor	Mechanism
Valgus >12°	Stretch beyond tolerance
Flexion contracture >10°	Stretch → ischaemia
Lumbar laminectomy	**Double crush**
Epidural anaesthesia	↓ proprioception → unprotected compression
✗ Wrong option: **female gender**	**Not a risk factor**

Palsy in recovery remove dressing, **flex knee** · observe **3 months** · then NCS / exploration · explore at **posterior border of biceps femoris**

Severe valgus (>15°) + FFD → ✓ **prophylactic CPN release** – fascial septum between **peroneus longus + EDL** (anterior/lateral compartments, below knee)
✗ Wrong options: MCL pie-crust (varus) · less distal femur (resect more for FFD) · medial opening HTO (worsens valgus) · CR implant (needs more constraint)

Flexion contracture

Order	Posterior release
1	**Posterior femoral + tibial osteophytes**
2	**Posterior capsule** (stripped short distance from condyles)
3	**Additional distal femoral resection** (raises joint line → mid-flexion instability; try soft tissue first)
4	**Gastrocnemius origins** (medial + lateral)

Releases at 90° flexion popliteal artery falls posteriorly · do not remove **more tibia** for contracture → changes joint line → **patella alta**

Extra-articular deformity & constraint

>20° coronal deformity ✗ Wrong options: correctable with cuts + balancing → ✓ **extra-articular osteotomy** (staged or concomitant)

– – Active space – –

Femoral varus 28° after old distal femur fracture → ✓ distal femoral lateral closing-wedge osteotomy
✗ Wrong options: cuts + balancing · medial closing wedge (worsens varus) · HTO (wrong bone) · hinged TKA



Femoral extra-articular deformity correction

Kinematic alignment

Varus knee, kinematic → ✓ varus tibial cuts + valgus femoral cuts · equal aseptic loosening, satisfaction equal/better, ROM equal/greater, same reoperation
✗ Wrong options: ↑ loosening · ↓ satisfaction · ↓ ROM · ↑ reoperation

X vs Y

	Varus knee	**Valgus knee**
Tight side	**Medial**	**Lateral**
Bone deficiency	Posteromedial **tibia** (tibia-driven)	**Lateral femoral condyle** (femur-driven)
Key error	–	**Femoral IR** with posterior referencing
Nerve risk	Low	**Peroneal palsy**
Release for flexion tightness	Anterior sMCL, pes	**Popliteus**
Release for extension tightness	POL, posterior capsule, semimembranosus	**ITB**

How it's asked

"Peroneal palsy risk factors – EXCEPT?"	**→ female gender**
"Isolated popliteus release appropriate when?"	**→ valgus knee tight in flexion**
"Severe valgus – pathology encountered?"	**→ lateral femoral hypoplasia**
"Valgus knee, IM femoral guide, posterior referencing – watch for?"	**→ femoral component rotation**
"Popliteus release – effect?"	**→ increases lateral flexion space**
"Balanced in extension, lateral tight in flexion?"	**→ release anterior popliteus**

"Valgus + 15° FFD — consider?"	→ **release fascial septum between peroneus longus and EDL**
"3° ER guide off posterior condyles left IR femur — prevention?"	→ **add 6° more ER**
"28° femoral varus deformity?"	→ **distal femoral lateral closing-wedge osteotomy**
"Blount's varus TKA — true?"	→ **constrained implants may be needed**
"Kinematic alignment in varus knee?"	→ **varus tibial + valgus femoral cuts**
"POL primary function?"	→ **resists internal rotation**
"Soft-tissue balancing — true?"	→ **tight extension space: release posterior capsule then semimembranosus** (TO)
"Pie-crusting used for?"	→ **soft-tissue balancing in TKR** (TO)
"FFD 15°, medial tight in extension, flexion balanced — release?"	→ **POL** (TO)

– – Active space – –

TKA EXTENSOR MECHANISM RUPTURE

– Recon · 42 / 65 –

1 Basics

Definition periprosthetic **quadriceps tendon** rupture, **patellar fracture** or **patellar tendon** rupture · arguably the most devastating TKA complication

Epidemiology

Injury	Incidence
Overall EM failure	**0.1–2.5%** (↑ in revision)
Patellar tendon rupture	**0.17%** · ✓ **most common type** · ↑ **females**
Quadriceps rupture	**0.1%** (OB: ~1%)
Patellar fracture	**0.05%** unresurfaced · **0.2–21%** resurfaced · ↑ **men**

Age mean 65–75

Risk factors

Category	Factors
Multiply operated knee	**3–4 prior surgeries** (TKA, revision, realignment, HTO)
Systemic	**Renal disease**, **DM**, **RA**, PVD, obesity, chronic steroids
Drugs	**Testosterone replacement** · **fluoroquinolones**
Technical	Stiff knee + forced **patellar eversion** · **residual patella <12 mm** · **lateral release** (superior lateral geniculate a.) · **V-Y turndown** · **component malrotation** (IR femur/tibia), **lateralised button** · **joint-line elevation** (max acceptable **4 mm**)
Implant	**Single-peg** > tri-peg patella fracture risk · **thick anterior femoral flange**

Anatomy & blood supply

Quadriceps tendon rectus femoris superficial (continues over patella), VM + VL deep, VI deepest · **watershed 1–2 cm proximal to superior pole** = commonest rupture site

Patellar tendon inferior pole → tubercle · **4–6 mm thick, 5 cm long**

Supply **geniculate anastomotic ring** (SM, SL, IM, IL) · **inferior geniculates + recurrent anterior tibial a.** → patellar tendon · ring disruption → **patellar AVN → stress fracture**

PF forces walking **0.5× BW** · stairs **3.1× BW** · squatting **7× BW** · patella ↑ quad mechanical advantage **30–50%**

2 Mechanism

Iatrogenic	**tendon avulsion** from tubercle in forced flexion of stiff knee · excess lateral release · V-Y turndown · **MUA**
Traumatic	direct blow (patella fracture) · eccentric load on weak EM · fall · hyperflexion

3 Presentation

Key sign	✓ **inability to straight-leg raise / active extension** · passive ROM preserved · extensor lag · palpable defect · effusion · pop
Patellar height	**patella alta** = **patellar tendon** rupture · **patella baja** = **quadriceps** rupture

4 Diagnosis & Classification

Key radiology findings

Modality	Use
X-ray (AP, lateral, Merchant)	**Insall–Salvati >1.2 → alta (PT rupture)** · **<0.8 → baja (QT rupture)** · normal 0.8–1.2 (lateral, 30° flexion) · posterior tibial subluxation · avulsions · compare old films for loosening, joint line
CT	**Component rotation** · patellar fragmentation + bone stock · loosening
MRI (MARS)	✓ **Modality of choice** for extent of tendon tear (partial vs complete) · **early patellar AVN**
US	Bedside continuity, dynamic gap · no metal artefact, cheap

Genes & lab markers

Always exclude PJI before reconstruction	ESR, CRP, aspiration (cell count, Gram, culture) · alpha-defensin / leukocyte esterase · BMP, CBC, **HbA1c** for optimisation

Ortiguera & Berry (periprosthetic patellar fracture)

Type	EM	Component	Treatment
I	**Intact**	**Well fixed**	**Non-op** (good–excellent)
II	**Disrupted**	Any	**ORIF** (adequate bone) ± EM reconstruction
IIIa	Intact	**Loose**, bone **≥10 mm**	ORIF + **revise patellar component**
IIIb	Intact	**Loose**, bone **<10 mm**	**Patellectomy** or component removal (patelloplasty)

Criteria	EM integrity · patellar component fixation · bone stock

– – Active space – –

Differential

Condition	Distinguisher
PJI	↑ ESR/CRP, positive aspirate
Component loosening	Pain, **no extensor lag**
Patellar clunk	Painful clunk **30–60°** flexion → extension · no lag · arthroscopic debridement
Maltracking	Tilt/subluxation on Merchant · **J-sign** · extension intact
Arthrofibrosis	↓ passive **and** active ROM, no lag

5 Management

EM disruption after TKA

- Partial, active extension kept / lag <20° undisplaced → **immobilizer 6 wk**
- Acute (<2 wk) quad tendon, intact patella → **primary repair (± augment)**
- Complete / chronic (>6 wk) / patellar tendon → **reconstruction**
  (fresh-frozen allograft or synthetic mesh, tensioned in full extension)
- Multiple failed reconstructions ± infection → **knee arthrodesis**

Nonoperative management indications

Immobilizer 6 wk — **partial rupture with active extension** · **type I** patellar fracture · non-displaced fracture with intact EM · **lag <20°** · active PJI / unfit · low-demand elderly

Technique — extension splint/cylinder cast · **drop-lock brace** if no intention of healing (only for those **unfit for reconstruction**) · WBAT in brace · X-ray every 4–6 wk · ROM after 6–8 wk · ~**69%** of patellar fractures managed conservatively

Partial quadriceps rupture, weak active extension → immobilizer 4–6 wk → ✓ **minor lag with acceptable flexion**
✗ Wrong options: ↓ flexion · re-rupture · 20° lag · early failure

Appropriate non-op → ✓ **chronic displaced patella nonunion with only 5° lag**
✗ Wrong options: acute PT rupture + dehiscence (I&D, poly exchange, repair) · QT rupture with 45° lag · PT rupture 30° lag (reconstruct) · partial QT + coronal instability (address instability)

Primary repair

Indication — acute **<2 wk**, intact patella + component, good tissue · **quad tendon repair <2 wk may equal allograft**

Avoid — **chronic >6 wk** · poor bone stock · PJI · **patellar tendon** (largely abandoned)

Repair	Complication rate
Patellar tendon	**63%**
Quadriceps tendon	**25%**

Technique — #5 non-absorbable, anchors/tunnels · **allograft augmentation** advised · knee in **full extension** · cast 6–8 wk

Why fails approach + dissection destroy **inferior genicular** pedicles to the tendon

– – Active space – –

Reconstruction – allograft vs synthetic mesh

Standard of care for tendinous disruption – ✓ **reconstruction**
✗ Wrong options: primary repair (high re-rupture/lag)

Option	Key points
Achilles allograft (calcaneal block)	PT rupture with intact patella mobile within **3–4 cm** of joint line · chronic retracted QT · block in tibial trough
Whole EM allograft	**Deficient/absent patella**, failed prior reconstruction · needs **5 cm** tibial bone + **5 cm** quad tendon · allograft patella **not resurfaced**
Synthetic (Marlex polypropylene, LARS)	PT insufficiency with intact patella/component/QT · **immunocompromised** · tunnelled lateral to host tendon, distal end in tibial trough with screw/ cement, covered by host flaps
Medial gastrocnemius flap	EM rupture + **poor anterior soft-tissue cover** · ↑ vascularity
Semitendinosus autograft	Option

Allograft rules ✓ **fresh-frozen** (✗ Wrong options: freeze-dried (↑ failure)) · ✓ **maximal tension in full extension** (✗ Wrong options: minimal tension → lag)

Mesh vs allograft	Finding
Cost	✓ **Mesh lower**
Failure / revision / infection / function	✓ **Equivalent** (OB219493, OB217844 – 10–20° lag either)
Extensor lag + walking aid	**Mesh smaller lag + fewer walking aids** (OB219933) · equivalent (other OB) – read the options
Disease transmission / availability	Mesh: none / readily available
Wound complications	Not shown lower with mesh



Mesh reconstruction of the extensor mechanism

Chronic PT rupture, 60° lag, patella alta → ✓ bone-tendon allograft reconstruction
✗ Wrong options: hamstring-augmented repair · therapy · arthrodesis · orthotics

Chronic QT rupture 4 mo, active patient → ✓ EM reconstruction
✗ Wrong options: repair via tunnels/anchors · hinged revision (fails without EM)

– – Active space – –

QT rupture (baja), 30° lag, infection excluded → ✓ mesh reconstruction
✗ Wrong options: drop-lock + PT · lower joint line (for true baja) · Krackow repair · spacer

Component malrotation & revision

60° lag, failed PT repair, femur IR 9°, tibia IR 12° → ✓ revise femoral + tibial components + EM allograft tensioned tightly in full extension · CT for rotation pre-op
✗ Wrong options: PT · brace · poly exchange + hamstring · poly exchange + allograft

Postoperative protocol

After mesh/allograft reconstruction	✓ **long-leg cast in full extension 6–12 wk**, then gradual ROM in hinged brace · Orthobullets: cylinder cast **6–8 wk** ✗ Wrong options: splint in 15° flexion · brace 2–4 wk · 4 months (stiffness) · immobilizer 4 wk

ORIF of patella

Indication	type II with adequate bone · type IIIa (revise component) · screws or tension band · complications: AVN, **nonunion (high)**, wire migration to popliteal fossa, infection

Knee arthrodesis salvage

Indication	✓ **recurrent EM failure after multiple reconstructions + recurrent infection** · active PJI precluding reimplantation · massive bone loss (or AKA)
Technique	**long IM nail** preferred · ex-fix if infected/bone loss · position **0–15° flexion, 5–7° valgus**, neutral rotation
Not arthrodesis	failed DAIR → **2-stage** · loose distal femoral periprosthetic fracture in elderly → **DFR** · aseptic loosening + metaphyseal loss → **cones/sleeves** · flexion instability → **revision with posterior femoral augments**

6 Complications

Complication	Rate / note
Extensor lag >30°	**Most common, 45%**
Re-rupture	**25%** · ↑ repair alone, ↑ freeze-dried
Infection	**23%** after reconstruction → 2-stage
Stiffness	ROM **75–90°** after 6–8 wk immobilisation
Instability / walking aid	Common
Patellar AVN	Lateral release, V-Y

Prognosis functional success **50–68%** at 3–5 yr · **54%** 5-yr implant survival · good: acute <2 wk, fresh-frozen graft, intact component, good bone, no infection

X vs Y

	Patellar tendon rupture	Quadriceps tendon rupture
Incidence	**Commonest** (0.17%), ♀	0.1%
Patellar height	**Alta** (I-S >1.2)	**Baja** (I-S <0.8)
Primary repair	**63%** complications — abandoned	25% · OK if <2 wk
Reconstruction	Achilles / whole EM / mesh	Achilles / whole EM / mesh

How it's asked

"Pop at PT 3 mo, 60° lag, alta — treatment?"	**→ bone-tendon allograft reconstruction**
"Chronic QT rupture after fall, active male — best long-term?"	**→ extensor mechanism reconstruction**
"Mesh vs allograft — what is decreased?"	**→ cost**
"PT rupture after hyperflexion — true about treatment?"	**→ synthetic mesh equivalent to allograft**
"Baja, 30° lag, negative aspirate — treatment?"	**→ tendon reconstruction with interwoven mesh**
"Allograft vs synthetic — true?"	**→ failure rates equivalent**
"Mesh vs allograft, 40° lag with alta — true?"	**→ mesh has lower cost**
"Polypropylene graft — true?"	**→ less cost than allograft**
"Post-op protocol after mesh reconstruction?"	**→ long-leg cast in full extension 6–12 wk, then gradual ROM**
"Partial QT tear, immobilizer 6 wk — outcome?"	**→ minor extension lag with acceptable flexion**
"Which scenario suits non-op?"	**→ chronic displaced patella nonunion with 5° lag**
"60° lag, failed PT repair, IR components?"	**→ revise both components + EM allograft tensioned in full extension**
"Knee arthrodesis optimal in?"	**→ recurrent EM disruption with recurrent PJI**

– – Active space – –

TKA HETEROTOPIC OSSIFICATION

– Recon · 43 / 65 –

1 Basics

Definition bone in soft tissues (not periosteum) after surgery, with or without a precipitating event · distinct from **myositis ossificans** (post-traumatic muscle injury)

Incidence **1–42%** (Orthobullets) · **15–50%** (OB) – read the options · **<1%** clinically relevant · much **less** frequent / problematic than after THA

Risk factors

✓ Risk factor	Not a risk factor
Hypertrophic arthrosis (pre-op osteophytes, often post-traumatic)	✓ **Incorrect component sizing**
Male gender	**Valgus deformity**
Obesity	–
History of trauma	–
Periosteal stripping of anterior femur	–
Post-op **effusion / hematoma**	–
Post-op **forced manipulation**	–

3 Presentation

Usually asymptomatic · rarely: ↓ motion, pain, **quadriceps snapping**, **PF instability**

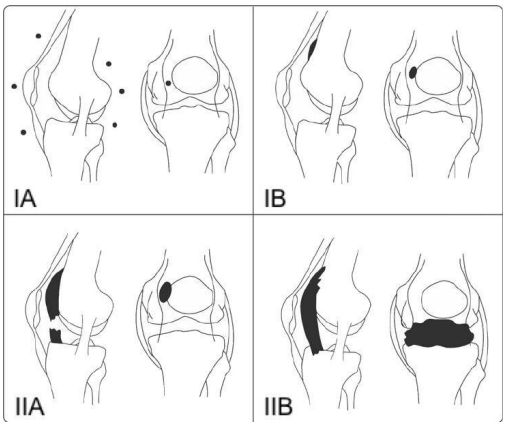
4 Diagnosis & Classification

Key radiology findings

AP + lateral ossification along **anterior distal femur** and in **quadriceps mechanism** · visible **3–4 wk** · matures **1–2 yr**

Furia & Pellegrini classification

Type	Location	Size
IA	Island in **suprapatellar** soft tissue	**≤5 cm**
IB	Island in suprapatellar soft tissue	**>5 cm**
IIA	**Contiguous with anterior distal femur**	**≤5 cm**
IIB	Contiguous with anterior distal femur	**>5 cm**



Furia & Pellegrini HO classification

5 Management

Observe	most cases (rarely affects outcome)
Excise	**quadriceps snapping** · **PF tracking difficulty** · **PF instability**
Prophylaxis	**radiation** + **NSAIDs** used in high-risk **THA** · no evidence in TKA

How it's asked

"HO risk factors – EXCEPT?"	→ **incorrect sizing of femoral/tibial components**
"Indication for HO excision after TKA?"	→ **quadriceps snapping / PF tracking problems / PF instability**

– – Active space – –

TKA IN PATELLA BAJA (PRIOR HTO)

– Recon · 44 / 65 –

1 Basics

Definition TKA in a knee with patella lowered relative to its normal position (patella infera), classically after **HTO**

Causes of acquired baja

Proximal tibial osteotomy ✓ **most frequent complication of both lateral closing-wedge and medial opening-wedge HTO** · patellar tendon **shortening** during osteotomy + post-op **scarring** · ↑ with post-op immobilisation · patellar height ↓ 89% (Insall–Salvati) / 76.3% (Blackburne–Peel) after HTO
✗ Wrong options: fracture · peroneal palsy · compartment syndrome · infection

Others **tibial tubercle slide/transfer** · proximal tibial trauma · **joint-line elevation at TKA** · **ACL reconstruction**

Prior procedure	Converting to TKA →
Medial opening-wedge HTO	✓ **Patella baja**
Lateral closing-wedge HTO	**Baja** · leg **shortening**
Lateral distal femoral (closing) osteotomy	**Patella alta**
Medial UKA / PF arthroplasty	not associated

Other HTO-to-TKA problems proximal tibial surface shifted relative to canal · retained hardware · old incisions · scar · **valgus** more frequent after HTO
✗ Wrong options: severe varus · collateral instability · alta

Pseudo-baja from joint line elevation

Mechanism elevated joint line with normal tendon length = "equivalent" of baja

Insall–Salvati 0.95 → 0.76 after TKA + lateral release → ✓ excessive distal femoral resection + lateral release
✗ Wrong options: notching · pre-op baja · tendon released off tubercle · excess tibial resection (lowers joint line)

3 Presentation

Symptoms ✓ **anterior knee pain** (commonest after lateral closing-wedge HTO) · **stiffness**
✗ Wrong options: lengthening · quad weakness (pre-existing) · infection (0–4%)
✗ Wrong options: laxity

Exam **mechanical block to flexion** (patella impinges on tibia) · ↓ patellar glides · pain with compression

Closing-wedge HTO, burning anterior pain at 9 mo, ↓ patellar glides → ✓ patella infera
✗ Wrong options: alta · recurrent medial overload · nonunion

4 Diagnosis & Classification

Key radiology findings

Views AP, lateral, **lateral at 30° flexion** · patella sits distal to trochlea

Index	Baja	Normal	Alta
Insall–Salvati (tendon ÷ patella length)	✓ **<0.8** (e.g. **0.7**)	**0.8–1.2** (~1:1)	>1.2
Caton–Deschamps	–	–	**1.4** → alta
Blackburne–Peel	–	**0.6** normal	–

Distractors TT-TG **22 mm** → instability · patellar translation **>20 mm** abnormal (↓ after HTO scarring)

INDEX	MEAN NORMAL	PATELLA ALTA	PATELLA BAJA
INSALL-SALVATI	1.0	>1.2	<0.8
MODIFIED INSALL-SALVATI	1.25	>2.0	
BLACKBURN-PEEL	0.6	>1.0	<0.5
CATON-DESCHAMPS	1	>1.3	<0.6

Patellar height measurements

5 Management

Non-op activity modification, PT · mild symptoms, younger

TKA severe impingement, older with OA

Techniques for baja at TKA

Severity	Technique
Mild	**Patellar button placed superiorly** (smaller dome) + trim inferior patellar bone
Moderate	✓ **Lower joint line** – **distal femoral augments** + **lower (more) proximal tibial cut** · may need revision system
Moderate	**Tibial tubercle transfer cephalad** – difficult near cemented tray · unpredictable healing, extensor lag
Severe	**Patellectomy** (partial to **10–12 mm**) + **cruciate-substituting** design

Avoid anything that **raises joint line** (thicker poly, under-resecting tibia) · **inferior** button · **caudad** tubercle transfer
✗ Wrong options: downsizing femur (no effect)

HTO + baja – intra-op technique → ✓ add distal femur augments

– – Active space – –

X vs Y

	True baja	Pseudo-baja
Cause	Short / scarred **patellar tendon** (HTO, tubercle surgery, ACLR)	**Elevated joint line** (excess distal femoral cut + thick poly)
Insall–Salvati	**<0.8**	Normal tendon ratio, patella low relative to joint line
Fix	Lower joint line, superior button, tubercle transfer, patellectomy	Restore joint line

How it's asked

"HTO 10 yr ago, TKA planned — likely complicating factor?"	**→ patella baja**
"Baja commonly encountered converting which procedure?"	**→ medial proximal tibial opening-wedge osteotomy**
"Most common complaint after lateral closing-wedge HTO?"	**→ anterior knee pain**
"HTO 5 yr ago, converting to TKA — patellar problem?"	**→ patella baja**
"Most frequent complication of closing and opening HTO?"	**→ patella baja**
"Closing-wedge HTO, anterior burning pain at 9 mo?"	**→ patella infera (baja)**
"Opening HTO 15 yr ago — consistent finding?"	**→ Insall–Salvati 0.7**
"I-S 0.95 → 0.76 after TKA + lateral release?"	**→ excessive distal femoral resection + lateral release**
"HTO with baja — intra-op technique?"	**→ add distal femur augments**

– – Active space – –

TKA INSTABILITY

– Recon · 45 / 65 –

1 Basics

Epidemiology common cause of **early** TKA failure · **10–20%** of revisions · causes: ligament dysfunction, component malposition, poly wear

Types **extension (varus-valgus)** · **flexion (AP)** · **mid-flexion** · **genu recurvatum** · **global, multiply operated**

Valgus TKA instability (TO)

Most common complication of valgus TKR

✓ **tibiofemoral instability** (TO)
✗ Wrong options: peroneal palsy · maltracking · recurrent valgus (second)

Valgus TKR complication (TO)	Rate
Tibiofemoral instability	**2–70%**
Recurrent valgus	**4–38%**
Stiffness needing MUA	1–20%
Wound problems	4–13%
Patellar stress fracture / ON	1–12%
Patellar tracking	2–10%
Peroneal palsy	**1–4%**

3 Presentation

History prior operations · original implant · connective tissue / inflammatory disease · DM, **Charcot** · trauma

Exam **varus/valgus thrust** in gait · laxity through ROM · ✓ **flexion instability test = posterior sag at 90°** · strength · extensor competence · patellar tracking

Flexion instability symptoms instability **without frank giving way**, worst **descending stairs** · **recurrent (often bloody) sterile effusions** · diffuse **peripatellar + pes anserinus** tenderness (anterior soft-tissue strain) · quad weakness

– – Active space – –

4 Diagnosis & Classification

Key radiology findings

View	Assesses
Weight-bearing AP	Joint-line symmetry
Full-length AP	Mechanical alignment
Lateral	**Tibial slope**, **tibial subluxation**, recurvatum
CT	**Component rotation**

Instability	Radiographic cause
Extension	**Excess distal femoral resection** · oversized femoral component
Flexion	**Over-resected posterior condyles** · **undersized femur** · **↑ tibial slope**
Mid-flexion	**Anterior or proximal femoral component** (joint-line change)

Genes & lab markers

Rule out infection CBC, ESR, CRP + **aspiration** (cell count, culture) · bloody aspirate with negative markers fits flexion instability

5 Management

Extension (varus-valgus) instability

Type	Cause	Treatment
Symmetric	**Excess distal femoral resection** → flexion/extension mismatch	**Distal femoral augments** · **upsizing poly fails** (tightens both gaps)
Asymmetric (commoner)	Uncorrected coronal deformity / ligament imbalance	Release contracted side · ligament insufficient → **VVC (varus-valgus constrained)**

MCL injury & constraint

MCL injury causes medial retractors · subperiosteal elevation (tibial sleeve avulsion) · **saw** during tibial / posterior condylar cut

Intra-op midsubstance MCL transection → ✓ suture repair + CR or PS implant + hinged brace 6 wk · alternative **unlinked constrained (CCK)** ± repair
✗ Wrong options: CR/PS without repair · PS + immobilizer · hinge (not for simple collateral rupture)

Hinge indications moderate–severe instability · **absent collateral(s)** · severe bone loss · large deformities

Charcot knee (diabetic) excellent ROM, **ligamentous laxity** + bone loss → ✓ **semiconstrained prosthesis prevents instability** · augments for bone loss
✗ Wrong options: stiffness (not a problem)



Unlinked constrained (CCK) implant

Flexion instability & gap mismatch

Definition	**flexion gap > extension gap** · PS knee can **"jump the post"** → dislocation
Femoral cam jump (TO)	**specific to PS knees** · ✓ **loose flexion gap** · varus/valgus stress in flexion opens gap → cam rotates in front of post · **popliteus over-release** may cause · closed reduction: **90° flexion + anterior drawer** (clunk) · definitive: **revision** (TO) ✗ Wrong options: tight flexion gap

Loose flexion gap, normal extension gap (PS) → ✓ **posterior knee dislocation** (cam locked behind/in front of post)
✗ Wrong options: MUA · hyperextension (loose extension gap) · anterior dislocation · patella fracture (<13 mm patella)

Recurrent dislocation of PS mobile-bearing revision → ✓ **loose flexion gap** (undersized / anteriorised femur, excess distal augmentation, collateral + posterior capsule insufficiency)
✗ Wrong options: loose extension gap (V/V instability) · malrotation (patellar/rotational) · design

Popliteus injury	popliteus = **flexion stabiliser** · cut while resecting **posterolateral femoral condyle** → flexion laxity → PS dislocation ✗ Wrong options: too little slope · posterior femoral translation · over-recessed PCL (PCL usually removed in PS) · joint line



PS knee – cam jumped post

– – Active space – –

Femoral cuts & flexion gap

Cut	Effect
Anterior	Does not affect flexion gap · under-resect → oversize → **PF stuffing** · over-resect → **notching** · good cut = "grand piano"
Posterior condylar	✓ **Sets flexion gap** · over-resection → **flexion instability** (line 2)
Anterior / posterior chamfers	Fit only
Distal femoral	**Sets extension gap** · = component thickness · extra for flexion contracture
Tibial slope	↑ slope → **↑ flexion gap** (line 5)

Flexion instability in PS knee — **anterior tibial translation at 90°** pathognomonic · no posterior subluxation (cam-post)

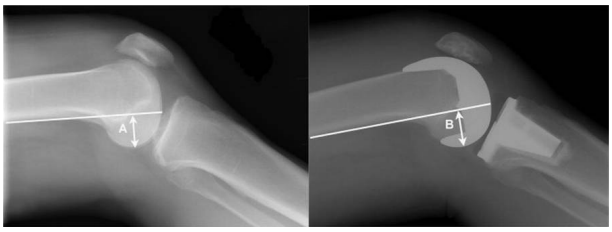
Flexion instability causes (OB) — ↑ posterior femoral resection · **under-resected distal femur** (joint line distalised) · **excess tibial slope** · PCL incompetence (CR)

Cause	Fix
Over-resected posterior condyles	**Posterior femoral augments**
Undersized femur	**Upsize femoral component**
Excess tibial slope	**↓ slope** + consider **PS**
PCL insufficiency after CR	**Convert to PS**

Flexion instability revision (to PS) → ✓ additional distal femoral resection + posterior femoral augments + upsize poly
✗ Wrong options: downsize femur + distal augments + thinner poly · ER tibial tray + thicker poly · isolated thicker poly (over-tightens extension) · IR tray + ↑ slope

Non-op — quad strengthening, bracing · revision for flexion instability shows **least improvement** of any revision indication

CR knee, instability on stairs, effusions, posterior tibial subluxation → ✓ PCL insufficiency
✗ Wrong options: patellar tendon rupture (no active extension) · joint line · ACL (sacrificed) · loosening



Femoral cutting block – posterior condylar cut

Mid-flexion instability

Features — poorly understood · **joint-line modification** (elevation) · instability/ malrotation between **45–90°** · contributors: sagittal femoral design, **anterior MCL attenuation**, tibiofemoral geometry

Joint-line elevation threshold (TO, FNB 2024) — options >2 / >4 / >6 / >8 mm · **answer not provided by the source** · Orthobullets: elevation **>8 mm** → motion problems incl. mid-flexion instability (TKA axial alignment)

Treatment — usually **full revision** · restore joint line · equalise gaps

Genu recurvatum

Associations	**poliomyelitis** (quad weakness → walks locked in hyperextension, ankle equinus) · **RA** · **Charcot** · fixed valgus + ITB contracture
Treatment	**long-stemmed PS or VVC** · **rotating hinge = salvage** (residual hyperextension → early failure)

Global, multiply-operated instability

Features	flexion + extension + V/V laxity · bone loss · recurvatum gait
Treatment	**VVC minimum** · usually **hinge** ± augments, sleeves, cones · **endoprosthesis** if severe bone loss

Component malrotation

Valgus knee, femur IR 8°, tibia IR 4°, dislocated patella + coronal instability → ✓ revise femoral + tibial components into ER
✗ Wrong options: PT · brace · thicker liner · lateral release

X vs Y

	Flexion instability	Extension (V/V) instability	Mid-flexion
Gap	Flexion > extension	Extension loose	45–90°
Cause	Posterior condyle over-resection, small femur, ↑ slope, PCL loss	Excess distal cut (symmetric) · imbalance (asymmetric)	Joint-line elevation
Fix	Posterior augments, upsize femur, ↓ slope, PS	Distal augments · release · VVC	Revision, restore joint line
Thicker poly alone	No (over-tightens extension)	No (symmetric)	–

How it's asked

"Femoral cam jump – EXCEPT?"	**→ associated with tight flexion gap** (TO)
"CR TKA, instability on stairs, posterior tibial subluxation?"	**→ PCL insufficiency**
"PS TKA, anterior translation at 90° – over-resection at?"	**→ posterior femoral condylar cut (line 2)**
"PS mobile-bearing revision, recurrent dislocation?"	**→ loose flexion gap**
"Loose flexion gap, normal extension gap (PS) – risk?"	**→ posterior knee dislocation**
"Medial + lateral laxity at 90°, bloody negative aspirate – which cut?"	**→ excess posterior tibial slope (line 5)**
"Flexion instability revision to PS – additional strategy?"	**→ more distal femoral resection + posterior augments + thicker poly**
"Intra-op MCL transection, coronal instability?"	**→ suture repair + PS implant + hinged brace**

– – Active space – –

"Charcot knee – semiconstrained prosthesis prevents?"	→ **instability**
"PS knee dislocation on lateral X-ray – cause?"	→ **popliteus injury cutting posterolateral condyle**
"IR femur 8°, tibia 4°, dislocated patella?"	→ **revise femoral and tibial components**
"Most common complication of valgus TKR?"	→ **tibiofemoral instability** (TO)

– – Active space – –

TKA KINEMATIC ALIGNMENT

– Recon · 46 / 65 –

1 Basics

Definition — TKA that restores the patient's **pre-arthritic limb alignment + joint lines** by **resurfacing** distal femur and proximal tibia · ✓ **"ligament-sparing"** – bone cuts, not releases, recreate native laxity

Origin (TO) — **Howell and Hull (2008)**, "custom-fit positioning" · driven by **~20%** dissatisfaction after mechanical alignment (TO)

Individualised approaches (TO) — ✓ **kinematic** + **functional** · fixed approaches = **mechanical** + **anatomical**
✗ Wrong options: anatomical as individualised

Anatomic rationale

HKA axis (varus / neutral / valgus) and **joint-line obliquity** (varus / neutral / valgus) are **independent → 9 combinations**

Only ~15% of knees have 180° HKA with joint line perpendicular → MA alters **~85%** of patients

Native alignment (TO / OB)

Measure	Value
Distal femur	**0–4° valgus** (to mechanical axis) · **~9° valgus** to vertical
Femoral anatomic vs mechanical axis	**4–6°** valgus (OB) · **6°** (TO)
Proximal tibia	**1–5° varus** · ~3° varus to midline
Overall limb	Slight **varus ~3°**
Joint line	**~3° varus** (flexion + extension)
Tibial slope	**1–9°** posterior

Foundational principles

Arthritis — minimal **femoral** bone wear · minimal ligament contracture/attenuation · PCL usually intact · varus/valgus deformity = **cartilage loss + tibial wear**

Balance — laxity least in **extension** · in flexion **lateral more lax than medial** · avoid joint-line elevation/depression (baja/alta)

The three kinematic axes of the femur

Axis	Description
Flexion–extension (cylindrical)	Transverse axis across distal femur about which tibia flexes · **oriented differently from TEA** (Orthobullets) · **same as TEA** (TO) – read the options
Internal–external rotation	Longitudinal axis in **medial compartment** (**medial pivot**) · tibial IR in flexion → lateral condyle **roll-back**
Patellar tracking	Isometric axis across **medial condyle** · analogous to **Schöttle point** (MPFL)

– – Active space – –

All three parallel or perpendicular to the native femoral joint line → resurfacing femur restores them

4 Diagnosis & Classification

CPAK classification & constitutional alignment (TO)

Constitutional alignment
✓ **arithmetic HKA (aHKA)** (TO)
✗ Wrong options: mechanical HKA (changed by arthritis) · LDFA · MPTA alone

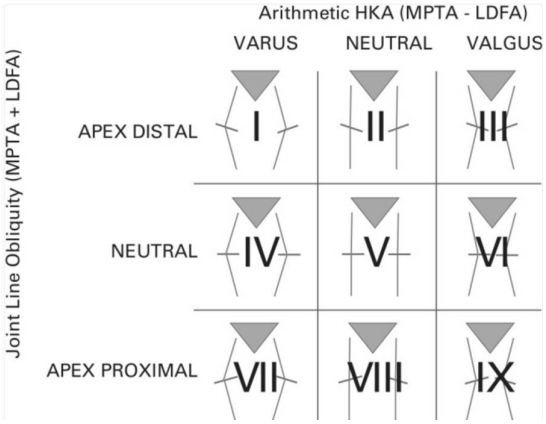
Formula (TO)	Meaning
aHKA = MPTA – LDFA	**Negative = varus**, positive = valgus · unaffected by joint-space narrowing/subluxation · ignores JLCA
JLO = MPTA + LDFA	**180° = neutral · >180° apex proximal · <180° apex distal**
mHKA	Angle between femoral + tibial mechanical axes
mLDFA	Lateral angle, femoral mechanical axis vs distal femoral joint line
mMPTA	Medial angle, tibial mechanical axis vs proximal tibial joint line

Boundaries (TO)	Neutral	Varus / apex distal	Valgus / apex proximal
aHKA	**0 ± 2°**	**< –2°**	**> +2°**
JLO	**180 ± 3°**	**<177°** (apex distal)	**>183°** (apex proximal)

CPAK (TO)	aHKA	JLO	Note
I	Varus	Apex distal	KA better from outset
II	**Neutral**	**Apex distal**	✓ **Commonest (~40%) → anatomic alignment**
III	Valgus	Apex distal	Constitutional valgus
IV	Varus	Neutral	KA better from outset
V	**Neutral**	**Neutral**	✓ **Best balanced by mechanical alignment** · only **~15%**
VI	Valgus	Neutral	Constitutional valgus
VII–IX	Varus / neutral / valgus	Apex proximal	–

CPAK I / IV under MA need varus recuts or extensive **MCL release** (TO)

CPAK III / VI lateral femoral + tibial deficiency, femoral/tibial ER deformity, lateral contracture, later MCL attenuation (TO)



CPAK matrix

5 Management

Surgical technique

Femoral target ✓ **femoral component condyle thickness − 1 mm (saw kerf) − 2 mm (cartilage loss)**

Tibial balancing after femoral resurfacing, adjust tibial **varus/valgus + slope** · no soft-tissue releases · no PCL sacrifice/recession · TO: only **tibial cut** used to balance (plus osteophytes, posterior capsule off femur, tibial ML shift)

Gap	Shape
Extension	**Rectangular** (equal, negligible V/V laxity)
Flexion	**Trapezoidal** (lateral progressively looser)

Patella resect component thickness − 1 mm kerf − **4 mm** cartilage · leave **≥13 mm** bone

Distal femoral cut angle

LDFA 81°, anatomic–mechanical difference 6° → femur in **9° valgus** at joint line → ✓ **9° valgus** to cut parallel to native joint line (KA) · MA would use **6°** · KA femoral component **2–4° more valgus**, tibia more **varus**
✗ Wrong options: 0°, 3°, 6° valgus, 3° varus

– – Active space – –

Component positioning goals

Target	Mechanical alignment	Kinematic alignment
Femoral flexion	Similar	Similar
Distal femoral cut	⊥ femoral mechanical axis	∥ distal articular surface (after wear correction)
Posterior femoral cut	**3° ER** relative to posterior condyles	**Neutral (0°)** relative to posterior condyles
Tibial cut	⊥ tibial mechanical axis	∥ articular surface (after wear correction)
Tibial slope	**2–7°** posterior	∥ **medial plateau**
Tibial axial rotation	**Aligned with tibial tubercle**	∥ **long axis of lateral plateau**
Balancing	Ligament releases	Bone cuts (tibia)

NOT a KA goal (TO) ✓ **tibial rotation aligned with tubercle** (that is MA)

Methods of performing KA

Method	Pros	Cons
Caliper-verified	Any instruments/implant · cuts within **±0.5 mm** of target	Sagittal femoral flexion hard with IM rod · **tibial recuts** common · sizing hard pre-op
Patient-specific guides	Detailed planning · few trays	Recuts still needed · advanced imaging · closed vendor
Navigation	Planning + intra-op adjust, avoid recuts	Imaging · closed vendor · cost
Robotics	Planning + intra-op adjust	Imaging · closed vendor · equipment burden · cost

KA versus MA

Varus knees varus tibial cuts / residual varus ✓ **not associated with bony or hardware failure**

Valgus knees (KA view) **no hypoplastic LFC** · natural valgus joint line with comparable condyles · lateral cartilage + tibial wear (contrast: conventional teaching = LFC hypoplasia)

Rotation **ER of femur/tibia not needed** for tracking · many native trochleae are **IR relative to TEA** · tracking from correct HKA, JLO, joint-line height + tubercle IR in flexion

Implant **medial-pivot** design needed for normal kinematics · concern about shear on **PS cam-post**

Outcomes (OB) loosening, reoperation equal · satisfaction + ROM equal or better

Controversies (TO) needs PSI/technology · ACL + menisci sacrificed · non-weight-bearing kinematics differ · no limits on outliers → restrict to mild deformity · few long-term data

Related techniques

Technique	Principle
Restricted KA	Limits on component + HKA varus/valgus · safe zone (TO): **LDFA + MPTA 86–93°**, final **mHKA −5° varus to +4° valgus** · minor releases rather than exceed limits
Functional alignment	Adjust femoral + tibial cuts in all 3 planes for symmetric/asymmetric gaps · no releases · e.g. tibia up to **3° varus** for tight medial extension · **femoral IR** for tight lateral flexion · limb within **0 ± 3°**
Inverse KA	Predetermined tibial cut → femur gap-balanced · no releases
Anatomical (Hungerford 1982)	Tibia **3° varus** + femur extra 3° valgus (**7–10°** total) · for **CR** · risks asymmetric poly wear, LCL/popliteus damage (TO)
Mechanical (Freeman & Swanson 1973)	Both components ⊥ mechanical axis · releases balance gaps · varus knee: more lateral tibia + more distal/posterior MFC removed → MCL tight · **15–20%** remain symptomatic (TO)

X vs Y

	Mechanical	Kinematic
Target	Neutral 0° HKA, horizontal joint line	Patient's pre-arthritic HKA + JLO
Balancing	**Soft-tissue release**	**Bone cuts** (tibia)
Femoral rotation	3° ER to posterior condyles / TEA	0° to posterior condyles
Best CPAK	**V**	I, III, IV, VI
Implant	Any	Medial pivot preferred

How it's asked

"LDFA 81°, cut parallel to native joint line – guide setting?"	→ **9° of valgus**
"CPAK type best balanced by MA?"	→ **CPAK V** (TO)
"Constitutional alignment best defined by?"	→ **arithmetic HKA** (TO)
"Commonest CPAK type?"	→ **neutral aHKA + apex distal JLO (type II)** (TO)
"Individualised approach to alignment?"	→ **kinematic alignment** (TO)
"NOT a KA positioning goal?"	→ **tibial rotation aligned with tibial tubercle** (TO)
"Kinematic vs mechanical in varus knee – result?"	→ **varus tibial + valgus femoral cuts**

– – Active space – –

TKA METAL HYPERSENSITIVITY

– Recon · 47 / 65 –

1 Basics

Definition allergic reaction to metallic TKA components → **persistent pain + stiffness · rare**

Type IV hypersensitivity

Mechanism ✓ **Type IV delayed-type, T-cell (cell-mediated)** hypersensitivity

Allergen **nickel** (in **cobalt-chromium** alloys) · metal ions/wear lodge in tissue, **bind proteins** → initiate reaction · may initiate **early loosening**

Gell–Coombs	Mediator	Example
I	IgE	Anaphylaxis, bee sting
II	IgG (antibody, cytotoxic)	Haemolytic anaemia, Goodpasture, Graves
III	Immune complex	SLE, serum sickness, PAN
✓ **IV**	**T cells, delayed**	**Metal allergy**
V	IgG stimulatory	–

3 Presentation

History dermatitis, eczema, atopy · **intolerance to cheap jewellery**

Symptoms **vague persistent pain** · **persistent effusion / swelling** · **stiffness** · document ROM

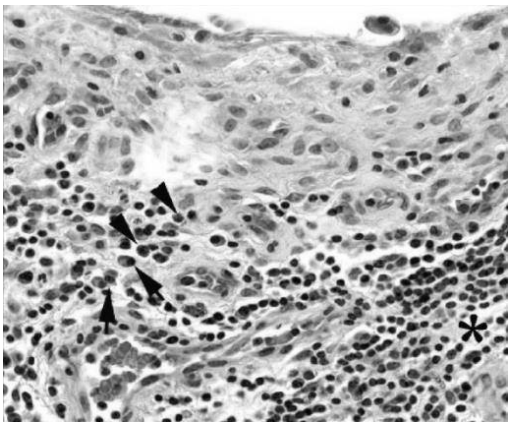
4 Diagnosis & Classification

Genes & lab markers

Test	Note
Patch test	Limited evidence of direct link
Lymphocyte transformation test (LTT)	–
T-lymphocyte-rich immunohistopathology	Synovium
Infection work-up	✓ **Must be negative** (alpha-defensin, synovial fluid)

Diagnostic criteria (controversial) **positive patch test + positive immunohistology + relief after exchange** (strict) · or **diagnosis of exclusion** after infection + aseptic loosening ruled out

– – Active space – –



Metal hypersensitivity synovial histology

5 Management

Implant exchange — persistent symptoms affecting quality of life · **hypoallergenic femoral component** (e.g. nickel-free) + **all-polyethylene tibia** if possible · chronic inflammatory **synovitis** at surgery

How it's asked

"Jewellery intolerance — Gell–Coombs type of metal hypersensitivity?"	→ **IV (delayed type)**
"Painful TKA, negative infection work-up, jewellery reaction — pathophysiology?"	→ **type IV delayed T-cell mediated**

– – Active space – –

TKA OTHER COMPLICATIONS

1 Basics

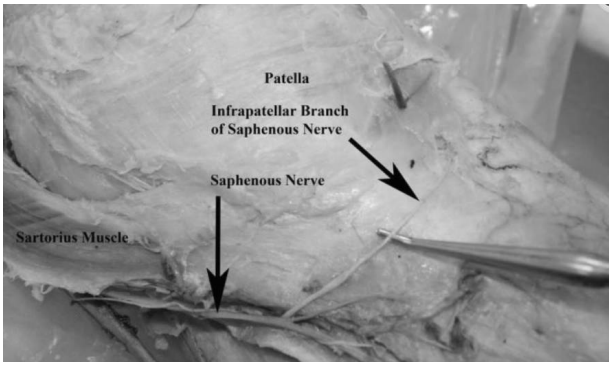
Covered elsewhere	neurovascular (vascular injury, peroneal nerve) · extensor mechanism (patellar loosening, clunk, maltracking, EM rupture) · PJI · periprosthetic fracture · metal hypersensitivity · wounds · tibiofemoral instability · stiffness

Popliteus snapping

Presentation	**posterolateral** pain, popping, **palpable snapping at posterolateral joint line** · no effusion, normal gait · ~6 mo after TKA
Cause	✓ **popliteal snapping syndrome** · tendon subluxes over a **retained lateral femoral condyle osteophyte**
Traps	✗ Wrong options: **patellar clunk** (PS knees, patella jumps flexion → extension, anterior) · tight PCL · retained PMMA · tight lateral retinaculum

Infrapatellar saphenous neuroma (TO)

Nerve	✓ **infrapatellar branch of saphenous nerve (IPBSN)** (TO) ✗ Wrong options: superomedial, inferomedial, inferolateral genicular
Anatomy (TO)	saphenous n. exits **subsartorial canal → infrapatellar branch** (fat pad + anteromedial knee) + **descending branch** (anteromedial leg/foot to 1st MT)
Causes (TO)	**midline/parapatellar TKA approach** · more often ACL reconstruction, arthroscopy, tibial nailing
Features (TO)	sharp **medial** knee pain · ↓ ROM · medial tenderness · **Tinel +**
Treatment (TO)	conservative → **neuroma excision ± nerve graft**



Infrapatellar branch of saphenous nerve

X vs Y

	Popliteus snapping	Patellar clunk
Site	**Posterolateral** joint line	**Anterior** (suprapatellar nodule)
Cause	**Retained LFC osteophyte**	Fibrous nodule at quad tendon–patella junction
Implant	Any	**PS** knees

How it's asked

"Posterolateral snapping 6 mo after TKA, no effusion?"	→ **popliteal snapping syndrome**
"Nerve causing painful neuroma after TKR?"	→ **infrapatellar branch of saphenous nerve** (TO)

– – Active space – –

TKA PATELLAR MALTRACKING

– Recon · 49 / 65 –

1 Basics

Epidemiology common cause of dissatisfaction · ✓ **most common reason for secondary surgery after TKA** · historically main cause of revision

Principle anything that **↑ Q angle** → lateral pull → subluxation/dislocation

Causes of lateral maltracking (TO)

✓ Causes (↑ Q angle)	Not a cause
Femoral component IR (missed **hypoplastic LFC** in valgus knee)	✓ **Medial placement / translation of patellar button** (TO) – this is the correct position
Medial shift of femoral component (moves trochlea medially)	Excess femoral ER (helps tracking; may loosen medial flexion gap)
Tibial component IR	Joint-line elevation (→ mid-flexion instability)
Lateral placement / centralisation of patellar button	Femoral component in extension (→ notching)
Flexed femoral component	KA vs MA per se (debated)
Residual valgus	–
Asymmetric patellar resection / thickness	–
Pre-op: excess femoral anteversion, tibial IR · design · extensor imbalance · anterior femoral component · patellar height	–

Rules for tracking

Femur **lateralise** · avoid IR · ER helps

Tibia **lateralise** · avoid IR · centre of tray on **medial ⅓ of tibial tubercle** · slight **ER** → net medialisation of tubercle

Patella ✓ **medialise button** (replicates median eminence) – the only implant placed medially · total thickness ~**21–22 mm** native/composite

3 Presentation

Symptoms subluxation sensation · frank dislocation · **peripatellar / anterolateral pain** · **limited flexion** · stiffness

Sign ✓ **J-sign** (subluxed in extension, reduces as it engages trochlea in flexion)

4 Diagnosis & Classification

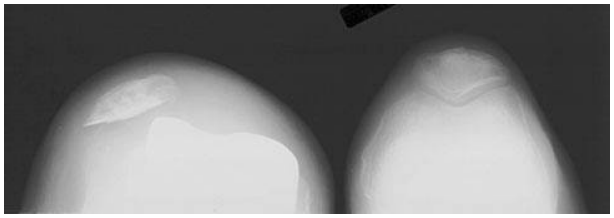
– – Active space – –

Key radiology findings

View	Assesses
AP	Femoral component placement, Q angle
Lateral	Femoral size + rotation
Tangential (Merchant)	**Patellar subluxation**, button placement, resection angle, **tilt**
CT	✓ **Best for rotational malalignment**

Findings lateral subluxation · ↑ Q angle · anterior femoral component · asymmetric resection · lateral button · lateral patellar osteophyte

Most important test for multifactorial subluxation → ✓ axial CT from supracondylar femur to below tibial tubercle
✗ Wrong options: long-leg AP · fluoroscopy · bone scan · MRI (artefact)



Merchant view – subluxed patellar component

5 Management

Rule address the **cause** · component malalignment first

Component internal rotation – CT evaluation & revision

Subluxed patella on Merchant, femoral IR → ✓ revision with ER of femoral component + lateral release (± tibial revision if IR, soft-tissue realignment)
✗ Wrong options: brace · tantalum patella lateralised · more tibial IR · femoral IR

CR TKA, anterolateral pain, J-sign, normal X-ray + labs, CT → revision → ✓ internal rotation of components
✗ Wrong options: joint-line elevation · infection · osteolysis · clunk (PS knees)

Tray centred medial to medial ⅓ of tubercle → ✓ patellar maltracking (IR tray)
✗ Wrong options: flexion instability (gaps balanced) · osteolysis (HXLPE) · fracture (22 mm thick) · clunk (PS)

Intra-op lateral subluxation despite lateral release

✓ **Slight ER of tibial component** (medialises tubercle)
✗ Wrong options: femoral IR · anterior tibial translation · fixed vs mobile bearing (no difference) · thicker button (overstuffs)

Hypoplastic lateral condyle

Valgus knee, patella tracks laterally on trials → ✓ failure to recognise hypoplastic LFC → femoral IR with posterior referencing
✗ Wrong options: KA · raised joint line · excess ER · femur in extension

– – Active space – –

Patellofemoral arthroplasty failure

Most common early failure	✓ **patellar instability / maltracking** (malalignment, soft tissue, malposition) · **late** failure = **tibiofemoral OA progression (~25%)** · loosening **<1%** cemented, higher cementless

X vs Y

Error	Effect on Q angle / tracking
Femur IR / femur medialised	↑ Q → lateral maltracking
Tibia IR	↑ Q → lateral maltracking
Button lateral / central	↑ Q
Button medial	✓ ↓ Q (correct)
Femur ER / lateralised	✓ Better tracking

How it's asked

"Subluxed patellar component 3 mo post-op — best course?"	→ **revision with lateral release + femoral ER**
"Intra-op lateral subluxation despite lateral release?"	→ **slight ER of tibial component**
"Multifactorial patellar subluxation — most important test?"	→ **axial CT supracondylar femur to below tibial tubercle**
"CR TKA, J-sign, CT → revision — cause?"	→ **internal rotation of components**
"Tray medial to medial ⅓ of tubercle — complication?"	→ **patellar maltracking**
"Valgus knee, patella tracks laterally on trials — error?"	→ **failure to recognise hypoplastic LFC**
"Most common early failure of PF arthroplasty?"	→ **patellar instability/maltracking**
"Lateral subluxation after TKR — cannot be the reason?"	→ **medial placement of patellar component** (TO)
"Lateral maltracking caused by all EXCEPT?"	→ **medial translation of patellar component** (TO)

TKA PATELLAR PROSTHESIS LOOSENING

– Recon · 50 / 65 –

1 Basics

Incidence **0.6–5%** (OB: 0.6–4.8%) · ↑ with **metal-backed** patellae

Mechanism **maltracking** or **PF overstuffing** → high shear (↑ PFJRF at high flexion) · **subclinical infection**

Risk factors **metal backing** · obesity · **lateral release** · **joint-line elevation** · **flexion >100°** · poor bone stock · **asymmetric resection** · **small pegs** (inadequate fixation) · **secondary osteonecrosis** + osteolysis

Patellar blood supply & AVN

Supply **anastomotic ring** · ✓ **medial-sided vessels contribute more** than lateral

Step	Vascularity disrupted
Medial arthrotomy	**0–35%**
Fat pad excision	**0–16%** (anterior tibial recurrent a. + inferior ring, not medial/lateral vessels)
Lateral release	**53–75%** (superior + inferior lateral geniculates)
✓ **Medial arthrotomy + lateral release**	**50–100%** (OB option) · **up to 75%** (Orthobullets/OB text)
Lateral meniscectomy	Does not disrupt ring

Patellar ON incidence up to **33% at 1 yr**, **10% at 10 yr** · direct link unproven

Protect leave **≥15 mm medial cuff** · lateral release **distal to superolateral border**, **proximal to lateral joint line**, **≥15 mm lateral** to patella · care converting **lateral UKA → TKA** (prior lateral arthrotomy)

3 Presentation

Symptoms persistent knee pain · swelling · pain with flexion · **no extensor lag** (vs EM rupture) · may see **subcutaneous component** through capsular defect

4 Diagnosis & Classification

Key radiology findings

View	Finding
Lateral	**Loss of PF radiolucent space** – patellar bone touching femoral component · dissociated component
Tangential	Dissociated component superficial to patella · **thinning of polyethylene**

– – Active space – –

Aspirate 700 WBC, 55% PMN, no lag, component migrated subcutaneously → ✓ implant loosening
✗ Wrong options: infection · EM rupture (no lag) · fracture (would give haemarthrosis) · overstuffing (cause not diagnosis)



Loose patellar component migrated subcutaneously

5 Management

Revise patellar component · if revision fails → **resect component** (patelloplasty)

How it's asked

"10 yr TKA, swelling, no lag, low WBC, component migrated — diagnosis?"	**→ implant (patellar component) loosening**
"Patellar blood supply — accurate statement?"	**→ medial arthrotomy + lateral release can disrupt 50–100%**
"Which side contributes more to patellar supply?"	**→ medial**

TKA PATELLOFEMORAL ALIGNMENT

– Recon · 51 / 65 –

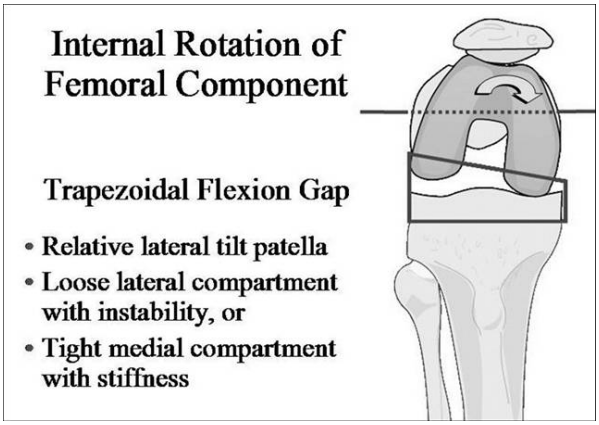
1 Basics

Key fact	✓ **abnormal patellar tracking = most common complication of TKA** · PF complications (subluxation, clunk, wear/loosening, fracture) occur in ~**30%**
Most important variable	**preserve normal Q angle** (ASIS → patella centre vs patella centre → tibial tuberosity)
↑ Q angle	↑ lateral subluxation force → pain, mechanical symptoms, wear, dislocation · ↑ need for **lateral release**

Errors that ↑ Q angle

Component	Bad: ↑ Q angle (maltracking)	Good: ↓ Q angle (good tracking)
Femur	**Internal rotation** · **medialisation**	**External rotation** · **lateralisation**
Tibia	**Internal rotation** · **medialisation**	**External rotation** · **lateralisation** (with adequate ER)
Patella	**Lateral placement**	**Medial / central** placement
PF space	↑ composite thickness (under-resection), **anterior femoral translation** → tight lateral retinaculum (medial approach)	–

Lateral release rates (OB) IR femur: **24% varus, 33% valgus** · ER off TEA: **7% varus, 29% valgus**



Femoral IR and lateral patellar subluxation

- - Active space - -

4 Diagnosis & Classification

Key radiology findings

Malrotation ✓ **CT scan of the knee**
✗ Wrong options: 3-joint standing film · fluoroscopic EUA · 45° flexed PA · MR arthrogram

Merchant / tangential lateral tilt + subluxation

5 Management

Femoral rotation landmarks (posterior condylar axis, hypoplastic lateral condyle)

Axis	Definition	Use
AP (Whiteside's line / trans-sulcus)	Deepest trochlear groove → top of intercondylar notch	**Perpendicular** = neutral rotation · fewer maltracking reoperations in valgus knees (Whiteside & Arima)
Transepicondylar (TEA)	Medial ↔ lateral epicondyle · ⊥ **Whiteside's line** · parallel to cut tibia	Posterior cut ∥ TEA → **rectangular flexion gap** · more reliable than PCA
Posterior condylar (PCA)	Tips of posterior condyles · ✓ **~3° IR to TEA** (range **1–10°**)	Cut in **3° ER** from PCA

Posterior condylar axis normal relation → ✓ 3° internally rotated to TEA
✗ Wrong options: parallel · ⊥ Whiteside · 3° or 7° ER

Hypoplastic LFC + posterior referencing → ✓ femoral component IR → lateral tilt, maltracking, anterior pain in flexion · fix: add ER, posterior lateral augments, check against TEA / Whiteside
✗ Wrong options: tibial rotation errors

Femoral IR also → tight medial / loose lateral flexion gap → stiffness
✗ Wrong options: joint-line change · poly thickness change · ↓ Q

Gap-balanced flexion (tensioner) – varus (A) vs valgus with hypoplastic LFC (B) → ✓ B needs more ER relative to PCA · posterior **medial** resection thicker than lateral when ER correct · B would need posterior **lateral** augments if any
✗ Wrong options: more IR to trans-sulcus

Valgus knee with lateral tilt after TKA → ✓ posterior condyles used as rotation reference
✗ Wrong options: excess ER · femoral lateralisation · gap balancing (not linked) · TEA matching

Component internal rotation & maltracking

Most common intra-op error causing maltracking → ✓ femoral component IR
✗ Wrong options: gap balancing vs measured resection · tibial ER · femoral lateralisation · over-resected patella (**under**-resection overstuffs)

Optimise tracking → ✓ ER of femoral and tibial components · medialise patellar button

↓ Need for lateral release → ✓ femoral ER · **✓ tibial ER** (medialises tubercle)
✗ Wrong options: mobile bearing (no benefit) · anterior tray (force not tracking) · CR device

Lateral subluxation after medial approach → ✓ femoral IR (also tibial IR, ↑ composite patellar thickness, anterior femoral translation)
✗ Wrong options: tibial ER · posterior femoral translation · ↓ patellar thickness · lateral tibia

Patellar tilt – all cause EXCEPT → ✓ lateralisation of the femoral prosthesis

Patellar component & tibial position (medialisation, Q angle)

Patella — centred or ✓ **medialised** (replicates median eminence) · ideal **40–45%** (medial edge → central peg ÷ resected width) · uncovered **lateral facet** → excise to avoid lateral facet syndrome · or **oval** button with apex medial

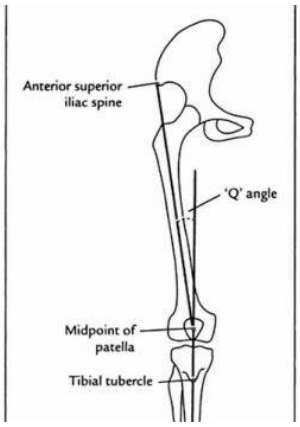
Most risky for maltracking → ✓ lateral placement of patellar component
✗ Wrong options: resection thickness · CR · medial placement · pre-op tilt

Decreases Q angle → ✓ medialisation of patellar component
✗ Wrong options: medialise femur/tibia · IR femur/tibia

Tibia — **neutral rotation**, centred over **medial ⅓ of tibial tubercle** (junction medial–middle third) · may leave posteromedial tibia uncovered, posterolateral overhang · tibial IR = relative tubercle ER → ↑ Q

Tray at medial/middle ⅓ with 10° ER, still lateral tilt → ✓ lateralise the tibial component
✗ Wrong options: medialise femur · lateralise button · IR femur/tibia

Tibial size — not related to tracking



Q angle and component position

Tourniquet effect on tracking

Lateral subluxation with perfect components → ✓ deflate tourniquet and re-evaluate before lateral release
✗ Wrong options: lateral release first · revise components

Patellar resurfacing vs patelloplasty

Absolute indications to resurface — **inflammatory arthritis** (✓ consistent benefit) · **patellar maltracking** · **PF OA as main indication** · severe patellar deformity

– – Active space – –

Strategies always · never (± **patelloplasty**: excise marginal osteophytes, reshape) · **selective**

Resurfaced vs not	Finding
Anterior knee pain	**Less with resurfacing** (Orthobullets; OB4743) · **no difference** (OB215498) – read the options
Revision / secondary surgery	**Unresurfaced → ↑ secondary resurfacing (8.7%)** · secondary resurfacing gives **inferior** results
Tibial/femoral revision	No difference
Fracture, AVN, tendon injury	No difference
Infection	No difference
Satisfaction / function	Similar · TKA improves function either way
Metal-backed button	Not lower loosening – **higher**

Patelloplasty EXCEPT → ✓ "no difference in anterior knee pain" = false

"Patella-friendly" trochlea **thinner anterior flange** · anatomic groove

Late instability – realignment

Instability after fall, components correctly ER on CT, tiny patellar fragment → ✓ lateral retinacular release + proximal realignment (excise fragment)
✗ Wrong options: ORIF + brace · tibial revision · distal tubercle medialisation (not first line) · patellar revision

Rule malpositioned components → **revise** · well-positioned → **soft-tissue procedures** (lateral release first)

X vs Y

	Posterior condylar axis	TEA	Whiteside's line
Relation	3° IR to TEA	Reference	⊥ TEA
Pitfall	**Hypoplastic LFC → IR**	Epicondyles hard to palpate	Trochlear dysplasia

How it's asked

"Femoral IR results in?"	**→ increased need for lateral release**
"Decreases need for lateral release?"	**→ external rotation of femoral component**
"Patellar tilt – all cause EXCEPT?"	**→ lateralisation of femoral prosthesis**
"Valgus knee – error causing lateral patellar instability?"	**→ femoral component IR**
"Femoral IR may lead to?"	**→ net lateral patellar tilt + lateral subluxation**
"Risk for lateral maltracking?"	**→ IR of tibial component**
"Most common intra-op error causing maltracking?"	**→ IR of femoral component**

"Lateral subluxation after medial parapatellar TKA — cause?"	→ **IR of femoral component**
"Optimal implant position for tracking?"	→ **ER of femoral and tibial components**
"Enhance tracking and minimise lateral release?"	→ **ER of tibial component**
"Missed hypoplastic LFC + posterior referencing?"	→ **IR of femoral component**
"Normal posterior condylar axis relation?"	→ **3° IR to TEA**
"Valgus knee, posterior referencing — error?"	→ **IR of femoral component**
"Valgus knee, lateral tilt on Merchant — technique?"	→ **posterior condyles used as rotation reference**
"Varus (A) vs valgus (B), gap-balanced?"	→ **B needs more ER relative to PCA**
"MOST places knee at risk of maltracking?"	→ **lateral placement of patellar component**
"Aids correction of tracking?"	→ **medialisation of patellar component**
"Decreases Q angle?"	→ **medialisation of patellar component**
"Tray ER 10°, still lateral tilt?"	→ **lateralise the tibial component**
"Malrotation best diagnosed by?"	→ **CT scan of the knee**
"Perfect components, patella subluxes — next?"	→ **re-evaluate after tourniquet deflation**
"Patelloplasty vs resurfacing — EXCEPT?"	→ **no difference in anterior knee pain**
"True about the patella in TKA?"	→ **resurfacing benefits inflammatory arthritis**
"Post-fall instability, components well rotated?"	→ **lateral release + proximal realignment**

– – Active space – –

TKA PERIPROSTHETIC FRACTURE

– Recon · 52 / 65 –

1 Basics

Sites **distal femur** · **proximal tibia** · **patella** · overall prevalence **0.39%** (OB)

Decision drivers **fracture location** · **implant stability** · **bone stock** · comorbidity · implant design (CR / open-box PS / closed-box PS / stemmed)

Risk factors & intraoperative fractures

Category	Risk factors
Bone quality	**Osteopenia** · age **>70** · **female** · **chronic steroids** · **RA** · stress-shielding
Stress risers	Screw holes · **osteolysis** · stiffness · **revision TKA** · navigation pin sites
Neurological	**Epilepsy** · **Parkinson's** · cerebellar ataxia · myasthenia · **polio** · CP
Femur-specific	Modulus mismatch · **rotationally constrained** components · **anterior notching** – debatable
Not risk factors	**Male gender** · **anterior femoral notching** (TO: controversial; >3 mm ↓ strength in biomechanics, not clinically)

Notching conflict **not a risk factor** (TO) · **listed as risk** (OB210125, OB220387) · **debatable** (Orthobullets) – read the options

Notching (TO lecture notes) notching (and pistol-grip stems) act as **stress risers** · **PS > CR** fracture risk (box cut) · other TO risks: **mini-incision**, size mismatch, uncemented

Intra-op fractures ✓ **medial femoral condyle commonest** (incomplete sagittal cut between notch and MFC) → then **lateral condyle → supracondylar femur → proximal tibia** · ↑ in **PS (box cut)** · **femoral > tibial** · patella rare · most need **fixation** (wires/screws/plates) · occur in exposure/bone prep > trialing

Navigation / robotics ✓ **femoral shaft fracture through tracker pin sites** = complication unique to navigation (TO) · ✓ **transcortical / bicortical diaphyseal, large pins** ↑ risk · safer: **two 3 mm unicortical metaphyseal pins**, self-drilling/self-tapping · fractures within ~**3 months**; thermal necrosis · 5 mm pin removes **2.78×** bone of 3 mm · drill holes ↓ bending strength up to **40%** ✗ Wrong options: PS implant risk

4 Diagnosis & Classification

Key radiology findings

X-rays diagnosis · **CT** – bone stock, **loosening**, planning · identify **CR (no box, notch recess) vs PS (box)** on X-ray

Supracondylar periprosthetic femur within **15 cm** of joint line or **5 cm** of top of a stemmed component

Acceptable outcome ROM **>90°** · shortening **<2 cm** · V/V **<5°** · F/E malalignment **<10°**

Lewis & Rorabeck (distal femur) — most used

Type	Displacement	Component
I	**Non-displaced**	Intact
II	**Displaced**	**Intact**
III	Displaced	**Loose / failing**

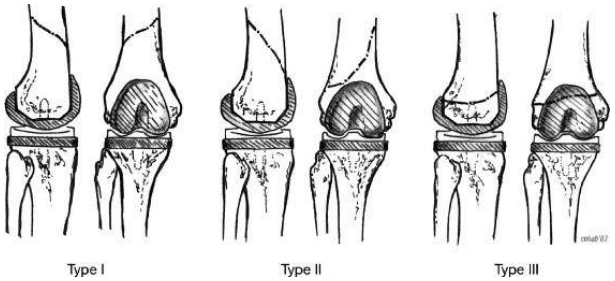
Su (distal femur) (TO)

Type	Location	Treatment
I	**Proximal** to femoral component	**Retrograde or antegrade IMN** / plate
II	Starts at **proximal edge of component**, extends proximally	**Retrograde IMN** or fixed-angle plate
III	Any part **distal to upper edge of anterior flange**	**Fixed-angle/locked plate** or **revision / DFR** if poor bone · no rIMN

Su III (TO notes) distal to component → **often needs revision stem** · stable prosthesis + distal fracture → **long-stem femoral component ± plate** (TO) · fixed-angle plate (Orthobullets) — read the options

Interprosthetic (THR + TKR) **Su type I** for the TKR, Vancouver C for the THR → **long bridging plate ± graft** (TO)

Su used for ✓ **periprosthetic femur** (TO)
✗ Wrong options: patella · humerus · tibia



Su classification

Other femoral classifications

System	Types
Neer (1967)	I non-displaced (<5 mm/<5°) · II >1 cm (IIa lateral, IIb medial shaft displacement) · III displaced + comminuted
DiGioia & Rubash (1991)	I extra-articular non-displaced · II extra-articular displaced (>5 mm/>5°) · III lost cortical contact / angulated 10°, ± intercondylar/T
Chen (1994)	I non-displaced · II displaced/comminuted

– – Active space – –

Felix (tibia)

Type	Location	Subtype
I	**Tibial plateau** (implant interface)	**A** stable component
II	**Adjacent to stem** (meta-diaphysis)	**B loose** component
III	**Shaft distal** to component	**C intra-operative**
IV	**Tibial tubercle**	–

Felix treatment (TO) **IB, IIB → revision** (most postoperative I/II are B) · IIA, IIIA, IVA → usual fracture principles · IC → intra-op fixation · IIC → graft / immobilise / restricted WB · IIIC → conventional

Felix by type (TO notes) **IA** → ORIF / non-op · **IB** → long-stem revision · **II** stable → ORIF, loose → revision · **III** → ORIF **± dual plating** if fixation poor · **IV** (tubercle) → **screws / tension band**

Unified Classification System (TO) A apophyseal · B implant bed · C clear of bed · D between two arthroplasties · E two bones, one arthroplasty · F faces unresurfaced implant

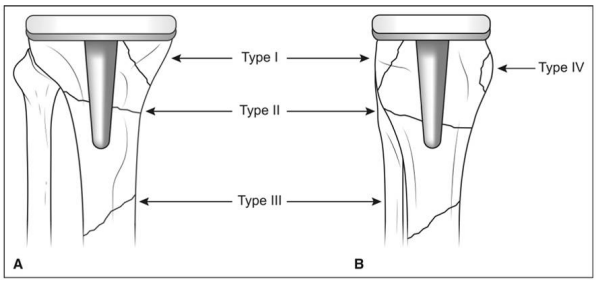


Illustration of the Felix classification system for periprosthetic tibial fracture.

Felix tibial classification

Patellar classifications

Ortiguera & Berry	EM	Component	Treatment
I	Intact	**Stable**	**Cast/brace in extension**
II	**Disrupted**	Stable or loose	**ORIF** (good bone, ± mesh augment) · partial patellectomy + repair · patellectomy needs allograft/ mesh reconstruction
IIIa	Intact	**Loose**, bone **≥10 mm**	Component **revision** or resection arthroplasty
IIIb	Intact	Loose, **<10 mm** / comminuted	**Component removal + patelloplasty** or patellectomy

Goldberg	Description
I	Not involving implant/cement interface or quad mechanism
II	Involves interface and/or quad mechanism
IIIA / IIIB	Inferior pole **with / without** patellar ligament rupture
IV	Fracture-dislocation

Ortiguera and Berry Classification of Periprosthetic Patellar Fractures	
Type of Fracture	Characteristics
I	Stable implant, intact extensor mechanism
II	Disrupted extensor mechanism
IIIa	Loose patellar component, reasonable bone stock
IIIb	Loose patellar component, poor bone stock

Ortiguera and Berry classification

5 Management

Periprosthetic distal femur fracture

- Non-displaced + stable component → **cast / brace**
- Displaced + stable component → **fix**
  (rIMN if CR / open box + Su I–II; locked plate if closed box, stem, Su III)
- Loose component / poor bone → **revision (long stem) or DFR**
  - Elderly, very distal, comminuted → **distal femoral replacement**

Supracondylar femur – locked plating vs retrograde nail

Retrograde IMN needs **CR or open-box PS** (8–12 mm recess) · Su I/II · ≥**2 distal interlocks**, end cap · insert **deep** (no patellar abrasion) · not with **closed-box PS**, **stemmed** femur, loose component, ipsilateral long antegrade/ cephalomedullary nail or hip stem (stress riser), insufficient distal bone · not Su III (distal to flange) · ~**15% revision** (nonunion)

rIMN malunion femoral component forces **posterior start point → ✓ extension (recurvatum) deformity** (OB) · **valgus + recurvatum** (TO) · distal **10° bend** nails improve alignment · hyperextension tolerated, valgus poorly tolerated (TO)

Locked plate polyaxial screws around lugs/stems · independent of design + distal bone · **MIPO/submuscular** · not non-locked condylar buttress (no varus resistance; nonunion up to **50%**) · blade plate/DCS hard around PS

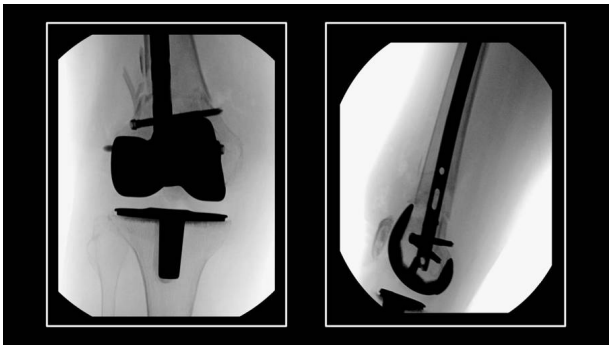
Plate vs rIMN	Finding
Union / nonunion	✓ **Equivalent** · similar secondary surgery
Malunion	↑ **with IMN** (plating ↓ malunion) · ✓ **IMN higher malunion** than ORIF or DFR
Weight bearing	**IMN faster** to full WB
Soft tissue	IMN less injury

Nonunion risk
✓ **extensile lateral approach > submuscular** · MIPO ↑ malunion (Orthobullets)
✗ Wrong options: no BMP (not approved) · no lag/cerclage · plate vs antegrade nail · titanium vs steel

Commonest reason ORIF fails
✓ **nonunion (persistent fracture line)** (~10% revised) → revision ORIF or DFR
✗ Wrong options: aseptic loosening · PE wear · infection · dislocation

– – Active space – –

Nail + lateral plate vs dual plating ✓ **earlier weight bearing** · dual plating stiffer (axial + torsion) + more medial dissection · infection same



Posterior start point — extension deformity

Scenario	Answer
Far-distal fracture, well-fixed CR, community ambulator	✓ **distal femoral locking plate** (✗ Wrong options: cast · buttress plate · DFR · antegrade nail)
PS TKA, displaced fracture just above, osteopenic, well fixed	✓ **MIPO locked plating** (✗ Wrong options: cast/brace · rIMN (PS) · DFR in 62 yo)
Su I, CR knee, well fixed	✓ **retrograde IMN** (✗ Wrong options: ex-fix · recon plate · stem revision · antegrade)
Stemmed revision femur, well fixed	✓ **lateral locked plate** (✗ Wrong options: rIMN/antegrade (stem blocks))
PS box, 5 days post-op, well fixed	✓ **PS design → lateral locking plate**
Bridge plating vs rIMN	✓ **similar nonunion, ↓ malunion**

Loose components or poor bone – revision / distal femoral replacement

Loose femoral component → ✓ **revision TKA** (long stem or DFR) · loose + medial condyle fracture → **ORIF + femoral revision**
✗ Wrong options: nail · plate · cast · ex-fix

DFR indications **elderly / sedentary** · **loose or malpositioned** component · **poor bone stock** · very distal / comminuted (Su III) · requires **rotating hinge** (collaterals lost) · tibia often revised too

DFR vs ORIF ✓ **similar complication rate** · similar reoperation, function, return to ambulatory status · **immediate weight bearing** · shorter surgery · ROM similar or **slightly less** with DFR
✗ Wrong options: better outcomes · higher reoperation · greater ROM

84 yo walker user, very distal comminuted → ✓ **DFR** (best chance ambulatory at 1 yr)
✗ Wrong options: plate · dual plate · nail · ex-fix (seeding risk)

Distal to flange, poor bone, medial condyle free → ✓ **revision arthroplasty only (DFR)**



Distal femoral replacement

Tibial periprosthetic fracture

Incidence **0.4–1.7%** · risks: prior **TTO**, **loosening**, malposition, **long-stemmed** tibial components

Situation	Treatment
Non-displaced, stable	Cast/brace (non-WB)
Unstable fracture, stable component	**ORIF** / IMN
✓ **Displaced + loose tibial component** (Felix IIB)	**Revision with long stem bypassing fracture** (TO) (✗ Wrong options: locking plate, LCP, LISS, ex-fix, IMN, brace)
Loose + poor bone, prior pain	✓ **ORIF + revision TKA** (✗ Wrong options: cast · strut graft · IMN · proximal tibial replacement (salvage))

Patella fractures

Incidence resurfaced **0.2–21%** (Orthobullets) · **0.68–5.2%** (OB) · unresurfaced **0.05%**

Risks patellar ON · asymmetric / over-resection · central **single peg** · **uncemented** · **metal backing** · **inset** component · lateral release · ↑ BMI

Scenario	Answer
Displaced, **no SLR**, well-fixed component, good bone	✓ **ORIF** (✗ Wrong options: patellectomy · allograft · component revision · brace)
Transverse, **active extension intact**, well fixed	✓ **long-leg cast / brace in extension** (✗ Wrong options: immediate ROM · component removal · ORIF)
Displaced mid-patella, no SLR	✓ **ORIF** (✗ Wrong options: hinged ROM · 2 wk immobilise · 6 wk cast (lag) · partial patellectomy (poles/comminution))
Type II at 10 wk, stable button, good bone	✓ **ORIF** (± mesh augmentation)
Intact EM, stable component	✓ **knee brace locked in extension / cylinder cast**
Non-displaced transverse, SLR intact	✓ **immobilise in extension 4–6 wk**

Type II/III outcomes **50% complications**, **20–40% reoperation** · **total patellectomy** = salvage, **>50% loss of extension strength**

– – Active space – –

Extensor mechanism failure

Chronic EM disruption after revision (6 mo, wide pole diastasis) → ✓ extensor mechanism allograft
✗ Wrong options: fusion (salvage) · patellectomy + repair · ORIF · cast

X vs Y

	CR femoral component	PS open box	PS closed box / stemmed
rIMN possible	Yes (notch recess)	Yes (8–12 mm)	No → **plate**

	Lewis–Rorabeck	Su
Based on	Displacement + **component stability**	**Fracture location** vs component

How it's asked

"Complication unique to navigated TKR?"	→ **femoral shaft fracture** (TO)
"Not a risk factor for periprosthetic fracture?"	→ **anterior femoral notching** (TO)
"Risk factors for supracondylar fracture — EXCEPT?"	→ **male gender**
"True about intra-op fractures?"	→ **medial femoral condyle most common**
"Robotic TKA, femur fracture at 3 wk — greatest risk?"	→ **transcortical tracking pins**
"Most common intra-op fracture?"	→ **medial femoral condyle**
"Malunion after rIMN for Su II?"	→ **valgus and recurvatum** (TO)
"Far-distal fracture, well-fixed implants?"	→ **distal femoral locking plate**
"Nonunion after locked plating — risk?"	→ **extensile lateral approach (vs submuscular)**
"PS TKA, displaced fracture, well fixed?"	→ **minimally invasive plate osteosynthesis**
"Bridge plating vs rIMN?"	→ **similar nonunion, decreased malunion**
"Su I, CR, well fixed?"	→ **retrograde IMN**
"Stemmed femur, well fixed?"	→ **ORIF with lateral plate**
"Design + treatment 5 days post-op?"	→ **PS; ORIF with lateral locking plate**
"Nail-plate vs dual plating?"	→ **earlier weight bearing**
"Locked plate ORIF failure at 1 yr — commonest finding?"	→ **persistent fracture line (nonunion)**
"CR knee, considering rIMN?"	→ **higher potential for extension deformity**
"Revised stemmed femur, well fixed, adequate bone?"	→ **lateral locked plating**
"IMN vs ORIF vs DFR — difference?"	→ **IMN higher malunion**

"Malalignment after rIMN?"	→ **extension deformity**
"Plate vs IMN?"	→ **union rate equivalent**
"Loose femoral component after fall?"	→ **revision TKA**
"Loose femoral component + medial condyle fracture?"	→ **ORIF + femoral component revision**
"Elderly, loose component, poor bone?"	→ **revision with distal femoral replacement**
"84 yo walker user, very distal fracture — best ambulation at 1 yr?"	→ **conversion to DFR**
"Tibial fracture, prior activity pain, loose implant?"	→ **ORIF with revision TKA**
"Fracture distal to flange, poor bone?"	→ **revision arthroplasty only**
"DFR vs ORIF?"	→ **similar complication rate**
"Felix 2B tibial fracture?"	→ **revision with long stem** (TO)
"Tibial fracture at stem with loose component?"	→ **long-stem tibial revision bypassing fracture**
"Patella fracture, no SLR, well-fixed button?"	→ **ORIF**
"Transverse patella fracture, active extension intact?"	→ **long-leg cast**
"Displaced patella fracture, no SLR, 1 yr post-op?"	→ **ORIF**
"Type II patellar fracture, stable button, good bone?"	→ **ORIF**
"Intact EM, stable button — initial management?"	→ **brace locked in extension or cylinder cast**
"Non-displaced transverse patella, SLR intact?"	→ **immobilise in extension 4–6 wk**
"Chronic EM disruption after revision?"	→ **extensor mechanism allograft**
"Su classification used for?"	→ **periprosthetic femur fracture** (TO)

– – Active space – –

TKA PERONEAL NERVE PALSY

– Recon · 53 / 65 –

1 Basics

Incidence **0.3–2%** (Orthobullets) · **0.3–1.3%** (OB) · **0.3–9.5%** after valgus TKR (TO)

Prognosis **≥50%** improve/recover without further treatment

Risk factors (valgus & flexion contracture)

✓ Risk factor	Note
✓ **Valgus + flexion contracture**	**Highest risk** (✗ Wrong options: valgus only · varus ± FFD · FFD only)
Valgus >12° · correction of 20° valgus	Nerve tethered at fibular head → stretch
FFD >10–15°	Stretch
Tourniquet >120 min	–
Post-op epidural analgesia	Unprotected limb → compression
Aberrant retractor	–
Pre-op neuropathy (central or peripheral) · lumbar laminectomy (double crush)	–
✗ Wrong option: **excessive medial release**	**Not** a risk factor

Lateral OA + FFD >15° + valgus >12° – counsel for → ✓ CPN injury (TO)
✗ Wrong options: DVT · instability · quadriceps contracture

Large valgus + FFD – higher risk than typical for → ✓ peroneal nerve injury
✗ Wrong options: DVT · wound · loosening · quad rupture

Anatomy & release risk

At tibial cut level CPN lies ✓ **superficial to (on) lateral head of gastrocnemius** (protected by it) · in interval between **biceps femoris and lateral gastrocnemius**
✗ Wrong options: deep to arcuate ligament · on popliteus · on bone

Distance from posterolateral tibia **9–15 mm** (Orthobullets) · mean **14.9 mm**, never <**9 mm**, ↑ in larger legs (OB MRI) · as close as **7–9 mm in extension** (cadaver)

Greatest risk ✓ **release of posterolateral capsule with knee extended**
✗ Wrong options: posterior capsule · PLC in flexion · ITB (anterior to nerve)

Danger zone (TO) PLC release in triangle of **popliteus, tibial cut surface, posterior ITB fibres** · ITB pie-crusting = **safe zone** (TO)

3 Presentation

Signs no ankle dorsiflexion or toe extension · ↓ sensation **dorsum of foot + first web space**

5 Management

Foot drop after TKA

- Recovery room → **remove dressings + flex knee 20–30°**
- Complete foot drop → **AFO + PT (keep passive DF)**
- No recovery at 3 mo → **EMG; consider decompression / exploration (3–6 mo)**
- Persistent palsy, tibialis posterior 5/5 → **tibialis posterior transfer to dorsum**

Immediate management

✓ **Unwrap compressive dressings + flex knee**
✗ Wrong options: AFO acutely · immediate EMG · exploration · reassurance

Late management

EMG	after **3 months** if no improvement
Exploration / neurolysis	no recovery at **3–6 months** · explore at posterior border of biceps femoris

Tendon transfer for persistent palsy

Failed AFO + neurolysis, 18 mo, passive DF intact, inversion 5/5 → ✓ tibialis posterior transfer to dorsum (through interosseous membrane → **lateral cuneiform**)
✗ Wrong options: continue AFO · repeat neurolysis · peroneus tertius ± TAL (same anterior compartment, paralysed)

Valgus TKR technique (TO)

Tibial cut	referenced **6–8 mm off medial compartment** after removing lateral osteophytes (TO)
Femur	IM entry **more medial** · distal valgus cut **3°** (not 5–7°) · lateral distal resection minimal (1–2 mm) or none · medial ≤10 mm (usually 7–8) (TO)
Rotation	hypoplastic LFC → use **~6° ER** or anterior referencing · Whiteside line unreliable (trochlear dysplasia) → **TEA** or parallel to tibial cut (TO)

Lateral release orders (TO)

Author	Sequence
Ranawat	**PCL** → PLC (at tibial cut, cautery) → ITB + LCL pie-crust · popliteus preserved
Favorito	**LCL** (tightest) → popliteus → PLC → ITB → lateral gastrocnemius
Whiteside	Tight in **flexion → LCL, popliteus** (near epicondyle) · tight in **extension → ITB, PLC** · LHG only for FFD

X vs Y

	ITB release	PLC release (extension)
Nerve risk	**Safe zone** (ITB anterior to nerve)	✓ **Danger zone**

– – Active space – –

How it's asked

"Lateral OA, FFD >15°, valgus >12° — counsel for?"	→ **CPN injury** (TO)
"Deformity with most risk of peroneal palsy?"	→ **valgus + flexion contracture**
"Peroneal palsy risk factors — EXCEPT?"	→ **excessive medial release**
"Large valgus + FFD — higher risk of?"	→ **peroneal nerve injury**
"Recovery room, cannot dorsiflex?"	→ **unwrap dressings and flex the knee**
"Where is CPN at tibial cut level?"	→ **superficial to lateral head of gastrocnemius**
"Lateral tight in flexion + extension — greatest nerve risk during?"	→ **posterolateral capsule release with knee extended**
"Persistent foot drop at 18 mo, inversion 5/5?"	→ **tibialis posterior transfer to dorsum of foot**

TKA POLYETHYLENE WEAR & MANUFACTURING

– Recon · 54 / 65 –

1 Basics

Definition	macroscopic premature PE failure from excess loading + mechanical loosening
Catastrophic failure	typical of **TKA** (vs **osteolytic** failure in THA)
Primary variables	**PE thickness** · **articular geometry** · **kinematics** · **sterilisation** · **manufacturing** · **surgical technique** (tight flexion gap ↑ sliding wear; tight PCL + anterior tibial slope ↑ stress)
Treatment	revision TKA or isolated **PE exchange** if implants stable · X-ray: narrowing of tibiofemoral implant interface

Polyethylene thickness

Rule	keep thinnest PE **>8 mm** · <8 mm → loads exceed PE yield strength (**12–20 MPa**) → catastrophic failure (older PE data)
Labelled vs true	"8 mm" insert may be **4–5 mm** PE on ~**3 mm** tray · deeper tibial cut avoids thin insert · young + active + thin PE = ↑ risk

Articular surface design and geometry

	Congruent ("dished")	Flat
Rollback	Sacrificed	↑ rollback, ↑ flexion
Anatomy	Less anatomic	More anatomic, PCL sparing
Contact area	↑	↓
Contact stress	↓	↑ → catastrophic wear

Best	**biplanar congruency** (coronal + sagittal) · maximise contact area, minimise contact stress

Insert conformity & stability

Anterior-stabilised (AS) vs CR	✓ **↑ contact area + ↑ stability** in PCL-deficient knee · highly conforming + **large anterior lip** blocks anterior femoral translation · no box cut

Kinematics

Sliding wear	**ACL sacrificed + PCL retained** · worst in **CR + flat insert** · least in **PS or AS + congruent insert**

Varus alignment → catastrophic failure · **excess rollback** → dyskinetic sliding → cracking

Solutions	correct varus · congruent design · ↑ posterior slope · **PS** for incompetent PCL · posterior contact point

– – Active space – –

2 Mechanism

Sterilisation – gamma in air & oxidation

Gamma radiation commonest sterilisation · breaks C–H bonds → **free radicals**

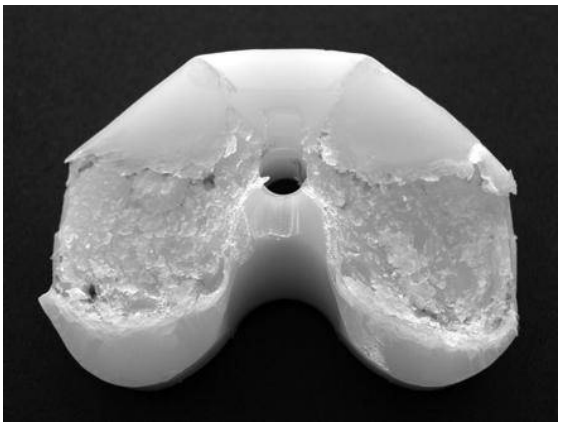
Environment	Pathway	Result
Air (O_2 rich)	**Oxidation** (chain scission)	✓ **Early failure**: **subsurface delamination**, **pitting**, **cracking**, fatigue · ↑ wear + **osteolysis** · abandoned
Inert (argon, nitrogen, vacuum)	**Cross-linking**	↑ resistance to **adhesive + abrasive wear** · ↓ ductility + fatigue resistance · ✓ **↓ pre-implantation oxidation**
Ethylene oxide / gas plasma / H_2O_2	No radicals, **no cross-linking**	No oxidation, but no wear benefit

Oxidised PE mechanics ✓ **↑ elastic modulus + ↓ strength** (and ↓ ductility) → brittle

Gamma in nitrogen produces ✓ **free radicals**
✗ Wrong options: annealing · gas plasma · uncross-linking · remelting

Shelf life gamma-in-air packaged PE stored **5 yr** → ✓ **substantially oxidised** (still sterile, no creep) · oxidation ∝ time exposed to O_2 · (✗ Wrong options: OR FiO_2 matters (no radiation))

Doses sterilisation **2.5–4 Mrad** (conventional 2.5 in air) · HXLPE **5–10 Mrad** (5–15 Mrad) in inert environment, separate earlier step · little benefit **>10 Mrad**



Gamma irradiation in air – oxidation

Cross-linking & thermal processing (annealing vs remelting)

Most improvement in wear ✓ **gamma irradiation in argon (inert)** · ✓ **radiation** has most positive effect on wear resistance
✗ Wrong options: ram bar · gas plasma · EtO · air

↑ Irradiation dose ✓ **↑ volumetric wear resistance** · ↓ tensile strength, fracture toughness, crack resistance · ✓ principal negative: **↓ fatigue strength**
✗ Wrong options: yellow · unstable · absorbs fluid

Radiation + heating ✓ **↓ wear but ↑ fracture risk** (↓ ductility) · forms interpenetrating network · fewer, smaller particles vs smooth counterface · **worse vs rough counterface** (↓ abrasive resistance)

Free radicals removed by ✓ **thermal processing** (or **vitamin E**; ✗ Wrong options: vitamin C)

	Remelting	Annealing
Temperature	**Above** melting point	**Below** melting point
Free radicals	✓ **Most removed** (largest reduction)	**More remain** → oxidation risk
Crystallinity	**Partial crystalline → amorphous**	Preserved
Mechanical properties	↓	✓ **Better maintained**
Wear	**Equivalent**	Equivalent
In vivo oxidation	Seen in both	Seen in both

Annealed vs remelted XPE → ✓ more free radicals, better mechanics, equivalent wear

Preferred annealing (better mechanics) (OB8309) · remelting most effective radical removal

HXLPE vs conventional (THA RCT) head penetration **0.003 vs 0.051 mm/yr** · ✓ **superior wear** · ↓ osteolysis · larger head: volumetric wear equal or slightly higher

Manufacturing – compression moulding

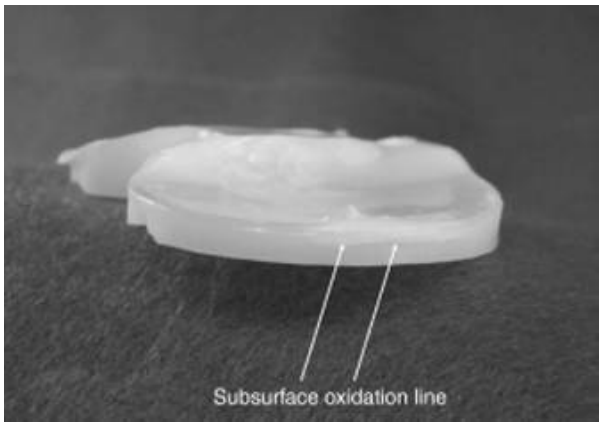
Method	Notes
Ram bar extrusion + machining	Powder pushed through heated barrel → rods → machined · variable quality
Calcium stearate additive	**Fusion defects** · ↓ mechanics
Sheet compression moulding	Sheets up to 8" → machined
✓ **Direct compression moulding (net shape)**	Powder → final mould · **no machining**, high gloss · **lowest fatigue crack formation + propagation** · wear **~50%** of machined (Orthobullets/Miller) · wear **equal** to ram extrusion (Comprehensive Ortho Review) – read the options · slow, costly

Machining shear stretches chains **1–2 mm** below surface (amorphous > crystalline) → radiation-susceptible → **white band of subsurface oxidation** → delamination + fatigue cracks

Crystallinity ideal **~50%** · **>70%** ↑ failure

"Perfect storm" for catastrophic wear metal-backed tray + **thin PE** · **flat coronal** design · **CR + flat insert** · **ram bar + calcium stearate** · **gamma in air** · **machined surface**

– – Active space – –



Subsurface white band of oxidation

X vs Y

	Oxidised PE	Cross-linked PE
Made by	Gamma **in air**	Gamma in **inert** gas / vacuum
Wear	↑ delamination, pitting	↓ adhesive + abrasive
Mechanics	↑ modulus, ↓ strength	↓ ductility, ↓ fatigue strength

How it's asked

"Catastrophic delamination at 3 yr – cause?"	**→ gamma irradiation in air**
"Gamma in air vs inert – result?"	**→ accelerated wear and failure in air**
"Gamma in nitrogen results in?"	**→ free radicals**
"Gamma causes rapid wear when combined with?"	**→ irradiation in air**
"Gamma in air degrades wear because of?"	**→ oxidation**
"Properties after PE oxidation?"	**→ ↑ elastic modulus, ↓ strength**
"Greatest PE oxidation?"	**→ gamma radiation in air**
"Gamma-in-air cup stored 5 yr – why not implant?"	**→ may be substantially oxidised**
"Processing step → ↑ wear + osteolysis?"	**→ gamma irradiation in air**
"Inert vs air sterilisation – true?"	**→ decreased pre-implantation oxidation**
"Irradiation in O_2-rich environment – effect?"	**→ early failure**
"MOST improvement in wear?"	**→ gamma irradiation in argon**
"Radiation then heating – effect?"	**→ lower wear but ↑ fracture risk**
"Annealed vs remelted XPE?"	**→ more free radicals, better mechanics, equivalent wear**
"↑ Irradiation dose improves?"	**→ volumetric wear**
"Principal negative of high cross-linking?"	**→ loss of fatigue strength**

"Most positive effect on wear resistance?"	→ **radiation**
"Partial crystalline → amorphous?"	→ **remelting**
"Free radicals removed by?"	→ **thermal processing**
"Largest reduction of free radicals?"	→ **remelting**
"Annealed HXLPE in THA – counsel?"	→ **superior wear to conventional PE**
"Annealed vs remelted – difference?"	→ **annealing changes mechanical properties less**
"Lowest fatigue crack susceptibility?"	→ **direct compression moulding**
"Anterior-stabilised vs CR?"	→ **increased contact area, increased stability**

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TKA POSTOPERATIVE INPATIENT MANAGEMENT

– Recon · 55 / 65 –

1 Basics

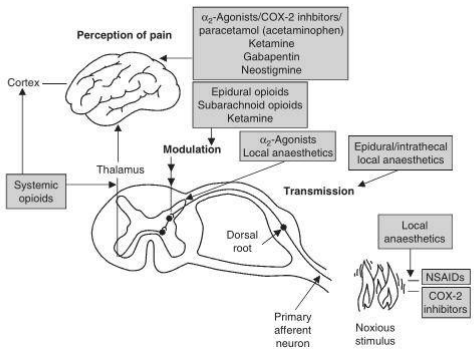
Team surgeon, PT, OT, case manager, nursing, patient + family

5 Management

Pain management

Phase	Measure
Pre-op	NSAIDs + opioids immediately before surgery ↓ post-op pain
Intra-op	**Regional (spinal/epidural) preferred** over GA · **peripheral nerve blocks** · **periarticular multimodal injection** (low risk)
Post-op	**Multimodal oral therapy = gold standard**

Drug class	Mechanism
Opioids	**Mu agonist** → neuronal hyperpolarisation, ↓ excitability
NSAIDs	Inhibit **COX-1 + COX-2** → ↓ PGs, TXA, AA mediators
Selective COX-2	Block AA → PG precursors · ↓ GI effects · may impair bone healing
Gabapentin / pregabalin	↓ hyperexcitability of voltage-dependent **Ca^{2+} channels** · pregabalin better oral bioavailability
SNRIs	Inhibit serotonin + noradrenaline reuptake (CNS)



Multimodal analgesia – pathway targets

Regional anaesthesia – adductor canal block

ACB vs femoral nerve block

✓ **ACB → earlier ambulation** · **equivalent analgesia** · ACB = **sensory** block (saphenous n. + nerve to vastus medialis, distal to quad motor branches) · FNB → **quad weakness, falls, delayed ambulation**
✗ Wrong options: FNB better pain control · ACB blocks quads · longer spinal → earlier ambulation · CPM → early ambulation

Tourniquet use

Tourniquet-less TKA

✓ **↑ intra-op blood loss** (consistent) · no difference in 1-yr flexion, operative time, DVT · thigh pain with tourniquet · ~**90%** of surgeons use one

Physical therapy & ROM

Start therapy on day of surgery

✓ **↓ length of stay** (AAOS **strong**) · ↓ pain, ↑ function vs POD1 (moderate)

Activity	Flexion needed
Swing phase of gait	65°
ADLs	90°
Stairs	95°
Rising from chair	105°

Continuous passive motion



CPM – no long-term benefit

CPM

↑ **early** flexion only (2 wk) · ✓ **no difference in outcomes at 1 yr** · ✓ **no long-term ROM difference** · (✗ Wrong options: ↓ contracture · ↓ MUA · ↑ analgesic need · ↓ VTE (insufficient evidence) · ↑ passive flexion at 6 mo · ↓ SSI) · some studies ↓ LOS · Salter 1970

– – Active space – –

AAOS guidelines (early rehab, drains)

Recommendation (AAOS knee OA surgical CPG)	Strength
✓ **Rehab on day of TKA ↓ LOS**	**Strong**
CPM does not improve outcomes	**Strong** (against CPM)
No surgical navigation (no outcome/ complication difference)	**Strong**
No difference with/without tourniquet	**Strong**
No difference patellar resurfacing vs not	**Strong**
✓ **Drains – no difference in complications/ outcomes** (↑ cost, ↑ LOS)	**Strong** (OB213939) · **Moderate** against (2022 CPG, OB219801) – read the options
Cryotherapy devices do not improve outcomes	**Moderate**
Pre-op PT improves pain/function	**Limited**
Tourniquet ↓ short-term function	**Limited**
Robotic vs conventional – no difference	**Limited**
Simultaneous bilateral TKA – avoid in high cardiopulmonary risk	**Consensus**
PCA	No recommendation

Post-operative anaemia

Tourniquet ↓ intra-op blood loss · **earlier discharge to rehab** → better outcomes

Outpatient TKR pathway (TO)

Components (TO)	**TXA** · periarticular injection (liposomal/plain bupivacaine ± ketorolac, morphine, dexamethasone) · **ACB + iPACK** (infiltration between popliteal artery and capsule) → best early function, ↓ LOS vs FNB · **PT at 2 h**, **WB at 6 h** · minimally invasive · ✓ **no tourniquet** (TO)
NOT a component (TO)	✓ **tourniquet**
Discharge criteria (TO)	walk safely with 2 crutches, stairs, home exercises, self-care, **≥90° flexion** · no brace
Patient selection (TO)	**age <75**, **BMI <35**, no anaemia, COPD, CHF, cirrhosis, VTE, spinal stenosis, BPH, chronic narcotics · primary TKA, first/second case · RAPT **>10**, near hospital, private insurance

Drain management

Drains	no difference in complications or outcomes · ↑ cost

Discharge

Destination	Criteria
Home	Medically stable · **80–90° active flexion** · **ambulate 75–100 ft** · stairs
Rehab / SNF	Earlier discharge → better outcomes

X vs Y

	Adductor canal block	Femoral nerve block
Nerves	Saphenous + nerve to VM	Femoral (motor + sensory)
Quads	Spared	**Weak** → falls
Ambulation	**Earlier**	Delayed
Analgesia	Equivalent	Equivalent

How it's asked

"CPM after TKA — true?"	→ **no difference in clinical outcomes at 1 yr**
"CPM in uncomplicated TKA — true?"	→ **no long-term difference in ROM**
"Highest-strength AAOS recommendation?"	→ **rehab on day of TKA reduces LOS**
"Patient requests — strong evidence against?"	→ **drain placement**
"Moderate AAOS recommendation against?"	→ **intra-articular closed suction drain**
"ACB vs FNB — accurate?"	→ **ACB gives earlier ambulation**
"Tourniquet-less TKA — true?"	→ **increased intra-op blood loss**
"Not a component of outpatient TKR?"	→ **tourniquet** (TO)

– – Active space – –

TKA POSTOPERATIVE REHABILITATION & OUTPATIENT MANAGEMENT

– Recon · 56 / 65 –

1 Basics

Phases — **inpatient extended care** (rehab/SNF) · **outpatient care** · earlier discharge to rehab → better outcomes · home discharge criteria as in hospital

5 Management

Rehabilitation (cryotherapy, home PT)

Outpatient PT — **2–3×/week for ≥2 weeks** · ✓ **closed-chain concentric** exercises · crutches → cane → unassisted

Self-directed home PT vs formal outpatient PT — ✓ **similar clinical outcomes** (ROM, MUA rates) + **cost savings** in selected (motivated, equipped) patients · up to ⅓ need formal PT later
✗ Wrong options: ↑ cost · ↓ ROM · 50% need OPPT · OPPT saves money

Modality	Point
Cryotherapy	Local temperature ∝ synovial **PGE2** · ✓ **better ROM at discharge** · ↓ blood loss but not transfusion · (✗ Wrong options: pain, swelling, analgesia · long-term ROM) · AAOS: moderate evidence no outcome benefit
Aquatic	Buoyancy ↓ compressive load + gives resistance
Balance training	Proprioception, postural control
NMES	Overrides CNS-related activation deficits

Opioid use risk factors

↑ post-op opioid use — ✓ **cigarette smoking** · **pre-op opioid** (chronic use) or **benzodiazepine** use · **↑ BMI** · psychiatric comorbidity · **younger age (<55)** ·
✗ Wrong options: valgus alignment · robotic TKA may ↓ need

Return to sport

Preferred — **low-impact closed chain**: elliptical, **biking**, **golf** · **running discouraged** (impact ↓ implant longevity)

Golf after TKA — ✓ **handicap change <5 strokes** · play frequency usually ↑ · drive length ↓ **~12 yd after TKA** · **↑ ~3 yd after THA** · lead (opposite-handed) knee stressed
✗ Wrong options: unable to return · fewer rounds · >20 yd longer

Driving

– – Active space – –

Side	Return
Right TKA	**4 weeks**
Left TKA	**<4 weeks**

How it's asked

"Cryotherapy after TKR — demonstrated effect?"	**→ improved ROM at discharge**
"Self-directed home PT vs outpatient PT?"	**→ similar clinical outcomes**
"Golf after TKA — true?"	**→ handicap change <5 strokes**
"Variable linked to ↑ post-op opioid use?"	**→ cigarette smoking**

– – Active space – –

TKA PROSTHESIS DESIGN

– Recon · 57 / 65 –

1 Basics

History

Era	Milestone
19th c	Soft-tissue interposition
1950s	**Walldius** – first **hinged** knee
1958	**MacIntosh & McKeever** – acrylic tibial plateau
1960s	**Gunston** – first cemented surface arthroplasty
1970	**GUEPAR** hinge (↑ motion, ↓ bone loss) · first-gen hinges → **high loosening from over-constraint**
~1973	**Total condylar** – first to resurface all 3 compartments · PCL sacrificing, no cam-post → flexion ~**90°**
Later	**Duo-patellar (kinematic) = CR** · **Insall–Burstein PS** · LCS rotating platform · medial pivot · ultracongruent (TO)

Concepts in prosthetic design

Femoral rollback ✓ **posterior translation of the femur with flexion** → ↑ quad lever arm + flexion by **preventing posterior impingement** in deep flexion (TO) · native: controlled by **ACL + PCL** · CR: native PCL · PS: **tibial post contacts femoral cam**

Constraint V/V + F/E stability despite **ligament laxity or bone loss** · more constraint → more force to fixation → **aseptic loosening**

Least → most constrained
Bicruciate-retaining (keeps ACL; ↑ op time + complications) → **CR** → **PS** → **VVC / CCK (non-hinged)** → **rotating hinge**

Constraint classification (TO)

Class (TO)	Ligament status	Designs
Unconstrained	Collaterals **+ PCL intact**	**CR**
Semi-constrained	Collaterals intact, **PCL removed**	CS, **PS**, **UC**
Constrained	**Collaterals deficient**	CCK / hinge

Ultra-congruent (deep dish) substitutes PCL **without cam–post** (TO) · rollback produced by PCL / cam–post / deep dish (TO)

Design limitations (TO) total condylar (Insall + Ranawat): cupped femur, flat tibia, **flexion < 90°, no rollback** · duo-patellar: **dependent on intact PCL** · Insall–Burstein PS: needs **femoral notch (box) cut** · LCS rotating platform: complex technique · medial pivot: closest to native kinematics

Modularity metal tray + modular insert, augments, stems · customise intra-op · but ↑ **osteolysis** + **backside wear** (micromotion insert–tray) · more expensive

2 Mechanism

Kinematics (medial pivot, paradoxical motion)

Native knee ✓ **femur pivots about a medial axis** in flexion (lateral condyle rolls back more: larger radius / convex lateral plateau) · flexion → **femoral ER (tibial IR)** + rollback · terminal extension → **femur IR, lateral condyle translates anteriorly** (screw-home = tibial ER last 15°) · lateral condyle posterior **12.7 mm** vs medial **2.9 mm** (TO)

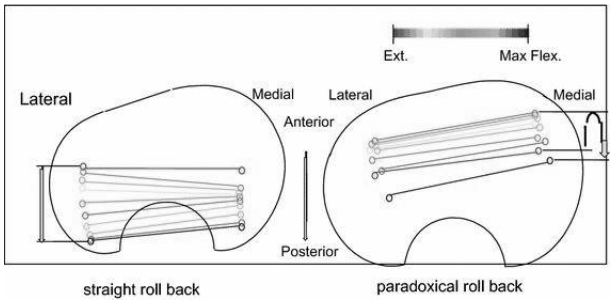
Paradoxical translation ✓ **anterior translation of tibiofemoral contact point from extension → flexion** · commonest in **CR** (no ACL, posterior resting point; PCL not tight until ~**45°**) · also PS before cam engages · CR shows it in first **~40°**

Medial-pivot / medial-stabilised / medial congruent (MC) ✓ **ball-and-socket medial** (**spherical single-radius MFC**), less congruent lateral → lateral rollback · asymmetric liner · raised anterior lip · higher PROMs in some studies

NOT true of medial pivot (TO) ✓ **"spherical variable radius MFC"** (it is **single** radius)

Flexion → extension in medial-congruent design → ✓ medial pivot, femur internally rotates + translates anteriorly

Insert truths ✓ **standard CR can show paradoxical anterior femoral translation** · AS lip prevents femur rolling **forward** when PCL weak (✗ Wrong options: replaces ACL) · **dual-pivot** = lateral pivot early, **medial pivot deep** · mobile bearing (✗ Wrong options: medial pivot) has insert–tray motion · PS **promotes** rollback (✗ Wrong options: minimises)



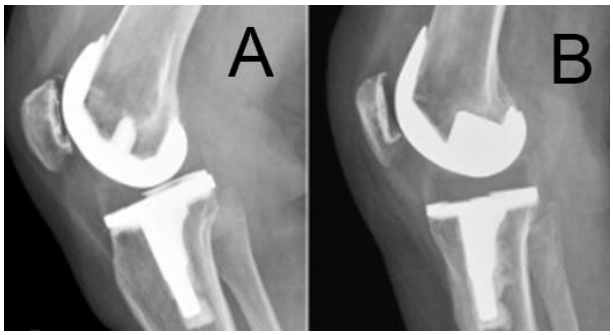
Paradoxical anterior translation

4 Diagnosis & Classification

Key radiology findings

CR vs PS ✓ **PS shows box / cam outline** on lateral X-ray · CR has none (notch recess) (TO: A is CR, B is PS)

– – Active space – –



CR (A) vs PS (B) — box visible in PS

5 Management

Cruciate-retaining vs posterior-stabilised

	CR	PS (cruciate-substituting)
Needs	**Intact functional PCL**	PCL **sacrificed** → flexion gap ↑
Indications	Minimal bone loss/laxity · **varus <10°**, **valgus <15°**	**Patellectomy** · **inflammatory arthritis** (late PCL rupture) · deficient PCL · severe deformity (allows **lateralisation** of components → better tracking) (TO)
X-ray	**No box**	**Box / cam**
Advantages	No post impingement/dislocation · less distal femur cut · **proprioception** · closer kinematics (debated)	Easier balance without PCL · arguably more ROM · easier exposure · ✓ **tolerates joint-line change better**
Disadvantages	Tight PCL → **accelerated PE wear** · loose PCL → **flexion instability** · ↑ PF contact pressure · **does not tolerate joint-line alteration** (TO)	**Cam jump** · **tibial post wear** · ✓ **patellar clunk** · **more bone resected** · ↑ **intra-op fracture** (box)
Survivorship / flexion	**Equal**	**Equal**
Cementless	Available	Available

Low-volume surgeons (<20 TKA/yr) use PS (CR balancing demanding) (TO)

PCL resection simulates → ✓ excessive posterior femoral resection (↑ flexion gap)
✗ Wrong options: distal resection (loose extension) · augments/oversize (tight flexion)

CR — recreate PCL tension EXCEPT → ✓ increasing posterior tibial slope (loosens flexion, under-tensions PCL) · do: match AP femoral size, conservative distal cut + soft-tissue release, anatomic rotation (femoral IR / tibial ER alter tension most), resect only what is replaced

PCL injured intra-op, gaps balanced → CR with conforming (UC/AS) insert → ✓ no difference in functional outcomes vs PS
✗ Wrong options: flexion instability · paradoxical translation (only with standard insert) · clunk · PJI

Patellectomy → ✓ PS (AP instability with weak extensor) · gold standard, though newer CR data similar
✗ Wrong options: UKA · CR · constrained

Best CR candidate → ✓ OA with 10° valgus
✗ Wrong options: RA · patellectomy · bicruciate rupture · steroid-treated SLE

PS vs CR – true → ✓ higher patellar clunk
✗ Wrong options: less bone · better survivorship · fewer fractures · cementless only

Flexed femoral component + excess tibial slope (PS) → ✓ excessive tibial post wear (anterior post impingement → osteolysis)
✗ Wrong options: recurvatum · coronal instability · clunk · patellar instability



PS knee – cam and box on lateral

PS complications

Complication	Mechanism	Treatment
Cam jump	**Loose flexion gap** or hyperextension	Closed reduction (**anterior drawer at 90°**) → revise flexion gap
Post wear	Femoral flexion, slope, hyperextension	Avoid excess distal resection, flexion, slope
Patellar clunk	Scar nodule (posterior distal quad tendon / superior pole) catches in box ~**30–45°** extension · small femur, flexed femur, undersized button · within 12 mo	Observe or **arthroscopic/open excision** · longer trochlear groove designs ↓

Constraint selection (VVC, hinge)

Design	Mechanism	Indications
CCK / VVC (constrained non-hinged)	**Large tall tibial post in deep femoral box**, no axle · V/V + rotational stability (partial) · functions as PS otherwise	**MCL or LCL attenuation/deficiency** · **V/V laxity >5 mm** · **flexion gap laxity** · **moderate bone loss in neuropathic** knee · **severe deformity (>20°) with collateral incompetence** · irreparable intra-op MCL · post-polio (some) · stems advised
Rotating hinge	**Linked** components (axle) · tibial bearing rotates on yoke (↓ constraint) · **extension stop**	✓ **Global ligament deficiency** · **hyperextension instability / recurvatum (>5°)** (polio, tumour) · tumour resection · **massive bone loss in neuropathic joint** · **gross flexion–extension mismatch (>15 mm)** despite releases/upsizing · chronic MCL loss in **elderly low-demand** · periprosthetic fracture / DFR

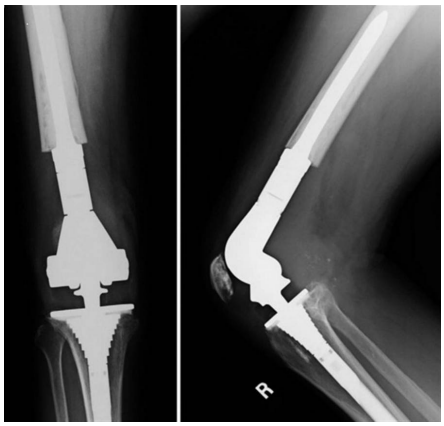
CCK cannot control hyperextension → ✓ only hinge for hyperextension deformity (TO) · hinge **can** be used in primary TKR (TO) · CCK is not "deep dish + small post"

CCK identification picture of large post = **CCK** (TO) · CCK = "semi-hinge", low-budget back-up for on-table collateral transection (TO) · TO also states on-table **MCL transection → hinge ideal** vs OB: repair + CR/PS + brace or CCK – read the options

– – Active space – –

Both constrained ↑ femoral bone resection · ✓ **aseptic loosening** from constraint (first-gen **GUEPAR hinge** → high loosening)

Scenario	Answer
Charcot valgus, still medial lax after ITB/popliteus/LCL release + thicker poly	✓ **convert to constrained TKA** (✗ Wrong options: lateral retinacular release ± MPFL · MCL imbrication · valgus recut)
Diabetic **Charcot knee** – initial implant	✓ **hinge / constrained** (✗ Wrong options: CR · PS · fusion · UKA)
Charcot, PS used → instability, maltracking, AP laxity	✓ **should have used hinged TKA**
Charcot valgus with bone loss – counsel	✓ **↑ constraint needed for global instability** (↑ loosening risk; cones + long stems)
84 yo, grossly attenuated MCL (valgus)	✓ **VVC** (96% at 10 yr) (✗ Wrong options: MCL repair/scaffold + PS · LCL release · lateral UKA)
Obese, **MCL-deficient** revision	✓ **constraint resisting varus–valgus loads**
Absolute indication for non-hinged constrained	✓ **varus >20° with LCL incompetence** (✗ Wrong options: arthrodesis conversion · hyperextension (hinge) · repairable MCL · UKA revision)
Post-polio, hyperextension 15°, global instability	✓ **hinged TKA** (✗ Wrong options: CR + reconstruction · PS · DFO · AKA)
Polio, quad weakness, valgus + recurvatum	✓ **rotating hinge** (extension stop) (✗ Wrong options: PS with under-resection · VVC)
CCK kept available because	✓ **27° valgus** (MCL likely incompetent) (✗ Wrong options: RA · 22° recurvatum (hinge) · LLD · patellar dislocation)
Strongest hinge indication	✓ **aseptic revision with >15 mm F/E gap mismatch despite femoral upsizing** (✗ Wrong options: 15° deformity + erosion (wedges) · after dynamic spacer (collaterals intact → CCK) · baja)
Elderly RA, 20° FFD, valgus laxity, + posterior drawer, low demand	✓ **cemented rotating hinge + metaphyseal cones** (✗ Wrong options: press-fit · MCL reconstruction)
Implant for polio genu recurvatum	✓ **rotating hinge** (✗ Wrong options: intra-op LCL/MCL disruption · diabetic neuropathy · aseptic loosening)
CCK on X-ray – indication	✓ **MCL insufficiency + 13° valgus** (✗ Wrong options: hyperextension · 6° valgus with 2 mm laxity (CR) · unresurfaced patella · PCL over-resection (UC or PS))
Rotating hinge figure – constrained motions	✓ **partial V/V + partial IR/ER** · use least constraint that gives stability
Overconstrained, high loosening	✓ **GUEPAR (first-gen hinge)**



Rotating hinge

Mobile bearing & polyethylene conformity

Mobile bearing
PE **rotates on tibial baseplate** · PCL removed · ↑ contact area → theoretically ↓ **PE wear + ↓ bone–implant stress** · relative indication: young active · ✓ **no improvement in survivorship** vs fixed (flexion 112° vs 108°; recovery, strength, wear not proven)
✗ Wrong options: 15° more flexion · faster recovery

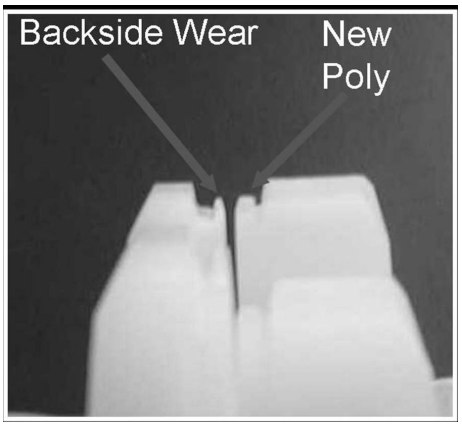
Bearing spin-out
✓ mobile-bearing-specific · **loose flexion gap** → tibia rotates behind femur · lateral X-ray without cam outline (TO) · closed reduction at 90° + anterior drawer → revise flexion gap

↑ Conformity (fixed bearing)
✓ ↓ **contact stress** in PE · ✓ ↓ **femoral rollback** · ↑ shear to fixation
✗ Wrong options: ↑ cracking · ↑ delamination · ↓ loosening · PF stability change

Modular vs all-polyethylene tibia

	Modular (metal tray + insert)	**All-polyethylene (monoblock)**
Wear	✓ **More backside wear** (articular side unchanged)	No backside wear → ✓ **less osteolysis**
Cost	Higher	✓ **Less expensive**
Flexibility	Poly exchange possible · sizing after cementing	Size chosen before cementing · full tibial revision if problem
Outcomes	–	**Equivalent function** (2–10 yr) · equivalent loosening (Orthobullets) · recent: better survival, ↓ infection, instability, loosening, fracture (OB) · OK in BMI <37.5
Load distribution	More even (biomechanics)	–
Prep / bone removal / stability	Same	Same

– – Active space – –



Backside wear on modular insert

Materials & component geometry (TO)

Item	TO fact
Femoral component	**CoCr** (strength + polish) · zirconium / ceramic = hypoallergenic
Tibial tray	**Titanium** – modulus close to bone
PE	**6th-generation cross-linked UHMWPE + vitamin E** → best wear / longevity
Single-radius femur	Constant curvature → consistent rollback, **greater flexion**
Multi-radius (J-curve) femur	Mimics native knee · smooth ligament tension
Deep trochlear groove	Better **patellar tracking**
Symmetric tibia	Easier rotational alignment, less anatomic
Asymmetric tibia	Matches anatomy, **better coverage**
Insert anterior bevel	Prevents **patellar tendon impingement**
Size matching	Femoral + tibial sizes may differ by **± 1 size only**

Posterior condylar offset (TO) posterior femoral cortex → posterior condyle · **↑ offset = ↑ flexion**

High-flex design (TO) **high posterior condylar offset** · slope **~7°** · deep trochlea · bevelled anterior insert · balanced gaps → **≥ 130° flexion**

Bone cement (TO) **Simplex = white** (standard PMMA) · **Palacos = green** (faster setting, high visibility) · antibiotic cement only if infection risk

Fixation – cemented vs cementless, cement viscosity

Most consistent tibial fixation ✓ **cement metaphysis + keel**
✗ Wrong options: cement plateau only + press-fit keel → ↑ loosening

Cementless trabecular/porous ingrowth · ✓ **equivalent functional outcomes** (RCTs to 2 yr) · **shorter** op time, ↑ cost · no difference in femoral loosening, DVT, Hb drop · ideal: ✓ **young, active, obese** (e.g. 52 yo, BMI 44)
✗ Wrong options: **elderly** (esp. elderly white females), **osteoporosis**, **irradiated bone**, **inflammatory arthritis**, metabolic bone disease

High-viscosity cement ✓ **longer working time** · shorter mixing/waiting · bone penetration equal or less · loosening data conflicting (possibly ↑) · pulmonary complications same

Patellar resurfacing

Routine resurfacing vs retention ✓ ↓ **revision risk** (retention: **20–30%** ↑ revision in registries; 5.7% revised for anterior pain)
✗ Wrong options: no complications · ↑ anterior pain · ↓ fracture

TO lecture notes **do not resurface routinely in OA** · resurface for **RA / crystal arthropathy (gout)** or severely damaged cartilage · routine resurfacing ↓ revision (current chapter) – read the options

Gender-specific implants

Women with standard components ✓ **greater implant survivorship** than men · no evidence of inferior outcomes · female designs: narrower ML, thinner anterior flange, ↑ trochlear angle

Metal allergy

Nickel allergy → ✓ **oxidised zirconium femur + titanium (or all-poly) tibia** · CoCr + stainless contain nickel · **titanium** unsuitable as articulating femur (metallosis)

Patient-specific instrumentation

PSI custom guides from CT/MRI · ✓ **no IM femoral hole needed** (also navigation) → theoretical ↓ fat embolism (TO) · claimed ↓ time, blood loss, trays (TO)

Only proven benefit (OB) ✓ **fewer instrument trays** · (✗ Wrong options: cost · alignment · function · pain/satisfaction) · custom guides → **no difference in ROM, alignment, outcome scores**

Highly congruent liners & minimum thickness

Highly congruent (medial-pivot) liners medial concavity + lateral translation · equivalent short/mid-term outcomes + survival

Minimum PE insert thickness ✓ **8 mm** (TO)

X vs Y

	Cam jump (PS)	**Bearing spin-out (mobile)**
Cause	Loose flexion gap / hyperextension	Loose flexion gap
X-ray	Box visible	No cam outline
Reduction	90° + anterior drawer	90° + anterior drawer
Definitive	Revise flexion gap	Revise flexion gap

	CCK / VVC	**Rotating hinge**
Link	None	**Axle**
Hyperextension	Not controlled	✓ Extension stop
Collaterals	Attenuated / one deficient	**Globally absent**

– – Active space – –

How it's asked

"CR vs CS – true?"	→ **CR does not tolerate joint-line alteration** (TO)
"Post-op X-rays A and B?"	→ **A is CR, B is PS** (TO)
"Prior patellectomy, degenerative knee – implant?"	→ **posterior stabilised TKA**
"Who benefits most from CR TKA?"	→ **OA with 10° valgus**
"Patellectomy, stable ligaments – most predictable implant?"	→ **PCL-substituting knee**
"Recreate PCL tension – EXCEPT?"	→ **increasing posterior tibial slope**
"PCL resection simulates?"	→ **excessive posterior femoral resection**
"PS vs CR – true?"	→ **higher incidence of patellar clunk**
"Flexed femur + excess slope – predisposes to?"	→ **excessive tibial post wear**
"PCL injured, CR with conforming insert – outcome?"	→ **no difference vs PS**
"Constrained TKR – true?"	→ **only hinge for hyperextension deformity** (TO)
"Severe valgus – prosthesis allowing component lateralisation?"	→ **PS** (TO)
"Implant in picture?"	→ **CCK type** (TO)
"Charcot valgus, still lax after releases?"	→ **convert to constrained TKA**
"Rotating hinge – constrained motions?"	→ **partial V/V + partial IR/ER**
"Charcot knee – initial implant?"	→ **hinge (Figure B)**
"High loosening from overconstraint?"	→ **GUEPAR hinge (Figure C)**
"Charcot, PS used, unstable – should have used?"	→ **hinged TKA**
"84 yo, attenuated MCL?"	→ **varus–valgus constrained TKA**
"Obese MCL-deficient revision – essential feature?"	→ **constraint resisting varus–valgus loads**
"Absolute indication for constrained non-hinged?"	→ **varus >20° with LCL incompetence**
"Post-polio, hyperextension, global instability?"	→ **hinged TKA**
"Polio recurvatum – technique?"	→ **rotating hinge**
"CCK available – clinical factor?"	→ **27° valgus**
"Strongest rotating-hinge indication?"	→ **revision with >15 mm F/E mismatch despite upsizing**
"Elderly RA, MCL + PCL incompetent?"	→ **cemented rotating hinge + metaphyseal cones**
"Indication for hinge on X-ray?"	→ **poliomyelitis with genu recurvatum**
"CCK on X-ray – indication?"	→ **MCL insufficiency + 13° valgus**

"Charcot valgus – discuss?"	→ **increased constraint for global instability**
"Lateral view – what complication?"	→ **bearing spin-out in mobile-bearing TKA** (TO)
"Modular vs monoblock tibia?"	→ **more backside wear with modular**
"Design that reduces PE wear + interface stress?"	→ **mobile-bearing TKA**
"↑ Conformity – effect?"	→ **decreased contact stress**
"Mobile vs fixed bearing?"	→ **no improvement in survivorship**
"↑ Conformity – fixed bearing effect?"	→ **decreased femoral rollback**
"All-poly vs metal-backed tibia?"	→ **less expensive**
"Monoblock advantage?"	→ **less osteolysis**
"True about femoral rollback?"	→ **posterior femoral translation preventing posterior impingement** (TO)
"Medial pivot – NOT true?"	→ **spherical variable-radius MFC** (TO)
"Normal knee kinematics extension → flexion?"	→ **femur pivots about a medial axis**
"Medial-congruent design, flexion → extension?"	→ **medial pivot, femur IR + anterior translation**
"Paradoxical translation?"	→ **anterior contact-point translation with flexion**
"Insert facilitating lateral condyle rollback?"	→ **medial congruent**
"Tibial insert designs – true?"	→ **standard CR can cause paradoxical anterior translation**
"Most consistent tibial fixation?"	→ **cement metaphysis and keel**
"Cementless TKA – true?"	→ **equivalent functional outcomes**
"High vs low viscosity cement?"	→ **longer working time**
"Ideal cementless candidate?"	→ **52 yo active male, BMI 44**
"Routine patellar resurfacing associated with?"	→ **decreased risk of revision**
"Women with standard components?"	→ **greater implant survivorship**
"Nickel allergy – implant?"	→ **titanium tibia + oxidised zirconium femur**
"IM femoral hole not needed in?"	→ **patient-specific instrumentation** (TO)
"PSI vs standard – true?"	→ **fewer instrument trays**
"Custom cutting guides – outcome difference?"	→ **no difference in ROM, alignment or scores**
"Minimum PE insert thickness?"	→ **8 mm** (TO)
"Increasing posterior condylar offset – effect?"	→ **↑ flexion** (TO)

– – Active space – –

"Deep-dish insert substitutes PCL how?" → **congruency, no cam–post** (TO)

– – Active space – –

TKA REVISION

– Recon · 58 / 65 –

1 Basics

Causes of failure (loosening, wear, osteolysis, metallosis)

Cause	Note
✓ **Infection**	**Most common overall** (national databases, TO) · ✓ **most common early (<2 yr)** · always exclude first · **neutrophil**-predominant
Aseptic loosening	**2nd overall** · ✓ **most common late (>2 yr)** (OB5949: late = **>10 yr**) · **tibial > femoral** · femoral lysis hidden (posterior condyles) → oblique views, **serial X-rays**
Instability	**~8%** · MCL/LCL incompetence · flexion instability (PCL attenuation in CR, unbalanced gap, ↑ slope, small femur, femur in extension)
Periprosthetic fracture	**~5%** (3–5%) · supracondylar femur commonest
Arthrofibrosis	**~5%** (0–3% OB) · **fibroblasts**
PF maltracking	**Component malposition**
Joint-line problems · patellar clunk (arthroscopic excision) · metal hypersensitivity (**lymphocytes, plasma cells**)	–

TO top four **infection → aseptic loosening → periprosthetic fracture → instability** (TO)

Trends ✓ **PE wear no longer the major cause** (HXLPE) · instability/malalignment share ↓ · stiffness common early
✗ Wrong options: infection commonest late · loosening commonest early

– – Active space – –

Osteolysis PE, PMMA, metal particles → **macrophages + giant cells** · femur: condyles near collateral attachments · tibia: periphery + cancellous access channels · commonest in **uncemented** + **backside wear** · modern cementless ↓ aseptic revision in morbidly obese

Scenario	Answer
15 yr, pain + instability, osteolysis + loosening, infection excluded	✓ **revision TKA** (✗ Wrong options: protected WB · bisphosphonate · observe · liner exchange + graft)
15 yr, cemented PS, HXLPE 13 mm — likely cause of revision	✓ **aseptic loosening**
16 yr, well-defined femoral metaphyseal lysis, medial narrowing	✓ **particle-mediated osteolysis** (✗ Wrong options: metastasis · stress shielding · sepsis · sarcoma)
13 yr, metal-backed patella, squeaking/grinding	✓ **extensive metallosis** (worn-through PE → metal-on-metal, black tattooing)
12 yr, femoral migration into varus, normal labs	✓ **polyethylene wear** (commonest contributor to late lysis/ loosening)
6 yr, tibia 5° varus, medial PE narrowing, lateral laxity	✓ **revise all components + soft-tissue balancing** (✗ Wrong options: poly exchange · isolated tibial (PCL then incompetent))
Commonest early revision — cell lineage	✓ **neutrophil** (✗ Wrong options: macrophage (PE lysis) · lymphocyte/plasma (metal allergy) · fibroblast (arthrofibrosis))

3 Presentation

History original indication · pre-op ROM, ambulation · infection, thrombophlebitis, falls · **THA** · implant records

Temporal pattern pain **since index** vs new pain (acute vs chronic infection) · **weight-bearing pain → mechanical**

Exam gait (stiff-legged, no full extension in stance) · ROM · skin, effusion, warmth (**infection vs CRPS**) · ligaments · patellar tracking

4 Diagnosis & Classification

Key radiology findings

Test	Use
Serial AP + lateral	Timeline of migration/loosening
Weight-bearing films	Asymmetric wear
Skyline	Patellar tracking
Standing long-leg	Alignment
AP pelvis	Exclude hip pathology
CT	Component rotation (vs femoral neck version) · bone defects
Bone scan	**Positive up to 2 yr** after TKA · positive = nonspecific (loosening, infection, stress fracture) · ✓ **negative rules out loosening** · diffuse uptake → **CRPS**

Genes & lab markers

CBC, ESR, CRP + aspiration (cell count, culture) before any revision

Antibiotics already started → ✓ stop antibiotics and repeat aspiration
✗ Wrong options: continue 2 wk · add gram-negative cover · revise · PT

Painful nonunion after periprosthetic ORIF with effusion + loosening screws → ✓ inflammatory labs + aspiration first
✗ Wrong options: PT · bone stimulator · revision ORIF / DFR before infection excluded

AORI (Anderson) bone defect classification

Type	Description	Treatment
1	Minor defects, **intact metaphysis**, stability not compromised	**Cement** or **impaction allograft** (no metaphyseal augmentation)
2A	Metaphyseal damage, **one** condyle/plateau	Cement, **augments**, small graft
2B	Metaphyseal damage, **both** condyles/ plateaus	Cement, augments, small graft
3	**Massive** loss of condyle/plateau ± **collateral / patellar tendon** attachments	**Bulk allograft**, custom, megaprosthesis, **tantalum cones**, **sleeves**, **rotating hinge**

AORI by size (TO notes) **I < 5 mm, contained** → cement / graft · **II 5–15 mm, uncontained** → structural graft / augments · **III > 15 mm, uncontained** → sleeves / cones / hinge

Anderson Orthopaedic Research Institute (AORI) Classification

	Description	*Treatment*
Type 1	Minor bone defects with intact metaphyseal bone that do not compromise stability	Cement fill or impaction allograft
Type 2A	Metaphyseal bone damage that involves 1 femoral condyle or tibial plateau	Cement fill, augments, small bone graft
Type 2B	Metaphyseal bone damage that involves both femoral condyles or tibial plateaus	Cement fill, augments, small bone graft
Type 3	Massive bone loss comprising a large portion of condyle/plateau, and can involve the collateral ligaments/patellar tendon	Bulk allografts, custom implants, megaprosthesis, porous tantalum, metaphyseal sleeves, rotating hinge

AORI classification

5 Management

Goals & steps

Goals extract with minimal bone loss · restore bone · ✓ **restore joint line** · balance ligaments · stable implants · soft-tissue cover

Sequence extensile exposure → remove implants → **tibia first** (establish tibial joint line **1.5–2 cm above fibular head**, use contralateral X-ray) → femur matched to tibia → **flexion–extension gaps** → **medial–lateral gaps** → PF tracking (patella **>12 mm**)

– – Active space – –

Exposure (rectus snip, tibial tubercle osteotomy)

Rectus (quad) snip extension of MPP superolaterally across rectus tendon · ✓ **no difference in outcomes** vs MPP · ✓ **enhances exposure without changing rehab or outcomes** · **most common** extra-exposure method
✗ Wrong options: lateral arthrotomy (devascularises patella) · full MCL release · patellectomy · mobile bearing

TTO **8–10 cm** medial osteotomy hinged on lateral periosteum · ✓ **patella baja** (contracted tendon; prevents avulsion; allows **proximal transfer**) · well-fixed long stems/sleeves/cones · RA, valgus, anteromedial scarring · ✓ **very low nonunion (~100% union)** · fixed with screws or **16-gauge wires/cerclage** (long stems do not preclude) · OK in infected/staged cases · ROM restricted 4–6 wk · used if snip insufficient (V-Y → lag)
✗ Wrong options: well-fixed standard baseplate alone · slope · CR · BMD predicts

Figure	TTO needed?
✓ **Patella baja**	**Yes**
Posterior dislocation, PT avulsion (alta), spin-out, recurvatum	No

Post-HTO baja, PJI explant, flexion 65°, stiff after release → ✓ TTO (best exposure, preserves patellar supply)
✗ Wrong options: snip + lateral release (lateral geniculate) · V-Y turndown (blood supply) · snip alone (insufficient) · modified V-Y

Bone loss (augments, cones, sleeves, stems)

Stems load-share to diaphysis · bypass metaphyseal defects

	Press-fit diaphyseal stem	Cemented stem
Pros	Scratch fit, aids alignment, no cement to remove later	**Bowed femur**, antibiotic delivery, **osteopenic bone**
Cons	No ingrowth · **iatrogenic fracture** · not for bowed femur · ✓ **end-of-stem pain** (modulus mismatch)	Harder future revision

Shin pain at stem tip after hybrid (press-fit) stem → ✓ fully cemented stem would have prevented
✗ Wrong options: hinge · spacer · teriparatide · short press-fit

Medial baseplate overhang with long stem → pain → ✓ offset tibial stem
✗ Wrong options: ingrowth augment · no stem · allograft · cement

– – Active space – –

Defect	Option
Cavitary <1 cm	**Cement** (structurally better than graft)
Cavitary >1 cm, contained	**Metaphyseal sleeve**
Large, uncontained / asymmetric	✓ **Trabecular (tantalum) cone** + press-fit diaphyseal stem (intact diaphysis)
Small asymmetric uncontained	Metal augment
Small symmetric contained	Impaction graft
Large symmetric contained + poor diaphysis	Sleeve + **cemented** stem
Elderly, massive tibial loss incl. posterior cortex	✓ **metal-augmented revision tibia** (structural allograft for younger) (✗ Wrong options: hinge · ICBG · cement)

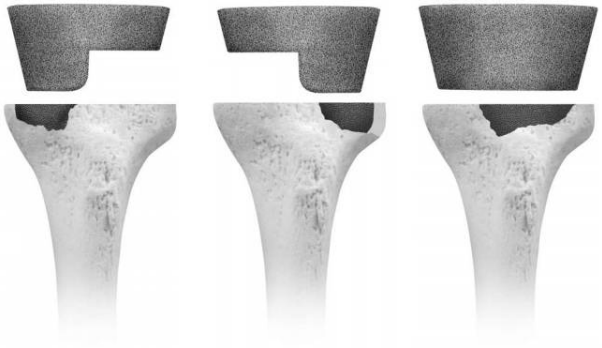
	Metaphyseal sleeves	Trabecular metal cones	Structural allograft
Data	Mid–long term good	Short–mid term good	Mid–long term OK
Link to implant	✓ **Morse taper** (modular, attached to stem/tray)	**Independent**, unitised with cement	Cemented
Vendor	**Manufacturer-specific**	✓ **Any system**	–
Defect	**Contained only** (✗ Wrong options: large uncontained)	✓ **Uncontained** OK	Custom shape
Other	Simple, acts as cutting guide · expensive, hard to remove	**Brittle**, can be cemented to bone if scratch fit poor · soft-tissue irritant · expensive	Time, disease transmission, **resorption**, infection

Porous cones vs cortical allograft → ✓ higher long-term survival
✗ Wrong options: implant-specific · ductile Morse taper (sleeves) · cannot be cemented · AORI 1

Cone purpose → ✓ ↑ metaphyseal contact for osseointegration
✗ Wrong options: antibiotics · diaphyseal fit · no cement · easier future revision

Stage 2, massive metaphyseal defects, intact collaterals, healed TTO → ✓ uncemented porous metaphyseal sleeves + uncemented stems
✗ Wrong options: impaction graft · allograft without stem/cement · "diaphyseal" defects

Sleeve relative contraindication → ✓ large uncontained metaphyseal defect
✗ Wrong options: hinge · stem · RA · capacious diaphysis (sleeve helps)



Trabecular metal cone

– – Active space – –

Constraint in revision (VVC, hinge)

Implant	Indication
CR	Intact PCL (keep PS available)
PS	PCL deficiency
Constrained non-hinged	V/V instability, LCL/MCL attenuation (MCL controversial — post may break), flexion laxity (tall post)
Rotating hinge	**Global ligament deficiency** · flexion laxity with mismatch · multiply revised / post-traumatic · ✓ **hyperextension (polio)** — absolute · tumour / infection resection · Charcot (relative) · **collateral origin bone loss**
DFR	Salvage: fracture, massive bone loss

Rule **least constraint** that gives stability

Scenario	Answer
1990 TKA, two-component osteolysis + instability	✓ **revise femur + tibia with stems/augments** (✗ Wrong options: tibia only ± LCL reconstruction · immobiliser · no stems)
Irreparable saw MCL transection, valgus laxity >1 cm (RA)	✓ **unlinked VVC** (least interface stress) (✗ Wrong options: PS fixed/mobile · CR · hinge)
80 yo, MCL deficient + anterior drawer + no SLR (multiplanar)	✓ **hinged revision** + EM repair (✗ Wrong options: poly upsize · MCL reconstruction · femoral only · high-post)
Absolute indication for rotating hinge	✓ **hyperextension instability** (✗ Wrong options: LCL attenuation (CCK) · EM deficiency · PCL deficiency (PS) · contained condylar defect (AORI 2A))
Stage 2 — strongest hinge indication	✓ **medial femoral cortical loss involving medial epicondyle** (MCL origin) (✗ Wrong options: cancellous loss (cones/sleeves) · cortical loss sparing collaterals (augments))



Rotating hinge revision

Flexion instability

Revision options ✓ **upsize femoral component** · ↓ tibial slope · more distal femur + thicker poly · correct rotation
✗ Wrong options: ↑ slope · more posterior femur · more tibia (both gaps) · distal augments (extension only)

Infection-related (cement, host, workup)

Stage 2 for MRSA → ✓ **pre-mixed gentamicin cement + hand-mixed vancomycin** · pre-mixed better elution + mechanics · FDA: antibiotic cement approved only for **stage 2 of two-stage**; primary use off-label
✗ Wrong options: plain cement

Failure of repeat two-stage → ✓ **MSIS host grade C** (none cleared) → consider **fusion / amputation** · also MRSA, polymicrobial
✗ Wrong options: WBC >25k · anaerobic cocci (low virulence) · semi-constrained · static spacer

Revision infection risk **4–7%** (double primary) · ↑ with MSIS C · PJI 1–2% primary, up to 4% revision

Cementless TKA indications

Best evidence → ✓ **younger, obese males** (↓ tibial aseptic loosening, HA-coated; short–mid term)
✗ Wrong options: osteopenic · RA · elderly ♀ · elderly smokers

Wound & extensor problems

Skin incorporate prior scars · supply medial → lateral edge hypoxic → **use most lateral incision** · full-thickness defect → **medial gastrocnemius flap**

iNPWT ✓ **revision TKA** clearly benefits (↓ wound complications)
✗ Wrong options: UKA · primary TKA · primary THA any approach

EM disruption Achilles bone-block allograft (residual lag common) · **mesh** may give better mid-term function + survival

Other pain worse than primary (activity pain ~6 mo) · stiffness · peroneal palsy (valgus + FFD correction)

Distal femoral replacement outcomes

DFR for fracture around primary components → ✓ **lower revision rates** than DFR after prior revision components or failed ORIF

X vs Y

	Early failure (<2 yr)	Late failure (>2 yr)
Commonest	**Infection**	**Aseptic loosening**
Cells	Neutrophils	Macrophages (wear)

	Sleeve	Cone
Defect	Contained	**Uncontained**
Attachment	Morse taper, vendor-specific	Independent, any vendor

How it's asked

"Oblique rectus snip vs MPP in revision?"	→ **no difference in outcomes**
"Enhance exposure without changing rehab?"	→ **extensile rectus snip**
"Which X-ray most likely needs TTO?"	→ **patella baja (Figure D)**

– – Active space – –

"Post-HTO baja, PJI explant — best exposure preserving patellar supply?"	→ **TTO (Fig D)**
"Radiographic predictor of needing TTO?"	→ **patella baja**
"True about TTO?"	→ **very low nonunion rates at 1 yr**
"Stage 2, metaphyseal defects, intact collaterals?"	→ **uncemented metaphyseal sleeves + uncemented stems**
"Medial tibial overhang pain with long stem — prevention?"	→ **offset tibial stem**
"84 yo, massive tibial loss — management?"	→ **metal-augmented revision tibia**
"Relative contraindication to metaphyseal sleeve?"	→ **large uncontained metaphyseal defect**
"Tantalum cone + press-fit stem — appropriate for?"	→ **large asymmetric uncontained metaphyseal defect, intact diaphysis**
"Highly porous cones — true?"	→ **higher long-term survival than cortical allograft**
"End-of-stem shin pain — prevention?"	→ **fully cemented stem**
"Purpose of metaphyseal cone?"	→ **increase metaphyseal contact for osseointegration**
"Two-component osteolysis + instability?"	→ **revise both with stems and/or augments**
"Irreparable MCL transection — least interface stress?"	→ **unlinked VVC**
"Elderly, MCL deficient, EM deficient, multiplanar instability?"	→ **hinged revision**
"Absolute indication for rotating hinge?"	→ **hyperextension instability**
"Stage 2 — strongest hinge indication?"	→ **cortical loss involving medial epicondyle**
"Most common cause of TKR revision?"	→ **infection** (TO)
"15 yr, osteolysis + loosening, infection negative?"	→ **revision TKA**
"Late revision at 15 yr — likely cause?"	→ **aseptic loosening**
"Well-defined femoral lysis at 16 yr?"	→ **particle-mediated osteolysis**
"Metal-backed patella, squeaking at 13 yr — finding?"	→ **extensive metallosis**
"Varus tibia, medial PE wear, lax laterally?"	→ **revise all components + balance**
"Femoral migration into varus at 12 yr, normal labs?"	→ **polyethylene wear**
"Revision causes vs historical?"	→ **PE wear no longer the major cause**
"Commonest early revision — cell lineage?"	→ **neutrophil**
"Flexion instability revision?"	→ **upsize femoral component**
"Stage 2 MRSA — cement?"	→ **pre-mixed gentamicin + hand-mixed vancomycin**

"Failure of second two-stage — risk factor?"	→ **MSIS type C host**
"Nonunion after periprosthetic ORIF, effusion — next?"	→ **inflammatory labs + aspiration**
"Possible PJI, already on antibiotics, normal aspirate, high CRP?"	→ **stop antibiotics and repeat aspiration**
"Best evidence for cementless TKA?"	→ **younger obese males**
"Procedure clearly benefiting from iNPWT?"	→ **revision TKA**
"DFR for fracture around primary vs prior ORIF?"	→ **lower revision rates**

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TKA SAGITTAL PLANE BALANCING

– Recon · 59 / 65 –

1 Basics

Gap definitions (TO)

Extension gap	✓ **distal femoral cut ↔ proximal tibial cut** (in extension) (TO)
Flexion gap	**posterior femoral cut ↔ proximal tibial cut** (at 90°) (TO)
Goal	gaps **rectangular and equal** (TO)

Rules

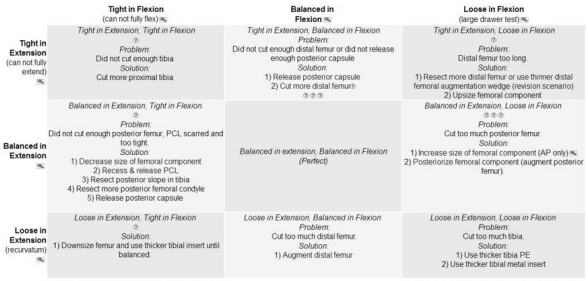
Change	Affects
Distal femoral cut / distal augment	**Extension gap only**
Posterior femoral cut / femoral AP size / posterior augment / AP position	**Flexion gap only**
Tibial cut / poly thickness	**Both gaps**
Tibial slope	**Flexion gap** (↑ slope opens flexion)
Posterior capsule release	Extension
PCL recession / release (CR)	Flexion

Symmetric problem (both tight / both loose) → **tibia / poly** · **asymmetric problem** → **femur**

Femoral AP position	**anterior** shift → ↑ flexion gap · **posterior** shift → ↓ flexion gap (and tightens posterior tissues) · proximal/distal shift → extension gap
Downsizing femur	changes **AP dimension only** (no proximal–distal change) · with anterior referencing → removes more posterior condyle → opens flexion
Gap-balancing technique (TO)	remove osteophytes → precise tibial cut → **release ligaments before femoral cuts**

Sagittal balancing grid (TO + OB)

	Tight in flexion	Balanced in flexion	Loose in flexion
Tight in extension	✓ **Cut more proximal tibia** (or thinner poly)	✓ **Cut more distal femur** · release posterior capsule	✓ **Cut more distal femur + thicker poly** · or more distal femur + **upsize femur**
Balanced in extension	✓ **Downsize femur** (same poly) · ↑ **posterior slope** · CR: **recess / cut PCL** · anteriorise femur	Done	✓ **Upsize femur ± posterior augments** · move femur posteriorly · more distal femur + thicker poly · less slope + thicker poly
Loose in extension	✓ **Downsize femur + thicker poly**	**Distal femoral augment, same poly** (OB) · downsize femur + thicker poly or more slope + thicker poly (TO) – read the options	✓ **Thicker poly**



Gap balancing grid

Tight in extension – resect distal femur

Flexion contracture, flexion balanced → ✓ resect more distal femur (or release posterior capsule)
✗ Wrong options: PCL recession · ↑ slope (flexion problems) · smaller femur · more tibia (loosens flexion) · thinner poly (both) · posterior augments (tighten flexion)

Pre-op 25–95°, after capsule release + osteophytes still 20° short, flexion fine → ✓ resect 2 mm more distal femur
✗ Wrong options: 2 mm tibia · downsize + thicker poly · ↑ slope · femur 4 mm posterior (tightens flexion + posterior tissues)

Flexion contracture steps (OB)
1: ML balance + osteophytes + distal femur over-resected **2 mm** · 2: **posterior capsule release** · 3: distal femur up to **4 mm** more · 4: **hamstring tenotomy** · <15° → steps 1–2 suffice · steps 3–4 for some >35° · residual contracture >15° less likely to improve

Flexion contracture treatments EXCEPT → ✓ downsize femoral component (flexion gap only)

Revision: 2 mm more tibia for 10° contracture → ✓ flexion instability (common pitfall; cut distal femur instead)

Loose in flexion – upsize femur & posterior augments

Causes
excess posterior femoral resection (✓ loose flexion gap) · ↑ slope · undersized femur · **PCL resection** · popliteus injury

– – Active space – –

Loose flexion, balanced extension — EXCEPT (TO) → ✓ downsize femur + thicker spacer · valid: more distal femur + thicker spacer · upsize femur, same spacer · less slope + thicker spacer (TO)

Cruciate-sacrificing, flexion instability, extension fine → ✓ upsize femur + posterior augments (or move femur posterior, or thicker poly then fix tight extension)
✗ Wrong options: downsize femur/tibia · upsize tibia · femur anterior

Loose flexion, stable extension → ✓ move femoral component posterior
✗ Wrong options: thicker poly alone · downsize · anterior + distal augment · ER both

Revision: upsized femur + posterior augments because → ✓ balanced in extension, loose in flexion

CR TKA, AP instability at 90°, femur anterior, liftoff → ✓ revise to larger PS implant
✗ Wrong options: NSAID · brace · PT · thicker poly

Revision with distal augment → dislocated at 1 mo → ✓ flexion gap instability (distal augments treat extension; thin patella / EM weakness aggravates)
✗ Wrong options: joint line · PT rupture · tibial IR · femoral hyperextension

Robotic plan 8 mm, 13 mm posterior condyle removed → ✓ difficulty rising from a chair · flexion instability: recurrent effusions, **quad weakness**, stairs, ↑ (not ↓) terminal flexion
✗ Wrong options: hyperextension · ↓ flexion · coronal instability in extension · hamstring weakness

Tight in flexion – downsize femur or increase slope

Downsizing femur helps → ✓ tight flexion + normal extension (TO) · keep anterior cut (avoid notching) → removes more posterior condyle

PS revision, full extension, tight flexion → ✓ downsize femoral component (TO)
✗ Wrong options: flex femur (tightens flexion) · thinner tibia (both) · more distal femur (extension)

CR knee, tight flexion gap (TO) → ✓ cut PCL (convert to PS) · other: ↑ slope (limited), downsize femur (may need anterior referencing) (TO)
✗ Wrong options: more tibia (opens extension too) · upsize femur · distal femur

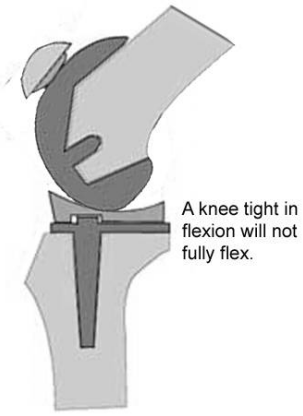
Anterior referencing, 8 mm posterior cut, still tight flexion (TO) → ✓ downsize implant (TO)
✗ Wrong options: distal femur · ↓ slope · thinner poly

Full extension, flexion only to 90° → ✓ downsize femur
✗ Wrong options: more tibia / thinner poly (loosen extension) · distal augment · distal femur

IM referencing, stable extension, won't flex past 90° → ✓ downsize femur · also **PCL recession**, **↑ slope**
✗ Wrong options: thinner insert · posterior augments · distal femur · medial tibial reduction osteotomy (valgus laxity)

CR, recut tibia with ↑ slope because → ✓ balanced extension, tight flexion (slope opens flexion via rollback)

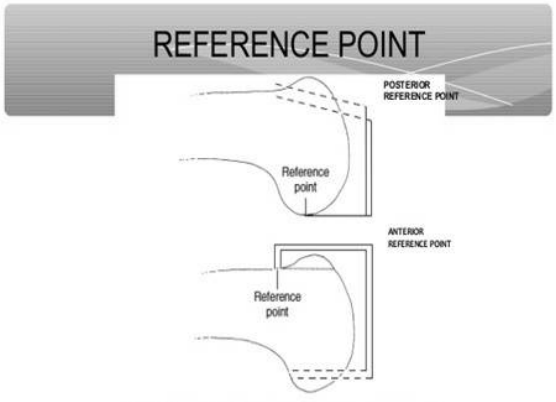
Tibial tray lift-off in flexion = tight flexion · easy extraction at 90° = loose flexion



Tight in flexion — downsize femur

Femoral referencing (TO)

	Posterior referencing	Anterior referencing
Fixed	**Posterior cut** (~8–9 mm) → flexion gap fixed	**Anterior cut** (no notching)
Sequence	Distal cut → size → anterior/posterior cuts → chamfers + notch	Anterior cut first (Whiteside rotation) → size
Size change alters	**Anterior** resection	**Posterior** resection (flexion gap)
Between sizes	**Choose larger** (smaller → notching)	**Choose smaller** (wider flexion gap, better flexion; rare mid-flexion instability)



Posterior vs anterior referencing

Choice of referencing (TO) **posterior = standard in varus** knees · **anterior preferred in valgus / RA** knees · posterior → upsize (avoid notching) · anterior → downsize (avoid overstuffing)

Tight in both – resect tibia

Cannot extend + tray lifts off past 90° → ✓ resect more proximal tibia (or thinner poly)
✗ Wrong options: distal augment · distal femur only · downsize femur only · thicker poly

– – Active space – –

Loose in extension – distal femoral augment

Balanced flexion, loose extension → ✓ distal femoral augment, same poly
✗ Wrong options: distal resection · downsize + thinner poly · tibial resection · augment + thicker poly

Combined gap mismatch

Tight extension + loose flexion (TO) → ✓ distal femoral cut + thicker poly
✗ Wrong options: downsize + slope · downsize + thicker poly · tibia + thicker poly

Post-op flexion contracture + loose flexion (revision) → ✓ resect more distal femur + upsize femur
✗ Wrong options: thicker poly (worsens extension) · more tibia · anterior femur + thinner poly · distal femur + tibia

CR TKA, recurvatum + flexion only 80° (loose extension, tight flexion) → ✓ downsize femur + thicker insert
✗ Wrong options: semi-constrained · more tibia · anteriorise femur alone · ↑ slope alone

Extension gap > flexion gap – release? (TO, FNB 2024)	options anterior sMCL, posterior capsule, POL, deep MCL · **answer not provided by the source** · TO rule (chapter 41): **anterior half of sMCL + pes → flexion space**; POL, posterior capsule, semimembranosus → extension space
Revision joint-line elevation	23 mm spacer, full extension, stable flexion, **patella impinging on poly → ✓ upsize femur + distal and posterior augments** (lower joint line, shrink flexion gap) · error: femoral component seated on residual bone → too small + too proximal ✗ Wrong options: proceed · tendon Z-lengthening · posterior augments only · distal augments only

PCL deficiency in CR knee – anterior femoral subluxation (TO)

CR TKR, anterior femoral subluxation on X-ray → ✓ PCL deficiency (TO)
✗ Wrong options: ACL deficiency · PCL impingement · PLC instability

CR contraindications (TO)	PCL insufficiency · posterolateral instability · **flexion >20°**, **coronal >15°** deformity · inflammatory arthritis + EM deficiency (now relative) · prior ORIF/HTO/UKA (poor ligaments → PS easier) · bone defects needing augments/stems · **excessive native slope** (flexion tight → convert to PS) (TO)
CR technique (TO)	avoid over-resecting distal femur (CR tends to flexion tightness) · choose **smaller** femur between sizes (PS: **larger**) · PCL inserts **5–10 mm** below tibial surface → protect (osteotome / bone block) · tight PCL → release / convert to PS (TO)
Kinematics (TO)	CR: anterior femoral translation **30–60°** WB · PS: constant contact, rollback 60–90° NWB · both show early paradoxical translation · clinical scores, ROM, mid-term survival, quad recovery **no different** · **bicruciate-retaining** preserves screw-home partially (TO)

Flexion deformity by degree (TO)

Fixed flexion	Management (TO)
< 15°	**Posterior capsule release**
15–30°	+ **2 mm extra distal femoral cut**
> 30°	**Two-stage** or **constrained** prosthesis

Tibial cut (TO) — **perpendicular to mechanical axis**, **8–10 mm** (implant thickness) · measured resection: bone removed = implant thickness

Posterior tibial slope (TO) — native ~**7°** (promotes rollback) · ideal **5–7°** = **3–5° bone cut + 2° built into insert** · **> 10°** → risk of posterior fracture (TO)

Genu recurvatum correction (TO)

Steps (TO) — **minimal distal femoral resection** · **move joint line distally** · ✓ **smaller femoral component** (anterior referencing → bigger flexion gap) · thicker poly fills both gaps · **appropriate constraint**

EXCEPT (TO) → ✓ one size bigger femoral component

X vs Y

	Downsizing femur	More distal femur	More proximal tibia
Flexion gap	↑	–	↑
Extension gap	–	↑	↑
Fixes	Tight flexion	Tight extension / FFD	Both tight

How it's asked

"CR TKA, 15° FFD, flexion balanced?"	**→ resect more distal femur**
"Stable in flexion, cannot fully extend?"	**→ resect more distal femur**
"Flexion contracture techniques – EXCEPT?"	**→ downsize femoral component**
"Pre-op 25–95°, still 20° short, flexion balanced?"	**→ resect 2 mm more distal femur**
"Cannot fully extend after releases, flexion balanced – next?"	**→ resect additional distal femur**
"Loose flexion, balanced extension – EXCEPT?"	**→ downsize femur + thicker spacer** (TO)
"Cruciate-sacrificing, flexion instability, extension fine?"	**→ upsize femur + posterior augments**
"Excessive posterior femoral resection leads to?"	**→ loose flexion gap**
"Loose flexion, stable extension – option?"	**→ move femoral component posterior**
"Revision upsized femur + posterior augments because?"	**→ balanced extension, loose flexion**
"CR TKA, AP instability at 90°?"	**→ revise to larger PS implant**
"Distal augment, dislocation at 1 mo?"	**→ flexion gap instability**

– – Active space – –

"Excess posterior condyle cut by robot — symptom?"	→ **difficulty rising from a seated position**
"Downsizing femur helps when?"	→ **tight flexion, normal extension** (TO)
"PS revision, full extension, tight flexion?"	→ **downsize femoral component** (TO)
"Tight flexion gap in CR knee?"	→ **cut PCL** (TO)
"Anterior referencing, 8 mm posterior cut, still tight in flexion?"	→ **downsize implant** (TO)
"Full extension, flexion to 90° only?"	→ **downsize femoral component**
"IM referencing, won't flex past 90°?"	→ **downsize femoral component**
"Recut tibia with more slope because?"	→ **balanced extension, tight flexion**
"Cannot extend + tray lift-off in flexion?"	→ **resect more proximal tibia**
"Balanced flexion, loose extension?"	→ **distal femoral augment, same poly**
"Tight extension + loose flexion?"	→ **distal femoral cut + thicker poly** (TO)
"Revision: FFD + loose flexion?"	→ **more distal femur + upsize femur**
"2 mm more tibia for revision contracture — effect?"	→ **flexion instability**
"Recurvatum + flexion to 80° after CR TKA?"	→ **downsize femur + thicker insert**
"Revision: patella impinging on 23 mm spacer?"	→ **upsize femur + distal and posterior augments**
"CR TKR, anterior femoral subluxation?"	→ **PCL deficiency** (TO)
"Recurvatum correction — EXCEPT?"	→ **one size bigger femoral component** (TO)
"Extension gap refers to?"	→ **distance between distal femoral and proximal tibial cuts** (TO)

– – Active space – –

TKA STIFFNESS

– Recon · 60 / 65 –

1 Basics

Definition ✓ **flexion <90°** or **flexion contracture 10–15°** · poor function + satisfaction

Incidence **1.3–12%** (up to 6%; arthrofibrosis ~1%)

Risk factors

Phase	Factors
Pre-op	✓ **Poor pre-op ROM (most important)** · **patella baja** · **younger age (<55)** · smoking · comorbidity (DM) · low pain tolerance · **prior surgery** (distal femur ORIF, plateau ORIF, multi-ligament surgery → 2–3× arthrofibrosis) · post-traumatic OA · socioeconomic
Technical	**Overstuffed PF joint** · **malrotation** · **tight flexion/extension gaps** · **joint-line elevation** · **extensor mechanism tightened at closure** (close in flexion) · **tight PCL** (CR) · oversized components · anterior tibial slope · inadequate resections
Post-op	Delayed rehab · poor effort · **infection** · HO · hamstring spasm (usually resolves by **6 mo**)
Not	Skin staples · SNF discharge · CPM use · PS vs CR · ipsilateral THA · no tourniquet

Predictors of postoperative ROM

Most important determinant of final ROM ✓ **pre-op ROM** (TO, OB) · also intra-op ROM, flexion arc, lower tibiofemoral angle
✗ Wrong options: CPM/rehab · slope + capsular release · balancing

Gap errors (OB7594) **excess distal femoral resection + thick poly → tight flexion → loss of flexion** · **inadequate distal resection + retained osteophytes + tight posterior capsule → flexion contracture**
✗ Wrong options: the reversed pairings

Post-traumatic OA (prior plateau ORIF) TKA → ✓ limited knee flexion / stiffness needing MUA · also wound problems
✗ Wrong options: worse function (gain larger) · more loosening · HO · fatal PE

Prior distal femur ORIF → ✓ arthrofibrosis 10–85° at 3 mo
✗ Wrong options: good pre-op ROM · PS · ipsilateral THA · no tourniquet

Functional ROM requirements

Activity	Flexion
Stairs	**90°** (TO, OB) · 95° (Orthobullets)
Rise from chair	✓ **95–100°** (TO) · 105° (Orthobullets) – read the options
Lifting an object	**≥120°**
Squat, kneel, cross-legged	**110–165°**
Efficient gait	Near-full extension

Minimum flexion to rise from chair (TO) → ✓ 95°

– – Active space – –

3 Presentation

Difficulty kneeling · compare with pre-op ROM in records

4 Diagnosis & Classification

Key radiology findings

X-ray / CT exclude malposition, malalignment, oversizing, notching

Genes & lab markers

ESR / CRP ± aspiration **exclude infection** before MUA

5 Management

Stiff TKA (flexion <90°)

- Always rule out infection (ESR, CRP, aspirate)
- <12 wk → **manipulation under anaesthesia**
- >12 wk / failed MUA → **arthroscopic (or open) lysis + MUA**
- Identifiable technical cause → **revision TKA**

Manipulation under anaesthesia – indications & timing

Indication **flexion <90° within 12 weeks** despite PT
✗ Wrong options: 110° threshold

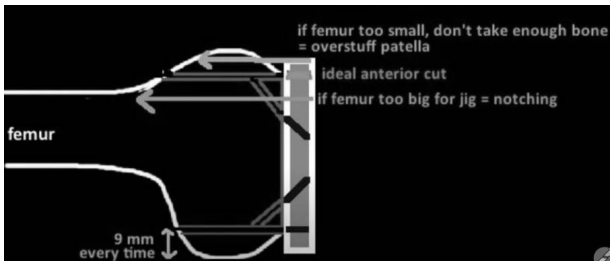
Timing ✓ **best before 12 weeks / within 3 months** (6 wk no better than 12 wk; max gains 12–14 wk) · late (>3 mo) less effective, more risk

Loss of flexion responds better than loss of extension

Gains average **67° → 111°** · early MUA +**20–30°** · resume aggressive PT immediately

Risks **fracture** (<1–2%), **extensor mechanism disruption** · ✓ **notching** (undersized femur with **posterior referencing** → anterior over-resection; **3 mm** notch ↑ fracture risk; anterior femur stressed most in extension)
✗ Wrong options: unequal gaps · ↑ offset · femoral IR · press-fit

Scenario	Answer
ROM 0–60° at **2 mo** (pre-op 5–100°)	✓ **best MUA indication** (✗ Wrong options: 0–60° at 8 mo · 30–120° (extension loss))
6 wk, good intra-op motion, poor flexion	✓ **MUA** (✗ Wrong options: resurface patella · thinner poly · arthroscopy (>3 mo) · CPM)
0–80° at 6 wk – MUA best before	✓ **12 weeks**
10–70° at 10 wk, normal ESR/CRP	✓ **MUA** (✗ Wrong options: aspiration (labs normal) · steroid + PT · poly exchange · revision)



Notching – fracture risk during MUA

Late stiffness – workup & revision

Deteriorating motion 9 mo after revision → ✓ investigate for PJI
✗ Wrong options: PT · MUA · revision · resection

9 mo, 5–80°, painful, CR well aligned, infection negative → ✓ arthroscopic lysis of adhesions + manipulation
✗ Wrong options: dynamic splint · open release · MUA alone (late) · revision to CS

Revision for stiffness	only with **identifiable technical cause** · results modest (arc +~20°; 40% good–excellent; 66% satisfied) · ✓ **main post-op focus = regain near full extension** (more satisfying than arc gain) ✗ Wrong options: 75–80% reach 0–90° · pre-op arc irrelevant · CR→PS much better · MUA reliable after revision
Other options	open adhesiolysis ± poly exchange, quadricepsplasty, component revision

CPM

✓ **May improve early ROM, not final ROM** · no benefit in pain, ambulation, extension · cost questioned

X vs Y

	< 12 weeks	> 12 weeks
First choice	**MUA**	**Arthroscopic lysis + MUA**
Risk	Low	↑ fracture, tendon rupture

How it's asked

"When is MUA best indicated?"	→ **0–60° at 2 months**
"6 wk, poor flexion despite aggressive rehab?"	→ **manipulation under anaesthesia**
"MUA gives largest flexion gain if before?"	→ **12 weeks**
"10–70° at 10 wk, normal labs?"	→ **manipulation under anaesthesia**
"True about MUA?"	→ **best results within 3 months**
"Fracture during MUA – technical error?"	→ **undersized femur with posterior referencing (notching)**
"Most important factor for final ROM?"	→ **preoperative ROM** (TO)

– – Active space – –

"ROM after TKA – true?"	→ **pre-op ROM is the principal predictor**
"TKA after plateau ORIF – increased risk of?"	→ **limited knee flexion**
"10–85° at 3 mo – which history?"	→ **prior distal femur ORIF**
"Deteriorating motion after revision?"	→ **investigate for infection**
"Stiff revision candidate – counsel?"	→ **focus on regaining near-full extension**
"9 mo, 5–80°, painful, infection negative?"	→ **arthroscopic lysis + manipulation**
"Routine CPM outcome?"	→ **may improve early ROM, not final ROM**
"Minimum flexion to rise from chair?"	→ **95°** (TO)

TKA TEMPLATING

– Recon · 61 / 65 –

1 Basics

Definition	anticipating implant **size + position** pre-op · predicts sizes needed in OR · reliable starting point
Accuracy	**manual 92–100%** within ± one size · **automatic only 51%**
Steps	radiographs → analysis → check **scale (20% magnification)** → femur → tibia

4 Diagnosis & Classification

Key radiology findings

View	Role
AP weight-bearing knee	Required · bony defects + osteophytes easiest here
✓ **Lateral knee**	**Most important view for templating** · slope, patellar height
Patellofemoral (sunrise)	Not needed for templating · tilt/shift
Full-length hip-to-ankle WB AP	Optional · rule out **extra-articular deformity**, coronal laxity, plan cuts vs mechanical axis

Radiographic analysis

Step	Assessment
1	**Mechanical axis** (hip-to-ankle) – neutral bisects knee centre
2	Coronal deformity – **tibiofemoral angle**
3	**Femoral resection (valgus cut) angle** = mechanical vs anatomic femoral axis
4	**Tibial cut** ⊥ mechanical axis
5	Bony defects + osteophytes (AP)
6	**Tibial slope** (lateral)
7	**Patellar height** (lateral) – **baja** → harder exposure
8	Patellar shift/tilt (skyline)

– – Active space – –



Valgus cut angle

Alignment strategies

	Mechanical alignment	Kinematic alignment
Goal	Neutral axis **0 ± 3°**	Restore pre-arthritic joint line + kinematics
Femur	⊥ femoral mechanical axis (**5–7° valgus** to anatomic)	Cylindrical axis of posterior condyles · **0.7–1.8° more valgus**
Tibia	⊥ tibial mechanical axis (0°), avoid accidental varus	Pre-arthritic joint line · **1.2–2.2° more varus**
Slope	–	Native · ~**1.2° more**
Rotation	**TEA** or **3° ER** to posterior condyles	–
Soft tissue	Releases anticipated for rectangular gaps	Few/no releases · equal medial/lateral resections

5 Management

Template the femur

Size on lateral restore **posterior condylar offset** · avoid **notching**

AP ML position, no overhang (overhang → consider downsizing)

Pitfall ✓ **oversizing the femur** · femur **less accurate** than tibia

Template the tibia

Size on lateral no overhang · match native slope · confirm on AP

Pitfall **equally likely to over- or undersize** · intra-op tendency to **downsize**



Tibial template — lateral

Extra-articular deformity

Healed femoral fracture with hardware, mechanical axis ~3° valgus, 14° deformity not affecting joint → ✓ routine knee prosthesis · avoid hardware removal with **extramedullary jigs / navigation**
✗ Wrong options: custom components · special IM jigs · hinge · osteotomy (only for severe deformity)

X vs Y

	Femoral templating	Tibial templating
Accuracy	Lower	Higher
Error	**Oversizing**	Over = under · surgeons downsize

How it's asked

"Prior femoral fracture, axis 3° valgus — restore mechanical axis with?"	**→ routine knee prosthesis**
"Most important view for TKA templating?"	**→ lateral knee**
"Manual vs automatic templating accuracy?"	**→ 92–100% vs 51% (± one size)**

– – Active space – –

TKA VASCULAR INJURY & BLEEDING

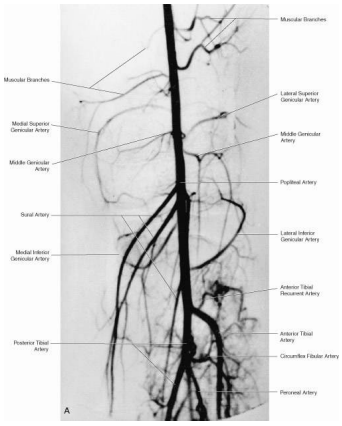
– Recon · 62 / 65 –

1 Basics

Incidence	**0.017–0.2%** · rare, devastating · commonest = **direct popliteal artery injury**
Risk factors	**revision** · **peripheral vascular disease** (consider TKA without tourniquet)

Popliteal artery anatomy & retractor injury

Origin	continuation of **SFA** at **adductor hiatus** · anchored by adductor magnus insertion
At the joint	✓ **lateral** structure · directly behind **posterior capsule** · posterior to **posterior horn/root of lateral meniscus** · ✓ **posterolateral to PCL tibial insertion** · with popliteal vein + tibial nerve
Relations	✓ **anterior to popliteal vein** · **9 mm** posterior to posterior tibial plateau at **90° flexion** ✗ Wrong options: 15 mm · posterior/medial/lateral to vein
Branches	above joint: **medial + lateral sural**, cutaneous, **middle genicular** · at joint: **medial + lateral geniculars** · exits under **soleus** tendon · divides into **anterior + posterior tibial** at distal popliteus
Flexion vs extension	furthest from plateau at **60–90° flexion** · **hyperflexion** displaces it posteriorly ✓ · **hyperextension brings artery closer** + traction → injury during cuts



Popliteal artery relations

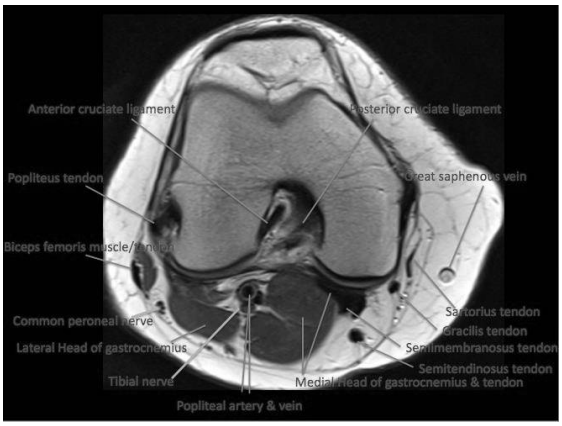
2 Mechanism

Direct laceration	sharp dissection in posterior compartment
Blunt	**posterior retractor lateral to PCL** or **>1 cm** into posterior soft tissue · hyperextension traction
Thrombosis	

Safe practice single-prong posterior retractor **medial**, never lateral to PCL, ≤1 cm deep

– – Active space – –

Retractor	Structure at risk
✓ **Posterior, lateral to PCL / posterolateral to PCL insertion**	**Popliteal artery** (✗ Wrong options: biceps · lateral meniscus root · semimembranosus · peroneal n.)
Posterior, 1 cm, **medial** to PCL	Safe
Lateral, around fibular head 3 cm	**Common peroneal nerve** (within 3 cm of posterior fibular head)
Medial along tibial plateau 3 cm	**Saphenous nerve** stretch (2–3 cm posterior to medial tibia)
Lamina spreader medial	Not linked



Posterior retractor lateral to PCL

3 Presentation

Intra-op brisk bleeding · loss of pulses

Post-op ischaemia · sensory change · **absent distal pulses** · **skin mottling** · severe pain + global motor loss

4 Diagnosis & Classification

Key radiology findings

Intra-op angiogram via vascular surgery

Post-op **CT angiogram** for ischaemic signs
✗ Wrong options: MR angiogram (delay)

5 Management

Suspected popliteal injury

- Intra-op bleeding / pulse loss → **immediate vascular consult ± angiogram**
- Post-op absent pulses → **back to OR, vascular consult, intra-op arteriogram**
  - Stent / bypass / endarterectomy ± prophylactic fasciotomy

– – Active space – –

Management of absent pulses

Asymmetric absent pulses + poor refill post-op → ✓ return to OR + vascular consult + intra-op arteriogram (localise before exploring; stent possible)
✗ Wrong options: observe 4 h · compartment pressures · MRA · blind exploration

Valgus TKA, 5 h, pain, no DF/PF, asymmetric pulses after flexing + loosening dressing → ✓ vascular surgery consultation
✗ Wrong options: serial exams + EMG · ACB · peroneal exploration (global deficit) · thrombectomy without vascular decision

Options (vascular)	observation · **stent** · **bypass** · **endarterectomy** · **prophylactic fasciotomy** (reperfusion compartment syndrome)
Complications	compartment syndrome · amputation · mortality · ↑ LOS + cost

Drain & transfusion

Closed suction drain in primary TKA → ✓ ↑ transfusion
✗ Wrong options: dehiscence · ↓ infection · ↓ hematoma return · ↓ pain

X vs Y

	THA vascular risk	TKA vascular risk
Vessel	External iliac (anterosuperior screw), obturator (TAL retractor), femoral (anterior retractor)	**Popliteal artery** (posterolateral to PCL)
Protective position	–	**Flexion**

How it's asked

"Associated with closed suction drain?"	**→ increased transfusion**
"Popliteal artery anatomy?"	**→ anterior to popliteal vein, 9 mm behind posterior tibial plateau at 90°**
"Sharp Hohmann posterolateral to PCL – injury?"	**→ popliteal artery**
"Absent pulses in PACU, blunt popliteal injury – retractor?"	**→ posterior retractor 1 cm deep, lateral to PCL**
"Asymmetric absent pulses after TKA – next?"	**→ OR + vascular consult + intra-op arteriogram**
"Valgus TKA, global motor loss, asymmetric pulses after flexing knee?"	**→ vascular surgery consultation**

TKA WOUND COMPLICATIONS

– Recon · 63 / 65 –

1 Basics

Risk factors (diabetes, obesity, smoking, nutrition)

Category	Factors
Systemic	**Diabetes mellitus** · vascular disease · **inflammatory arthritis (RA)** · immunosuppressives / **steroids** · **tobacco** · renal + liver disease · **poor nutrition** (albumin <3.5 g/dL, TLC <1,500/µL) · perioperative **anaemia** · **morbid obesity (BMI >40)**
Local	**Previous incisions** · knee deformity · skin adhesions · poor local blood supply · traumatic arthritis
Technique	**Large subcutaneous flaps** · short incision → vigorous skin traction · not preserving subcutaneous fat layer · long surgical time · long tourniquet time (Orthobullets)
Post-op	**Haematoma** · infection · **early (first 4 days) flexion >40°** in a high-risk knee

Diabetes ✓ **most strongly associated** with early wound complication needing surgery (Galat: > inflammatory arthritis, smoking, PVD, steroids, BMI >30, previous open surgery)

Tourniquet Orthobullets lists long tourniquet time · OB: tourniquet use + duration **not correlated** with wound complications – read the options
✗ Wrong options: "tourniquet 70 min" as strongest factor

Previous knee surgery **open** surgery = risk · esp. if a more **medial** incision used → **lateral wound necrosis** (may need gastrocnemius flap)
✗ Wrong options: arthroscopy

DVT prophylaxis **aspirin** → **fewer** wound complications than LMWH / NOACs
✗ Wrong options: aspirin as risk factor

Early surgical intervention for wound → ↑ major subsequent surgery (**5.3%**) + deep infection (**7.0%**) → aim for primary healing at index surgery

Modifiable before surgery smoking cessation · nutritional optimisation · stop harmful immunomodulators · glucose control · weight loss

Genes & lab markers – nutrition

Marker	Malnutrition cut-off
Albumin	**<3.5 g/dL**
Total lymphocyte count (TLC)	**<1,500 /µL** (cells/mm³)
Prealbumin	**<15 mg/dL**
Transferrin	**<200 mg/dL**

TLC + prealbumin for a chronic wound → ✓ **assess nutrition**
✗ Wrong options: rule out infection · detect inflammation (ESR/CRP) · immune status (complement/immunoglobulins) · anaemia (Hb, Hct, RBC)

– – Active space – –

Smoking

Smoker vs non-smoker TKA → ✓ higher readmission · smoking = single greatest predictor of **30-day readmission + reoperation** after outpatient TJA · ↑ deep + superficial SSI · ↑ **MUA** · ↑ wound breakdown, loosening
✗ Wrong options: better SF-36 · better pain relief · no difference in SSI or MUA

Cessation	**4–6 wk before and after** TJA → complications ↓ **50%** (↑ blood O_2-carrying capacity)
MUA risk factors (Issa)	diabetes · hypercholesterolaemia · preop ROM **<100°** · TKA for osteonecrosis · smoking

Obesity

BMI >40 TKA → ✓ more post-op radiolucent lines · ↑ complications · ↑ revision, inferior survivorship · poorer function
✗ Wrong options: no difference in function / complications / revision

AAOS 2016 knee OA surgical guideline

Strength	Statement
Strong	Obese patients **less improvement** after TKA · **no difference** in pain/function with **patellar resurfacing** · **no difference CR vs PS** · **periarticular + peripheral nerve** anaesthesia ↓ pain + opiates
Moderate	**Diabetes** ↑ complications · **neuraxial** anaesthesia improves perioperative outcomes + complications
Limited	✓ **Hepatitis C / cirrhosis ↑ complications**

3 Presentation

History	multiply operated knee · prior infection
Symptoms	persistent knee pain · **persistent drainage beyond 7 days** (early post-op period)
Exam	wound breakdown · erythema · warmth · drainage · **peri-incisional eschar**



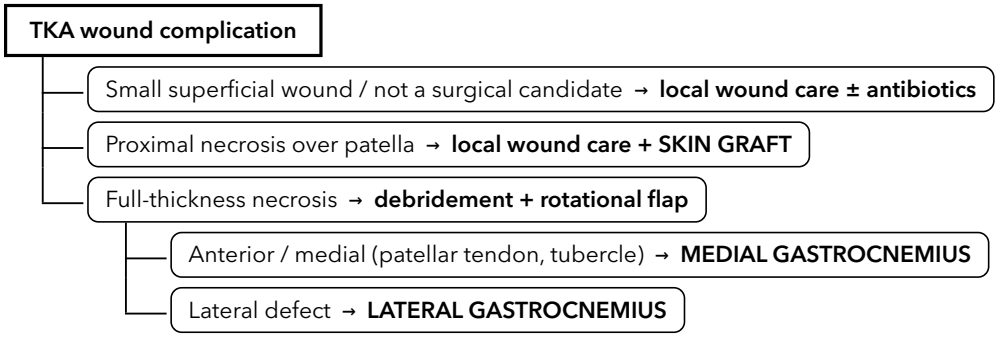
Full-thickness peri-incisional necrosis after TKA

4 Diagnosis & Classification

Key radiology findings

Test	Role
Aspiration	Exclude **deep infection** before wound surgery
Radiographs	Help rule out deep infection – look for **bone resorption**
Bone scan (radionuclide)	Infection suspected but **aspiration + serology not confirmatory**
Angiography	**Sural artery flow** in question **before gastrocnemius flap**

5 Management



Operative criteria – incision planning

Rule	Value
Skin bridge between incisions	**>5–6 cm**
Crossing a previous incision	Never at an angle **<60°**
Which old incision	✓ **Most lateral** usable – anterior knee superficial supply mainly from **medial geniculates**

Soft tissue coverage - gastrocnemius flap

Exposed patellar tendon after superficial debridement, aspiration negative → ✓ medial gastrocnemius flap + skin graft
✗ Wrong options: split-thickness graft alone · wet-to-dry Dakin's · VAC to secondary intention · latissimus free flap

Ries — necrosis over **patella** → local care + **skin graft** · over **tibial tubercle / patellar tendon** → ✓ **muscle flap** (prevents extensor mechanism disruption + deep infection) · all TKAs salvaged

Flap artery — medial gastrocnemius = ✓ **medial sural artery** · lateral gastrocnemius = **lateral sural artery**
✗ Wrong options: inferomedial geniculate (medial plateau + patella) · recurrent anterior tibial (patellar anastomosis) · profunda femoris (thigh) · descending branch of circumflex femoral (joins patellar anastomosis, not below knee)

– – Active space – –

Reach proximally based → rotates up to **15 cm above the joint line** · low failure rate

Muscle flap types (Markovich) **prophylactic** (before reconstruction) · **treating** (infected prosthesis, deficient cover — least successful) · **salvage** (early dehiscence / necrosis)



Medial gastrocnemius rotational flap

6 Complications

Haemarthrosis – commonest early complication (TO)

Commonest post-op complication of knee arthroplasty → ✓ **haemarthrosis** (TO) · a little over **1%** · esp. after **lateral retinacular release** or **liberal synovectomy**
✗ Wrong options: infection · compartment syndrome · tourniquet palsy

Wound complication sequelae deep infection · extensor mechanism disruption (exposed tendon) · ↑ reoperation + cost

X vs Y

	Medial gastrocnemius flap	Lateral gastrocnemius flap
Artery	**Medial sural**	**Lateral sural**
Defect covered	**Anterior + medial** (patellar tendon, tubercle)	**Lateral**

Test ordered	What it assesses
TLC, albumin, prealbumin, transferrin	**Nutrition**
ESR, CRP	Inflammation / infection
Complement, immunoglobulins	Immune status
Hb, Hct, RBC	Anaemia

How it's asked

"Exposed patellar tendon after debridement of TKA skin necrosis — next?"	→ **medial gastrocnemius flap + skin graft** ✗ Wrong options: STSG · VAC · Dakin's · latissimus
"Artery of the rotational flap for distal medial necrosis?"	→ **medial sural artery**

"Comorbidity most associated with early wound complication needing surgery?"	→ **type 2 diabetes** ✗ Wrong options: arthroscopy · aspirin · BMI 31 · tourniquet 70 min
"BMI >40 effect on TKA?"	→ **more post-op radiolucent lines**
"Active smoker vs non-smoker after TKA?"	→ **higher readmission risk**
"Purpose of TLC + prealbumin for chronic TKA wound?"	→ **assess nutrition status**
"AAOS 2016 statement WITHOUT strong recommendation?"	→ **hepatitis C / cirrhosis ↑ complications** (limited)
"Commonest post-op complication of knee arthroplasty?"	→ **haemarthrosis** (TO)
"Multiple previous incisions — which to use?"	→ **most lateral; bridge >5–6 cm; cross at ≥60°**

– – Active space – –

– – Active space – –

UNICOMPARTMENTAL KNEE REPLACEMENT

– Recon · 64 / 65 –

1 Basics

Epidemiology

Incidence **5%** of knee arthroplasties · **medial** compartment most common

Classic candidate older (**>60**), lower-demand, thin (**<82 kg**) · only **6%** of patients meet these criteria with no contraindication · indications expanding to younger, heavier, more active patients

Implant types – fixed vs mobile bearing

	Fixed-bearing (Miller-Galante, Zuk)	**Mobile-bearing (Oxford)**
Design	PE **locked** in tray · **flat** tibial surface	Fully **congruent, unconstrained** PE glides between components
Contact	**Flat-on-round** → point loading, high contact stress	Large contact area → low stress
PE wear	Theoretically ↑ (fails in **later** years)	↓ · excellent survivorship to 2nd decade
Dislocation	**Impossible**	✓ **Bearing dislocation (spin-out)** + impingement (fails in **early** years)
Technique	Easier · forgiving of laxity / small gap mismatch	**Technically demanding** · flexion gap must **equal** extension gap · stability from **MCL** tension
ACL deficiency	Controversial	**Contraindicated** (spin-out)
Loosening	Component loosening more common	Poor bone preparation / thermal necrosis

History (TO) **McKeever (1950s)** tibial hemiarthroplasty · **Marmor (1970s)** added femoral resurfacing = **first cemented UKA** · **Oxford** = mobile bearing, **lowest wear** · fixed bearing (Sigma) = easier technique, less dislocation

Design (TO) femur: **lug + fin** fixation, wide posterior condyle, anterior taper (avoid patellar impingement) · tibia: **metal-backed PE ≥ 6 mm**, **75–80% coverage**

Mobile-bearing specific disadvantage → ✓ higher risk of bearing dislocation (TO)
✗ Wrong options: PCL sacrifice · limited femoral sizing · ↑ PE wear

After UKA + twisting injury – true? → ✓ mobile-bearing UKA has more dislocation (TO)
✗ Wrong options: lateral UKA more linear wear · medial more dislocation · fixed less linear wear

Anteromedial OA (AMOA) concept (TO)

AMOA	wear confined to **anterior + middle thirds** of the medial tibial plateau · needs a **functional ACL** + competent **MCL** · varus is **intra-articular + correctable** once cartilage height restored ✓ ✗ Wrong options: justifies fixed bearing · eliminates PF symptoms · allows PCL sacrifice
Why	intact ACL keeps femur anterior on tibia in flexion · ACL deficient → tibia subluxes anteriorly → femur articulates further back → **posterior tibial wear**
True AMOA	**bone-on-bone in extension** · joint space partly preserved in **deep flexion** (posterior cartilage spared) · **full-thickness lateral cartilage**

Advantages

vs TKA	vs Osteotomy (HTO)
✓ **Faster rehab + recovery** · ↓ blood loss · ↓ morbidity · ↓ cost · ✓ **↓ PJI + wound complications** · ↓ MI, PE, VTE, mortality · ✓ **near-native kinematics** (ACL, PCL, other compartments kept) · ✓ **better ROM** · smaller incision → less pain, shorter stay · bone stock preserved	Faster rehab · better cosmesis · ↑ initial success · fewer short-term complications · **lasts longer** · **easier conversion to TKA**

Disadvantages vs TKA	**higher revision rate** · lower survivorship · ↑ aseptic loosening · **technically more demanding** · **OA progression** (unique) · **2×** early (90-day) periprosthetic fracture

4 Diagnosis & Classification

Key radiology findings

Lateral compartment	confirm **full-thickness cartilage** on **valgus stress view** (TO)
Lateral view of medial OA	anterior / central tibial wear = AMOA ✓ · ✓ **posterior-third tibial wear = ACL incompetent** (TO) → contraindication to standard (mobile-bearing) UKA ✗ Wrong options: PCL incompetent · global OA · MCL contracted
Mobile bearing	contains a **radiographic marker** → shows bearing dislocation
Stress fracture	X-ray may be normal · ✓ **bone scan uptake under the tibial component**
PFA work-up	standing AP + lateral · **Rosenberg** (45° flexion PA) · **skyline** (Merchant) · standing full-leg

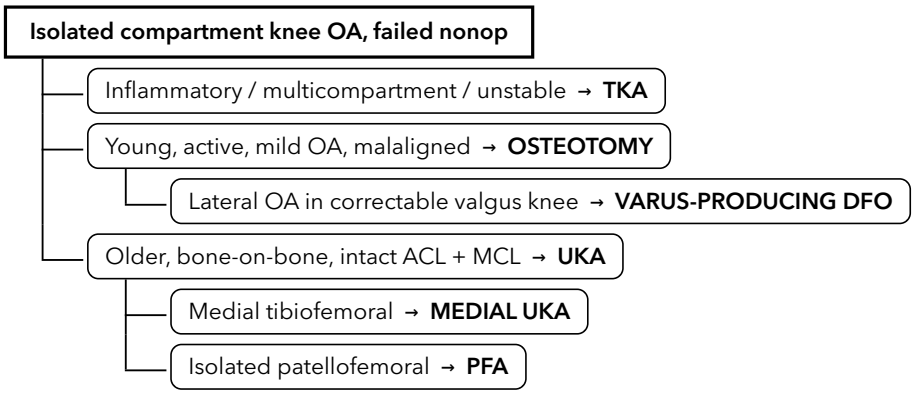
– – Active space – –

Diagnostic criteria – Kozinn & Scott vs Oxford (TO)

Criterion	Kozinn & Scott 1989 (fixed bearing)	Oxford (mobile bearing)
Age & weight	**>60 yr, <82 kg (180 lb)**	**No limits**
Activity	Low demand / sedentary	No restriction
Deformity	**Varus <10°, valgus <5°**	Varus intra-articular + **manually correctable to neutral**
Flexion	**>90°, contracture <5°**	**>90°, contracture <15°**
Ligaments	Intact ACL	Intact **ACL + competent MCL**
PF joint	No PFJ arthritis	Ignored unless **severe lateral facet grooving + bone loss + anterior pain**
Lateral compartment	Pristine	**Full-thickness cartilage** (valgus stress view)

No longer a barrier (modern / Oxford) → ✓ high BMI (TO)
✗ Wrong options: lateral cartilage loss · fixed flexion deformity · inflammatory arthritis (still barriers)

5 Management



Indications & contraindications

Lecture-note thresholds (TO)

varus **< 5°**, **weight < 80 kg**, BMI **> 30** = CI, fixed varus **> 10°** + MCL contracture = CI · Kozinn & Scott **< 82 kg, varus < 10°** – read the options · HTO → younger, heavier, **varus > 10°** · UKA → older, lower demand, varus **< 5°** (TO)

Indications (TO) isolated medial or lateral OA · **osteonecrosis** · age **>60** · weight **<82 kg** · deformity **<15°**, passively correctable to neutral · flexion contracture **<5°** · ROM **>90°**

– – Active space – –

Contraindication	Threshold / note
✓ **Inflammatory arthritis** (RA, psoriatic)	**Absolute** — involves all compartments → progression (TO, OB)
Fixed varus	**>10°**
Fixed valgus	**>5°** → ✓ fixed valgus **10°** = absolute CI (TO)
Flexion contracture	**>5–10°** (Orthobullets) · **>10°** (TO list) · **≥15°** (TO) · **15°** = CI (OB) · >5° relative (OB) — read the options
Arc of motion	**<90°**
ACL deficiency	**Absolute for lateral UKA** · controversial for medial fixed-bearing · **contraindicated for mobile-bearing** (spin-out) · TO: **relative** — reconstruct ACL at same sitting
Previous meniscectomy, other compartment	e.g. complete **lateral** meniscectomy for a medial UKA
Tricompartmental OA	Diffuse / global pain
Opposite-compartment cartilage erosion	Weight-bearing area
Grade IV PF chondrosis	Anterior knee pain · exposed subchondral bone under patella
Young, high-activity, heavy labourer	Relative
Joint subluxation ≥5 mm	Relative (OB)
Obesity	Relative historically (BMI >30, OB) · no longer a barrier (TO, OB)

NOT contraindications **SONK / osteonecrosis of MFC** (UKA works: 90% good at 5.5 yr, **96%** survivorship at 12 yr) · prior **ACL reconstruction** with functioning graft (**grade 1A Lachman**) · prior **medial meniscal transplant** · **contralateral knee OA** · asymptomatic / medial PF chondromalacia · **age >70** · age <60 with modern implants · **prior HTO** · previous trauma / plateau ORIF (scar only) · **fixed varus 5°** · **9° varus** · BMI 36, weight 80 kg · **Q-angle 13°** in a male (normal)
✗ Wrong options: traps

ACL found torn at arthrotomy for planned UKA (55 yr) → ✓ reconstruct ACL + proceed with UKA (TO)
✗ Wrong options: continue + intervene later · convert to TKR and tell after · simply fixed-bearing UKA

Most important relative CI to mobile-bearing UKA → ✓ prior failed ACL reconstruction (ACL deficiency → spin-out)
✗ Wrong options: medial PF OA · age >70 · flexion contracture 10° (OK if <15°) · previous HTO

Relative CI → ✓ Outerbridge IV MFC + grade IIb Lachman (non-functioning ACL)
✗ Wrong options: meniscal transplant · ACLR with grade Ia Lachman · Outerbridge IV medial alone (that is the indication)

– – Active space – –

UKA vs HTO – patient selection

	UKA	HTO or UKA	HTO
Age	>55 yr	55–65 yr	<65 yr
Activity	Low demand	Moderately active	Active
BMI	<30	Any	–
Alignment	0–5°	5–10°	5–15°
AP instability	None–grade I	None–grade I	Any
ML instability	None–grade I	None–grade I	None–grade II
ROM	Arc 90°, FC <5°	Arc 100°, FC <5°	Arc 120°, FC <5°
Arthrosis	Any	Ahlbäck II	Ahlbäck I–II

HTO prerequisites (TO) <60 yr, active · varus **5–15°** · < grade 3 medial involvement · intact lateral + PF · flexion **>120°** · no laxity · expanded: posterolateral laxity, varus thrust, ACL deficiency (slope change)

Kissing bone-on-bone lesion, good ligaments, one compartment → UKA over HTO · **KL 4 + other compartments / instability / gross malalignment → TKA**

UKA older, less active, **minimal varus, more severe OA**, little instability · **HTO** – younger, active, **milder OA, more malalignment**, AP instability

40 yr, lateral OA after lateral meniscectomy, correctable 5° valgus, stable → ✓ varus-producing DFO · osteotomy CIs: inflammatory, cruciate deficiency, FFD **>15°**, flexion **<90°**
✗ Wrong options: valgus HTO · TKA · arthroscopic debridement · AMZ

68 yr, isolated medial Outerbridge IV, worse after arthroscopy → ✓ medial UKA
✗ Wrong options: opening-wedge HTO (younger, milder) · bicompartmental (medial + PF) · PFA · lateral UKA

71 yr, BMI 28, mild varus, stable, wants to kneel → ✓ UKA – best ROM
✗ Wrong options: HTO · CR / PS / rotating-platform TKA (equivalent ROM to each other)

Technique

Alignment ✓ **undercorrect mechanical axis by 2–3°** · overcorrection overloads unresurfaced compartment → OA progression · Orthobullets: correct varus to **1–5° valgus**
✗ Wrong options: overcorrect into valgus 2–3°

Steps remove peripheral + **notch osteophytes** · minimal bone resection · **avoid extensive releases** (never release the **MCL**) · avoid edge loading · correct ML position → no tibial spine impingement · **avoid a varus tibial cut** (↑ loosening) · care with proximal tibial **guide pins** (stress fracture)

Gap balancing (TO) femoral jig on tibial spacer · spacer moves with **light two-finger resistance** · no overcorrection, **no ligament release** · e.g. bone loss 2 mm + cartilage 3 mm = 5 mm; 8 mm implant → resect **3 mm** bone

Gaps (mobile bearing) flexion gap set by **posterior femoral cut** · flexion = extension gap · tibial slope parallels native **3–7°**

Bearing unstable in mid-flexion, tight in full extension → ✓ inadequate posterior femoral resection (TO)
✗ Wrong options: tibial over-resection · excess slope · MCL over-release (lax in both flexion + extension)

Small incision without patellar dislocation → ✓ faster recovery (2× standard UKA, 3× TKA)
✗ Wrong options: conversion option · better positioning · better ultimate ROM · ↑ 10-yr survival

– – Active space – –

Patellofemoral arthroplasty – indications & failure (TO)

Ideal PFA candidate — **<65 yr**, debilitating **isolated PF OA**, no patellar malalignment · pain localised to PFJ, failed NSAIDs / injection · flexion **120°**, FFD **<10°** · not if **valgus >8°** or **varus >5°** unless osteotomy first

Not an indication → ✓ BMI >30 (implant overload) (TO) · inflammatory arthropathy = CI · **tibiofemoral OA** = principal CI

	Inlay (1st gen, resurfacing)	Onlay (2nd gen, anterior cut)
Trochlea	Replaces cartilage only; sits in native trochlea	Uses TKA anterior femoral cut; replaces **entire anterior trochlea**
Main problem	**Patellar instability** (short-term failure)	Instability minimised

Commonest failure (overall + long-term) → ✓ progression of tibiofemoral OA (TO, OB) · up to **22%** at 5–15 yr
✗ Wrong options: PF instability (inlay short-term) · patellar tendon rupture · aseptic loosening (<0.5% at 7 yr)

Early PFA complications — higher than TKA · persistent anterior pain (extensor malalignment, **oversized femoral component**) · catching / snapping · extensor rupture

TF progression after PFA → convert to TKA · PFA **not** superior to medial/lateral UKA
✗ Wrong options: add medial UKA

6 Complications

Failure modes (OA progression, loosening, bearing dislocation)

Timing	Commonest cause
Early (<5 yr)	✓ **Aseptic loosening** – **25–45.3%** of early failures (✗ Wrong options: OA progression · PE wear · stress fracture ~2%)
Mid (5–10 yr)	**OA progression (38%)**
Late (>10 yr)	✓ **OA progression (40%)** · then aseptic loosening **29%**, PE wear **10%**
Overall (TO)	Contralateral arthritis **39.5%** · aseptic loosening **25.4%** · instability **15.3%** · PE wear **14%** · arthrofibrosis **2.5%** · pain **1.8%** · infection **1.5%**

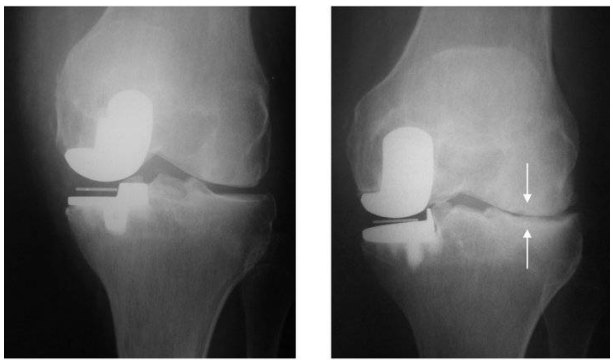
Late (>5 yr) causes (Orthobullets) — OA progression (idiopathic, **over-correction**, more with mobile-bearing) · component failure (**under-correction** overload) · component loosening (fixed-bearing) · **patellar impingement** on femoral component · PE wear

Unique to UKA vs TKA → ✓ OA progression
✗ Wrong options: periprosthetic fracture · PJI · liner dissociation · PE wear (all shared)

Price & Svard (682 Oxford UKA) — 29 revisions: **10 OA progression** · 9 loosening · 5 infection · 2 bearing dislocation · 3 unexplained pain

Survivorship — fixed-bearing 10 yr **87.4–96%** (failure ~**1%/yr** first decade) → 2nd decade **79–90%** · mobile-bearing **93% at 15 yr** · lateral = medial long-term · UKA revision results worse than TKA revision

– – Active space – –



Overcorrection → lateral compartment OA after medial UKA

Aseptic loosening

Risk — **all-PE tibia** · **fixed-bearing** · component malposition (**varus tibia**) · poor initial fixation · weight **>82 kg** · tibial side more common

43 yr, BMI 43, loose varus tibial component at 1 yr, normal ESR + CRP → ✓ revise UKA to TKA (no aspiration needed when markers normal)
✗ Wrong options: aspiration · explant + spacer · tibial-only or both-component UKA revision

Bearing dislocation (mobile bearing)

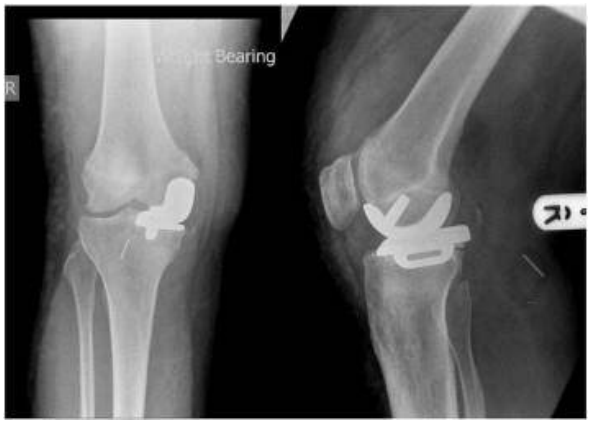
Cause — almost always **technical error**: **gap mismatch** · **MCL damage / release** · impingement by retained **cement or osteophytes** · component malalignment

Technical error	Effect
✓ **Flexion gap 1–2 mm looser than extension** (bearing sized to tight extension gap)	**Posterior spin-out in deep flexion** (squatting) – most classic
Flexion gap tighter than extension	Stiffness + deep-flexion pain · no dislocation
Retained **anterior** femoral / notch osteophytes	Push bearing **posteriorly** in **terminal extension**
Retained **posterior** condylar bone	Pushes bearing **anteriorly** in deep flexion
MCL injury / pie-crusting	Medial opening → bearing escapes
Excess posterior slope (>10°)	Bearing slides posteriorly in flexion
Too little / reverse slope	Pushes bearing anteriorly
Internally rotated tibial tray	Edge-loading, twists out

Presentation — pain, giving way on stairs, **clicking**, effusion, ↓ ROM, stable ligaments, normal markers → ✓ **bearing dislocation**
✗ Wrong options: PJI · ACL rupture · lateral OA · lateral meniscal tear

Treatment — closed reduction · open exploration + **bearing exchange** · convert to fixed UKA · convert to TKA

Anteromedial knee pain after mobile-bearing medial UKA → ✓ anterior impingement (TO) – technique / malposition
✗ Wrong options: bearing dislocation · loosening · instability



Dislocated mobile bearing after medial UKA

Medial vs lateral UKA (TO)

Lateral UKA	traditionally ↑ dislocation + failure · LCL allows **6.7 mm** lateral distraction in flexion (vs **2 mm** medial) · lateral tibial condyle **convex** (medial **flat**) · lateral femoral condyle subluxes posteriorly in high flexion · newer designs → **equivalent** survival, pain relief, function

Tibial plateau stress fracture

Early / intra-op complication (TO)	**tibial plateau fracture** · mid/late: OA progression, PE wear, subsidence, loosening, bearing dislocation (mobile), **PF OA from overcorrection**
Features	**always tibia** · high activity + heavy patient (**>200 lb**) · **pain-free interval → spontaneous activity pain** · **blood on aspiration** (RBC 45,000, WBC 6,500) · normal X-ray + **bone scan uptake under component** ✗ Wrong options: infection (WBC >20,000, ↑ ESR/CRP) · loosening · crystals · synovial entrapment · pes bursitis · CRPS
Technical risk factors	guide pin **penetrating posterior tibial cortex** · **peripheral medial cortical pin** · **re-drilling** guide holes (≥3 holes) · **undersized tibial component** · **vertical sagittal saw cut too deep** past the horizontal cut (TO)

NOT a cause of the post-op stress fracture → ✓ excessive impaction force → intra-operative fracture, seen at once

Rate	~**2%** of UKA failures · nondisplaced → protected weight bearing · displaced → ORIF or revision



Medial tibial plateau stress fracture under UKA

– – Active space – –

X vs Y

	UKA	TKA
Recovery / rehab	✓ **Faster**	Slower
Kinematics	✓ **Near-native**	Altered
ROM	**Better**	Less
Blood loss, pain meds, stay	Less	More
PJI, wound, stiffness	**Lower**	Higher
VTE, MI, mortality	Lower	Higher
Early periprosthetic fracture (90 d)	**2× higher**	✓ Lower
Revision / survivorship	**Higher revision**, lower survival	✓ Lower revision, better survival
Aseptic loosening	Higher	Lower
OA progression	**Yes** – commonest revision cause	No
Technical demand	**More**	Less
Mobile bearings	Successful	Successful

	Early failure (<5 yr)	Late failure (>10 yr)
Commonest	**Aseptic loosening**	**OA progression**
Typical design	Mobile bearing (dislocation, impingement)	Fixed bearing (PE wear)

How it's asked

"Contraindication to UKR?"	→ **RA with medial compartment involvement** (TO) ✗ Wrong options: osteonecrosis · medial OA · previous patellectomy
"Absolute contraindication to UKA?"	→ **fixed valgus 10°** (TO) ✗ Wrong options: FC 10° · fixed varus 10° · ACL deficiency
"ACL found torn at UKA arthrotomy – next?"	→ **reconstruct ACL + proceed with UKA** (TO)
"Posterior-third medial tibial wear on lateral X-ray – implication?"	→ **ACL incompetent** (TO)
"Traditional criterion no longer a barrier (Oxford)?"	→ **high BMI** (TO)
"Significance of AMOA?"	→ **deformity is correctable** (TO)
"All are contraindications to medial UKA EXCEPT?"	→ **osteonecrosis of the MFC**
"Strongest contraindication to UKA, 65 yr active woman?"	→ **rheumatoid arthritis** ✗ Wrong options: BMI ≥30 · fixed varus 5° · FC 5° · contralateral OA

"Failed UKA at 4 yr — preoperative factor?"	→ **inflammatory arthritis** ✗ Wrong options: old ACLR · SONK · 80 kg · 5° varus
"Relative contraindication to UKA?"	→ **Outerbridge IV MFC + grade IIb Lachman**
"Not a UKA candidate?"	→ **50 yr with psoriatic arthritis** ✗ Wrong options: BMI 36 · 9° varus · Q-angle 13° · prior medial meniscectomy
"Most important relative CI to mobile-bearing UKA?"	→ **prior failed ACL reconstruction**
"58 yr, old plateau ORIF, grade 1A Lachman, FC 15° — CI?"	→ **flexion contracture**
"Lateral OA, 40 yr, correctable 5° valgus?"	→ **varus-producing DFO**
"Isolated medial Outerbridge IV, 68 yr — best treatment?"	→ **medial UKA**
"Wants maximum ROM / kneeling, 71 yr?"	→ **UKA**
"Advantage of small incision without patellar eversion?"	→ **improved rate of recovery**
"Benefit of UKA over TKA?"	→ **faster rehabilitation** ✗ Wrong options: better 1-yr outcomes · better 10-yr survival · better alignment
"True of UKA vs TKA?"	→ **UKA more closely replicates normal kinematics** ✗ Wrong options: TKA higher reoperation · mobile bearings fail in one
"True of UKA?"	→ **lower PJI risk than TKA** ✗ Wrong options: mobile-bearing for ACL deficient · age <45 · overcorrect 2–3° · stress fracture commonest failure
"Potential advantage of UKA over TKA?"	→ **faster rehab + recovery** ✗ Wrong options: lower revision · less loosening · higher survivorship · less demanding
"Risk after UKA but not TKA?"	→ **osteoarthritis progression**
"After TKA (vs UKA) expect?"	→ **decreased risk of early periprosthetic fracture** ✗ Wrong options: reduced mortality · higher revision · lower PJI · shorter stay
"Most likely reason for revision of medial UKA?"	→ **progression of osteoarthritis**
"Most common cause of late (>10 yr) mobile-bearing UKA failure?"	→ **progression of OA**
"Most common cause of early UKA failure?"	→ **aseptic loosening**
"Most accurate statement on onlay PFA?"	→ **commonest long-term failure = OA progression to other compartments**

– – Active space – –

"Loose varus tibial UKA component, BMI 43, normal ESR + CRP?"	→ **revise UKA to TKA**
"Giving way, clicking, effusion 12 months after medial UKA?"	→ **bearing dislocation**
"Anteromedial pain after mobile-bearing UKA?"	→ **anterior impingement** (TO)
"Posterior bearing dislocation in deep flexion — technical error?"	→ **flexion gap 2 mm looser than extension** (TO)
"Bearing unstable mid-flexion, tight in extension?"	→ **inadequate posterior femoral resection** (TO)
"Pain 8 months after UKA, normal X-ray, uptake on bone scan?"	→ **stress fracture**
"Heavy patient, 4 months after UKA, bloody aspirate WBC 6,500?"	→ **stress fracture**
"Errors causing post-op tibial stress fracture EXCEPT?"	→ **excessive force impacting the tibial component**
"Not an indication for PFA?"	→ **BMI >30 in a 50-year-old** (TO)
"Commonest cause of PFA failure overall?"	→ **progression to tibiofemoral OA** (TO)

WEAR & OSTEOLYSIS BASIC SCIENCE

– Recon · 65 / 65 –

1 Basics

Overview

Osteolysis	**histiocytic (macrophage) response to wear debris** → bone loss · same biology as aseptic loosening, faster + focal
4 steps	(1) **particulate debris formation** → (2) **macrophage-activated osteoclastogenesis + osteolysis** → (3) **prosthesis micromotion** → (4) **particulate debris dissemination**
Aseptic loosening	✓ commonest cause of **late** THA failure · supra-acetabular osteolysis usually **>10 yr** after index surgery · ↓ since HXLPE
Burden (TO)	aseptic osteolysis = **most common cause of THR failure, ≈ 70% of revisions** · tribology = friction, lubrication + wear of surfaces in relative motion · septic vs aseptic osteolysis

Closest correlate of osteolysis → ✓ **linear wear rate** of the bearing couple · activity, cup position, PE oxidation, bearing all act by ↑ wear
✗ Wrong options: weight · head material · offset

Wear mechanisms (adhesion, abrasion, backside wear)

Mechanism	What happens	Key fact
Adhesive	Surfaces bond under load → weaker material (PE) pulled off	✓ **Most common / most important** in THA osteolysis (TO)
Abrasive (two-body)	Harder, rough surface cuts softer – **cheese grater**	Rough head scratches PE · ploughing, cutting, fragmentation
Third-body	Trapped particles between bearing surfaces	Type of abrasive wear · **PMMA** mainly (bone, metal beads, HA, Al oxide, $BaSO_4$, taper corrosion debris)
Fatigue / delamination	Cyclic subsurface stress → microcracks, pits → large particles	✓ **TKA** PE failure (TO) – max stress just **below** surface
Fretting	Small oscillation between surfaces not meant to move	**Backside wear** (shell–liner, tray–insert)
Corrosive	Sliding in corrosive / oxidative environment	–

Hip vs knee PE — hip: ✓ **adhesion + abrasion** at the surface · knee: **fatigue cracking + delamination**
✗ Wrong options: fretting · crevice corrosion (under screw heads) · oxidative for hip

Third-body wear – NOT true → ✓ **"particle size more important than hardness"** (TO) · hardness / composition, concentration matter more · implant hardness **<0.8×** particle → severe abrasion · **>1.2×** → resistant · vitamin E PE has better surface properties

– – Active space – –

Wear modes (McKellop) **1**: bearing surfaces as intended · **2**: bearing vs non-bearing surface · **3**: third body between bearing surfaces · **4**: two non-bearing surfaces

Wear stages primary (run-in) → secondary (steady state; most of life) → tertiary (rapid failure)

Determinants of wear → wear factor, **sliding distance**, **load** – ✓ **all independent determinants** (TO)

PE cup wear pattern sequence → ✓ foldings first (loading force) → **ripples** (loading-shear; adhesive; slowly worn cups) → **fibrils** (abrasive; rapidly worn cups) (TO)
✗ Wrong options: fibrils · ripples · ditches

Optimally positioned cup – wear site → ✓ anterosuperior quadrant (TO) · excess inclination ↑ edge wear · greatly anteverted cup → posterior (neck impingement in extension)

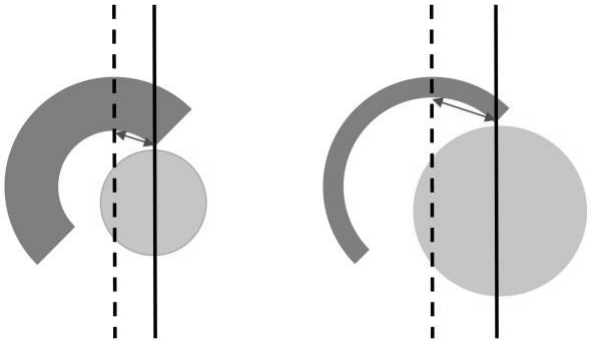
Linear vs volumetric wear

	Linear wear	Volumetric wear
Definition	**Head penetration into the liner** (thickness lost)	Total volume removed → ✓ **number of particles**
Measure	**Livermore** – shortest distance head centre → acetabular surface	**$V = \pi r^2 w$** ($\propto$ **radius²** of head)
Main factor	Orthobullets: independent of head size · TO: **22 mm > 32 mm** – read the options	✓ **Head size** · **total contact area** (TO)
Failure	Wear-through of PE cup	**Osteolysis**
Threshold	**>0.1 mm/yr** → osteolysis + loosening (rare below)	–

Head size **22 mm**: ↑ linear, ↓ volumetric, ↑ central wear + dislocation · **32 mm**: ↓ linear, ↑ volumetric, thinner PE, ↑ ROM · **28 mm** = compromise → most systems (TO)

Conventional PE 32 mm heads significantly more volumetric wear than 22 or 26/28 mm · **1 mm ↑ head size → ~10% ↑ volumetric wear**

HXLPE wear **independent of head size 22–46 mm** → 32 vs 28 mm = ✓ **equivalent wear** · head **>40 mm** no proven longevity gain
✗ Wrong options: ↑ dislocation · ↓ ROM · ↓ osteolysis · ↑ periprosthetic fracture



Jump distance rises with head size

Bearing wear rates

Bearing	Wear rate / particles
Conventional UHMWPE	**0.1–0.2 mm/yr** (✗ Wrong options: 0.01–0.02)
HXLPE	Lower · smaller particles · penetration **0.003 vs 0.051 mm/yr** (McCalden)
Ceramic-on-ceramic	✓ **Lowest** of any couple – **0.5–2.5 µm / component / yr**
Ceramic-on-PE	0–150 µm
Metal-on-metal	**2.5–5.0 µm/yr** · **nanometre** particles, smaller than MoP

Order **CoC < MoM < MoP** (TO)

TO lecture rates PE **0.1–0.5 mm/yr** = **90% of debris volume** (vs 0.1–0.2 mm/yr above – read the options) · metal **10–100× less** · ceramic **1000× less** · cement debris rare (old cemented hips) · **adhesive volumetric wear = prime cause of THR failure** · delamination commoner in **TKR**

Conflict OB3311 calls **0.02–0.04 mm/yr** "conventional THA" wear (so 0.03 mm/yr ≈ conventional) · Orthobullets/OB218172: conventional UHMWPE **0.1–0.2 mm/yr** – read the options

New liner at 0.03 mm/yr (RSA, 4 yr) → ✓ osteolysis + loosening minimal if rate continues
✗ Wrong options: greater than conventional · positioning irrelevant · significantly lower than conventional · better satisfaction

Risk factors for wear

Patient **age <50** · **male** · high activity · high BMI

Implant / surgical PE thickness **<6 mm** · head **>32 mm** · ↑ contact stress · **cup inclination >45°** · component malalignment
✗ Wrong options: >6 mm

TKA, PE gamma-sterilised in air ↑ **shelf age** · younger patient · male · **rough tibial baseplate**
✗ Wrong options: **PCL-retaining design** (not a risk)

Particle type & size

Commonest particle ✓ **UHMWPE** (up to **90%** of debris volume; colourless, mostly **spheroid**) · also PMMA, Al oxide, Zr oxide, CoCr, Ti, third body

Size implicated ✓ **<1 µm (submicron)**, **0.1–1 µm** · mean PE **0.5 µm**, metal **0.7 µm** (Maloney) · most stimulatory **0.2–7/8 µm** · most particles **<5 µm**, random shape (TO) · active **0.5–10 µm** (TO)
✗ Wrong options: 10, 100, 1000 µm

Size vs type **size more important than type** for biological activity (Orthobullets)

Severity of response ∝ → ✓ particle number (dose-dependent)
✗ Wrong options: size · composition · molecular weight

- - Active space - -

2 Mechanism

Osteolysis biology (macrophages, RANKL, cytokines, particle size)

Submicron wear particle

- Macrophage phagocytosis / toll-like receptor contact → ↑ **NF-κB**
  - TNF-α, IL-1, IL-6, IL-8, PGE2, MCP-1 → ↑ **RANK + RANKL**
- Osteoclast precursor + RANKL (from osteoblasts) → **OSTEOCLAST BONE RESORPTION**
  - Osteolysis → micromotion → more wear → **LOOSENING**

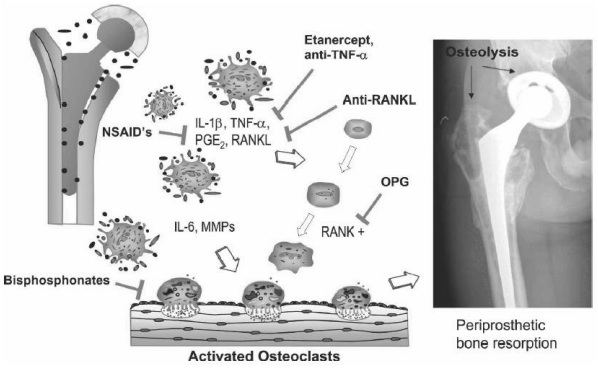
Bearing / process	Cell driving response
PE debris (MoP, CoP)	✓ **Macrophage (histiocyte)** – also fibroblasts, giant cells
Metal-on-metal	✓ **Lymphocyte (T cell)** – type IV hypersensitivity, ALVAL
Infection	Neutrophil
Eosinophilic granuloma	Langerhans-type histiocytes (not wear)

Promote osteolysis	Inhibit osteolysis
✓ **TNF-α** (key; from macrophages) · **IL-1** · **IL-6** · IL-8 · **PGE2** (COX-2) · PDGF · **RANKL** · **VEGF** · MCP-1 (recruits monocytes) · osteoclast activating factor · acid phosphatase · GM-CSF → ROS (H_2O_2, oxide radicals)	**OPG** · ✓ **interferon-γ** · **IL-4** · **IL-10** · calcitonin

Pro-osteolytic target	Inhibitor
RANKL	**Denosumab** (mAb) · OPG
TNF-α	**Etanercept** (decoy receptor)
COX-2 → PGE2 → IL-6	**Celecoxib** / NSAIDs
IL-6	**Tocilizumab** (mAb)
IL-1	**Anakinra** (receptor antagonist) (✗ Wrong options: "IL-10, anakinra") (IL-10 itself inhibits osteolysis)
Osteoclast	Bisphosphonates

RANK–RANKL–OPG RANKL made by **osteoblasts** (✗ Wrong options: osteoclasts) · RANK on **osteoclast precursors** · **OPG binds RANKL** (✗ Wrong options: RANK) · interleukins from **macrophages** (✗ Wrong options: osteoclasts) · RANK/RANKL transcription ↑ (✗ Wrong options: ↓) · TNF-α ↑ RANK

UHMWPE-induced osteolysis → ✓ **both VEGF and RANKL ↑** (VEGF enhances RANK/RANKL)



Cytokine cascade and inhibitors in periprosthetic osteolysis

Micromotion & dissemination

Micromotion — osteolysis → micromotion → ↑ particle wear + loosening · joint-fluid pressure → bone resorption

Effective joint space — anywhere joint fluid reaches bone · fluid + particles follow **pressure gradients / path of least resistance** · ↑ **hydrostatic pressure** (inflammation) spreads debris · above cup, **through screw holes**, down the stem · cementless cup → **expansile** lysis from implant–bone interface

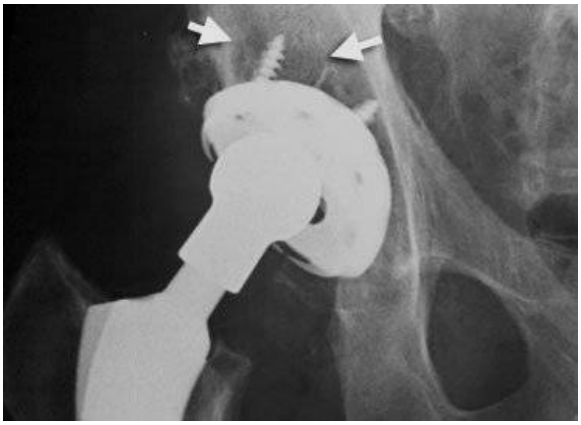
Effective joint space ↑ by (TO) — **non-circumferential porous coating** · **excess screw holes** in shell · poor cement seal / gaps → prevent with circumferential coating, few screws, perfect press-fit or sealed mantle (Harris concept)

Pelvic lysis around cup screws 10 yr after THA → ✓ PE particles tracking through the effective joint space

✗ Wrong options: osteosarcoma · galvanic corrosion · sepsis · occult fracture

Systemic — ↑ serum metal ions with MoM · **no clear link** to neoplastic, toxic or metabolic disease

✗ Wrong options: ↑ sarcoma



Osteolysis behind the cup via screw holes

Osseointegration & micromotion (TO)

Micromotion	Tissue formed
✓ **<20 µm**	Predominantly **bone** (ideal)
40–150 µm	Bone + fibrous tissue
>150 µm	**Fibrous tissue**

– – Active space – –

Osseointegration (Albrektsson 1981)	lamellar bone on implant without fibrous tissue · **4–12 wk**, continues up to **3 yr**
Engh signs of osseointegration	no radiolucent lines around porous surface · **spot welds** (new bone bridging endosteum → porous coating)

3 Presentation

Osteolysis	often **clinically silent** until pathological fracture · **start-up pain** with loosening · groin pain (cup) · thigh pain (stem)
Labs	aseptic: **ESR + CRP normal** · **urine N-telopeptide ↑**

4 Diagnosis & Classification

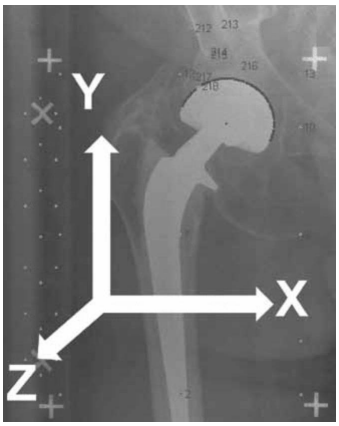
Genes & lab markers

Marker	Finding
✓ **Urinary N-telopeptide**	↑ – type I collagen breakdown = bone turnover (✗ Wrong options: ↓ in osteolysis)
Serum ALP, urine hydroxyproline, deoxypyridinoline	↑ in high turnover (Paget's)
ESR / CRP	**Normal** in aseptic osteolysis
Urine cAMP ↑	Pseudohypoparathyroidism type 2 (wrong option)
Urine phosphoethanolamine	Hypophosphatasia (wrong option)
Urine Bence Jones protein	Myeloma (wrong option)
Serum bone sialoprotein	Mineralised tissue component (wrong option)
✓ **Serum cobalt >1 ppb on two samples**	Diagnostic of **MACC-associated ALTR** (✗ Wrong options: >0.5 ppb) · Co ↑ more than Cr
MoM serum ions ↑	**Cup abduction >55°** · **smaller component**

Key radiology findings

Modality	Finding / role
Plain X-ray	Monitors progression · ✓ **underestimates** size, misses small lesions · femur: **endosteal scalloping** = hallmark (TO) · round lysis **behind cup / around screws**
Livermore method	Max head penetration (manual)
Computer-assisted	2-D radiographs → 3-D wear estimate
✓ **RSA**	**Most accurate + precise** wear / migration measure · **tantalum beads** in bone · paired radiographs → 3-D position
✓ **CT**	**Preferred for extent** of osteolysis · 3-D lesion + component position · surgical planning

Best study of a new bearing's wear → ✓ prospective trial with RSA
✗ Wrong options: serial X-rays · annual CT · edge-detection radiographs



Radiostereometric analysis

Loosening signs – cemented femoral stem (TO)

Sign	Note
Radiolucency stem–cement, **Gruen zone 1** (superolateral)	Debonding, early deformation
Radiolucency cement–bone	Loosening
Subsidence of stem ± mantle	Loosening
Stem shifts into **varus**	Loosening
Cement fragmentation, **zone 7** (superomedial)	Loosening
Cement fracture near tip, **zone 4**	Loosening
Stem **deformation / fracture**	Loosening
✗ Wrong option: **parallel radiodense line around stem**	**Not** loosening

Where cemented loosening starts (TO) femur: **stem–cement** interface · acetabulum: **bone–cement** interface · significant lucency **> 2 mm** or progressive · zones: **Gruen 1–7** (femur), **DeLee + Charnley I–III** (acetabulum)

Histology (TO) many **osteoclasts**, few osteoblasts, **PE-laden macrophages** · TNF-α, IL-1, IL-6, PGE2 → RANKL · **MSC → osteoblast differentiation suppressed**

Engh – cementless stem (TO)	X-ray
Bone ingrowth	Stable, no subsidence, **no radiolucent lines**
Stable fibrous ingrowth	**Parallel** radiolucent lines ± subsidence
Unstable	**Divergent** lines + **cortical hypertrophy at stem tip**

Monitoring tool (TO)	Use
RSA	**Gold standard** (research) · micromotion **≤ 1 mm** via **tantalum beads**
EBRA	Digital follow-up of migration on plain films
DEXA	Periprosthetic bone density over time

– – Active space – –

Barrack cementation grade (TO)

Grade	Finding
A	Complete fill – **white-out**, no radiolucency
B	Radiolucency **≤50%** of cement–bone interface
C	✓ Radiolucency **50–99%** or incomplete mantle
D	**100%** radiolucency and/or **no cement beyond stem tip**

Cementless stem loosening

	Osteolysis	Stress shielding (TO)
Cause	Inflammatory reaction to debris	**Altered load transfer**
Site	Periprosthetic zones	**Proximal (calcar / GT)**
X-ray	Lucency ± sclerotic rim	**Thinning of proximal cortex**

Subsidence + thick distal pedestal + no calcar stress shielding, 10 yr → ✓ femoral loosening (failed ingrowth)
✗ Wrong options: acetabular / femoral osteolysis alone · acetabular loosening (radiolucent lines + cup migration)

5 Management

Highly cross-linked polyethylene

UHMWPE resin (MW 2–6 million)
- Shaped → **COMPRESSION MOULDING / RAM EXTRUSION / ISOSTATIC PRESSING**
- Gamma or e-beam 5–10 Mrad, no oxygen → **CROSS-LINKS + FREE RADICALS**
  - Free radicals removed → **REMELT / ANNEAL / VITAMIN E**
- Terminal sterilisation → **GAMMA, E-BEAM, ETHYLENE OXIDE, GAS PLASMA**

Property	HXLPE vs conventional
Wear resistance (adhesive + abrasive)	✓ ↑ (TKA in vitro: wear ↓ **30–80%**)
Particle size	**Smaller** (more bioreactive)
Particle number	Fewer (process dependent)
Tensile strength	↓ (✗ Wrong options: ↑)
Fatigue strength / crack-propagation resistance	↓
Fracture toughness	↓
Ductility / elongation to break	↓
10-yr outcome	✓ **↓ wear + ↑ survival** (✗ Wrong options: no difference)
Head size effect	None, **22–46 mm**

– – Active space – –

PE generation (TO)	Process	Note
1st (1990s)	Gamma cross-linking + **annealing / remelting**	↓ wear · ↓ toughness
2nd (2000s)	**Sequential annealing** / **vitamin E**	↑ oxidation resistance, better strength · expensive
3rd (2010s)	Low-O2 processing, solid-state deformation	Wear **~0.02 mm/yr** · long-term data pending (TO – verify: compression moulding / ram extrusion are consolidation methods for all UHMWPE, not a generation)

Other materials (TO) **PEEK** under evaluation · **polyacetal + PTFE abandoned** (poor wear) · prevention: **3rd/4th-generation cementing**, PE **≥ 6 mm**, head **22–46 mm** (> 46 mm → thinner PE → ↑ wear), ceramic or MoP (avoid MoM) · drugs: **bisphosphonates**, **denosumab** (investigational), BMP-2

Toughness affects mechanical strength, **not** wear rate

Conventional PE – the false statement → "conventional generates smaller particles" is false (conventional = **larger**) · true: higher head penetration, more adhesive wear, ↑ toughness, ↑ tensile strength

Free-radical step	Effect
Remelting (above **135 °C**)	Radicals **virtually eliminated** · **↓ crystallinity** → ↓ mechanical properties
Annealing (just below melt)	Less effective on radicals · crystallinity preserved
Vitamin E (α-tocopherol) soak / blend – 2nd gen	Scavenges radicals without remelting · ↑ oxidation + delamination resistance
Sequential irradiation + annealing ×3 – 2nd gen	↓ radicals, good mechanics
✗ Wrong option: **ethylene oxide / gas plasma**	Terminal **sterilisation** only
✗ Wrong option: **electron beam / gamma irradiation**	**Creates** cross-links + free radicals

HXLPE insert – false → ✓ **"sterilized by ether"** (TO) (✓ true: compression moulding, vitamin E, less delamination)

Gamma in air (historic) → oxidation → early failure

Young active THA patient asks about longevity → ✓ **HXLPE liners successful at 15 yr** in her age group
✗ Wrong options: dual mobility for longevity (it is for stability) · 40 mm head · liner <3 mm (use **≥5–6 mm**) · vertical cup

– – Active space – –

Stem design, stress shielding & fatigue failure (TO)

Distal femoral osteolysis prevention → ✓ **proximal circumferential porous coating** (proximal **33–40%** of stem; seals interface)
✗ Wrong options: long stem · collared · fully cemented · diaphyseal fill · extended offset

Coating	Osteolysis site
Fully (extensively) coated	✓ **Proximal** (often no loosening)
Proximal circumferential	**Proximal**
Patch-coated	**Distal** – fluid reaches diaphysis

Stress shielding — stiff stem carries load → proximal bone resorption (Wolff) · ✓ **cemented < uncemented** · **fully (HA) coated > proximally coated** · extensively coated: **25–30%** show shielding, calcar rounding, distal **spot welds**
✗ Wrong options: fully coated ↓ shielding

Stem material to ↓ stress shielding → ✓ **titanium** (low modulus, highest fatigue strength) · CoCrMo **~10×** stiffer than femur · alumina **~19×** · flexible, **short, hollow** stems ↓ shielding

Bending cantilever fatigue → stem fracture → ✓ **proximal loosening of a cemented stem** with distal stem fixed (TO)

Cemented component loosening (TO)

Where loosening starts — cemented **cup → cement–bone** · cemented **stem → cement–prosthesis** (mechanical) · site unspecified → ✓ **any site** (TO)

Stem technique errors — soft cancellous bone left on medial neck · thin / incomplete mantle · **removing all trabecular bone** · too little cement · no pressurisation · **stem motion while curing** · non-neutral stem · voids
✗ Wrong options: **H_2O_2 canal preparation** (not a cause)

Cup technique errors — poor superior/posterior support · cartilage / soft tissue left, too few anchor holes · no pressurisation · cement not around entire cup · movement while curing · **undersized cup** in a large mantle · **malposition** (horizontal, not anteverted)
✗ Wrong options: **serial reaming + removing all cartilage** (correct technique)

Causes of aseptic loosening (THR) — wear debris (major) · lack of primary osseointegration · insufficient press-fit
✗ Wrong options: **decreased BMD** (TO)

Clinical osteolysis & polyethylene failure

Aseptic loosening TKA – ESR 3, CRP 0.1, aspirate 157 WBC, 18% PMN → ✓ **one-stage revision**
✗ Wrong options: DAIR + PE exchange · spacer · two-stage · antibiotics

Osteolysis — revise if **painful** · asymptomatic → monitor

TKA 16 yr, distal femoral lysis → ✓ **macrophage activation by PE particles**
✗ Wrong options: PE toughness · more with HXLPE · infection · immunosuppression

TKA 18 yr, no joint space, metal debris, well-fixed → ✓ **polyethylene failure** (metal-on-metal articulation)
✗ Wrong options: loose components · particle osteolysis

Knee backside wear (TO)

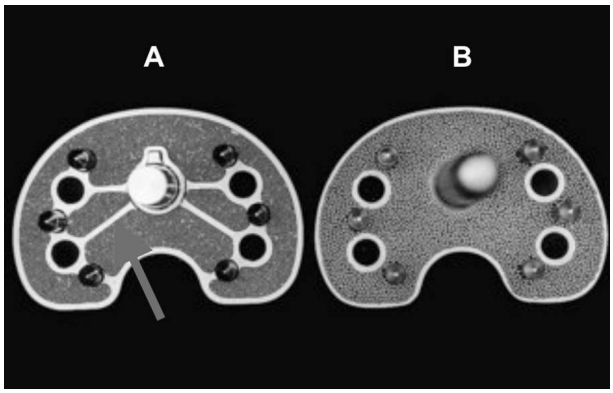
Backside wear — micromotion tray ↔ insert → osteolysis **under the tibial tray + around screw holes**

Cemented TKA, lysis under baseplate, no articular wear, no stem → ✓ backside fretting PE debris
✗ Wrong options: stress shielding (no stem) · articular PE or metal debris · cement debris

↓ Backside wear — **polished tray** · **robust locking mechanism** (most important — no micromotion) · **monoblock** · not **rough tray** (↑ wear) · obesity ↑ · modularity ↑

Least backside wear, same insert → ✓ cruciate-retaining (flat, less conforming, no cam-post) · most → **highly constrained** · PS intermediate

Cementless tibia with patch porous coating + smooth tracks → ✓ ↑ metaphyseal osteolysis · continuous porous coating seals metaphysis
✗ Wrong options: early loosening · hypersensitivity · fracture · instability



Patch (A) vs continuous (B) porous coating of tibial tray

6 Complications

Metal-on-metal debris & lymphocyte response

MoM particles — **nanometre**, ✓ **smaller than MoXLPE** · ↓ volumetric wear vs MoP · **↑ serum ions** · wear ↓ after **year 1** (steady state) · volumetric wear > metal-on-ceramic · no ↑ sarcoma
✗ Wrong options: lower than MoXLPE

Response — ✓ **lymphocyte (T cell)** · particles + native proteins → **haptens → type IV delayed hypersensitivity** · perivascular lymphocytes + **ALVAL** (aseptic lymphocyte-dominated vasculitis-associated lesion) = pseudotumour histology

Lymphocyte-driven soft-tissue reaction, well-positioned THA → ✓ metal-on-metal
✗ Wrong options: MoP, CoP (macrophage) · CoC (lowest wear)

MoM failure — **2–3×** higher than MoP · **4–5%** aseptic loosening at 6 yr

↑ serum metal ions (MoM resurfacing) → ✓ smaller implant diameter · cup abduction >55° (edge loading)
✗ Wrong options: smaller abduction · anteversion 10–20° · function scores · preop OA severity

MOMHRA pseudotumour mechanism → ✓ edge loading + loss of fluid-film lubrication (malpositioned cup) · incidence **<1%**, usually asymptomatic
✗ Wrong options: lymphocyte reactivity to nickel (no difference) · third body · P. acnes

ALTR fluid — watery, **yellow / grey**, **basic pH**, **low cell count**

– – Active space – –

Titanium as bearing surface — high failure: poor wear resistance + **notch sensitivity**

Taper corrosion (MACC) & trunnionosis (TO)

Trunnionosis — wear at the femoral **head–neck** interface · risk: **modularity**, **large CoCr heads** (↑ torque), **taper design**, mixed alloys, MoM · **small head** is not a risk

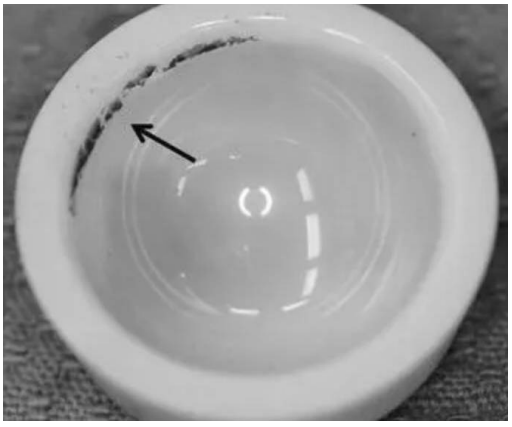
MACC — mechanical abrasion + chemical corrosion at head–neck → ions + debris → **ALTR** · ✓ **also in MoP THR** (≥1 CoCr component) · **Co ↑ > Cr** · **Co >1 ppb ×2** diagnostic · **MRI** for pseudotumour
✗ Wrong options: only MoM · Co >0.5 ppb · more with small heads

Ceramic damage (stripe wear)

Stripe wear — CoC · ✓ **lift-off separation** of head (edge loading in gait / deep flexion)
✗ Wrong options: "unique to metal-on-poly"
✗ Wrong options: third body · neck–socket impingement · insertion · infection

"Lead pencil" metal marks on ceramic head — → ✓ **recurrent dislocations** (metal transferred from shell) → ↑ head roughness + PE wear
✗ Wrong options: infection · LLD · ceramic fracture · ALVAL

CoC debris — results from malposition / instability rather than causing failure



Stripe wear on ceramic head

X vs Y

	PE osteolysis	**ALVAL / pseudotumour**	**Stress shielding**
Bearing	MoP, CoP	MoM, MACC (CoCr head)	Stiff stem
Cell	**Macrophage**	**Lymphocyte**	– (Wolff's law)
Particle	Submicron PE	Nanometre metal + ions	–
Labs	ESR/CRP normal · NTx ↑	**Co, Cr ↑**	–
Imaging	Expansile lysis, scalloping · CT	MRI	Proximal bone loss, calcar rounding

	22 mm head	28 mm head	32 mm head
Linear wear	**Highest**	Intermediate	Lowest
Volumetric wear	Lowest	Intermediate	**Highest**
Failure mode	Wear-through	–	Osteolysis
Stability / ROM	Lowest	Better	Best

	Hip PE	Knee PE
Wear mechanism	**Adhesion + abrasion** (surface)	**Fatigue + delamination** (subsurface)
Backside wear	Shell–liner	Tray–insert → lysis under tray

How it's asked

"VEGF and RANKL in UHMWPE osteolysis?"	→ **both increased**
"Pelvic lysis at cup screws 10 yr after THA?"	→ **PE particles tracking through the effective joint space**
"Particle size implicated in osteolysis?"	→ **<1 µm (submicron)**
"Distal femoral lysis 16 yr after TKA – pathogenesis?"	→ **macrophage activation by PE particles**
"OPG binds to?"	→ **RANK-ligand** ✗ Wrong options: RANK
"Marker of type I collagen breakdown / bone turnover?"	→ **increased urinary N-telopeptide**
"Molecule associated with macrophage-induced osteolysis?"	→ **IL-1** ✗ Wrong options: BMP-7 · IL-10 · SOX-9 · OPG
"Promoter + inhibitor pairs – EXCEPT?"	→ **IL-10, anakinra**
"Severity of response to PE debris proportional to?"	→ **particle number**
"Supra-acetabular lysis 20 yr after THA – process?"	→ **macrophage activation by particulate debris** ✗ Wrong options: lymphocytes (MoM) · trunnion wear · stress shielding
"Osteolysis 10 yr after THA – true?"	→ **RANK activity increased** ✗ Wrong options: X-ray accurate · NTx ↓ · TNF-α ↓ · ESR/CRP ↑
"Osteolysis 16 yr after THA – true?"	→ **TNF-α released by macrophages**
"Cytokine that inhibits osteolysis?"	→ **interferon-γ**
"Well-fixed THA with peri-acetabular lysis – cause?"	→ **macrophage activation by 0.1–1 µm particles**
"HXLPE vs standard PE – NOT true?"	→ **increased tensile strength** (TO)
"Free radicals in HXLPE reduced by all EXCEPT?"	→ **ethylene oxide treatment** (TO)
"Gold standard for detecting implant micromotion?"	→ **radiostereometric analysis (RSA)** (TO)

– – Active space – –

"Divergent radiolucent lines + cortical hypertrophy at stem tip (Engh)?"	→ **unstable cementless stem** (TO)
"Not a process to reduce free radicals?"	→ **electron beam irradiation** (TO)
"HXLPE TKA insert — EXCEPT?"	→ **sterilized by ether** (TO)
"HXLPE liner vs conventional?"	→ **increased resistance to abrasive wear / improved wear resistance** ✗ Wrong options: larger particles · ↑ ductility · ↑ crack resistance
"Conventional poly — incorrect statement?"	→ **generates smaller wear particles**
"32 vs 28 mm head on HXLPE?"	→ **equivalent wear rate**
"Other property change with cross-linking (TKA)?"	→ **diminished fatigue strength**
"True of THA bearings?"	→ **XLPE ↓ wear + ↑ 10-yr survival vs UHMWPE**
"Young active THA patient — longevity statement?"	→ **HXLPE successful at 15 yr**
"Not true about PE wear in THA?"	→ **32 mm head has less volumetric wear than 22 mm** (TO)
"Not a determinant of wear?"	→ **all are independent determinants** (TO)
"Commonest PE failure mechanism in TKR?"	→ **fatigue wear** (TO)
"True about bearing wear in THA?"	→ **CoC has the lowest wear rate** (TO) ✗ Wrong options: stripe wear in MoP · abrasive wear = head penetration
"Initial wear pattern of acetabular cup?"	→ **foldings** (TO)
"Wear quadrant of a well-positioned cup?"	→ **anterosuperior** (TO)
"22 mm vs 32 mm head?"	→ **higher linear wear rate** (TO)
"Third-body wear — NOT true?"	→ **particle size more important than hardness** (TO)
"Factor most responsible for volumetric wear?"	→ **total contact area** (TO)
"Reduces osteolysis around distal femoral stem?"	→ **proximal circumferentially coated ingrowth stem**
"Wear-related TKA failure risks (gamma in air) EXCEPT?"	→ **posterior cruciate-retaining design**
"Primary PE wear mechanism in the hip?"	→ **adhesion and abrasion**
"Osteolysis most closely correlated with?"	→ **linear wear rate**
"Lysis under cemented tibial baseplate, no articular wear?"	→ **backside fretting PE debris**
"Patch porous-coated tibial tray risk?"	→ **metaphyseal osteolysis**
"Cell activated by MoM debris?"	→ **lymphocyte**
"MoM bearings — true?"	→ **smaller particles than metal-on-XLPE**

"↑ serum ions after MoM resurfacing?"	→ **smaller implant diameter**
"MOMHRA pseudotumour mechanism?"	→ **edge loading + loss of fluid-film lubrication**
"Lymphocyte-driven soft-tissue reaction, well-positioned THA?"	→ **metal on metal**
"Stripe on retrieved ceramic head?"	→ **lift-off separation during ROM**
"Pencil-lead marks on ceramic head?"	→ **recurrent dislocations**
"Most accurate study of a new bearing's wear?"	→ **prospective trial with radiostereometric analysis**
"Wear 0.03 mm/yr — implication?"	→ **minimal osteolysis + loosening if rate continues**
"Causes of aseptic loosening of THR EXCEPT?"	→ **decreased bone-mineral density** (TO)
"Loosening of cemented prosthesis begins at?"	→ **any site** (TO) (cup → cement–bone · stem → cement–prosthesis)
"Not suggestive of cemented stem loosening?"	→ **parallel radiodense line around stem** (TO)
"Not associated with cemented stem loosening?"	→ **canal preparation with hydrogen peroxide** (TO)
"Won't increase cemented cup loosening?"	→ **serial reaming + removing all cartilage** (TO)
"Barrack grade C?"	→ **radiolucency >50% (50–99%) of cement–bone interface** (TO)
"Aseptic TKA loosening, benign aspirate — next?"	→ **one-stage revision**
"Cell implicated in 18-yr THA osteolysis?"	→ **macrophage**
"TKA 18 yr, lost joint space + metal debris?"	→ **polyethylene failure**
"Thigh pain 10 yr after THA, subsidence + pedestal?"	→ **femoral loosening**
"Best design feature against distal stem osteolysis?"	→ **circumferential porous coating** (TO)
"True statement on stress shielding?"	→ **cemented stems less than uncemented** (TO)
"Osteolysis site with fully porous-coated stem?"	→ **around proximal stem** (TO)
"Stem material to reduce stress shielding?"	→ **titanium** (TO)
"How to avoid distal femoral osteolysis after THR?"	→ **proximally circumferentially coated stem** (TO)
"Ideal micromotion for osseointegration?"	→ **<20 µm** (TO)
"Bending cantilever fatigue + stem failure caused by?"	→ **proximal loosening of a cemented stem** (TO)
"Does NOT reduce TKA backside wear?"	→ **rough tibial tray surface** (TO)
"Least backside wear with same insert?"	→ **cruciate-retaining design** (TO)

– – Active space – –

"Not a risk factor for trunnionosis?"	→ **small head** (TO)
"True about MACC?"	→ **can occur in metal-on-polyethylene THR** (TO)